The Haldane–Rezayi zero modes are not a gapped ground-state manifold on the torus

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Abstract

We consider spin-$\frac{1}{2}$ electrons in the lowest Landau level of a rectangular torus pierced by $2N$ flux quanta, interacting through the hollow-core interaction (the pure V_1 pseudopotential acting on all pairs regardless of spin). In the sector with $N/2$ electrons of each spin the zero modes of this Hamiltonian form a ten-dimensional space, spanned by the Haldane–Rezayi states on the torus; a proof of this fact is included. We prove that the eleventh eigenvalue of the sector tends to zero as $N \to \infty$, at every fixed aspect ratio and uniformly over the rectangular tori whose shorter side exceeds a fixed length: the zero modes are not separated from the rest of the spectrum by a gap that is uniform in N. More generally, a patch of p eigenvalues at the bottom of the sector whose width is less than half of its gap to the rest, uniformly in N, has p divisible by four. The proof is a lowest-Landau-level version of the fractional index theorem of Bachmann, Bols, De Roeck and Fraas, applied to the magnetic translation by one orbital, with the gap assumed only inside the sector. Its main new ingredient is the torus analogue of the windowed-Fourier frame embedding of [OAI26], an isometric embedding of the lowest Landau level into a two-dimensional lattice fermion system in which the hollow-core Hamiltonian is a rapidly decaying interaction of uniform size and a Lieb–Robinson-type propagation bound with a polynomial light cone holds with constants independent of N, and in which every Landau-gauge orbital is exactly supported in a column of lattice sites. The result does not exclude a gapped phase whose ground-state space strictly contains the zero modes, and it is asymptotic: the constants are not explicit. At fixed circumference L_y, a regime complementary to the one above, a variational argument with spin twists, also given, bounds the eleventh eigenvalue by $C(L_y)/N$ for every even N, with $C(L_y)$ explicit.

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1 Introduction

The model. Consider N electrons of spin one half in the lowest Landau level of a flat torus pierced by $N_\phi = 2N$ flux quanta, so that the filling factor is one half, and let them interact through the *hollow-core* interaction: the two-body interaction which, for a pair of electrons in the plane, is the orthogonal projection onto the states of relative angular momentum one. In the language of pseudopotentials [Hal83] it is the interaction with $V_1 = 1$ and $V_m = 0$ for $m \neq 1$. A pair with odd relative angular momentum has an antisymmetric orbital wave function and therefore a symmetric spin wave function, so the interaction acts only on spin-triplet pairs; spin-singlet pairs, whose relative angular momentum is even, cost no energy even at contact, whence the name. The Hamiltonian H_N of this paper (Definition 2.7) is the periodization of this interaction on the torus, in second quantization on the lowest-Landau-level Fock space. It conserves the numbers $N_\uparrow$ and $N_\downarrow$ of electrons of each spin, commutes with spin rotations and with the magnetic translations of the torus, and is nonnegative. Energies are measured in units in which a planar pair of relative angular momentum one has energy one (Lemma 2.8); statements about gaps that are uniform in N refer to these units.

Haldane and Rezayi [HR88] proposed a spin-singlet incompressible state at filling one half; it has zero energy for this interaction [RR96, GFN97]. On the torus the zero modes form a degenerate space. At even N, in the sector $N_\uparrow = N_\downarrow = N/2$, the kernel of H_N is ten-dimensional at every aspect ratio, and at odd N each of the sectors $N_\uparrow - N_\downarrow = \pm 1$ contains exactly two zero modes. Read and Rezayi [RR96] wrote down the ten states (the count of ten was first published by Keski-Vakkuri and Wen [KVW93]), and their completeness is proved in Appendix C; see Theorem 2.10. Whether these zero modes are the ground states of a gapped phase has been debated. The count of ten was read as the topological ground-state degeneracy of a non-Abelian phase [KVW93, GFN97, Ino99]; in the description of the Haldane–Rezayi state by a non-unitary conformal field theory of central charge $c = -2$, the ten states were attributed to a logarithmic partner of the identity field [GFN97]. Read and Green [RG00] argued instead that the Haldane–Rezayi state is the critical point of a transition between weak- and strong-pairing phases of spin-singlet d-wave pairing, with gapless bulk excitations, and that both neighbouring gapped phases have eight states on the torus; general arguments that trial states built from non-unitary conformal field theories do not describe gapped phases are given in [Rea09a, Rea09b], and numerical studies of small systems [CRE19, NPBG23] are consistent with a vanishing gap. To our knowledge the question has not been settled rigorously.

The result. We prove that the zero modes are not separated from the rest of the spectrum by a gap that is uniform in the number of electrons.

Theorem (informal version of Theorem 8.1)**.** *Fix $\lambda > 0$, and let H_N be the hollow-core Hamiltonian on the rectangular torus with side lengths L_x, L_y, $L_x/L_y = \lambda$, pierced by $N_\phi = 2N$ flux quanta. For even N let $E_1(N) \le E_2(N) \le \cdots$ be the eigenvalues of H_N in the sector $N_\uparrow = N_\downarrow = N/2$, repeated according to multiplicity. Then $E_1(N) = \cdots = E_{10}(N) = 0$, and $E_{11}(N) \to 0$ as $N \to \infty$ through the even integers.*

In other words, there is no $\gamma > 0$ such that for infinitely many N the ten-dimensional space of zero modes is separated by γ from the rest of the spectrum in this sector. The theorem follows from a stronger statement (Theorem 8.1(i)): if for infinitely many even N the p lowest eigenvalues of the sector lie in an interval of length Δ and are separated from the rest of the sector by a gap $\gamma > 2\Delta$, with p, Δ and γ independent of N, then p is divisible by four. We call such eigenvalues a *uniformly gapped patch* of dimension p at the bottom of the sector: along an infinite set of N, the width of the patch is less than half of its gap to the rest of the sector, uniformly in N. The same condition for a general Hamiltonian and general sectors is Assumption 6.3, and the term is used only in this sense; every result stated below for such patches holds also when the width is merely less than the gap (Remark 6.20). Since ten is not divisible by four, the eleventh eigenvalue cannot stay away from zero. At odd N the same argument gives $E_3(N) \to 0$ in each of the sectors $N_\uparrow - N_\downarrow = \pm 1$, which contain two zero modes each (Corollary 8.2).

The index theorem of Bachmann, Bols, De Roeck and Fraas. Our tool is the index theorem of [BBDF21] for gapped quantum lattice systems with a conserved charge. Suppose that a charge-conserving Hamiltonian of finite range on a lattice has a patch of p eigenvalues at the bottom of its spectrum, separated from the rest by a gap, and that a locality-preserving, charge-conserving unitary U leaves the corresponding spectral subspace invariant. Then the charge transported by U across a cut, averaged over the patch, lies in $p^{-1}\mathbb{Z}$ up to errors that vanish faster than any power of the size of the system [BBDF21, Theorem 3.2]. When U is a lattice translation the transported charge is the average charge of a slab, and the theorem becomes a fractional Lieb–Schultz–Mattis theorem [BBDF21, Corollary 5.2], extending [LSM61, Osh00, Has04, NS07].

In our setting U is the magnetic translation Θ by one Landau-gauge orbital, which commutes with H_N, $N_\uparrow$ and $N_\downarrow$; the charge is the number of up-spin electrons in a band of N consecutive orbitals; and one application of Θ moves exactly one orbital's worth of up-spin charge across each of the two boundaries of the band (Lemma 7.1). By translation invariance every orbital carries, on average over a spectral patch of the sector $(N/2, N/2)$, the up-spin charge $(N/2)/N_\phi = 1/4$. The index theorem therefore requires that $p/4$ be an integer up to $O(N^{-\infty})$, hence exactly, since p does not depend on N. For the ten zero modes, $10/4 = 5/2$ is a half-integer.

Why the index theorem does not apply as written. Two features of the lowest Landau level stand in the way.

The lowest Landau level is not a lattice. The theorem of [BBDF21] is formulated on a finite graph with a finite-dimensional space at each vertex, for Hamiltonians that are sums of uniformly bounded terms of bounded range, and the locality of operators is measured in the graph. The natural basis of the lowest Landau level of the torus, the Landau-gauge orbitals ψ_j, $j \in \mathbb{Z}_{N_\phi}$, is localized in one direction only: ψ_j is a Gaussian of width one magnetic length about the line $x = 2\pi j/L_y$, extended along it. In this basis H_N is a ring of N_ϕ orbitals in which pairs of orbitals at index distance up to order L_y interact (Theorem 4.2(c)), and the velocity given by the standard

Lieb–Robinson bound for this chain, measured in orbitals per unit time, grows with L_y; a heuristic count, which we do not make precise, puts it of order L_y^2. (In the form of Theorem 4.2(c), the pair weights $2d\,e^{-\kappa^2 d^2/4}$ of the pairs that contain a given orbital sum to order κ^{-2}, the pair operators b_n^ν have norms at most of order κ^{-2}, and the prefactor is κ^3; so the part of H_N that does not commute with an even operator at one orbital has norm at most of order κ^{-1}, and its range is of order κ^{-1} orbitals, which gives a velocity of order $\kappa^{-2} \propto L_y^2$.) The two boundaries of the band in the argument of [BBDF21] are N orbitals apart, and at fixed aspect ratio N is of order L_y^2, so the time after which this bound no longer separates the two boundaries does not grow with N. The argument of [BBDF21], however, filters the dynamics over all times, and it needs the filtered operators to stay localized near the boundaries (Section 6).

The gap can only be assumed in a sector. The Fock space of the lowest Landau level contains every particle number, and at densities below one half the hollow-core Hamiltonian has large kernels; every state with at most one electron, for instance, is a zero mode. The theorem of [BBDF21] assumes a gap above the bottom of the spectrum on the whole Hilbert space. For H_N this assumption fails, and a gap can be assumed at most inside the sector $N_\uparrow = N_\downarrow = N/2$.

The lowest Landau level as a lattice. The main new ingredient of this paper is a way to regard the lowest Landau level of the torus as a two-dimensional lattice system in which the hollow-core Hamiltonian is a short-range interaction of uniform size. We construct in Section 3 vectors z_s of the lowest Landau level, indexed by the points s of a rectangular lattice Λ_N of *slots* whose spacings are of the order of the magnetic length, with the following properties.

(i) They form a tight (Parseval) frame: $\sum_s |z_s\rangle\langle z_s|$ is the identity on the lowest Landau level, exactly, for every N (Proposition 3.3).

(ii) Each z_s is localized near the centre r_s of its slot, with Gaussian decay across the orbitals and decay faster than any power along them, with constants independent of N (Lemma 3.5).

(iii) Each orbital ψ_j is a linear combination of the z_s in a single column of slots, two lattice spacings wide (Lemma 3.6).

The z_s are windowed Fourier transforms of the orbitals: smooth windows in the orbital index, about one magnetic length wide, multiplied by Fourier modes along the circumference. The lattice has about twice as many slots as there are orbitals. A redundant frame rather than an orthonormal basis is natural here: the lowest Landau level is a band of nonzero Chern number, and for such bands no orthonormal basis made of magnetic translates of finitely many functions (a system of composite Wannier functions) has finite second moments of the position [MPPT18]. By (i) the analysis map $\mathcal{S}f = (\langle z_s, f\rangle)_{s\in\Lambda_N}$ is an isometry, and $c(f) \mapsto \hat{c}(\mathcal{S}f)$ extends to an isometric embedding ι of the algebra of operators on the lowest-Landau-level Fock space into the algebra of a lattice fermion system on Λ_N with one spin-up and one spin-down mode per slot (Lemma 3.8). The physical Fock space is the subspace of the lattice Fock space in which the modes orthogonal to the range of $\mathcal{S}$ are empty, and ι intertwines the dynamics and the gauge transformations generated by $N_\uparrow$ and $N_\downarrow$.

In this lattice the hollow-core Hamiltonian is local. We show that $H_N = \int_{\mathbb{T}_N} h_R \,\mathrm{d}R$ exactly, where $h_R = 8\pi \sum_\nu B_\nu(R)^* B_\nu(R)$ and $B_\nu(R)$ annihilates a spin-triplet pair at the point R with relative angular momentum one (Theorem 4.2); and that $\iota(h_R)$ is approximated, within $C_k(1+r)^{-k}$ for every k, by operators acting only on the slots within distance r of R, with $\|h_R\|$ and C_k independent of N (Proposition 4.4). Lieb–Robinson bounds [LR72, NS06] hold for lattice fermions

with even interactions [NSY18]. We prove that the dynamics generated by $\iota(H_N)$ carries an operator supported in a set X of slots to within distance ρ of X, up to an error $C_k\|A\||X|\rho^{-k}$, whenever $\rho \geq C_{\mathrm{lc}}(1+|t|)^{\alpha}$, for any fixed $\alpha > 1$, with constants independent of N (Theorem 5.3). The two boundaries of the band are at distance of order $\sqrt{N}$ magnetic lengths, so the time up to which this bound keeps them apart now grows with N, and the filtered operators of the argument of [BBDF21] stay localized near the boundaries (Lemma 6.8, Remark 6.9). Property (iii) makes the band charges exactly supported in strips of slots, and conjugation by a band charge preserves strips exactly (Lemma 4.11); every distance used in the argument is a distance between strips.

We regard this embedding as a result of independent interest. It turns lowest-Landau-level Hamiltonians that are integrals of local terms into two-dimensional lattice fermion Hamiltonians to which the locality theory of lattice systems applies with constants independent of the size of the system. The device itself is not new: on the sphere, [OAI26, Section 3] builds a tight frame of the lowest Landau level from windowed Fourier modes of the orbitals, embeds the physical fermions by its analysis map into the Fock space of a larger set of fermionic modes, and proves there propagation and time-filter estimates with constants independent of the flux; it attributes the use of localized frames for the locality of fermions in the continuum to [BDN24]. The frame constructed here is the analogue on the torus. What the index argument needs from it beyond locality is that each Landau-gauge orbital is exactly supported in a column of slots, so that the band charges are exactly supported in strips and conjugation by them preserves strips.

The gap in a sector. The assumption of a gap on the whole Hilbert space enters the proof of [BBDF21] in two places: in the identity $[K, P] = [Q, P]$ for the generator K that dresses the charge Q, and in the clustering of the spectral patch P [BBDF21, Proposition 3.1]. We show that both remain true when the gap is assumed only inside a sector, provided they are applied to operators that conserve $N_\uparrow$ and $N_\downarrow$ (Lemma 6.7 and Proposition 6.10), and every operator to which the argument applies them does. The gap is thus assumed only where it can hold.

The general statement. The argument is not specific to the hollow-core interaction. Theorem 6.18 is an index theorem for any sequence of lowest-Landau-level Hamiltonians on rectangular tori which are integrals of terms that conserve $N_\uparrow$ and $N_\downarrow$, are uniformly bounded, and are rapidly local in the slot lattice (Definition 4.6); the hollow-core Hamiltonian belongs to this class (Corollary 4.8). For members of the class that are covariant under the magnetic translation it yields a constraint of Lieb–Schultz–Mattis type (Corollaries 7.2 and 7.3): in the sector $N_\uparrow = N_\downarrow = N/2$, a uniformly gapped patch at the bottom of the spectrum, in the sense defined above, has dimension divisible by four. In particular no Hamiltonian of the class has in this sector a uniformly gapped ground-state space of dimension ten. The proof of this constraint, the proof of Theorem 8.1, and Theorem 2.10 on the zero modes it uses have been formalized and verified in Lean 4 (Appendix E).

What is excluded, and what is not. The theorem excludes a gapped phase whose ground-state space is the space of Haldane–Rezayi zero modes, at any fixed aspect ratio of a rectangular torus, and uniformly over the aspect ratios of rectangular tori whose shorter side exceeds a fixed length (Corollary 8.4). This is the scenario in which the degeneracy ten of the zero modes would be the topological ground-state degeneracy of a gapped Haldane–Rezayi phase, as in [KVW93, GFN97, Ino99]. A uniformly gapped patch at the bottom of the sector, in the sense defined above, contains the ten zero modes, so its dimension is at least ten and, being divisible by four, at least twelve; in particular the eight-dimensional ground-state spaces of the gapped phases adjacent to the Haldane–Rezayi point in the picture of [RG00] are not ground-state spaces

of H_N either. The theorem does *not* exclude a gapped phase whose ground-state space strictly contains the zero modes and has dimension 12, 16, ...: in such a phase $E_{11}(N) \to 0$ would be a splitting inside the ground-state space, not the closing of a gap above it. The index theorem cannot exclude this, because the conclusions it yields here are divisibility conditions (Remark 8.7). To our knowledge no such phase has been proposed, and the expectation since [RG00] is that the hollow-core Hamiltonian has no gap at all; that stronger statement is not proved here. The result is also purely asymptotic. The constants in our estimates exist but are not explicit, and through the light cone of the propagation bound of Theorem 5.3 they are very large; the proof gives neither a rate for $E_{11}(N) \to 0$ nor a value of N beyond which $E_{11}(N)$ is below a prescribed number. At fixed circumference a separate, variational argument with spin twists bounds $E_{11}(N)$ by $C(L_y)/N$ for every even N and every circumference L_y, with $C(L_y)$ explicit (Theorem D.1); this complements the result above and, together with it, gives $E_{11}(N) \to 0$ uniformly over all rectangular tori (Corollary D.2).

Organization. Section 2 sets up the torus, its lowest Landau level, the Fock space and its sectors, and the hollow-core Hamiltonian, and quotes the theorem on its zero modes. Section 3 constructs the slot frame and the embedding ι and defines locality in the slot lattice. Section 4 proves the local form of the hollow-core Hamiltonian, its uniform size and rapid locality in the slot lattice and its covariance, introduces the class of Hamiltonians to which the index theorem applies, and describes the band charges in the slot lattice. Section 5 proves the locality estimates: a Lieb–Robinson bound, propagation from a set with a light cone independent of N, the filters, and the calculus of almost-supported operators. Section 6 states and proves the index theorem with the gap assumed in a sector, and Section 7 specializes it to the magnetic translation. Section 8 applies the result to the hollow-core Hamiltonian and discusses what it excludes. Appendix A reproduces the operator-theoretic lemmas of [BBDF21] that the proof uses, Appendix B states the Lieb–Robinson input from [NSY18] and derives the form we use, Appendix C proves the theorem on the zero modes, Appendix D proves the variational bound at fixed circumference, and Appendix F collects the notation.

Conventions. Inner products are linear in the second argument. Annihilation operators $c(f)$ are conjugate-linear in f. For $u \in \mathbb{R}$ and $P > 0$, $|u|_P$ denotes the distance from u to $P\mathbb{Z}$, and $(u)_+ = \max\{u, 0\}$. Constants denoted C, C', c may change from line to line but never depend on N; C_k denotes a constant that may depend on an index k. A sequence of operators or numbers A_N is $O(N^{-\infty})$ if for every k there is C_k with $\|A_N\| \le C_k N^{-k}$; we write $A \doteq B$ if $A - B = O(N^{-\infty})$, and $x \in \mathbb{Z} + O(N^{-\infty})$ if $x = x_N$ lies within $O(N^{-\infty})$ of an integer n_N that may depend on N.

2 Setting

We use units in which the magnetic field and the magnetic length are equal to one ($\hbar = e = 1$).

2.1 The torus and its lowest Landau level

Definition 2.1 (the torus and its lowest Landau level)**.** Let $N \in \mathbb{N}$ and $N_\phi := 2N$. Let $L_x, L_y > 0$ with $L_x L_y = 2\pi N_\phi = 4\pi N$, put $\kappa := 2\pi/L_y$, so that $L_x = \kappa N_\phi$, and let $\mathbb{T}_N := \mathbb{R}^2/(L_x\mathbb{Z} \times L_y\mathbb{Z})$, with points $r = (x, y)$, Lebesgue measure $\mathrm{d}r$, area $|\mathbb{T}_N| = L_x L_y$, and flat distance $d(r, r')$. Let

$$D_x := \partial_x, \qquad D_y := \partial_y - \mathrm{i}x, \qquad D := \tfrac{1}{2}(D_x - \mathrm{i}D_y) \tag{2.1}$$

be the covariant derivatives of the vector potential $(0, x)$, whose magnetic field is $\partial_x A_y - \partial_y A_x = 1$. The *lowest Landau level* $\mathfrak{h}_N$ is the space of smooth functions $f : \mathbb{R}^2 \to \mathbb{C}$ such that

$$f(x, y + L_y) = f(x, y), \qquad f(x + L_x, y) = e^{\mathrm{i}L_x y} f(x, y), \tag{2.2}$$

$$(D_x + \mathrm{i}D_y) f = 0, \tag{2.3}$$

with the inner product $\langle f, g\rangle := \int_{[0,L_x)\times[0,L_y)} \overline{f}\, g \,\mathrm{d}r$. By (2.2) the functions $|f|^2$ and $\overline{f}g$ are L_x- and L_y-periodic, so the integral is an integral over $\mathbb{T}_N$.

The conditions (2.2) are the magnetic periodicity conditions in this gauge without additional Aharonov–Bohm phases through the two cycles of the torus; all results of this paper concern these boundary conditions. The direction x will be called *transverse*: the Landau-gauge orbitals below are localized in x and extended in y.

Lemma 2.2 (Landau-gauge orbitals)**.** *For $n \in \mathbb{Z}$ let $\phi_n(x, y) := (L_y\sqrt{\pi})^{-1/2} e^{\mathrm{i}\kappa n y} e^{-(x-\kappa n)^2/2}$, and for $j \in \mathbb{Z}_{N_\phi}$ let*

$$\psi_j := \sum_{m\in\mathbb{Z}} \phi_{\tilde{\jmath}+mN_\phi}, \tag{2.4}$$

where $\tilde{\jmath} \in \mathbb{Z}$ is any representative of j.

(i) The series (2.4) converges, together with all its derivatives, locally uniformly on $\mathbb{R}^2$; ψ_j does not depend on the choice of $\tilde{\jmath}$, and $\psi_j \in \mathfrak{h}_N$.

(ii) $(\psi_j)_{j\in\mathbb{Z}_{N_\phi}}$ is an orthonormal basis of $\mathfrak{h}_N$. In particular $\dim \mathfrak{h}_N = N_\phi$.

(iii) For $f \in \mathfrak{h}_N$, $Df = D_x f = \partial_x f$, and Df satisfies (2.2). In particular $(D\phi_n)(x, y) = -(x - \kappa n)\,\phi_n(x, y)$.

Proof. (i) Every derivative of ϕ_n is ϕ_n times a polynomial in x and n, and

$$|\phi_n(x, y)| = (L_y\sqrt{\pi})^{-1/2} e^{-(x-\kappa n)^2/2};$$

these Gaussians are summable over $n \in \tilde{\jmath} + N_\phi\mathbb{Z}$ with any polynomial weight, locally uniformly in x. This gives the convergence, and replacing $\tilde{\jmath}$ by $\tilde{\jmath} + N_\phi$ only relabels the sum. Since $\kappa L_y = 2\pi$, each ϕ_n is L_y-periodic in y. Since $\kappa N_\phi = L_x$,

$$\phi_n(x + L_x, y) = (L_y\sqrt{\pi})^{-1/2} e^{\mathrm{i}\kappa n y} e^{-(x-\kappa(n-N_\phi))^2/2} = e^{\mathrm{i}L_x y}\phi_{n-N_\phi}(x, y),$$

and summing over $n \in \tilde{\jmath} + N_\phi\mathbb{Z}$ gives the second condition in (2.2) for ψ_j. Finally $(D_x + \mathrm{i}D_y)\phi_n = (-(x - \kappa n) + \mathrm{i}(\mathrm{i}\kappa n) + x)\phi_n = 0$, and (2.3) follows termwise.

(ii) Let $0 \le i, j < N_\phi$ be representatives. For $n, n' \in \mathbb{Z}$ the y-integral of $\overline{\phi_n}\phi_{n'}$ over $[0, L_y)$ is $L_y\delta_{nn'}(L_y\sqrt{\pi})^{-1} e^{-(x-\kappa n)^2}$. Hence $\langle\psi_i, \psi_j\rangle = 0$ for $i \ne j$, and

$$\langle\psi_j, \psi_j\rangle = \sum_{m\in\mathbb{Z}} \int_0^{L_x} \pi^{-1/2} e^{-(x-\kappa j-mL_x)^2}\,\mathrm{d}x = \pi^{-1/2}\int_{\mathbb{R}} e^{-u^2}\mathrm{d}u = 1.$$

For completeness, let $f \in \mathfrak{h}_N$. By the first condition in (2.2), $f(x, y) = \sum_{n\in\mathbb{Z}} f_n(x) e^{\mathrm{i}\kappa n y}$ with smooth f_n, the series converging with all derivatives. Condition (2.3) reads $f_n'(x) + (x - \kappa n) f_n(x) = 0$ for every n, so $f_n(x) = a_n e^{-(x-\kappa n)^2/2}$ with $a_n \in \mathbb{C}$. Comparing the coefficients of $e^{\mathrm{i}\kappa n y}$ in the

second condition of (2.2) gives $f_n(x+L_x) = f_{n-N_\phi}(x)$, that is $a_n = a_{n-N_\phi}$, because $x + L_x - \kappa n = x - \kappa(n - N_\phi)$. Hence $f = (L_y\sqrt{\pi})^{1/2} \sum_{j=0}^{N_\phi - 1} a_j \psi_j$.

(iii) Condition (2.3) says $D_y f = \mathrm{i} D_x f$, so $Df = \frac{1}{2}(D_x f - \mathrm{i} \cdot \mathrm{i} D_x f) = D_x f$. The operator ∂_x commutes with translations and with multiplication by $e^{\mathrm{i}L_x y}$, so $\partial_x f$ satisfies (2.2). The formula for $D\phi_n = \partial_x \phi_n$ is immediate. □

The orbital ψ_j is concentrated near the line $x = \kappa j$ modulo L_x:

$$|\psi_j(x,y)| \le (L_y\sqrt{\pi})^{-1/2} \sum_{m \in \mathbb{Z}} e^{-(x - \kappa\tilde{\jmath} - mL_x)^2/2}.$$

Note that Df is in general not in $\mathfrak{h}_N$; the derivative enters only through the evaluation of Df at a point.

Lemma 2.3 (evaluation vectors)**.** *For $R = (R_x, R_y) \in \mathbb{R}^2$ let*

$$e_R := \sum_{j \in \mathbb{Z}_{N_\phi}} \overline{\psi_j(R)}\, \psi_j, \qquad e_R^D := \sum_{j \in \mathbb{Z}_{N_\phi}} \overline{(D\psi_j)(R)}\, \psi_j \qquad \in \mathfrak{h}_N.$$

(i) $\langle e_R, f\rangle = f(R)$ and $\langle e_R^D, f\rangle = (Df)(R)$ for every $f \in \mathfrak{h}_N$. Moreover $e_{R+(0,L_y)} = e_R$ and $e_{R+(L_x,0)} = e^{-\mathrm{i}L_x R_y} e_R$, and likewise for e_R^D; in particular $|e_R\rangle\langle e_R|$ and $\|e_R\|$, $\|e_R^D\|$ are functions of $R \in \mathbb{T}_N$.

(ii) $\int_{\mathbb{T}_N} |e_R\rangle\langle e_R| \,\mathrm{d}R = \mathbf{1}_{\mathfrak{h}_N}$.

(iii) With $q_{pk}^2 := p^2 L_x^2 + k^2 L_y^2$ for $(p,k) \in \mathbb{Z}^2$,

$$\|e_R\|^2 = \frac{1}{2\pi} \sum_{(p,k)\in\mathbb{Z}^2} e^{-q_{pk}^2/4} \cos(kL_y R_x + pL_x R_y),$$

$$\|e_R^D\|^2 = \frac{1}{4\pi} \sum_{(p,k)\in\mathbb{Z}^2} \Big(1 - \frac{q_{pk}^2}{2}\Big) e^{-q_{pk}^2/4} \cos(kL_y R_x + pL_x R_y).$$

(iv) Let $L_{\min} := \min\{L_x, L_y\} \ge 4$. Then for every R,

$$|2\pi\|e_R\|^2 - 1| \le 4.2\, e^{-L_{\min}^2/4} \le 0.08, \qquad |4\pi\|e_R^D\|^2 - 1| \le 2.1\, L_{\min}^2 e^{-L_{\min}^2/4} \le 0.62.$$

Proof. (i) Since (ψ_j) is an orthonormal basis and the inner product is conjugate-linear in its first argument, $\langle e_R, f\rangle = \sum_j \psi_j(R)\langle\psi_j, f\rangle = f(R)$, and in the same way $\langle e_R^D, f\rangle = (Df)(R)$, because D is linear. The behaviour under $R \mapsto R + (L_x, 0)$ follows from (2.2): $\langle e_{R+(L_x,0)}, f\rangle = e^{\mathrm{i}L_x R_y} f(R) = \langle e^{-\mathrm{i}L_x R_y} e_R, f\rangle$ for every $f \in \mathfrak{h}_N$, and likewise for e_R^D, since Df satisfies (2.2) (Lemma 2.2(iii)). The y-periodicity is similar.

(ii) For $f, g \in \mathfrak{h}_N$, $\langle f, \int |e_R\rangle\langle e_R| g \,\mathrm{d}R\rangle = \int \langle f, e_R\rangle\langle e_R, g\rangle \mathrm{d}R = \int \overline{f(R)} g(R) \,\mathrm{d}R = \langle f, g\rangle$.

(iii) Write $R = (x, y)$. By (2.4), $\|e_R\|^2 = \sum_j |\psi_j(R)|^2 = \sum_{n\in\mathbb{Z}} \sum_{p\in\mathbb{Z}} \overline{\phi_n(R)} \phi_{n+pN_\phi}(R)$, the pairs (j,m) and (j,m') being relabelled by $n = \tilde{\jmath} + mN_\phi$ and $p = m' - m$. Using $\kappa N_\phi = L_x$ and $\frac{1}{2}(u^2 + (u - pL_x)^2) = (u - pL_x/2)^2 + p^2 L_x^2/4$ with $u = x - \kappa n$,

$$\|e_R\|^2 = \frac{1}{L_y\sqrt{\pi}} \sum_{p\in\mathbb{Z}} e^{\mathrm{i}pL_x y} e^{-p^2 L_x^2/4} \sum_{n\in\mathbb{Z}} e^{-(x - pL_x/2 - \kappa n)^2}.$$

By the Poisson summation formula, $\sum_n g(v-\kappa n) = \kappa^{-1}\sum_k \hat g(2\pi k/\kappa)e^{2\pi \mathrm{i} kv/\kappa}$ with $\hat g(\xi) = \int g(u)e^{-\mathrm{i}\xi u}\mathrm{d}u = \sqrt{\pi}e^{-\xi^2/4}$ for $g(u) = e^{-u^2}$. Here $2\pi/\kappa = L_y$, $L_y\kappa = 2\pi$, and $e^{-\mathrm{i}kL_y\cdot pL_x/2} = e^{-\mathrm{i}\pi kpN_\phi} = 1$ because N_ϕ is even. Hence $\|e_R\|^2 = (2\pi)^{-1}\sum_{(p,k)} e^{-q_{pk}^2/4}e^{\mathrm{i}(kL_yx+pL_xy)}$, and the sum is real, being symmetric under $(p,k)\mapsto(-p,-k)$. For e_R^D the same computation, with $D\phi_n = -(x-\kappa n)\phi_n$, replaces e^{-v^2} by $(v^2 - p^2L_x^2/4)e^{-v^2}$, $v = x - pL_x/2 - \kappa n$, because $(x-\kappa n)(x-\kappa n - pL_x) = v^2 - p^2L_x^2/4$. The Fourier transform of $v^2e^{-v^2}$ is $\sqrt{\pi}(\frac12 - \frac{\xi^2}{4})e^{-\xi^2/4}$, which gives the second formula.

(iv) The terms with $(p,k) = (0,0)$ give 1 in both formulas. For $(p,k)\neq(0,0)$ put $n := p^2+k^2 \geq 1$, so that $q_{pk}^2 \geq nL_{\min}^2 \geq 16$. Since $u\mapsto e^{-u/4}$ and, for $u\geq 4$, $u\mapsto \frac{u}{2}e^{-u/4}$ are decreasing, and $|1-u/2|\leq u/2$ for $u\geq 2$,

$$\big|2\pi\|e_R\|^2 - 1\big| \leq \sum_{(p,k)\neq 0} e^{-nL_{\min}^2/4} \leq e^{-L_{\min}^2/4}\sum_{(p,k)\neq 0} e^{-4(n-1)},$$

$$\big|4\pi\|e_R^D\|^2 - 1\big| \leq \sum_{(p,k)\neq 0} \frac{nL_{\min}^2}{2}e^{-nL_{\min}^2/4} \leq \frac{L_{\min}^2}{2}e^{-L_{\min}^2/4}\sum_{(p,k)\neq 0} n\,e^{-4(n-1)},$$

using $e^{-(n-1)L_{\min}^2/4}\leq e^{-4(n-1)}$. The first sum is $e^4\big((\sum_{p\in\mathbb{Z}} e^{-4p^2})^2 - 1\big)$, which is less than 4.08; the second is less than 4.15 (its terms with $n=1,2$ contribute $4+8e^{-4}$, and the rest less than 10^{-3}). This gives the bounds; the numerical values follow from $L_{\min}\geq 4$. □

2.2 Magnetic translations

For a function f on $\mathbb{R}^2$ let

$$(\theta_x f)(x,y) := e^{\mathrm{i}\kappa y}f(x-\kappa,y), \qquad (\theta_y f)(x,y) := f(x,y+L_y/N_\phi). \tag{2.5}$$

Lemma 2.4 (magnetic translations). *(i) θ_x and θ_y restrict to unitary operators on $\mathfrak{h}_N$, with $\theta_x\psi_j = \psi_{j+1}$ and $\theta_y\psi_j = e^{2\pi \mathrm{i} j/N_\phi}\psi_j$.*

(ii) On $\mathfrak{h}_N$, $\theta_y\theta_x = e^{2\pi\mathrm{i}/N_\phi}\theta_x\theta_y$ and $\theta_x^{N_\phi} = \theta_y^{N_\phi} = \mathbf{1}$.

(iii) θ_x and θ_y commute with D_x and D_y on smooth functions, hence with D.

(iv) $\theta_x e_R = e^{\mathrm{i}\kappa R_y}e_{R+\kappa\hat e_x}$ and $\theta_x e_R^D = e^{\mathrm{i}\kappa R_y}e^D_{R+\kappa\hat e_x}$ for every $R\in\mathbb{R}^2$, where $\hat e_x = (1,0)$.

Proof. (i) Directly from the definitions, $\theta_x\phi_n = \phi_{n+1}$ and $\theta_y\phi_n = e^{\mathrm{i}\kappa nL_y/N_\phi}\phi_n = e^{2\pi\mathrm{i}n/N_\phi}\phi_n$. Summing over $n\in\tilde\jmath + N_\phi\mathbb{Z}$ gives the action on ψ_j (the phase $e^{2\pi\mathrm{i}n/N_\phi}$ depends only on the class of n). Both operators therefore map the orthonormal basis (ψ_j) onto an orthonormal basis up to phases, so they restrict to unitary operators on $\mathfrak{h}_N$. (ii) follows from (i). (iii) For θ_y this is clear, since D_x and D_y do not involve y explicitly. For θ_x, $D_x\theta_x f = \theta_x D_x f$, and

$$\begin{aligned} D_y(\theta_x f)(x,y) &= e^{\mathrm{i}\kappa y}\big[(\partial_y f)(x-\kappa,y) + \mathrm{i}\kappa f(x-\kappa,y) - \mathrm{i}x f(x-\kappa,y)\big] \\ &= e^{\mathrm{i}\kappa y}(D_y f)(x-\kappa,y) = (\theta_x D_y f)(x,y). \end{aligned}$$

(iv) The inverse of θ_x on $\mathfrak{h}_N$ is $(\theta_x^{-1}f)(x,y) = e^{-\mathrm{i}\kappa y}f(x+\kappa,y)$, and $\theta_x^* = \theta_x^{-1}$ by (i). For $f\in\mathfrak{h}_N$, $\langle\theta_x e_R, f\rangle = (\theta_x^{-1}f)(R) = e^{-\mathrm{i}\kappa R_y}f(R+\kappa\hat e_x) = \langle e^{\mathrm{i}\kappa R_y}e_{R+\kappa\hat e_x}, f\rangle$. By (iii), $D\theta_x^{-1}f = \theta_x^{-1}Df$, and the same computation gives the formula for e_R^D. □

Only θ_x, the translation by one orbital across the cuts, enters the index theorem.

2.3 The spinful Fock space and its sectors

Definition 2.5 (Fock space, sectors, the many-body translation)**.** Let $\mathfrak{k}_N := \mathfrak{h}_N \otimes \mathbb{C}^2$, with the orthonormal basis $\psi_j \otimes \sigma$, $j \in \mathbb{Z}_{N_\phi}$, $\sigma \in \{\uparrow, \downarrow\}$, where $\uparrow, \downarrow$ denote the standard basis of $\mathbb{C}^2$. Let $\mathcal{H}_N := \mathcal{F}(\mathfrak{k}_N)$ be the fermionic Fock space over $\mathfrak{k}_N$ (of dimension 4^{N_ϕ}), with annihilation operators $c(f)$, $f \in \mathfrak{k}_N$, conjugate-linear in f and satisfying the canonical anticommutation relations $\{c(f), c(g)^*\} = \langle f, g\rangle$, $\{c(f), c(g)\} = 0$. Put

$$c_{j\sigma} := c(\psi_j \otimes \sigma), \qquad n_{j\sigma} := c_{j\sigma}^* c_{j\sigma}, \qquad N_\sigma := \sum_{j \in \mathbb{Z}_{N_\phi}} n_{j\sigma}, \qquad S_z := \tfrac{1}{2}(N_\uparrow - N_\downarrow).$$

For $\mathbf{n} = (n_\uparrow, n_\downarrow) \in \{0, \dots, N_\phi\}^2$ the *sector* $\mathcal{H}_N^{\mathbf{n}}$ is the joint eigenspace $\{\Psi : N_\uparrow \Psi = n_\uparrow \Psi,\ N_\downarrow \Psi = n_\downarrow \Psi\}$, and $\Pi^{\mathbf{n}}$ is the orthogonal projection onto it. An operator A on $\mathcal{H}_N$ is *sector-preserving* if $[A, N_\uparrow] = [A, N_\downarrow] = 0$, and *even* if it commutes with the parity $(-1)^{N_\uparrow + N_\downarrow}$. For a unitary u on $\mathfrak{k}_N$, $\mathcal{F}(u)$ denotes its second quantization, the unitary on $\mathcal{H}_N$ with $\mathcal{F}(u) c(f) \mathcal{F}(u)^* = c(uf)$ that fixes the vacuum. The *many-body magnetic translation* is $\Theta := \mathcal{F}(\theta_x \otimes \mathbf{1})^*$.

Lemma 2.6 (sectors and the many-body translation)**.** *(i) $\mathcal{H}_N = \bigoplus_{\mathbf{n}} \mathcal{H}_N^{\mathbf{n}}$, an orthogonal sum, and $\Pi^{\mathbf{n}} = \mathbf{1}_{\{n_\uparrow\}}(N_\uparrow)\, \mathbf{1}_{\{n_\downarrow\}}(N_\downarrow)$.*

(ii) An operator A is sector-preserving if and only if $A = \sum_{\mathbf{n}} \Pi^{\mathbf{n}} A \Pi^{\mathbf{n}}$, if and only if $[A, \Pi^{\mathbf{n}}] = 0$ for every $\mathbf{n}$. Sector-preserving operators form a unital $*$-subalgebra of the operators on $\mathcal{H}_N$, and they are even. If H is self-adjoint and sector-preserving, then $f(H)$ is sector-preserving for every function f on the spectrum of H, and for every $\mathbf{n}$ and every Borel set $B \subset \mathbb{R}$ the spectral projection of $H|_{\mathcal{H}_N^{\mathbf{n}}}$ onto B, extended by zero to $\mathcal{H}_N$, equals $\mathbf{1}_B(H)\Pi^{\mathbf{n}}$.

(iii) Θ is unitary and sector-preserving, $\Theta^{N_\phi} = \mathbf{1}$, and for all $f \in \mathfrak{k}_N$, j and σ,

$$\Theta^* c(f) \Theta = c((\theta_x \otimes \mathbf{1}) f), \qquad \Theta^* c_{j\sigma} \Theta = c_{j+1,\sigma}, \qquad \Theta^* n_{j\sigma} \Theta = n_{j+1,\sigma}.$$

Proof. (i) $N_\uparrow$ and $N_\downarrow$ are commuting self-adjoint operators with joint spectrum in $\{0, \dots, N_\phi\}^2$, and the $\Pi^{\mathbf{n}}$ are their joint spectral projections. (ii) A commutes with $N_\uparrow$ and $N_\downarrow$ if and only if it commutes with all their spectral projections, hence with every $\Pi^{\mathbf{n}}$; since $\sum_{\mathbf{n}} \Pi^{\mathbf{n}} = \mathbf{1}$ this is equivalent to $A = \sum_{\mathbf{n}} \Pi^{\mathbf{n}} A \Pi^{\mathbf{n}}$. These conditions are stable under sums, products and adjoints. A sector-preserving operator commutes with $N_\uparrow + N_\downarrow$, hence with $(-1)^{N_\uparrow + N_\downarrow}$. If H is self-adjoint and sector-preserving, then $H = \bigoplus_{\mathbf{n}} H|_{\mathcal{H}_N^{\mathbf{n}}}$, so $f(H) = \bigoplus_{\mathbf{n}} f(H|_{\mathcal{H}_N^{\mathbf{n}}})$; with $f = \mathbf{1}_B$ this gives the last claim. (iii) Θ is unitary because $\theta_x \otimes \mathbf{1}$ is. $\Theta^* c(f) \Theta = \mathcal{F}(\theta_x \otimes \mathbf{1}) c(f) \mathcal{F}(\theta_x \otimes \mathbf{1})^* = c((\theta_x \otimes \mathbf{1}) f)$, and with $f = \psi_j \otimes \sigma$ Lemma 2.4(i) gives the other two formulas. Summing them over j shows $\Theta^* N_\sigma \Theta = N_\sigma$. Finally $\Theta^{N_\phi} = \mathcal{F}((\theta_x \otimes \mathbf{1})^{N_\phi})^* = \mathbf{1}$ by Lemma 2.4(ii). □

The convention $\Theta^* n_{j\sigma} \Theta = n_{j+1,\sigma}$ is that of [BBDF21, Section 5.2] for the lattice translation.

2.4 The hollow-core Hamiltonian

For $a, c \in \mathbb{Z}_{N_\phi}$ the function $\overline{\psi_a}\, \psi_c$ is smooth on $\mathbb{T}_N$ by (2.2); let $\nabla^2 = \partial_x^2 + \partial_y^2$.

Definition 2.7 (the hollow-core Hamiltonian)**.** For $a, b, c, d \in \mathbb{Z}_{N_\phi}$ let

$$V_{abcd} := 4\pi \int_{\mathbb{T}_N} \overline{\psi_a} \psi_c\, \overline{\psi_b} \psi_d \, \mathrm{d}r + 4\pi \int_{\mathbb{T}_N} \overline{\psi_b} \psi_d\, \nabla^2 (\overline{\psi_a} \psi_c) \, \mathrm{d}r, \tag{2.6}$$

and let

$$H_N := \frac{1}{2} \sum_{\sigma,\sigma' \in \{\uparrow,\downarrow\}} \sum_{a,b,c,d \in \mathbb{Z}_{N_\phi}} V_{abcd}\, c^*_{a\sigma} c^*_{b\sigma'} c_{d\sigma'} c_{c\sigma} \tag{2.7}$$

on $\mathcal{H}_N$. Equivalently, by Parseval's identity on $\mathbb{T}_N$,

$$V_{abcd} = \frac{1}{|\mathbb{T}_N|} \sum_{q \in \frac{2\pi}{L_x}\mathbb{Z} \times \frac{2\pi}{L_y}\mathbb{Z}} 4\pi(1-|q|^2)\, \widehat{\overline{\psi_a}\psi_c}(-q)\, \widehat{\overline{\psi_b}\psi_d}(q), \qquad \hat{f}(q) := \int_{\mathbb{T}_N} f(r) e^{-\mathrm{i}q\cdot r}\, \mathrm{d}r,$$

that is, H_N is the second quantization, restricted to the lowest Landau level, of the pair interaction $V_{\mathbb{T}} = 4\pi(\delta_{\mathbb{T}} + \nabla^2 \delta_{\mathbb{T}})$, the periodization over $L_x\mathbb{Z} \times L_y\mathbb{Z}$ of the planar interaction $V := 4\pi(\delta + \nabla^2\delta)$, whose Fourier transform is $4\pi(1-|q|^2)$, 4π times the first Laguerre polynomial evaluated at $|q|^2$.

The operator H_N is self-adjoint, since $V_{abcd} = \overline{V_{cdab}}$, and it is sector-preserving, even, and invariant under global spin rotations. It commutes with Θ (Lemma 4.5), and $H_N = 8\pi \int_{\mathbb{T}_N} \sum_\nu B_\nu(R)^* B_\nu(R)\, \mathrm{d}R$ (Theorem 4.2(a)); in particular $H_N \geq 0$.

The normalization of (2.6) is the one in which a pair of electrons of relative angular momentum one in the plane has energy one. To state this precisely, consider two particles in the planar lowest Landau level for the same field, in the symmetric gauge: in complex coordinates $z_i = x_i + \mathrm{i}y_i$, a two-particle state is $\Psi(z_1, z_2) = G(z_1, z_2) e^{-(|z_1|^2+|z_2|^2)/4}$ with G entire and Ψ square-integrable. With $Z := (z_1+z_2)/2$ and $v := z_1 - z_2$ one has $|z_1|^2 + |z_2|^2 = 2|Z|^2 + |v|^2/2$, and we expand $G(Z + v/2, Z - v/2) = \sum_{m\geq 0} G_m(Z) v^m$. The component of relative angular momentum m is $\Psi_m := G_m(Z)\, v^m\, e^{-|Z|^2/2 - |v|^2/8}$; the Ψ_m are mutually orthogonal and $\Psi = \sum_m \Psi_m$. The quadratic form of V on such states is

$$\mathsf{V}(\Psi) := 4\pi \int_{\mathbb{R}^2} \Big(F_\Psi(R,0) + (\nabla_r^2 F_\Psi)(R,0) \Big) \mathrm{d}R, \qquad F_\Psi(R,r) := |\Psi(R + r/2, R - r/2)|^2,$$

the expectation of $V(r_1 - r_2)$ in the sense of distributions; it depends only on $|\Psi|^2$ and so is the same in every gauge.

Lemma 2.8 (V_1 units)**.** *For every planar two-particle lowest-Landau-level state Ψ, $\mathsf{V}(\Psi) = \|\Psi_1\|^2$. That is, on the planar two-particle lowest Landau level the interaction V is the orthogonal projection onto relative angular momentum one: its pseudopotentials are $V_m = \delta_{m1}$.*

Proof. In the coordinates $R = Z$ and $r = v$ (the change of variables $(r_1, r_2) \mapsto (R, r)$ has Jacobian one), $F_\Psi(Z, v) = e^{-|Z|^2} e^{-|v|^2/4} |\sum_m G_m(Z) v^m|^2$. At $v = 0$ this is $e^{-|Z|^2} |G_0(Z)|^2$. With $\nabla_v^2 = 4\partial_v \partial_{\bar{v}}$ and $e^{-v\bar{v}/4} |\sum_m G_m v^m|^2 = |G_0|^2 + G_0 \overline{G_1} \bar{v} + \overline{G_0} G_1 v + \big(|G_1|^2 - \tfrac{1}{4}|G_0|^2\big) v\bar{v} +$ (terms in $v^2, \bar{v}^2$, and of order $|v|^3$), one finds $(\nabla_v^2 F_\Psi)(Z, 0) = e^{-|Z|^2} \big(4|G_1(Z)|^2 - |G_0(Z)|^2\big)$. Hence

$$\mathsf{V}(\Psi) = 16\pi \int_{\mathbb{R}^2} |G_1(Z)|^2 e^{-|Z|^2} \mathrm{d}Z = \int |G_1(Z)|^2 e^{-|Z|^2} \mathrm{d}Z \int_{\mathbb{R}^2} |v|^2 e^{-|v|^2/4} \mathrm{d}v = \|\Psi_1\|^2. \qquad \square$$

Remark 2.9 (orbital form and normalization). Theorem 4.2(c) below writes H_N in the orbital basis: $H_N = (\kappa^3/\sqrt{2\pi}) \sum_{n \in \mathbb{Z}_{2N_\phi}} \sum_\nu (b_n^\nu)^* b_n^\nu$, with the pair operators b_n^ν of Definition 4.1. The operator $\sum_n \sum_\nu (b_n^\nu)^* b_n^\nu$ differs from H_N by the factor $\sqrt{2\pi}/\kappa^3$, which grows like $N^{3/2}$ at fixed aspect ratio. Every statement of this paper about gaps that are uniform in N refers to H_N in the normalization of Definition 2.7.

2.5 The zero modes

The kernel of H_N on the states with N electrons is known exactly. The proof of the following theorem is given in Appendix C.

Theorem 2.10 (the zero modes)**.** *Let $N \in \mathbb{N}$ and $L_x, L_y > 0$ with $L_x L_y = 4\pi N$, and let $\mathcal{K}_N$ be the kernel of H_N restricted to the states with N electrons, $\bigoplus_{n_\uparrow+n_\downarrow=N} \mathcal{H}_N^{(n_\uparrow,n_\downarrow)}$.*

(i) $\mathcal{K}_N \cap \mathcal{H}_N^{(n_\uparrow,n_\downarrow)} = \{0\}$ *whenever* $|n_\uparrow - n_\downarrow| \geq 2$.

(ii) If N is odd, $\mathcal{K}_N \cap \mathcal{H}_N^{((N+1)/2,(N-1)/2)}$ *and* $\mathcal{K}_N \cap \mathcal{H}_N^{((N-1)/2,(N+1)/2)}$ *are two-dimensional.*

(iii) If N is even, $\mathcal{K}_N \subset \mathcal{H}_N^{(N/2,N/2)}$ *and* $\dim \mathcal{K}_N = 10$.

Appendix C proves the theorem for tori of arbitrary modular parameter; we use it only for rectangular tori. The kernel in (iii) is spanned by the torus states of Read and Rezayi [RR96, Section V], and the kernel in (ii) by the corresponding states with one unpaired electron; we use only the dimensions. Since $H_N \geq 0$, the eigenvalues $E_1(N) \leq E_2(N) \leq \cdots$ of H_N in the sector $(N/2, N/2)$ at even N, repeated according to multiplicity, satisfy

$$E_1(N) = \cdots = E_{10}(N) = 0 < E_{11}(N), \tag{2.8}$$

and at odd $N \geq 3$ the eigenvalues in each of the sectors $((N \pm 1)/2, (N \mp 1)/2)$ satisfy $E_1(N) = E_2(N) = 0 < E_3(N)$. (The sector $(N/2, N/2)$ has dimension $\binom{2N}{N/2}^2 \geq 16$ for even $N \geq 2$; each of the two odd sectors has dimension $\binom{2N}{(N+1)/2}\binom{2N}{(N-1)/2}$, which is 2 at $N = 1$ and at least 3 for odd $N \geq 3$.)

Remark 2.11 (the Hamiltonian of Appendix C). In its own notation, Appendix C defines the hollow-core Hamiltonian as $\sum_{i<j} \mathcal{P}_{ij}$, where $\mathcal{P}$ is the sum over the lattice of periods of the translates of the planar projector onto relative angular momentum one, on the torus $\mathbb{C}/(L_1\mathbb{Z} + L_1\tau\mathbb{Z})$, with coordinates $X + \mathrm{i}Y$, in the Landau gauge $(-Y, 0)$. For $L_1 = L_y$ and $\tau = \mathrm{i}L_x/L_y$ this is our torus turned by a quarter turn. The beginning of Appendix C shows, using Theorem 4.2(c), that the corresponding unitary identification of the two lowest Landau levels, which takes the orbital ψ_j to the j-th Landau-gauge orbital of Appendix C, carries H_N, with the spin unchanged, to that operator. The identification preserves $N_\uparrow$ and $N_\downarrow$, so Theorem 2.10 holds for H_N as stated.

Remark 2.12 (spin). Since H_N commutes with the total spin and, by Theorem 2.10(i), no zero mode has $S_z = \pm 1$, every zero mode at even N is a spin singlet, and at odd N the kernel is a spin-$\frac{1}{2}$ doublet times the two centre-of-mass states (Remark C.20). Nothing in this paper uses this: the main theorems use only the dimensions in Theorem 2.10. The theorem as stated is formalized (Appendix E); this remark is not.

3 The lowest Landau level as a lattice: the slot frame

3.1 Construction

Definition 3.1 (admissible tori and the slot frame)**.** Fix once and for all a *profile* $\eta \in C^\infty(\mathbb{R})$ with $\eta > 0$ on $(-1, 1)$ and $\eta = 0$ on $\mathbb{R} \setminus (-1, 1)$, and a *target slot width* $\delta_0 > 0$ (for definiteness $\eta(u) = e^{-1/(1-u^2)}$ for $|u| < 1$, and $\delta_0 = \sqrt{\pi}$). Put

$$w(u) := \eta(u)\Big(\sum_{k\in\mathbb{Z}} \eta(u-k)^2\Big)^{-1/2}, \qquad L_0 := \max\{4,\, 3\pi/\delta_0\}, \qquad \delta_- := \tfrac{2}{3}\delta_0, \qquad \delta_+ := 2\delta_0.$$

A torus as in Definition 2.1 is *admissible* if $L_y \geq L_0$ and $L_x \geq 4\delta_0 + 4$. For an admissible torus let

$$n_x := \text{the integer nearest to } L_x/\delta_0, \qquad \delta_x := L_x/n_x, \qquad x_a := a\delta_x \ \ (a \in \mathbb{Z}),$$
$$n_y := \lceil 2\delta_x/\kappa \rceil, \qquad \delta_y := L_y/n_y, \qquad y_\ell := \ell\delta_y \ \ (\ell \in \mathbb{Z}),$$

and let the *slot lattice* be $\Lambda_N := \mathbb{Z}_{n_x} \times \mathbb{Z}_{n_y}$, with slots $s = (a, \ell)$, centres $r_s := (x_a, y_\ell) \in \mathbb{T}_N$ and transverse coordinates $x_s := x_a$. For $a, j \in \mathbb{Z}$ let $w_a(j) := w(\kappa j/\delta_x - a)$, supported in $I_a := \{j \in \mathbb{Z} : |\kappa j - x_a| < \delta_x\}$. The *slot frame* is the family

$$z_s := n_y^{-1/2} \sum_{j \in I_a} w_a(j)\, e^{-2\pi \mathrm{i} j\ell/n_y}\, \psi_{[j]} \in \mathfrak{h}_N, \qquad s = (a, \ell) \in \Lambda_N, \tag{3.1}$$

where $a \in \{0, \dots, n_x - 1\}$ is a representative and $[j]$ is the class of j in $\mathbb{Z}_{N_\phi}$. The *analysis map* is $\mathcal{S} : \mathfrak{h}_N \to \ell^2(\Lambda_N)$, $(\mathcal{S}f)_s := \langle z_s, f\rangle$; we write $\mathcal{S}$ also for $\mathcal{S} \otimes \mathbf{1}_{\mathbb{C}^2} : \mathfrak{k}_N \to \ell^2(\Lambda_N) \otimes \mathbb{C}^2$. Distances between slots, and between slots and points of $\mathbb{T}_N$, are the flat distances between the corresponding points: $d(s, s') := d(r_s, r_{s'})$, $d(s, R) := d(r_s, R)$, and $d(X, Y) := \min\{d(s, s') : s \in X,\ s' \in Y\}$ for $X, Y \subset \Lambda_N$.

Lemma 3.2 (consistency of the construction)**.** *Let the torus be admissible.*

(i) w is smooth, $w = 0$ on $\mathbb{R} \setminus (-1, 1)$, and $\sum_{k\in\mathbb{Z}} w(u - k)^2 = 1$ for every $u \in \mathbb{R}$.

(ii) $n_x \geq 4$, $\delta_x \in [\delta_-, \delta_+]$, and $\delta_x/\kappa \geq 1$: every window contains at least one orbital centre.

(iii) I_a contains fewer than $2\delta_x/\kappa + 1$ integers, and these lie in distinct classes of $\mathbb{Z}_{N_\phi}$. Moreover $w_{a+n_x}(j + N_\phi) = w_a(j)$, so that replacing the representative a of the first coordinate of $s = (a, \ell)$ by $a + mn_x$ in (3.1) multiplies z_s by the phase $e^{-2\pi \mathrm{i} m N_\phi \ell/n_y}$; no statement below depends on these phases.

Proof. (i) Let $\Sigma(u) := \sum_{k\in\mathbb{Z}} \eta(u - k)^2$. On every bounded interval only finitely many terms of this sum are not identically zero, so Σ is smooth; it is 1-periodic; and if k_0 is an integer nearest to u, then $\Sigma(u) \geq \eta(u - k_0)^2 \geq \min_{|t|\leq 1/2} \eta(t)^2 > 0$. Hence $w = \eta\, \Sigma^{-1/2}$ is smooth and vanishes wherever η does, in particular on $\mathbb{R} \setminus (-1, 1)$, and by the periodicity of Σ, $\sum_k w(u - k)^2 = \Sigma(u)^{-1} \sum_k \eta(u - k)^2 = 1$. In particular $0 \leq w \leq 1$. Since w is smooth and vanishes on $\mathbb{R} \setminus (-1, 1)$, so do all its derivatives (on the open set $|u| > 1$ trivially, at $u = \pm 1$ by continuity).

(ii) Let $v := L_x/\delta_0$. Admissibility gives $v \geq 4 + 4/\delta_0 > 4$, and $|n_x - v| \leq \frac{1}{2}$ (if v is a half-integer either nearest integer may be taken; only this inequality is used). Hence $n_x \geq 4$, and

$$\frac{\delta_x}{\delta_0} = \frac{v}{n_x} \in \Big[\frac{v}{v + 1/2}, \frac{v}{v - 1/2}\Big] \subset \Big[\frac{8}{9}, \frac{8}{7}\Big],$$

because $v \mapsto v/(v + \frac{1}{2})$ is increasing and $v \mapsto v/(v - \frac{1}{2})$ is decreasing on $(\frac{1}{2}, \infty)$, and $v \geq 4$. Thus $\delta_- < \frac{8}{9}\delta_0 \leq \delta_x \leq \frac{8}{7}\delta_0 < \delta_+$. Since $L_y \geq 3\pi/\delta_0$, $\delta_x/\kappa = \delta_x L_y/(2\pi) \geq \frac{8}{9} \cdot \frac{3}{2} = \frac{4}{3}$. The window $(x_a - \delta_x, x_a + \delta_x)$ has length $2\delta_x > \kappa$ and therefore contains a point of $\kappa\mathbb{Z}$.

(iii) An integer j lies in I_a if and only if it lies in the open interval of length $2\delta_x/\kappa$ centred at x_a/κ. If $j_1 < \cdots < j_p$ lie in this interval, then $p - 1 \leq j_p - j_1 < 2\delta_x/\kappa$, so $p < 2\delta_x/\kappa + 1$. Two elements of I_a differ by less than $2\delta_x/\kappa = 2N_\phi/n_x \leq N_\phi/2$ (recall $n_x\delta_x = L_x = \kappa N_\phi$), so they lie in distinct classes of $\mathbb{Z}_{N_\phi}$. Since $\kappa N_\phi/\delta_x = n_x$ and $x_{a+n_x} = x_a + L_x$, we have $w_{a+n_x}(j + N_\phi) = w_a(j)$ and $I_{a+n_x} = I_a + N_\phi$, and by induction $w_{a+mn_x}(j + mN_\phi) = w_a(j)$ and $I_{a+mn_x} = I_a + mN_\phi$ for all $m \in \mathbb{Z}$. Writing (3.1) with $a + mn_x$ in place of a, substituting $j' = j + mN_\phi$ with $j \in I_a$, and

using $\psi_{[j+mN_\phi]} = \psi_{[j]}$, we obtain $n_y^{-1/2}\sum_{j\in I_a} w_a(j)e^{-2\pi\mathrm{i}(j+mN_\phi)\ell/n_y}\psi_{[j]} = e^{-2\pi\mathrm{i}mN_\phi\ell/n_y}z_s$. The right side of (3.1) depends on ℓ only through its class modulo n_y, and r_s and x_s do not depend on the representatives, because $x_{a+n_x} = x_a + L_x$ and $y_{\ell+n_y} = y_\ell + L_y$. A change of these phases replaces $\mathcal{S}$ by $\mathcal{U}\mathcal{S}$, with $\mathcal{U}$ a diagonal unitary on $\ell^2(\Lambda_N)$. This changes neither $|z_s|$, $|Dz_s|$ and $|z_s\rangle\langle z_s|$ nor the supports of the vectors $\mathcal{S}f$, and $\hat{c}((\mathcal{U}g)\otimes\sigma)$ is a linear combination of the same operators $\hat{c}_{s\sigma}$ as $\hat{c}(g\otimes\sigma)$; these are the only ways in which the frame enters below. □

Proposition 3.3 (Parseval)**.** *For every admissible torus, $\sum_{s\in\Lambda_N}|z_s\rangle\langle z_s| = \mathbf{1}_{\mathfrak{h}_N}$. Equivalently $\mathcal{S}^*\mathcal{S} = \mathbf{1}$: the analysis map is an isometry, and $\mathcal{S}\mathcal{S}^*$ is the orthogonal projection of $\ell^2(\Lambda_N)$ onto* $\operatorname{ran}\mathcal{S}$*, of rank N_ϕ.*

Proof. Fix $a \in \{0,\dots,n_x-1\}$. Since w is real, (3.1) gives

$$\sum_{\ell=0}^{n_y-1}|z_{(a,\ell)}\rangle\langle z_{(a,\ell)}| = \sum_{j,j'\in I_a} w_a(j)w_a(j')\Big[\frac{1}{n_y}\sum_{\ell=0}^{n_y-1}e^{-2\pi\mathrm{i}(j-j')\ell/n_y}\Big]|\psi_{[j]}\rangle\langle\psi_{[j']}|.$$

The bracket is 1 if n_y divides $j-j'$ and 0 otherwise. For $j,j'\in I_a$ we have $|j-j'| < 2\delta_x/\kappa \le n_y$, so the bracket is $\delta_{jj'}$, and the sum equals $\sum_{j\in I_a} w_a(j)^2|\psi_{[j]}\rangle\langle\psi_{[j]}| = \sum_{j\in\mathbb{Z}} w_a(j)^2|\psi_{[j]}\rangle\langle\psi_{[j]}|$, because w_a vanishes off I_a. Now sum over a and group the $j\in\mathbb{Z}$ by their class $c\in\{0,\dots,N_\phi-1\}$, $j = c+mN_\phi$. By Lemma 3.2(iii), $w_a(c+mN_\phi) = w_{a-mn_x}(c)$, and as (a,m) runs over $\{0,\dots,n_x-1\}\times\mathbb{Z}$, $a-mn_x$ runs over $\mathbb{Z}$ exactly once. Hence the coefficient of $|\psi_c\rangle\langle\psi_c|$ is $\sum_{a'\in\mathbb{Z}} w(\kappa c/\delta_x - a')^2 = 1$ by Lemma 3.2(i), and $\sum_s|z_s\rangle\langle z_s| = \sum_c|\psi_c\rangle\langle\psi_c| = \mathbf{1}_{\mathfrak{h}_N}$ by Lemma 2.2(ii). Since $\mathcal{S}^*\mathcal{S}f = \sum_s z_s\langle z_s, f\rangle$, this is $\mathcal{S}^*\mathcal{S} = \mathbf{1}$. Then $\mathcal{S}\mathcal{S}^*$ is self-adjoint and $(\mathcal{S}\mathcal{S}^*)^2 = \mathcal{S}\mathcal{S}^*$, so it is an orthogonal projection; its range contains $\mathcal{S}\mathcal{S}^*\mathcal{S}\mathfrak{h}_N = \operatorname{ran}\mathcal{S}$ and is contained in $\operatorname{ran}\mathcal{S}$; and $\dim\operatorname{ran}\mathcal{S} = \dim\mathfrak{h}_N = N_\phi$, $\mathcal{S}$ being injective. The same holds for $\mathcal{S}\otimes\mathbf{1}_{\mathbb{C}^2}$. □

Proposition 3.4 (slot geometry)**.** *Let the torus be admissible.*

(i) $\pi/(\delta_x+\kappa/2) < \delta_y \le \pi/\delta_x$; in particular $\delta_y > \pi/(3\delta_0)$. The number of slots satisfies $2N_\phi \le |\Lambda_N| = n_xn_y \le (2+\kappa/\delta_x)N_\phi$.

(ii) (ball count) For all $R\in\mathbb{T}_N$ and $r\ge 0$, $\#\Lambda_r(R) \le (1+2r/\delta_x)(1+2r/\delta_y) \le C_{\mathrm{vol}}(1+r)^2$, where $\Lambda_r(R) := \{s\in\Lambda_N : d(s,R)\le r\}$ and C_{vol} depends only on δ_0.

(iii) (strip count) For every interval $I\subset\mathbb{R}/L_x\mathbb{Z}$ of length $|I|$, the strip $\mathrm{Str}(I) := \{s\in\Lambda_N : x_s\in I\}$ has at most $n_y(|I|/\delta_x+1) \le C_{\mathrm{str}}L_y(1+|I|)$ slots, with C_{str} depending only on δ_0.

Proof. (i) Since $2\delta_x/\kappa \le n_y < 2\delta_x/\kappa+1$ and $\kappa L_y = 2\pi$, we have $\delta_y = L_y/n_y \le \kappa L_y/(2\delta_x) = \pi/\delta_x$ and $\delta_y > \kappa L_y/(2\delta_x+\kappa) = \pi/(\delta_x+\kappa/2)$. By Lemma 3.2(ii), $\kappa\le\delta_x\le 2\delta_0$, so $\delta_x+\kappa/2 \le \frac{3}{2}\delta_x \le 3\delta_0$. Finally $n_x = L_x/\delta_x = \kappa N_\phi/\delta_x$, so $n_xn_y \ge n_x\cdot 2\delta_x/\kappa = 2N_\phi$ and $n_xn_y < n_x(2\delta_x/\kappa+1) = 2N_\phi + n_x = (2+\kappa/\delta_x)N_\phi$.

(ii) The lifts to $\mathbb{R}^2$ of the centre of $s=(a,\ell)$ are the points $((a+m_1n_x)\delta_x, (\ell+m_2n_y)\delta_y)$, $m_1,m_2\in\mathbb{Z}$, because $n_x\delta_x = L_x$ and $n_y\delta_y = L_y$. Hence the lifts of all slot centres form the lattice $\delta_x\mathbb{Z}\times\delta_y\mathbb{Z}$, and distinct slots have disjoint sets of lifts. Let $\tilde{R}\in\mathbb{R}^2$ be a lift of R. If $d(s,R)\le r$, some lift of r_s lies in the closed disc of radius r about $\tilde{R}$; choosing one such lift for each $s\in\Lambda_r(R)$ defines an injective map from $\Lambda_r(R)$ into the set of points of $\delta_x\mathbb{Z}\times\delta_y\mathbb{Z}$ in the square $[\tilde{R}_x-r,\tilde{R}_x+r]\times[\tilde{R}_y-r,\tilde{R}_y+r]$. A closed interval of length $2r$ contains at most $2r/\delta+1$ points of $\delta\mathbb{Z}$, so $\#\Lambda_r(R)\le(1+2r/\delta_x)(1+2r/\delta_y)$. By Lemma 3.2(ii) and (i), $\delta_x\ge\frac{2}{3}\delta_0$ and $\delta_y > \pi/(3\delta_0)$, so $1+2r/\delta_x \le \max\{1,3/\delta_0\}(1+r)$ and $1+2r/\delta_y \le \max\{1,6\delta_0/\pi\}(1+r)$; we take $C_{\mathrm{vol}} := \max\{1,3/\delta_0\}\max\{1,6\delta_0/\pi\}$.

(iii) Let I be the image of an interval $\tilde{I} \subset \mathbb{R}$ of length $|I|$. Every $a \in \mathbb{Z}_{n_x}$ with $x_a \in I$ has a lift in $\tilde{I} \cap \delta_x\mathbb{Z}$, and distinct classes have distinct lifts, so there are at most $|I|/\delta_x + 1$ such a, each contributing the n_y slots (a,ℓ), $\ell \in \mathbb{Z}_{n_y}$. Moreover $n_y < 2\delta_x/\kappa + 1 = \delta_x L_y/\pi + 1 \le (2\delta_0/\pi + \frac{1}{4})L_y$, by $\delta_x \le 2\delta_0$ and $L_y \ge 4$, and $|I|/\delta_x + 1 \le \max\{1, 3/(2\delta_0)\}(1 + |I|)$ by $\delta_x \ge \frac{2}{3}\delta_0$. So $C_{\mathrm{str}} := (2\delta_0/\pi + \frac{1}{4})\max\{1, 3/(2\delta_0)\}$ will do. □

Lemma 3.5 (uniform localization)**.** *For every $k \ge 0$ there is a constant C_k, depending only on k, η and δ_0, such that for every admissible torus, every slot $s = (a,\ell)$ and every $R = (R_x, R_y) \in \mathbb{R}^2$:*

(i) $|z_s(R)| \le C_k(1 + |R_y - y_\ell|_{L_y})^{-k} \exp(-(|R_x - x_a|_{L_x} - \delta_x)_+^2/8)$;

(ii) the same bound holds for $|(Dz_s)(R)|$;

(iii) $\|\mathbf{1}_{\Lambda_N \setminus \Lambda_r(R)}\, \mathcal{S} e_R\| \le C_k(1+r)^{-k}$ *and* $\|\mathbf{1}_{\Lambda_N \setminus \Lambda_r(R)}\, \mathcal{S} e_R^D\| \le C_k(1+r)^{-k}$ *for every $r \ge 0$, where $\mathbf{1}_X$ denotes the orthogonal projection of $\ell^2(\Lambda_N)$ onto the functions supported in X.*

Proof. It suffices to treat integer k, since $(1+t)^{-\lceil k\rceil} \le (1+t)^{-k}$ for $t \ge 0$. Fix $s = (a,\ell)$ with $a \in \{0, \dots, n_x - 1\}$ and $R \in \mathbb{R}^2$, and write

$$\Delta_x := |R_x - x_a|_{L_x}, \qquad \Delta_y := |R_y - y_\ell|_{L_y}, \qquad \xi := \kappa(R_y - y_\ell), \qquad \mathcal{I}_a := (x_a - \delta_x, x_a + \delta_x) \subset \mathbb{R}.$$

The only properties of the torus used below are those given by admissibility through Lemma 3.2(ii), namely $\delta_x/\kappa \ge 1$ and $\frac{2}{3}\delta_0 \le \delta_x \le 2\delta_0$, and $L_x \ge 4\delta_0 + 4$, which gives $L_x > 2\delta_x$ and $L_x \ge 4$.

Step 1: a representation. Let $q_0(u) := e^{-u^2/2}$ and $q_1(u) := ue^{-u^2/2}$. For $j \in \mathbb{Z}$, (2.4) with the representative j, together with $\kappa N_\phi = L_x$ and $\kappa y_\ell = 2\pi\ell/n_y$, gives

$$e^{-2\pi \mathrm{i} j\ell/n_y}\psi_{[j]}(R) = (L_y\sqrt{\pi})^{-1/2} \sum_{m\in\mathbb{Z}} e^{\mathrm{i} m L_x R_y}\, e^{\mathrm{i} j\xi}\, q_0(\kappa j + mL_x - R_x).$$

Since $D\phi_n = -(x - \kappa n)\phi_n$ and the series (2.4) may be differentiated termwise (Lemma 2.2), the same identity holds for $e^{-2\pi \mathrm{i} j\ell/n_y}(D\psi_{[j]})(R)$ with q_1 in place of q_0. For $q \in \{q_0, q_1\}$, $m \in \mathbb{Z}$ and $v \in \mathbb{R}$ let $R_x^{(m)} := R_x - mL_x$, $u_m(v) := \kappa v - R_x^{(m)}$,

$$g_m(v) := w(\kappa v/\delta_x - a)\, q(u_m(v)), \qquad S_m := \sum_{j\in\mathbb{Z}} g_m(j)\, e^{\mathrm{i} j\xi};$$

the sum is finite, since $g_m(j) = w_a(j) q(u_m(j))$ vanishes for $j \notin I_a$. Inserting the identity above into (3.1) gives

$$(n_y L_y \sqrt{\pi})^{1/2}\, z_s(R) = \sum_{m\in\mathbb{Z}} e^{\mathrm{i} m L_x R_y} S_m \qquad \text{for } q = q_0, \tag{3.2}$$

and the same expression for $(n_y L_y\sqrt{\pi})^{1/2}(Dz_s)(R)$ for $q = q_1$; the series converge absolutely, because of the Gaussian factors.

Step 2: summation by parts. For a finitely supported $g : \mathbb{Z} \to \mathbb{C}$ put $(\mathfrak{d}g)(j) := g(j) - g(j-1)$. Shifting the summation index gives $(1 - e^{\mathrm{i}\xi}) \sum_j g(j) e^{\mathrm{i} j\xi} = \sum_j (\mathfrak{d}g)(j) e^{\mathrm{i} j\xi}$, and by induction

$$(1 - e^{\mathrm{i}\xi})^k \sum_{j\in\mathbb{Z}} g(j) e^{\mathrm{i} j\xi} = \sum_{j\in\mathbb{Z}} (\mathfrak{d}^k g)(j)\, e^{\mathrm{i} j\xi}.$$

If g is the restriction to $\mathbb{Z}$ of a C^k function on $\mathbb{R}$, again denoted g, then the fundamental theorem of calculus and induction on k give $(\mathfrak{d}^k g)(j) = \int_{[0,1]^k} g^{(k)}(j - t_1 - \cdots - t_k)\, \mathrm{d}t_1 \cdots \mathrm{d}t_k$. We apply

this to $g = g_m$. By Lemma 3.2(i), w and all its derivatives vanish outside $(-1,1)$, so $g_m^{(k)}(v) = 0$ unless $\kappa v \in \mathcal{I}_a$. Hence $(\mathfrak{d}^k g_m)(j) = 0$ unless j lies in $\kappa^{-1}\mathcal{I}_a + [0,k]$, an open interval of length $2\delta_x/\kappa + k$ containing fewer than $2\delta_x/\kappa + k + 1$ integers, and $|(\mathfrak{d}^k g_m)(j)| \le \sup\{|g_m^{(k)}(v)| : \kappa v \in \mathcal{I}_a\}$.

Step 3: the Leibniz bound. Every derivative of q_0 and of q_1 is a polynomial times $e^{-u^2/2}$, so $\mu_n := \max_{i\in\{0,1\}} \sup_{u\in\mathbb{R}} e^{u^2/4}|q_i^{(n)}(u)|$ is finite for every n; let $C'_k := \max_{0\le i\le k}\|w^{(i)}\|_\infty \mu_{k-i}$, a number that depends only on k and η. Since $\frac{\mathrm{d}^i}{\mathrm{d}v^i} w(\kappa v/\delta_x - a) = (\kappa/\delta_x)^i w^{(i)}(\kappa v/\delta_x - a)$ and $\frac{\mathrm{d}^n}{\mathrm{d}v^n} q(u_m(v)) = \kappa^n q^{(n)}(u_m(v))$, the Leibniz rule gives

$$|g_m^{(k)}(v)| \le \sum_{i=0}^{k} \binom{k}{i} \Big(\frac{\kappa}{\delta_x}\Big)^i \kappa^{k-i} \|w^{(i)}\|_\infty \mu_{k-i}\, e^{-u_m(v)^2/4} \le C'_k\, \kappa^k \Big(1 + \frac{1}{\delta_x}\Big)^k e^{-u_m(v)^2/4}.$$

If $\kappa v \in \mathcal{I}_a$, then $|u_m(v)| \ge \varepsilon_m := \operatorname{dist}(R_x^{(m)}, \mathcal{I}_a)$.

Step 4: decay along the cuts. Steps 2 and 3 give

$$|1 - e^{\mathrm{i}\xi}|^k |S_m| \le \Big(\frac{2\delta_x}{\kappa} + k + 1\Big) C'_k\, \kappa^k \Big(1 + \frac{1}{\delta_x}\Big)^k e^{-\varepsilon_m^2/4},$$

and, for $k = 0$, $|S_m| \le (2\delta_x/\kappa + 1)\|w\|_\infty \mu_0 e^{-\varepsilon_m^2/4}$. Now $|1 - e^{\mathrm{i}\xi}| = 2|\sin(\xi/2)| \ge \frac{2}{\pi}|\xi|_{2\pi} = \frac{2}{\pi}\kappa\Delta_y$, because $|\sin t| \ge \frac{2}{\pi}|t|$ for $|t| \le \pi/2$ and $|\xi|_{2\pi} = \kappa |R_y - y_\ell|_{L_y}$ by $\kappa L_y = 2\pi$. With $1/\delta_x \le 3/(2\delta_0)$ and $\min\{1, \Delta_y^{-k}\} \le 2^k (1 + \Delta_y)^{-k}$ we obtain

$$|S_m| \le C''_k \Big(\frac{2\delta_x}{\kappa} + k + 1\Big)(1 + \Delta_y)^{-k} e^{-\varepsilon_m^2/4}, \qquad C''_k := 2^k \max\Big\{\|w\|_\infty \mu_0,\ \Big(\frac{\pi}{2}\Big)^k \Big(1 + \frac{3}{2\delta_0}\Big)^k C'_k\Big\}.$$

Step 5: the prefactor. Since $n_y \ge 2\delta_x/\kappa$ and $L_y = 2\pi/\kappa$, $n_y L_y \sqrt{\pi} \ge 4\pi^{3/2}\delta_x/\kappa^2$, and therefore, using $\kappa/\delta_x \le 1$ and $\delta_x \le 2\delta_0$,

$$(n_y L_y \sqrt{\pi})^{-1/2} \Big(\frac{2\delta_x}{\kappa} + k + 1\Big) \le \frac{\sqrt{\delta_x}}{2\pi^{3/4}} \Big(2 + (k+1)\frac{\kappa}{\delta_x}\Big) \le \frac{\sqrt{2\delta_0}}{2\pi^{3/4}}\,(k+3).$$

Step 6: the images. Choose $m_0 \in \mathbb{Z}$ with $|R_x^{(m_0)} - x_a| = \Delta_x \le L_x/2$. Since $\mathcal{I}_a$ is the open interval of radius δ_x about x_a, $\varepsilon_m = (|R_x^{(m)} - x_a| - \delta_x)_+$, so $\varepsilon_{m_0} = (\Delta_x - \delta_x)_+$. For $n := |m - m_0| \ge 1$, $|R_x^{(m)} - x_a| \ge nL_x - \Delta_x = (n-1)L_x + (L_x - \Delta_x)$, and $L_x - \Delta_x - \delta_x \ge (\Delta_x - \delta_x)_+$ because $L_x - \Delta_x \ge \Delta_x$ and $L_x - \Delta_x - \delta_x \ge L_x/2 - \delta_x > 0$. Hence $\varepsilon_m \ge (n-1)L_x + (\Delta_x - \delta_x)_+$, so $\varepsilon_m^2 \ge (n-1)^2 L_x^2 + (\Delta_x - \delta_x)_+^2$, and with $L_x \ge 4$,

$$\sum_{m\in\mathbb{Z}} e^{-\varepsilon_m^2/4} \le e^{-(\Delta_x - \delta_x)_+^2/4} \Big(1 + 2\sum_{n\ge 0} e^{-4n^2}\Big) \le 4\, e^{-(\Delta_x - \delta_x)_+^2/4}.$$

Proof of (i) and (ii). By (3.2) and Steps 4–6, for both choices of q,

$$|z_s(R)|,\ |(Dz_s)(R)| \ \le\ \frac{2\sqrt{2\delta_0}}{\pi^{3/4}}\,(k+3)\,C''_k\,(1+\Delta_y)^{-k} e^{-(\Delta_x - \delta_x)_+^2/4},$$

which is (i) and (ii) with $C_k := 2\sqrt{2\delta_0}\,\pi^{-3/4}(k+3)C''_k$, and with the stronger Gaussian factor $e^{-(\Delta_x - \delta_x)_+^2/4}$.

Proof of (iii). By Lemma 2.3(i), $(\mathcal{S}e_R)_s = \langle z_s, e_R\rangle = \overline{\langle e_R, z_s\rangle} = \overline{z_s(R)}$ and likewise $(\mathcal{S}e_R^D)_s = \overline{(Dz_s)(R)}$, so that $\|\mathbf{1}_{\Lambda_N\setminus\Lambda_r(R)}\mathcal{S}e_R\|^2 = \sum_{s:\, d(s,R)>r} |z_s(R)|^2$, and the same for e_R^D with Dz_s. The

flat distance between r_s and the image of R in $\mathbb{T}_N$ is $d(s,R) = (\Delta_x^2 + \Delta_y^2)^{1/2} \le \Delta_x + \Delta_y$. Let $\tilde{C}_{k'} := \sup_{t\ge 0}(1+t)^{k'} e^{-(t-2\delta_0)_+^2/8}$, which is finite and depends only on k' and δ_0. Since $(t-\delta_x)_+ \ge (t-2\delta_0)_+$ and $(1+\Delta_x)(1+\Delta_y) \ge 1 + d(s,R)$, (i) and (ii) of order k' give $|z_s(R)|, |(Dz_s)(R)| \le C_{k'}\tilde{C}_{k'}(1+d(s,R))^{-k'}$. Take $k' = k+2$ and let $n_0 := \lfloor r \rfloor$, so that $1 + n_0 \ge (1+r)/2$. Every s with $d(s,R) > r$ lies in one of the shells $\{n < d(s,R) \le n+1\}$, $n \ge n_0$, and by Proposition 3.4(ii) the nth shell has at most $C_{\mathrm{vol}}(2+n)^2 \le 4C_{\mathrm{vol}}(1+n)^2$ elements. Hence

$$\begin{aligned}\sum_{s:\, d(s,R)>r} |z_s(R)|^2 &\le 4C_{\mathrm{vol}}\, C_{k+2}^2 \tilde{C}_{k+2}^2 \sum_{n\ge n_0} (1+n)^{-2k-2} \\ &\le 8C_{\mathrm{vol}}\, C_{k+2}^2 \tilde{C}_{k+2}^2 (1+n_0)^{-2k-1} \le 2^{2k+4} C_{\mathrm{vol}}\, C_{k+2}^2 \tilde{C}_{k+2}^2 (1+r)^{-2k},\end{aligned}$$

where we used $\sum_{n\ge n_0}(1+n)^{-2k-2} \le (1+n_0)^{-2k-2} + \int_{n_0}^{\infty}(1+t)^{-2k-2}\mathrm{d}t \le 2(1+n_0)^{-2k-1}$. The same bound holds with Dz_s. This is (iii) with the constant $2^{k+2}C_{\mathrm{vol}}^{1/2} C_{k+2}\tilde{C}_{k+2}$, which depends only on k, η and δ_0; we rename the larger of the constants in (i)–(iii) C_k. □

The decay in (i) is Gaussian across the cuts and faster than any power along them, uniformly in N; the Gaussian decay across the cuts reflects the exact support property of the next lemma.

Lemma 3.6 (exact column support)**.** *For every admissible torus, $j \in \mathbb{Z}$ and $s = (a,\ell)$: $\langle \psi_{[j]}, z_s\rangle = n_y^{-1/2} w_a(j') e^{-2\pi \mathrm{i} j'\ell/n_y}$ if some $j' \in I_a$ lies in the class of j (it is then unique), and $\langle \psi_{[j]}, z_s\rangle = 0$ otherwise. Consequently $\mathcal{S}\psi_{[j]}$ is supported in the* column

$$\mathrm{Col}(j) := \{s \in \Lambda_N : |x_s - \kappa j|_{L_x} < \delta_x\} = \mathrm{Str}\big((\kappa j - \delta_x, \kappa j + \delta_x)\big),$$

a strip of width $2\delta_x$ containing the slots of at most two windows.

Proof. By (3.1), $\langle \psi_{[j]}, z_s\rangle = n_y^{-1/2}\sum_{j''\in I_a} w_a(j'') e^{-2\pi \mathrm{i} j''\ell/n_y}\langle \psi_{[j]}, \psi_{[j'']}\rangle$, and by Lemma 2.2(ii) the inner product $\langle \psi_{[j]}, \psi_{[j'']}\rangle$ is 1 if $[j''] = [j]$ and 0 otherwise. By Lemma 3.2(iii) at most one element j' of I_a lies in the class of j; this gives the formula. If $(\mathcal{S}\psi_{[j]})_s = \overline{\langle \psi_{[j]}, z_s\rangle} \ne 0$, there is $m \in \mathbb{Z}$ with $j' = j + mN_\phi \in I_a$, that is $|\kappa j + mL_x - x_a| < \delta_x$, so $|x_s - \kappa j|_{L_x} < \delta_x$ and $s \in \mathrm{Col}(j)$. The condition $|x_s - \kappa j|_{L_x} < \delta_x$ says that x_s lies in the image in $\mathbb{R}/L_x\mathbb{Z}$ of the interval $(\kappa j - \delta_x, \kappa j + \delta_x)$, which is the second description of $\mathrm{Col}(j)$. Finally, the lifts of the window centres form $\delta_x\mathbb{Z}$, an open interval of length $2\delta_x$ contains at most two points of $\delta_x\mathbb{Z}$, and since $2\delta_x < L_x$ distinct classes a with x_a in this arc have distinct lifts in the interval; so $\mathrm{Col}(j)$ consists of the slots of at most two windows. □

3.2 The slot Fock space and the embedding

Definition 3.7 (slot Fock space, local algebras, the embedding)**.** The *slot Fock space* is $\widehat{\mathcal{H}}_N := \mathcal{F}(\ell^2(\Lambda_N)\otimes\mathbb{C}^2)$, with annihilation operators $\hat{c}(g)$, $g \in \ell^2(\Lambda_N)\otimes\mathbb{C}^2$, and $\hat{c}_{s\sigma} := \hat{c}(\delta_s \otimes \sigma)$, where δ_s is the indicator of the slot s; $\widehat{N}_\sigma := \sum_s \hat{c}_{s\sigma}^*\hat{c}_{s\sigma}$. For $X \subset \Lambda_N$, $\mathrm{CAR}(X)$ is the unital $*$-algebra generated by the $\hat{c}_{s\sigma}$ with $s \in X$, and $\mathfrak{A}(X)$ is its even part, the elements commuting with $(-1)^{\widehat{N}_\uparrow + \widehat{N}_\downarrow}$. For $r \ge 0$ the *r-fattening* of X is $X_{(r)} := \{s \in \Lambda_N : d(s,X) \le r\}$. The *embedding* ι is the unital $*$-homomorphism from the operators on $\mathcal{H}_N$ to the operators on $\widehat{\mathcal{H}}_N$ determined by

$$\iota(c(f)) = \hat{c}(\mathcal{S}f), \qquad f \in \mathfrak{k}_N.$$

For an operator H on $\mathcal{H}_N$ we write $\widehat{H} := \iota(H)$.

Elements of $\mathfrak{A}(X)$ and of $\mathfrak{A}(Y)$ commute when $X \cap Y = \emptyset$, and an element of $\mathfrak{A}(X)$ commutes with every element of $\mathrm{CAR}(Y)$ in that case. Indeed, conjugation by $(-1)^{\widehat{N}_\uparrow + \widehat{N}_\downarrow}$ changes the sign of every $\hat{c}_{s\sigma}$ and of every $\hat{c}^*_{s\sigma}$, so it multiplies a monomial of degree n in these operators by $(-1)^n$; since $\mathrm{CAR}(X)$ is spanned by the monomials in the $\hat{c}_{s\sigma}, \hat{c}^*_{s\sigma}$ with $s \in X$, its even part $\mathfrak{A}(X)$ is spanned by those of even degree. By the canonical anticommutation relations the generators belonging to distinct slots anticommute, so a monomial of even degree in the generators of X commutes with every generator of Y, hence with $\mathrm{CAR}(Y)$.

Lemma 3.8 (properties of the embedding)**.** *Let the torus be admissible.*

(i) ι *is well defined, unital, injective and isometric, and it maps even operators to even operators. With* $\mathcal{F}(\mathcal{S}) : \mathcal{H}_N \to \widehat{\mathcal{H}}_N$ *the second quantization of the isometry* $\mathcal{S}$*, an isometry onto the subspace of* $\widehat{\mathcal{H}}_N$ *in which the modes orthogonal to* $\operatorname{ran}\mathcal{S}$ *are empty,* $\iota(A)\mathcal{F}(\mathcal{S}) = \mathcal{F}(\mathcal{S})A$ *for every operator* A *on* $\mathcal{H}_N$*.*

(ii) For every self-adjoint A *on* $\mathcal{H}_N$*,* $\iota(A)$ *has the same spectrum as* A *and* $\iota(f(A)) = f(\iota(A))$ *for every function* f *on that spectrum. In particular* $\iota(e^{\mathrm{i}tH}Ae^{-\mathrm{i}tH}) = e^{\mathrm{i}t\widehat{H}}\iota(A)e^{-\mathrm{i}t\widehat{H}}$*, and* ι *maps the spectral projections of* H *to those of* $\widehat{H}$*.*

(iii) For $\varphi = (\varphi_\uparrow, \varphi_\downarrow) \in \mathbb{R}^2$*,* $\iota\big(e^{\mathrm{i}(\varphi_\uparrow N_\uparrow + \varphi_\downarrow N_\downarrow)}Ae^{-\mathrm{i}(\varphi_\uparrow N_\uparrow + \varphi_\downarrow N_\downarrow)}\big) = e^{\mathrm{i}(\varphi_\uparrow \widehat{N}_\uparrow + \varphi_\downarrow \widehat{N}_\downarrow)}\iota(A)e^{-\mathrm{i}(\varphi_\uparrow \widehat{N}_\uparrow + \varphi_\downarrow \widehat{N}_\downarrow)}$*, and these gauge transformations map every* $\mathrm{CAR}(X)$ *and* $\mathfrak{A}(X)$ *onto itself. Consequently, if* A *is sector-preserving (respectively self-adjoint), then for every* $X \subset \Lambda_N$ *and every* $A' \in \mathfrak{A}(X)$ *there is* $A'' \in \mathfrak{A}(X)$ *that commutes with* $\widehat{N}_\uparrow$ *and* $\widehat{N}_\downarrow$ *(respectively is self-adjoint, respectively both, if* A *is both) and satisfies* $\|\iota(A) - A''\| \le \|\iota(A) - A'\|$*.*

Proof. Let $\mathfrak{r} := \operatorname{ran}\mathcal{S} \subset \ell^2(\Lambda_N)\otimes\mathbb{C}^2$, $n_1 := \dim \mathfrak{k}_N = 2N_\phi$ and $n_2 := 2|\Lambda_N|$. Choose an orthonormal basis $f_1, \dots, f_{n_1}$ of $\mathfrak{k}_N$ and put $g_i := \mathcal{S}f_i$ for $i \le n_1$. These are orthonormal by Proposition 3.3; complete them to an orthonormal basis $g_1, \dots, g_{n_2}$ of $\ell^2(\Lambda_N) \otimes \mathbb{C}^2$, so that $g_{n_1+1}, \dots, g_{n_2}$ is an orthonormal basis of $\mathfrak{r}^\perp$. Let Ω_0 and $\widehat{\Omega}_0$ be the vacua of $\mathcal{H}_N$ and $\widehat{\mathcal{H}}_N$, and let $\mathcal{F}(\mathfrak{r}^\perp)$ be the Fock space over $\mathfrak{r}^\perp$, with vacuum Ω'' and annihilation operators $a(e)$. For $\mathbf{m} = (m_1, \dots, m_{n_2}) \in \{0,1\}^{n_2}$ the vectors $\widehat{e}_{\mathbf{m}} := \hat{c}(g_1)^{*m_1} \cdots \hat{c}(g_{n_2})^{*m_{n_2}}\widehat{\Omega}_0$ form an orthonormal basis of $\widehat{\mathcal{H}}_N$ (the occupation-number basis), and likewise the vectors $e_{\mathbf{m}'} := c(f_1)^{*m_1} \cdots c(f_{n_1})^{*m_{n_1}}\Omega_0$, $\mathbf{m}' \in \{0,1\}^{n_1}$, for $\mathcal{H}_N$, and the vectors $e''_{\mathbf{m}''} := a(g_{n_1+1})^{*m_{n_1+1}} \cdots a(g_{n_2})^{*m_{n_2}}\Omega''$, $\mathbf{m}'' \in \{0,1\}^{n_2-n_1}$, for $\mathcal{F}(\mathfrak{r}^\perp)$. Let $\mathcal{V} : \widehat{\mathcal{H}}_N \to \mathcal{H}_N \otimes \mathcal{F}(\mathfrak{r}^\perp)$ be the unitary with $\mathcal{V}\widehat{e}_{(\mathbf{m}',\mathbf{m}'')} = e_{\mathbf{m}'} \otimes e''_{\mathbf{m}''}$. For $i \le n_1$, the canonical anticommutation relations give $\hat{c}(g_i)\widehat{e}_{\mathbf{m}} = 0$ if $m_i = 0$ and $\hat{c}(g_i)\widehat{e}_{\mathbf{m}} = (-1)^{m_1+\cdots+m_{i-1}}\widehat{e}_{\mathbf{m}-\mathbf{1}_i}$ if $m_i = 1$, with $\mathbf{1}_i$ the ith unit vector; $c(f_i)$ acts on the $e_{\mathbf{m}'}$ by the same formula, with the same sign, which involves only $m_1, \dots, m_{i-1}$. Hence $\mathcal{V}\hat{c}(g_i)\mathcal{V}^* = c(f_i) \otimes \mathbf{1}$ for $i \le n_1$, and since $\mathcal{S}f = \sum_i \langle f_i, f\rangle g_i$ and both c and $\hat{c}$ are conjugate-linear,

$$\mathcal{V}\hat{c}(\mathcal{S}f)\mathcal{V}^* = c(f) \otimes \mathbf{1} \qquad (f \in \mathfrak{k}_N). \tag{3.3}$$

(i) Define $\iota(A) := \mathcal{V}^*(A \otimes \mathbf{1})\mathcal{V}$. This is a unital, injective, isometric $*$-homomorphism, and $\iota(c(f)) = \hat{c}(\mathcal{S}f)$ by (3.3). It is the only unital $*$-homomorphism with this property, because the operators $c(f)$ generate all operators on $\mathcal{H}_N$ as a unital $*$-algebra: $|\Omega_0\rangle\langle\Omega_0| = \prod_{i=1}^{n_1}(\mathbf{1} - c(f_i)^*c(f_i))$, and the matrix units

$$|e_{\mathbf{a}}\rangle\langle e_{\mathbf{b}}| = c(f_1)^{*a_1} \cdots c(f_{n_1})^{*a_{n_1}}\,|\Omega_0\rangle\langle\Omega_0|\,\big(c(f_1)^{*b_1} \cdots c(f_{n_1})^{*b_{n_1}}\big)^* \qquad (\mathbf{a}, \mathbf{b} \in \{0,1\}^{n_1})$$

span the operators on $\mathcal{H}_N$. The number operator $\widehat{N}_\uparrow + \widehat{N}_\downarrow$ does not depend on the choice of orthonormal basis, so it equals $\sum_i \hat{c}(g_i)^*\hat{c}(g_i)$ and acts on $\widehat{e}_{\mathbf{m}}$ as multiplication by $\sum_i m_i$.

Therefore $\mathcal{V}(-1)^{\widehat{N}_\uparrow+\widehat{N}_\downarrow}\mathcal{V}^* = (-1)^{N_\uparrow+N_\downarrow}\otimes(-1)^{N''}$, with N'' the number operator of $\mathcal{F}(\mathfrak{r}^\perp)$, and ι maps even operators to even operators. The second quantization $\mathcal{F}(\mathcal{S})$ is determined by $\mathcal{F}(\mathcal{S})c(h_1)^*\cdots c(h_p)^*\Omega_0 = \hat{c}(\mathcal{S}h_1)^*\cdots\hat{c}(\mathcal{S}h_p)^*\widehat{\Omega}_0$; by (3.3), $\mathcal{V}\mathcal{F}(\mathcal{S})\Upsilon = \Upsilon\otimes\Omega''$ for every $\Upsilon\in\mathcal{H}_N$, so $\mathcal{F}(\mathcal{S})$ is a well-defined isometry with range $\mathcal{V}^*(\mathcal{H}_N\otimes\mathbb{C}\Omega'')$, the span of the $\widehat{e}_{(\mathbf{m}',0)}$. Since $\|\hat{c}(g_i)\Upsilon\|^2 = \langle\Upsilon,\hat{c}(g_i)^*\hat{c}(g_i)\Upsilon\rangle$ and $\hat{c}(g_i)^*\hat{c}(g_i)\widehat{e}_{\mathbf{m}} = m_i\widehat{e}_{\mathbf{m}}$, this span is the subspace on which every $\hat{c}(e)$, $e\in\mathfrak{r}^\perp$, vanishes: the subspace in which the modes orthogonal to $\operatorname{ran}\mathcal{S}$ are empty. Finally $\iota(A)\mathcal{F}(\mathcal{S})\Upsilon = \mathcal{V}^*(A\otimes\mathbf{1})(\Upsilon\otimes\Omega'') = \mathcal{V}^*(A\Upsilon\otimes\Omega'') = \mathcal{F}(\mathcal{S})A\Upsilon$.

(ii) $\iota(A)$ is unitarily equivalent to $A\otimes\mathbf{1}$, which has the spectrum of A, and $f(\iota(A)) = \mathcal{V}^*f(A\otimes\mathbf{1})\mathcal{V} = \mathcal{V}^*(f(A)\otimes\mathbf{1})\mathcal{V} = \iota(f(A))$. The two particular statements are the cases $f(x) = e^{\pm itx}$, together with the multiplicativity of ι, and $f = \mathbf{1}_B$ for Borel sets B.

(iii) Write $\varphi\cdot\mathbf{N} := \varphi_\uparrow N_\uparrow + \varphi_\downarrow N_\downarrow$ and $\varphi\cdot\widehat{\mathbf{N}} := \varphi_\uparrow\widehat{N}_\uparrow + \varphi_\downarrow\widehat{N}_\downarrow$. The anticommutation relations give $[N_\sigma, c(h\otimes\sigma')] = -\delta_{\sigma\sigma'}c(h\otimes\sigma')$ for $h\in\mathfrak{h}_N$, hence $e^{\mathrm{i}\varphi\cdot\mathbf{N}}c(h\otimes\sigma')e^{-\mathrm{i}\varphi\cdot\mathbf{N}} = e^{-\mathrm{i}\varphi_{\sigma'}}c(h\otimes\sigma')$, and in the same way $e^{\mathrm{i}\varphi\cdot\widehat{\mathbf{N}}}\hat{c}(g\otimes\sigma')e^{-\mathrm{i}\varphi\cdot\widehat{\mathbf{N}}} = e^{-\mathrm{i}\varphi_{\sigma'}}\hat{c}(g\otimes\sigma')$ for $g\in\ell^2(\Lambda_N)$. The maps $A\mapsto\iota(e^{\mathrm{i}\varphi\cdot\mathbf{N}}Ae^{-\mathrm{i}\varphi\cdot\mathbf{N}})$ and $A\mapsto e^{\mathrm{i}\varphi\cdot\widehat{\mathbf{N}}}\iota(A)e^{-\mathrm{i}\varphi\cdot\widehat{\mathbf{N}}}$ are unital $*$-homomorphisms which, since $\iota(c(h\otimes\sigma')) = \hat{c}(\mathcal{S}h\otimes\sigma')$, agree on the generators $c(h\otimes\sigma')$; by the uniqueness in (i) they are equal. Conjugation by $e^{\mathrm{i}\varphi\cdot\widehat{\mathbf{N}}}$ multiplies each $\hat{c}_{s\sigma}$ by a phase, so it maps $\mathrm{CAR}(X)$ into itself, and onto itself since its inverse is the conjugation with $-\varphi$; it commutes with the parity, so it maps $\mathfrak{A}(X)$ onto itself.

Now let $A'\in\mathfrak{A}(X)$. If A is sector-preserving, then A commutes with $e^{\mathrm{i}\varphi\cdot\mathbf{N}}$, hence $\iota(A)$ commutes with $e^{\mathrm{i}\varphi\cdot\widehat{\mathbf{N}}}$ for all φ. Since $\widehat{N}_\uparrow$ and $\widehat{N}_\downarrow$ have integer spectra, $e^{\mathrm{i}\varphi\cdot\widehat{\mathbf{N}}}$ is 2π-periodic in $\varphi_\uparrow$ and in $\varphi_\downarrow$; let

$$A'' := \frac{1}{4\pi^2}\int_{[0,2\pi]^2} e^{\mathrm{i}\varphi\cdot\widehat{\mathbf{N}}}A'e^{-\mathrm{i}\varphi\cdot\widehat{\mathbf{N}}}\,\mathrm{d}\varphi.$$

Then $A''\in\mathfrak{A}(X)$, a finite-dimensional space mapped into itself by each of these conjugations; $\|\iota(A)-A''\| = \|(4\pi^2)^{-1}\int e^{\mathrm{i}\varphi\cdot\widehat{\mathbf{N}}}(\iota(A)-A')e^{-\mathrm{i}\varphi\cdot\widehat{\mathbf{N}}}\,\mathrm{d}\varphi\| \le \|\iota(A)-A'\|$; and, by the invariance of the integral under translations of φ, A'' commutes with $e^{\mathrm{i}\varphi'\cdot\widehat{\mathbf{N}}}$ for all $\varphi'\in\mathbb{R}^2$, hence with $\widehat{N}_\uparrow$ and $\widehat{N}_\downarrow$ (differentiate at $\varphi'=0$). If A is self-adjoint, $A'' := \frac{1}{2}(A'+A'^*)\in\mathfrak{A}(X)$ satisfies $\|\iota(A)-A''\| = \|\frac{1}{2}(\iota(A)-A') + \frac{1}{2}(\iota(A)-A')^*\| \le \|\iota(A)-A'\|$. If A is both, apply the first construction and then the second; the self-adjoint part of an operator commuting with $\widehat{N}_\uparrow$ and $\widehat{N}_\downarrow$ commutes with them. □

The operators $\iota(A)$ are the *physical* operators. Spectra, traces, sectors and spectral projections are always those of operators on $\mathcal{H}_N$; operators in the algebras $\mathfrak{A}(X)$, which in general are not physical, appear only as approximants of physical operators.

3.3 Locality in the slot lattice

Definition 3.9 (strips, columns)**.** For an interval $I\subset\mathbb{R}/L_x\mathbb{Z}$, $\mathrm{Str}(I)$ is the strip of Proposition 3.4(iii), and I_ρ denotes the closed ρ-neighbourhood of I. For a set $\mathcal{B}\subset\mathbb{Z}_{N_\phi}$ of orbitals (a *band*), $\mathrm{Col}(\mathcal{B}) := \bigcup_{j\in\mathcal{B}}\mathrm{Col}(j)$.

Lemma 3.10 (strips and fattenings)**.** *Let the torus be admissible and let $I\subset\mathbb{R}/L_x\mathbb{Z}$ be an open interval of length at least $2\delta_x$. Then for every $\rho\ge 0$,*

$$\mathrm{Str}(I)_{(\rho)}\subset\mathrm{Str}(I_\rho)\subset\mathrm{Str}(I)_{(\rho+\delta_x)}.$$

Proof. Let I be the image of an open interval $(\underline{x},\overline{x})\subset\mathbb{R}$ with $\overline{x}-\underline{x}\ge 2\delta_x$ (if $\overline{x}-\underline{x}>L_x$, then I is the whole circle). For two slots, $d(s,s')^2 = |x_s-x_{s'}|_{L_x}^2 + |y_s-y_{s'}|_{L_y}^2$, where y_s denotes the second coordinate of r_s; in particular $|x_s-x_{s'}|_{L_x}\le d(s,s')$.

First inclusion: if $d(s,s') \le \rho$ for some $s' \in \mathrm{Str}(I)$, then $\mathrm{dist}(x_s, I) \le |x_s - x_{s'}|_{L_x} \le \rho$, so $x_s \in I_\rho$.

Second inclusion: we first show that every point x of the closure $\bar{I}$ is within distance δ_x of the centre $x_{a'}$ of some window with $x_{a'} \in I$. Let $\tilde{x} \in [\underline{x}, \overline{x}]$ be a lift of x. The half-open intervals $(\underline{x}, \underline{x} + \delta_x]$ and $[\overline{x} - \delta_x, \overline{x})$ are contained in $(\underline{x}, \overline{x})$, because $\overline{x} - \underline{x} > \delta_x$, and each contains a point of $\delta_x\mathbb{Z}$. Let $\underline{v}$ and $\overline{v}$ be the smallest and the largest point of $\delta_x\mathbb{Z} \cap (\underline{x}, \overline{x})$; then $\underline{v} \le \underline{x} + \delta_x$, $\overline{v} \ge \overline{x} - \delta_x$, and consecutive points of $\delta_x\mathbb{Z} \cap (\underline{x}, \overline{x})$ are δ_x apart. If $\tilde{x} \le \underline{v}$, then $\underline{v} - \tilde{x} \le \delta_x$; if $\tilde{x} \ge \overline{v}$, then $\tilde{x} - \overline{v} \le \delta_x$; otherwise $\tilde{x}$ lies between two consecutive points and is within $\delta_x/2$ of one of them. In each case some $v \in \delta_x\mathbb{Z} \cap (\underline{x}, \overline{x})$ satisfies $|v - \tilde{x}| \le \delta_x$, and its image is a window centre $x_{a'} \in I$ with $|x - x_{a'}|_{L_x} \le \delta_x$.

Now let $s = (a, \ell) \in \mathrm{Str}(I_\rho)$. The distance $\mathrm{dist}(x_s, I) \le \rho$ is attained at a point $x \in \bar{I}$, and with $x_{a'}$ as above, $s' := (a', \ell) \in \mathrm{Str}(I)$ and $d(s,s') = |x_a - x_{a'}|_{L_x} \le \rho + \delta_x$. Hence $s \in \mathrm{Str}(I)_{(\rho+\delta_x)}$. (The proof uses only $\overline{x} - \underline{x} > \delta_x$; the lemma therefore holds for open intervals of length greater than δ_x.) □

Definition 3.11 (almost supported operators)**.** Let $\mathcal{N} \subset \mathbb{N}$ be infinite, let $(\mathbb{T}_N)_{N\in\mathcal{N}}$ be admissible tori, and let $Z = (Z_N)_{N\in\mathcal{N}}$ with $Z_N \subset \Lambda_N$. A sequence $A = (A_N)_{N\in\mathcal{N}}$ of operators A_N on $\mathcal{H}_N$ is *almost supported in* Z, written $A \in \mathcal{A}_Z$, if there are $m \ge 0$ and, for every $k \in \mathbb{N}$, a constant C_k such that for all $N \in \mathcal{N}$ and $r \ge 0$ there is $A_{N,r} \in \mathfrak{A}((Z_N)_{(r)})$ with

$$\|\iota(A_N) - A_{N,r}\| \le C_k N^m (1+r)^{-k}.$$

Such an m is a *prefactor exponent* of A. A family of such sequences depending on a parameter is almost supported *uniformly* in the parameter if one m and one set of constants C_k serve for all values of the parameter. For an operator A on $\widehat{\mathcal{H}}_N$ and $X \subset \Lambda_N$, the *set norm* of order k is $\|A\|_{X;k} := \sup_{r\ge 0}(1+r)^k \inf\{\|A - A'\| : A' \in \mathfrak{A}(X_{(r)})\}$.

Since $\mathfrak{A}(X_{(r)})$ consists of even operators, only even operators can be almost supported in this sense; every operator to which the definition is applied below is even.

Remark 3.12 (comparison with [BBDF21])*.* In [BBDF21, Section 2.3.1 and Appendix C] the error is required to be $\|A_N\||Z_N|O(r^{-\infty})$. Since $\|A_N\||Z_N|$ is bounded by a power of N for every operator considered here, operators almost supported in their sense are almost supported in the sense of Definition 3.11. The relaxed form is needed because propagation from a strip produces factors of the size of the strip, which is of order L_y times its width, and because filters and products raise prefactors by fixed powers of N; the class of Definition 3.11 is closed under these operations (Section 5). Every almost-support estimate of the proof is finally used at a distance of order $\sqrt{N}$, where $N^m(1+r)^{-k} = O(N^{-\infty})$ for every fixed m (Lemma 5.13). At fixed aspect ratio the diameter of Λ_N, the distance between the two cuts and the square root of the number of slots are all of order $\sqrt{N}$, so $O(N^{-\infty})$ is the same as the $O(L^{-\infty})$ of [BBDF21] with L the diameter.

4 The Hamiltonian in the slot frame

4.1 Triplet pairs and the local form

Definition 4.1 (triplet pairs, local terms, centre pair operators)**.** The *triplet spinors* are the symmetric matrices $\chi^\nu = (\chi^\nu_{\sigma\sigma'})_{\sigma,\sigma'\in\{\uparrow,\downarrow\}}$, $\nu \in \{+,0,-\}$, with $\chi^+_{\uparrow\uparrow} = 1$, $\chi^-_{\downarrow\downarrow} = 1$, $\chi^0_{\uparrow\downarrow} = \chi^0_{\downarrow\uparrow} = 2^{-1/2}$, and all other entries zero. For $f, g \in \mathfrak{h}_N$ and $h, h' \in \ell^2(\Lambda_N)$ let

$$c^\nu(f,g) := \sum_{\sigma,\sigma'} \overline{\chi^\nu_{\sigma\sigma'}}\, c(f\otimes\sigma')\, c(g\otimes\sigma), \qquad \hat{c}^\nu(h,h') := \sum_{\sigma,\sigma'} \overline{\chi^\nu_{\sigma\sigma'}}\, \hat{c}(h\otimes\sigma')\, \hat{c}(h'\otimes\sigma),$$

so that $c^\nu(g,f) = -c^\nu(f,g)$ and $\iota(c^\nu(f,g)) = \hat{c}^\nu(\mathcal{S}f, \mathcal{S}g)$. For $R \in \mathbb{R}^2$ let

$$B_\nu(R) := c^\nu(e_R, e_R^D), \qquad h_R := 8\pi \sum_{\nu\in\{+,0,-\}} B_\nu(R)^* B_\nu(R).$$

For $n \in \mathbb{Z}$ let

$$b_n^\nu := \sum_{d\in\mathbb{N},\ d\equiv n \pmod 2} 2d\, e^{-\kappa^2 d^2/4}\, c^\nu\big(\psi_{[(n-d)/2]}, \psi_{[(n+d)/2]}\big),$$

an absolutely convergent sum with $b^\nu_{n+2N_\phi} = b^\nu_n$.

The antisymmetry follows from the symmetry of χ^ν and the anticommutation of the annihilation operators: $c^\nu(g,f) = -\sum_{\sigma\sigma'} \overline{\chi^\nu_{\sigma\sigma'}} c(f\otimes\sigma)c(g\otimes\sigma') = -c^\nu(f,g)$; in particular $c^\nu(f,f) = 0$. Both c^ν and $\hat{c}^\nu$ are conjugate-linear in each argument, and $\iota(c^\nu(f,g)) = \hat{c}^\nu(\mathcal{S}f,\mathcal{S}g)$ because $\iota(c(f\otimes\sigma)) = \hat{c}(\mathcal{S}f\otimes\sigma)$. The series defining b^ν_n converges absolutely since $\|c^\nu(f,g)\| \le \sqrt{2}\|f\|\|g\|$ (see the proof of Proposition 4.4), and $b^\nu_{n+2N_\phi} = b^\nu_n$ because $(n+2N_\phi \pm d)/2 = (n\pm d)/2 + N_\phi$.

By Lemma 2.3(i), $B_\nu(R+(0,L_y)) = B_\nu(R)$ and $B_\nu(R+(L_x,0)) = e^{2\mathrm{i}L_xR_y}B_\nu(R)$, so h_R is a function of $R \in \mathbb{T}_N$. By conjugate linearity, $c(f\otimes\sigma) = \sum_j \langle f,\psi_j\rangle c_{j\sigma}$ for $f \in \mathfrak{h}_N$; in particular $c(e_R\otimes\sigma) = \sum_j \psi_j(R)c_{j\sigma}$ and $c(e_R^D\otimes\sigma) = \sum_j (D\psi_j)(R)c_{j\sigma}$, so that $B_\nu(R) = \sum_{j,j'} \psi_j(R)\,(D\psi_{j'})(R)\,c^\nu(\psi_j,\psi_{j'})$. The functions ψ_j and $D\psi_{j'}$ are smooth on $\mathbb{R}^2$ (Lemma 2.2(i)), so $R \mapsto B_\nu(R)$ is a finite linear combination of fixed operators with smooth coefficients, and h_R is a finite linear combination of fixed operators whose coefficients are products of these coefficients and their complex conjugates; the map $R \mapsto h_R$ is therefore smooth. In the plane, the analogous operator $B_\nu(R)^*$ creates from the vacuum a spin-triplet pair of relative angular momentum one whose centre of mass is in the coherent state at R; this motivates the definition and is not used below. The operator b^ν_n annihilates triplet pairs of orbitals whose centres are symmetric about the half-integer orbital position $n/2$.

Theorem 4.2 (the exact local form)**.** *For every torus as in Definition 2.1:*

(a) $H_N = 8\pi \int_{\mathbb{T}_N} \sum_\nu B_\nu(R)^* B_\nu(R)\,\mathrm{d}R = \int_{\mathbb{T}_N} h_R\,\mathrm{d}R$, *as operators on* $\mathcal{H}_N$.

(b) For $R = (R_x,R_y) \in \mathbb{R}^2$,

$$B_\nu(R) = \frac{\kappa}{2L_y\sqrt{\pi}} \sum_{n\in\mathbb{Z}} e^{\mathrm{i}\kappa n R_y} e^{-(R_x-\kappa n/2)^2}\, b^\nu_n.$$

(c) $H_N = \dfrac{\kappa^3}{\sqrt{2\pi}} \displaystyle\sum_{n\in\mathbb{Z}_{2N_\phi}} \sum_\nu (b^\nu_n)^*\, b^\nu_n.$

The proof of (a) rests on the following identity for pair wave functions, in which the symmetric gauge centred at R is used; the proof of the theorem follows that of the lemma.

Lemma 4.3 (jets on the diagonal)**.** *Let* $U \subset \mathbb{R}^2\times\mathbb{R}^2$ *be an open neighbourhood of the diagonal, and let* $\Phi : U \to \mathbb{C}$ *be smooth and of the lowest Landau level in each variable,* $(D_{x_1} + \mathrm{i}D_{y_1})\Phi = (D_{x_2}+\mathrm{i}D_{y_2})\Phi = 0$, *where* D_{x_i}, D_{y_i} *are the covariant derivatives* (2.1) *in* $r_i = (x_i,y_i)$. *For* $R \in \mathbb{R}^2$ *and small* $r \in \mathbb{R}^2$ *let* $F_R(r) := |\Phi(R + r/2, R - r/2)|^2$.

(a) If $\Phi(r_1,r_2) = \Phi(r_2,r_1)$, *then* $F_R(0) + (\nabla^2 F_R)(0) = 0$ *for every* R.

(b) If $\Phi(r_1,r_2) = -\Phi(r_2,r_1)$, *then* $F_R(0) = 0$ *and* $(\nabla^2 F_R)(0) = 4|(D_1\Phi)(R,R)|^2$ *for every* R, *where* $D_1 := \frac{1}{2}(D_{x_1} - \mathrm{i}D_{y_1})$.

Proof. Fix $R = (R_x, R_y)$ and let $\varsigma_R(x,y) := \frac{1}{2}(x+R_x)(y-R_y)$. For a smooth function f of (x,y),

$$e^{-\mathrm{i}\varsigma_R} D_x (e^{\mathrm{i}\varsigma_R} f) = \Big(\partial_x + \frac{\mathrm{i}}{2}(y-R_y)\Big) f, \qquad e^{-\mathrm{i}\varsigma_R} D_y (e^{\mathrm{i}\varsigma_R} f) = \Big(\partial_y - \frac{\mathrm{i}}{2}(x-R_x)\Big) f,$$

the covariant derivatives of the symmetric gauge centred at R. In the complex coordinate $\omega := (x-R_x) + \mathrm{i}(y-R_y)$, with $\partial_\omega = \frac{1}{2}(\partial_x - \mathrm{i}\partial_y)$ and $\partial_{\bar\omega} = \frac{1}{2}(\partial_x + \mathrm{i}\partial_y)$, this reads

$$e^{-\mathrm{i}\varsigma_R}(D_x + \mathrm{i}D_y)(e^{\mathrm{i}\varsigma_R} f) = \Big(2\partial_{\bar\omega} + \frac{\omega}{2}\Big) f = 2e^{-|\omega|^2/4}\partial_{\bar\omega}\big(e^{|\omega|^2/4} f\big), \qquad e^{-\mathrm{i}\varsigma_R} D (e^{\mathrm{i}\varsigma_R} f) = \Big(\partial_\omega - \frac{\bar\omega}{4}\Big) f.$$

Let ω_1, ω_2 be the complex coordinates, in this sense, of r_1, r_2, and on (the image of) U let

$$\Xi(\omega_1, \omega_2) := e^{(|\omega_1|^2 + |\omega_2|^2)/4}\, e^{-\mathrm{i}\varsigma_R(r_1) - \mathrm{i}\varsigma_R(r_2)}\, \Phi(r_1, r_2).$$

Ξ is smooth, and the hypotheses on Φ give $\partial_{\bar\omega_1}\Xi = \partial_{\bar\omega_2}\Xi = 0$. For $r = (r_x, r_y)$ put $u := r_x + \mathrm{i}r_y$; the points $R + r/2$ and $R - r/2$ have the coordinates $\omega_1 = u/2$ and $\omega_2 = -u/2$. The function $\widetilde\Xi(u) := \Xi(u/2, -u/2)$ is defined and smooth near $u = 0$, and by the chain rule $\partial_{\bar u}\widetilde\Xi = \frac{1}{2}\partial_{\bar\omega_1}\Xi - \frac{1}{2}\partial_{\bar\omega_2}\Xi = 0$; so $\widetilde\Xi$ is holomorphic near 0, with $\widetilde\Xi(0) = \Xi(0,0)$ and $\widetilde\Xi'(0) = \frac{1}{2}(\partial_{\omega_1}\Xi - \partial_{\omega_2}\Xi)(0,0)$. Since $|\omega_1|^2 + |\omega_2|^2 = |u|^2/2$ and the phases have modulus one,

$$F_R(r) = e^{-u\bar u/4}\, \widetilde\Xi(u)\, \overline{\widetilde\Xi(u)}.$$

Now $\nabla^2 = 4\partial_u\partial_{\bar u}$, $\partial_{\bar u}\widetilde\Xi = 0$ and $\partial_u\overline{\widetilde\Xi} = 0$, and at $u = 0$ the function $e^{-u\bar u/4}$ equals 1, its first derivatives vanish, and $\partial_u\partial_{\bar u} e^{-u\bar u/4} = -\frac{1}{4}$. By the product rule, therefore, $\partial_u\partial_{\bar u} F_R$ at $u = 0$ equals $|\widetilde\Xi'(0)|^2 - \frac{1}{4}|\widetilde\Xi(0)|^2$, that is,

$$F_R(0) = |\widetilde\Xi(0)|^2, \qquad (\nabla^2 F_R)(0) = 4|\widetilde\Xi'(0)|^2 - |\widetilde\Xi(0)|^2. \tag{4.1}$$

(a) If Φ is symmetric, then so is Ξ, because the prefactor in its definition is symmetric in (r_1, r_2). Hence $\widetilde\Xi(-u) = \widetilde\Xi(u)$, $\widetilde\Xi'(0) = 0$, and (4.1) gives $F_R(0) + (\nabla^2 F_R)(0) = 0$.

(b) If Φ is antisymmetric, then so is Ξ. Hence $\widetilde\Xi$ is odd, $\widetilde\Xi(0) = 0$, and $(\partial_{\omega_2}\Xi)(0,0) = -(\partial_{\omega_1}\Xi)(0,0)$, so $\widetilde\Xi'(0) = (\partial_{\omega_1}\Xi)(0,0)$. By (4.1), $F_R(0) = 0$ and $(\nabla^2 F_R)(0) = 4|(\partial_{\omega_1}\Xi)(0,0)|^2$. On the other hand, by the second identity displayed above, applied in the variable r_1 (the factor $e^{-\mathrm{i}\varsigma_R(r_2)}$ does not depend on r_1),

$$e^{-\mathrm{i}\varsigma_R(r_1) - \mathrm{i}\varsigma_R(r_2)}(D_1\Phi)(r_1, r_2) = \Big(\partial_{\omega_1} - \frac{\bar\omega_1}{4}\Big)\Big(e^{-(|\omega_1|^2 + |\omega_2|^2)/4}\,\Xi\Big) = e^{-(|\omega_1|^2 + |\omega_2|^2)/4}\Big(\partial_{\omega_1}\Xi - \frac{\bar\omega_1}{2}\,\Xi\Big),$$

which at $r_1 = r_2 = R$, that is at $\omega_1 = \omega_2 = 0$, gives $|(D_1\Phi)(R,R)| = |(\partial_{\omega_1}\Xi)(0,0)|$. This proves (b). □

Proof of Theorem 4.2. Let Ω_0 be the vacuum of $\mathcal{H}_N$ and $\mathcal{H}_N^{(2)}$ the two-particle subspace, spanned by the vectors $c_i^* c_{i'}^* \Omega_0$. Indices $i = (j, \sigma)$ range over $\mathbb{Z}_{N_\phi} \times \{\uparrow, \downarrow\}$, which we order totally, and $c_i := c_{j\sigma}$.

Step 1: two-particle wave functions. For $\Upsilon \in \mathcal{H}_N^{(2)}$ let $\upsilon_{ii'} := 2^{-1/2}\langle \Omega_0, c_{i'} c_i \Upsilon\rangle$, so that $\upsilon_{i'i} = -\upsilon_{ii'}$ and $c_{i'} c_i \Upsilon = \sqrt{2}\, \upsilon_{ii'} \Omega_0$ (two annihilation operators map a two-particle vector into $\mathbb{C}\Omega_0$). For $r_1, r_2 \in \mathbb{R}^2$ and spins σ, σ' let

$$\Upsilon_{\sigma\sigma'}(r_1, r_2) := \sum_{j, j' \in \mathbb{Z}_{N_\phi}} \upsilon_{(j\sigma)(j'\sigma')}\, \psi_j(r_1)\, \psi_{j'}(r_2).$$

These functions are smooth, of the lowest Landau level and satisfy (2.2) in each variable, and $\Upsilon_{\sigma\sigma'}(r_1,r_2) = -\Upsilon_{\sigma'\sigma}(r_2,r_1)$. By the expansions of $c(e_R\otimes\sigma)$ and $c(e_R^D\otimes\sigma)$ given after Definition 4.1,

$$\langle\Omega_0, c(e_R\otimes\sigma')\, c(e_R^D\otimes\sigma)\Upsilon\rangle = \sum_{j,j'} \psi_{j'}(R)(D\psi_j)(R)\langle\Omega_0, c_{j'\sigma'}c_{j\sigma}\Upsilon\rangle = \sqrt{2}\,(D_1\Upsilon_{\sigma\sigma'})(R,R), \quad (4.2)$$

where D_1 is the operator D acting on the first variable.

Step 2: the quadratic form of H_N. By (2.7), since $(c_{b\sigma'}c_{a\sigma})^* = c^*_{a\sigma}c^*_{b\sigma'}$,

$$\langle\Upsilon, H_N\Upsilon\rangle = \frac{1}{2}\sum_{\sigma,\sigma'}\sum_{a,b,c,d} V_{abcd}\langle c_{b\sigma'}c_{a\sigma}\Upsilon, c_{d\sigma'}c_{c\sigma}\Upsilon\rangle = \sum_{\sigma,\sigma'}\sum_{a,b,c,d} V_{abcd}\,\overline{\upsilon_{(a\sigma)(b\sigma')}}\,\upsilon_{(c\sigma)(d\sigma')}.$$

Let $|\Upsilon|^2(r_1,r_2) := \sum_{\sigma,\sigma'}|\Upsilon_{\sigma\sigma'}(r_1,r_2)|^2$. Since

$$\sum_{a,b,c,d}\overline{\upsilon_{(a\sigma)(b\sigma')}}\,\upsilon_{(c\sigma)(d\sigma')}\,\overline{\psi_a(r_1)}\psi_c(r_1)\,\overline{\psi_b(r_2)}\psi_d(r_2) = |\Upsilon_{\sigma\sigma'}(r_1,r_2)|^2,$$

the definition (2.6) gives, after applying ∇^2 in r_1 to this identity and setting $r_1 = r_2 = r$,

$$\langle\Upsilon, H_N\Upsilon\rangle = 4\pi\int_{\mathbb{T}_N}\Big[|\Upsilon|^2(r,r) + (\nabla_1^2|\Upsilon|^2)(r,r)\Big]\mathrm{d}r,$$

where ∇_1^2 is the Laplacian in r_1. The function $|\Upsilon|^2$ is smooth, periodic in each variable with respect to $L_x\mathbb{Z}\times L_y\mathbb{Z}$ (the phases in (2.2) have modulus one), and symmetric: $|\Upsilon|^2(r_1,r_2) = |\Upsilon|^2(r_2,r_1)$.

Step 3: relative coordinates. Let $F_R(r) := |\Upsilon|^2(R+r/2, R-r/2)$, as a function of (R,r). Since $\partial_{r_1} = \frac{1}{2}\partial_R + \partial_r$ and $\partial_{r_2} = \frac{1}{2}\partial_R - \partial_r$ componentwise, $\nabla_1^2 + \nabla_2^2 = \frac{1}{2}\nabla_R^2 + 2\nabla_r^2$; and differentiating the symmetry relation twice gives $(\nabla_1^2|\Upsilon|^2)(R,R) = (\nabla_2^2|\Upsilon|^2)(R,R)$. Hence $(\nabla_1^2|\Upsilon|^2)(R,R) = \frac{1}{4}\nabla_R^2[F_R(0)] + (\nabla^2 F_R)(0)$. The function $R\mapsto F_R(0) = |\Upsilon|^2(R,R)$ is smooth and periodic, so the integral of its Laplacian over $\mathbb{T}_N$ vanishes, and

$$\langle\Upsilon, H_N\Upsilon\rangle = 4\pi\int_{\mathbb{T}_N}[F_R(0) + (\nabla^2 F_R)(0)]\mathrm{d}R.$$

Step 4: spin decomposition. Let χ^{sing} be the antisymmetric matrix with $\chi^{\mathrm{sing}}_{\uparrow\downarrow} = -\chi^{\mathrm{sing}}_{\downarrow\uparrow} = 2^{-1/2}$ and zero diagonal. The four matrices $\chi^+, \chi^0, \chi^-, \chi^{\mathrm{sing}}$ form an orthonormal basis of the complex 2×2 matrices for the Hilbert–Schmidt inner product. Let $f_\nu := \sum_{\sigma\sigma'}\overline{\chi^\nu_{\sigma\sigma'}}\Upsilon_{\sigma\sigma'}$ and $f_{\mathrm{sing}} := \sum_{\sigma\sigma'}\overline{\chi^{\mathrm{sing}}_{\sigma\sigma'}}\Upsilon_{\sigma\sigma'}$. Then $|\Upsilon|^2 = \sum_\nu|f_\nu|^2 + |f_{\mathrm{sing}}|^2$ pointwise; these functions are smooth and of the lowest Landau level in each variable; and by $\Upsilon_{\sigma\sigma'}(r_1,r_2) = -\Upsilon_{\sigma'\sigma}(r_2,r_1)$, the symmetry of χ^ν and the antisymmetry of χ^{sing}, $f_\nu(r_2,r_1) = -f_\nu(r_1,r_2)$ and $f_{\mathrm{sing}}(r_2,r_1) = f_{\mathrm{sing}}(r_1,r_2)$. Hence $F_R(r) = \sum_\nu|f_\nu(R+r/2,R-r/2)|^2 + |f_{\mathrm{sing}}(R+r/2,R-r/2)|^2$, and Lemma 4.3, applied to each f_ν and to f_{sing}, yields

$$\langle\Upsilon, H_N\Upsilon\rangle = 16\pi\sum_\nu\int_{\mathbb{T}_N}|(D_1 f_\nu)(R,R)|^2\,\mathrm{d}R.$$

(The integrand is periodic, since $D_1 f_\nu$ satisfies (2.2) in each variable by Lemma 2.2(iii).)

Step 5: the quadratic form of the local integral. For $\Upsilon\in\mathcal{H}_N^{(2)}$ the vector $B_\nu(R)\Upsilon$ lies in $\mathbb{C}\Omega_0$, so that $\langle\Upsilon, B_\nu(R)^*B_\nu(R)\Upsilon\rangle = |\langle\Omega_0, B_\nu(R)\Upsilon\rangle|^2$, and by (4.2)

$$\langle\Omega_0, B_\nu(R)\Upsilon\rangle = \sqrt{2}\sum_{\sigma,\sigma'}\overline{\chi^\nu_{\sigma\sigma'}}(D_1\Upsilon_{\sigma\sigma'})(R,R) = \sqrt{2}\,(D_1 f_\nu)(R,R).$$

Hence, with $H' := 8\pi \int_{\mathbb{T}_N} \sum_\nu B_\nu(R)^* B_\nu(R)\,\mathrm{d}R$ (the integrand is a continuous function on $\mathbb{T}_N$), $\langle \Upsilon, H'\Upsilon\rangle = 16\pi \sum_\nu \int_{\mathbb{T}_N} |(D_1 f_\nu)(R,R)|^2 \mathrm{d}R = \langle \Upsilon, H_N \Upsilon\rangle$.

Step 6: from two particles to the Fock space. For a pair $\mathbf{i} = (i, i')$ with $i < i'$ let $A_{\mathbf{i}} := c_{i'} c_i$. The vectors $A_{\mathbf{i}}^* \Omega_0 = c_i^* c_{i'}^* \Omega_0$ form an orthonormal basis of $\mathcal{H}_N^{(2)}$, and $A_{\mathbf{j}} A_{\mathbf{i}}^* \Omega_0$ is the multiple $\langle A_{\mathbf{j}}^* \Omega_0, A_{\mathbf{i}}^* \Omega_0\rangle \Omega_0 = \delta_{\mathbf{ji}} \Omega_0$ of the vacuum. Every product $c_{i_2} c_{i_1}$ is 0 or $\pm A_{\mathbf{i}}$ for some pair $\mathbf{i}$, and $B_\nu(R)$ is a linear combination of such products with continuous coefficients. Hence H_N and H' are both of the form $O = \sum_{\mathbf{i},\mathbf{j}} O_{\mathbf{ij}}\, A_{\mathbf{i}}^* A_{\mathbf{j}}$, and for such an operator $\langle A_{\mathbf{i}}^* \Omega_0, O\, A_{\mathbf{j}}^* \Omega_0\rangle = O_{\mathbf{ij}}$: the coefficient matrix of O is its matrix on $\mathcal{H}_N^{(2)}$. By Step 5 and polarization, $\langle \Upsilon, H_N \Upsilon'\rangle = \langle \Upsilon, H'\Upsilon'\rangle$ for all $\Upsilon, \Upsilon' \in \mathcal{H}_N^{(2)}$, so the coefficient matrices of H_N and H' coincide, and $H_N = H'$ on $\mathcal{H}_N$. This is (a).

(b) Unfolding the orbitals by (2.4), and using $D\phi_n = -(x - \kappa n)\phi_n$ (Lemma 2.2(iii)),

$$B_\nu(R) = \sum_{j_1, j_2 \in \mathbb{Z}} \phi_{j_1}(R)(D\phi_{j_2})(R)\, c^\nu(\psi_{[j_1]}, \psi_{[j_2]}) = -\sum_{j_1, j_2 \in \mathbb{Z}} (R_x - \kappa j_2)\, \phi_{j_1}(R)\phi_{j_2}(R)\, c^\nu(\psi_{[j_1]}, \psi_{[j_2]}),$$

an absolutely convergent double series. Put $n := j_1 + j_2$ and $d := j_2 - j_1$. Completing the square, $(R_x - \kappa j_1)^2 + (R_x - \kappa j_2)^2 = 2(R_x - \kappa n/2)^2 + \kappa^2 d^2/2$, so

$$\phi_{j_1}(R)\phi_{j_2}(R) = \frac{1}{L_y\sqrt{\pi}}\, e^{\mathrm{i}\kappa n R_y} e^{-(R_x - \kappa n/2)^2} e^{-\kappa^2 d^2/4},$$

and $-(R_x - \kappa j_2) = \frac{\kappa d}{2} - (R_x - \frac{\kappa n}{2})$. For fixed n, the exchange $j_1 \leftrightarrow j_2$ maps d to $-d$ and changes the sign of $c^\nu(\psi_{[j_1]}, \psi_{[j_2]})$. Hence the terms with the factor $R_x - \kappa n/2$, which does not change under the exchange, cancel in pairs (the term $d = 0$ vanishes because $c^\nu(\psi, \psi) = 0$), and the terms with the factor $\kappa d/2$, which changes sign, contribute twice their sum over $d > 0$:

$$\sum_{j_1 + j_2 = n} \frac{\kappa d}{2}\, e^{-\kappa^2 d^2/4} c^\nu\big(\psi_{[j_1]}, \psi_{[j_2]}\big) = \kappa \sum_{d \geq 1,\ d \equiv n \,(\mathrm{mod}\, 2)} d\, e^{-\kappa^2 d^2/4} c^\nu\big(\psi_{[(n-d)/2]}, \psi_{[(n+d)/2]}\big) = \frac{\kappa}{2}\, b_n^\nu.$$

This gives (b).

(c) By (b), for each ν,

$$B_\nu(R)^* B_\nu(R) = \frac{\kappa^2}{4\pi L_y^2} \sum_{n, n' \in \mathbb{Z}} e^{\mathrm{i}\kappa(n' - n)R_y} e^{-(R_x - \kappa n/2)^2 - (R_x - \kappa n'/2)^2} (b_n^\nu)^* b_{n'}^\nu,$$

absolutely and uniformly convergent on $[0, L_x] \times [0, L_y]$. The integral over $R_y \in [0, L_y)$ of $e^{\mathrm{i}\kappa(n'-n)R_y}$ is $L_y \delta_{nn'}$, since $\kappa L_y = 2\pi$. Writing $n = n_0 + 2N_\phi q$ with $n_0 \in \{0, \dots, 2N_\phi - 1\}$ and $q \in \mathbb{Z}$, we have $b_n^\nu = b_{n_0}^\nu$ and $\kappa n/2 = \kappa n_0/2 + q L_x$, so $\sum_{q \in \mathbb{Z}} \int_0^{L_x} e^{-2(R_x - \kappa n_0/2 - q L_x)^2} \mathrm{d}R_x = \int_{\mathbb{R}} e^{-2t^2} \mathrm{d}t = \sqrt{\pi/2}$. Hence

$$8\pi \int_{\mathbb{T}_N} B_\nu(R)^* B_\nu(R)\, \mathrm{d}R = 8\pi \cdot \frac{\kappa^2}{4\pi L_y} \sqrt{\frac{\pi}{2}} \sum_{n_0 \in \mathbb{Z}_{2N_\phi}} (b_{n_0}^\nu)^* b_{n_0}^\nu = \frac{\kappa^3}{\sqrt{2\pi}} \sum_{n_0 \in \mathbb{Z}_{2N_\phi}} (b_{n_0}^\nu)^* b_{n_0}^\nu,$$

using $L_y = 2\pi/\kappa$. Summing over ν and using (a) gives (c). □

4.2 The Hamiltonian in the slot lattice

Proposition 4.4 (uniform size and rapid locality)**.** *For every admissible torus and every $R \in \mathbb{R}^2$:*

(i) $\|B_\nu(R)\| \le \sqrt{2}\,\|e_R\|\|e_R^D\|$, *and* $\|h_R\| \le 32\pi\|e_R\|^2\|e_R^D\|^2 \le C_h$ *with* $C_h := 7/\pi$.

(ii) For $r \ge 0$ *let* $B_\nu^{(r)}(R) := \hat{c}^\nu(\mathbf{1}_{\Lambda_r(R)}\mathcal{S}e_R, \mathbf{1}_{\Lambda_r(R)}\mathcal{S}e_R^D)$ *and* $h_{R,r} := 8\pi\sum_\nu B_\nu^{(r)}(R)^* B_\nu^{(r)}(R)$. *Then* $h_{R,r} \in \mathfrak{A}(\Lambda_r(R))$ *is self-adjoint and commutes with* $\widehat{N}_\uparrow$ *and* $\widehat{N}_\downarrow$, *and for every* k

$$\|\iota(h_R) - h_{R,r}\| \le C_k^H (1+r)^{-k},$$

with C_k^H *depending only on* k, η *and* δ_0.

(iii) For each $r \ge 0$ *the map* $R \mapsto h_{R,r}$ *is Borel measurable.*

Proof. (i) For $f \in \mathfrak{k}_N$ the anticommutation relations give

$$\big(c(f)^*c(f)\big)^2 = c(f)^*(\|f\|^2 - c(f)^*c(f))c(f) = \|f\|^2 c(f)^*c(f),$$

because $c(f)^2 = 0$; so $c(f)^*c(f)$ is $\|f\|^2$ times an orthogonal projection, nonzero if $f \neq 0$, and $\|c(f)\| = \|f\|$. The same holds for $\hat{c}$. Hence $\|c(f\otimes\sigma')c(g\otimes\sigma)\| \le \|f\|\|g\|$, and since $\sum_{\sigma\sigma'}|\chi^\nu_{\sigma\sigma'}|$ is 1 for $\nu = \pm$ and $\sqrt{2}$ for $\nu = 0$,

$$\|c^\pm(f,g)\| \le \|f\|\|g\|, \qquad \|c^0(f,g)\| \le \sqrt{2}\|f\|\|g\|,$$

and likewise for $\hat{c}^\nu$. With $f = e_R$ and $g = e_R^D$ this gives the bound on $B_\nu(R)$, and $\|h_R\| \le 8\pi\sum_\nu\|B_\nu(R)\|^2 \le 8\pi(1+2+1)\|e_R\|^2\|e_R^D\|^2$. An admissible torus has $\min\{L_x, L_y\} \ge 4$, so by Lemma 2.3(iv) $2\pi\|e_R\|^2 \le 1.08$ and $4\pi\|e_R^D\|^2 \le 1.62$, and $32\pi\|e_R\|^2\|e_R^D\|^2 \le 4\cdot 1.08\cdot 1.62/\pi < 7/\pi$.

(ii) Let $X := \Lambda_r(R)$. For h supported in X, $\hat{c}(h\otimes\sigma) = \sum_{s\in X}\overline{h(s)}\hat{c}_{s\sigma} \in \mathrm{CAR}(X)$. Hence $B_\nu^{(r)}(R) \in \mathrm{CAR}(X)$, and $h_{R,r}$ is a linear combination of products of two creation and two annihilation operators of modes in X; it is self-adjoint, and even, since conjugation by the parity changes the sign of each $\hat{c}_{s\sigma}$; so $h_{R,r} \in \mathfrak{A}(X)$. The anticommutation relations give $[\hat{c}^*_{s\sigma}\hat{c}_{s\sigma}, \hat{c}(h\otimes\sigma')] = -\{\hat{c}^*_{s\sigma}, \hat{c}(h\otimes\sigma')\}\hat{c}_{s\sigma} = -\delta_{\sigma\sigma'}\overline{h(s)}\hat{c}_{s\sigma}$, hence $[\widehat{N}_\sigma, \hat{c}(h\otimes\sigma')] = -\delta_{\sigma\sigma'}\hat{c}(h\otimes\sigma')$. Each term of $B_\nu^{(r)}(R)$ contains the same number of annihilation operators of spin σ (two, one or none of each spin, for $\nu = +, 0, -$), so $[\widehat{N}_\sigma, B_\nu^{(r)}(R)]$ is this number times $-B_\nu^{(r)}(R)$; taking adjoints, $[\widehat{N}_\sigma, B_\nu^{(r)}(R)^*]$ is the same number times $B_\nu^{(r)}(R)^*$, and so $[\widehat{N}_\sigma, B_\nu^{(r)}(R)^* B_\nu^{(r)}(R)] = 0$.

For the estimate let $f := \mathcal{S}e_R$, $g := \mathcal{S}e_R^D$, $f_r := \mathbf{1}_X f$ and $g_r := \mathbf{1}_X g$. Since ι is a $*$-homomorphism, $\iota(B_\nu(R)) = \hat{c}^\nu(f,g)$ and $\iota(h_R) = 8\pi\sum_\nu \iota(B_\nu(R))^*\iota(B_\nu(R))$. By the identity $ab - a'b' = (a-a')b + a'(b-b')$, the bounds of (i), $\|g\| = \|e_R^D\|$ and $\|f_r\| \le \|f\| = \|e_R\|$ (Proposition 3.3), and Lemma 3.5(iii),

$$\|\iota(B_\nu(R)) - B_\nu^{(r)}(R)\| \le \sqrt{2}(\|f - f_r\|\|e_R^D\| + \|e_R\|\|g - g_r\|) \le \sqrt{2}(\|e_R\| + \|e_R^D\|)C_k(1+r)^{-k},$$

with C_k the constant of Lemma 3.5. Moreover $\|\iota(B_\nu(R))\|$ and $\|B_\nu^{(r)}(R)\|$ are at most $\sqrt{2}\|e_R\|\|e_R^D\|$, by (i). Writing $\iota(B_\nu)^*\iota(B_\nu) - B_\nu^{(r)*}B_\nu^{(r)} = \iota(B_\nu)^*(\iota(B_\nu) - B_\nu^{(r)}) + (\iota(B_\nu) - B_\nu^{(r)})^* B_\nu^{(r)}$, with $B_\nu = B_\nu(R)$ and $B_\nu^{(r)} = B_\nu^{(r)}(R)$, we obtain

$$\begin{aligned}\|\iota(h_R) - h_{R,r}\| &\le 8\pi\sum_\nu \|\iota(B_\nu(R)) - B_\nu^{(r)}(R)\|\cdot 2\sqrt{2}\|e_R\|\|e_R^D\| \\ &\le 96\pi\|e_R\|\|e_R^D\|(\|e_R\| + \|e_R^D\|)C_k(1+r)^{-k}.\end{aligned}$$

By Lemma 2.3(iv), $\|e_R\| < \frac{1}{2}$ and $\|e_R^D\| < \frac{2}{5}$, so the right side is at most $96\pi\cdot\frac{1}{5}\cdot\frac{9}{10}\,C_k(1+r)^{-k} < 55\,C_k(1+r)^{-k}$. Thus $C_k^H := 55\,C_k$ depends only on k, η and δ_0.

(iii) By Lemma 2.3(i), $\mathcal{S}e_{R+(L_x,0)} = e^{-\mathrm{i}L_xR_y}\mathcal{S}e_R$ and likewise for e_R^D, while $\mathcal{S}e_{R+(0,L_y)} = \mathcal{S}e_R$; since $\hat{c}^\nu$ is conjugate-linear in each argument, the phases cancel in $B_\nu^{(r)}(R)^*B_\nu^{(r)}(R)$, and $h_{R,r}$ is a function of $R \in \mathbb{T}_N$. For $Y \subset \Lambda_N$ let $\mathcal{D}_Y := \{R \in \mathbb{T}_N : \Lambda_r(R) = Y\} = \bigcap_{s\in Y}\{R : d(s,R) \le r\} \cap \bigcap_{s\notin Y}\{R : d(s,R) > r\}$, a Borel set, since $R \mapsto d(s,R)$ is continuous. The finitely many sets $\mathcal{D}_Y$ partition $\mathbb{T}_N$. On $\mathcal{D}_Y$, $h_{R,r} = 8\pi\sum_\nu \hat{c}^\nu(\mathbf{1}_Y\mathcal{S}e_R, \mathbf{1}_Y\mathcal{S}e_R^D)^*\hat{c}^\nu(\mathbf{1}_Y\mathcal{S}e_R, \mathbf{1}_Y\mathcal{S}e_R^D)$ is a linear combination of fixed operators whose coefficients are polynomials in the continuous functions $(\mathcal{S}e_R)_s = \overline{z_s(R)}$, $(\mathcal{S}e_R^D)_s = \overline{(Dz_s)(R)}$ and their complex conjugates. So $R \mapsto h_{R,r}$ is the sum over Y of $\mathbf{1}_{\mathcal{D}_Y}$ times continuous functions, and is Borel measurable. □

Thus the local terms of H_N have a size bounded independently of N, and in the slot lattice they are local up to errors that decay faster than any power of the distance, uniformly in N. In the language of lattice systems, $\widehat{H}_N = \int \iota(h_R)\,\mathrm{d}R$ is a rapidly decaying interaction of uniform size.

The constant $C_h = 7/\pi$ is not optimal, and the norm of h_R can be computed. Let $f, g \in \mathfrak{h}_N$ with $f \neq 0$, and write $g = \mu f + g^\perp$ with $\mu \in \mathbb{C}$ and $g^\perp \perp f$. Since c^ν is conjugate-linear in its second argument and $c^\nu(f,f) = 0$, we have $c^\nu(f,g) = c^\nu(f,g^\perp)$. If $g^\perp \neq 0$, let $\hat{f} := f/\|f\|$ and $\hat{g} := g^\perp/\|g^\perp\|$; then $c^\nu(f,g) = \|f\|\|g^\perp\|c^\nu(\hat{f},\hat{g})$, and the four vectors $\hat{f}\otimes\sigma$, $\hat{g}\otimes\sigma$ are orthonormal. By the factorization of the Fock space used in the proof of Lemma 3.8, with an orthonormal basis of $\mathfrak{k}_N$ that begins with these four vectors, $O := \sum_\nu c^\nu(\hat{f},\hat{g})^*c^\nu(\hat{f},\hat{g})$ is an operator on the sixteen-dimensional Fock space of the four modes, tensored with the identity, and its norm may be computed there. Each $c^\nu(\hat{f},\hat{g})$ lowers both $n_{\hat{f}} := \sum_\sigma c(\hat{f}\otimes\sigma)^*c(\hat{f}\otimes\sigma)$ and $n_{\hat{g}} := \sum_\sigma c(\hat{g}\otimes\sigma)^*c(\hat{g}\otimes\sigma)$ by one, so O commutes with them, and it vanishes unless $n_{\hat{f}}, n_{\hat{g}} \ge 1$. On the space $n_{\hat{f}} = n_{\hat{g}} = 1$, O is the orthogonal projection onto the span of the three orthonormal vectors obtained by applying the $c^\nu(\hat{f},\hat{g})^*$ to the vacuum. On the spaces $(n_{\hat{f}}, n_{\hat{g}}) = (2,1)$ and $(1,2)$, $O = \frac{3}{2}$: for instance, if $n_{\hat{f}} = 2$, then $c^\pm(\hat{f},\hat{g})^*c^\pm(\hat{f},\hat{g})$ are the occupation numbers of $\hat{g}\otimes\uparrow$ and of $\hat{g}\otimes\downarrow$, while $c^0(\hat{f},\hat{g})^*c^0(\hat{f},\hat{g})$ equals $\frac{1}{2}n_{\hat{g}}$, its spin-exchanging cross terms vanishing because both modes $\hat{f}\otimes\sigma$ are occupied; so $O = \frac{3}{2}n_{\hat{g}} = \frac{3}{2}$. On the one-dimensional space $n_{\hat{f}} = n_{\hat{g}} = 2$, $O = 3$, since each of the three operators $c^\nu(\hat{f},\hat{g})$ maps the fully occupied unit vector to a unit vector. Hence $\|O\| = 3$, and $\|\sum_\nu c^\nu(f,g)^*c^\nu(f,g)\| = 3\|f\|^2\|g^\perp\|^2$, which also holds if $g^\perp = 0$. With $f = e_R$ and $g = e_R^D$,

$$\|h_R\| = 24\pi\|e_R\|^2\Big(\|e_R^D\|^2 - \frac{|\langle e_R, e_R^D\rangle|^2}{\|e_R\|^2}\Big) \le 24\pi\|e_R\|^2\|e_R^D\|^2 < \frac{21}{4\pi}$$

on admissible tori, by Lemma 2.3(iv). Moreover $\langle e_R, e_R^D\rangle = \frac{1}{2}(\partial_{R_x} + \mathrm{i}\partial_{R_y})\|e_R\|^2$. Indeed, by Lemma 2.3(i) and Lemma 2.2(iii), $\langle e_R, e_R^D\rangle$ equals $\sum_j \psi_j(R)\overline{(\partial_x\psi_j)(R)}$, whose real part is $\frac{1}{2}\partial_{R_x}\sum_j|\psi_j(R)|^2$; and since $\partial_y\psi_j = \mathrm{i}\partial_x\psi_j + \mathrm{i}x\psi_j$ by (2.3),

$$\partial_{R_y}\sum_j|\psi_j(R)|^2 = 2\,\mathrm{Re}\sum_j\psi_j(R)\overline{(\partial_y\psi_j)(R)} = 2\,\mathrm{Im}\sum_j\psi_j(R)\overline{(\partial_x\psi_j)(R)}.$$

Differentiating the first formula of Lemma 2.3(iii) term by term (its constant term does not depend on R) gives $|\langle e_R, e_R^D\rangle| \le (4\pi)^{-1}\sum_v|v|e^{-|v|^2/4}$, the sum over the nonzero vectors v of the lattice $L_x\mathbb{Z}\times L_y\mathbb{Z}$, and by dominated convergence this tends to zero as $\min\{L_x, L_y\} \to \infty$. Together with Lemma 2.3(iv) this shows that $\|h_R\| \to 24\pi\cdot\frac{1}{2\pi}\cdot\frac{1}{4\pi} = \frac{3}{\pi}$, uniformly in R, as $\min\{L_x, L_y\} \to \infty$. Nothing below uses this value; any bound uniform over admissible tori serves as C_h.

Lemma 4.5 (covariance)**.** *For every torus as in Definition 2.1 and every $R \in \mathbb{R}^2$, $\Theta^*B_\nu(R)\Theta = e^{-2\mathrm{i}\kappa R_y}B_\nu(R+\kappa\hat{e}_x)$ and $\Theta^*h_R\Theta = h_{R+\kappa\hat{e}_x}$. Consequently, with $H_\mathcal{D} := \int_\mathcal{D} h_R\,\mathrm{d}R$ for a Borel set $\mathcal{D} \subset \mathbb{T}_N$, $\Theta^*H_\mathcal{D}\Theta = H_{\mathcal{D}+\kappa\hat{e}_x}$; in particular $[\Theta, H_N] = 0$.*

Proof. By Lemma 2.6(iii), $\Theta^* c(f)\Theta = c((\theta_x \otimes \mathbf{1})f)$ for $f \in \mathfrak{k}_N$. By Lemma 2.4(iv), $\theta_x e_R = e^{\mathrm{i}\kappa R_y} e_{R+\kappa\hat{e}_x}$ and $\theta_x e_R^D = e^{\mathrm{i}\kappa R_y} e^D_{R+\kappa\hat{e}_x}$, and c is conjugate-linear, so

$$\Theta^* c(e_R \otimes \sigma)\Theta = e^{-\mathrm{i}\kappa R_y} c(e_{R+\kappa\hat{e}_x} \otimes \sigma), \qquad \Theta^* c(e_R^D \otimes \sigma)\Theta = e^{-\mathrm{i}\kappa R_y} c(e^D_{R+\kappa\hat{e}_x} \otimes \sigma).$$

Since conjugation by Θ is multiplicative, $\Theta^* B_\nu(R)\Theta = e^{-2\mathrm{i}\kappa R_y} B_\nu(R+\kappa\hat{e}_x)$, and then

$$\Theta^* B_\nu(R)^* B_\nu(R)\Theta = \big(\Theta^* B_\nu(R)\Theta\big)^* \big(\Theta^* B_\nu(R)\Theta\big) = B_\nu(R+\kappa\hat{e}_x)^* B_\nu(R+\kappa\hat{e}_x),$$

so that $\Theta^* h_R \Theta = h_{R+\kappa\hat{e}_x}$. Conjugation by Θ is a linear map on a finite-dimensional space and commutes with the integral, and Lebesgue measure on $\mathbb{T}_N$ is invariant under translations; hence $\Theta^* H_{\mathcal{D}}\Theta = \int_{\mathcal{D}} h_{R+\kappa\hat{e}_x}\mathrm{d}R = \int_{\mathcal{D}+\kappa\hat{e}_x} h_R\,\mathrm{d}R = H_{\mathcal{D}+\kappa\hat{e}_x}$. For $\mathcal{D} = \mathbb{T}_N$, Theorem 4.2(a) gives $\Theta^* H_N \Theta = H_N$, and since Θ is unitary, $[\Theta, H_N] = 0$. □

4.3 A class of lowest-Landau-level Hamiltonians

Definition 4.6 (slot-local covariant Hamiltonians)**.** Let $\mathcal{N} \subset \mathbb{N}$ be infinite and let $(\mathbb{T}_N)_{N\in\mathcal{N}}$ be admissible tori. A sequence $H = (H_N)_{N\in\mathcal{N}}$ of self-adjoint operators on $\mathcal{H}_N$ satisfies:

(H1) (*slot locality of uniform size*) if $H_N = \int_{\mathbb{T}_N} h_R\,\mathrm{d}R$, where $R \mapsto h_R = h_{N,R}$ is a Borel measurable map from $\mathbb{T}_N$ to the self-adjoint, sector-preserving operators on $\mathcal{H}_N$ with $\sup_R \|h_R\| \le C_h$, and for every k there is C_k^H such that for all $N \in \mathcal{N}$, $R \in \mathbb{T}_N$ and $r \ge 0$ some $h_{R,r} \in \mathfrak{A}(\Lambda_r(R))$ satisfies $\|\iota(h_R) - h_{R,r}\| \le C_k^H (1+r)^{-k}$; the constants C_h and C_k^H do not depend on N;

(H2) (*covariance*) if, in addition, $\Theta^* h_R \Theta = h_{R+\kappa\hat{e}_x}$ for all $N \in \mathcal{N}$ and $R \in \mathbb{T}_N$.

Under (H1) the operators h_R are even (Lemma 2.6(ii)), and under (H2) $[\Theta, H_N] = 0$, by the translation invariance of Lebesgue measure. Condition (H1) is the lowest-Landau-level counterpart of the charge-conserving finite-range interactions of [BBDF21, Sections 2.1–2.2]: the terms h_R conserve the up-spin number $N_\uparrow$, which plays the role of the charge, and also $N_\downarrow$, which is needed because the gap will be assumed in a sector; finite range is replaced by rapid decay in the slot lattice.

Remark 4.7. A sum $\sum_i h_i$ of self-adjoint sector-preserving operators h_i that are uniformly bounded and rapidly local, in the sense of (H1), at points $R_i \in \mathbb{T}_N$ of which at most a fixed number lie in any disc of radius one, can be written as $\int h'_R\,\mathrm{d}R$ with $h'_R := (4/\pi)\sum_{i:\, d(R,R_i)<1/2} h_i$, which satisfies (H1); and (H2) is inherited when the family (R_i, h_i) is covariant. We do not use this.

Corollary 4.8 (the hollow-core Hamiltonian is in the class)**.** *For every infinite $\mathcal{N} \subset \mathbb{N}$ and admissible tori $(\mathbb{T}_N)_{N\in\mathcal{N}}$, the hollow-core Hamiltonians $(H_N)_{N\in\mathcal{N}}$ satisfy (H1), with the local terms h_R of Definition 4.1, $C_h = 7/\pi$ and the constants C_k^H of Proposition 4.4, and they satisfy (H2).*

Proof. By Theorem 4.2(a), $H_N = \int_{\mathbb{T}_N} h_R\,\mathrm{d}R$. The map $R \mapsto h_R$ is smooth (see the remarks after Definition 4.1), hence Borel measurable, and each h_R is self-adjoint. The computation in the proof of Proposition 4.4(ii), carried out on $\mathcal{H}_N$ instead of $\widehat{\mathcal{H}}_N$, gives $[N_\sigma, c(f\otimes\sigma')] = -\delta_{\sigma\sigma'} c(f\otimes\sigma')$ for $f \in \mathfrak{h}_N$; hence $B_\nu(R)$ lowers $(N_\uparrow, N_\downarrow)$ by $(2,0)$, $(1,1)$ or $(0,2)$, $[N_\sigma, B_\nu(R)^* B_\nu(R)] = 0$, and h_R commutes with $N_\uparrow$ and $N_\downarrow$, that is, it is sector-preserving. By Proposition 4.4(i), $\sup_R\|h_R\| \le 7/\pi$, and Proposition 4.4(ii) provides $h_{R,r} \in \mathfrak{A}(\Lambda_r(R))$ with $\|\iota(h_R) - h_{R,r}\| \le C_k^H(1+r)^{-k}$, where C_k^H depends only on k, η and δ_0. Neither constant depends on N, so (H1) holds; (H2) is Lemma 4.5. □

4.4 Band charges

Definition 4.9 (band charges)**.** For a band $\mathcal{B} \subset \mathbb{Z}_{N_\phi}$ let $Q_\mathcal{B} := \sum_{j\in\mathcal{B}} n_{j\uparrow}$.

Lemma 4.10 (elementary properties of band charges)**.** *The operators $Q_\mathcal{B}$ commute with one another and with $N_\uparrow$ and $N_\downarrow$; each has spectrum in $\{0,1,\dots,|\mathcal{B}|\}$, so that $\|Q_\mathcal{B}\| \le |\mathcal{B}|$ and $e^{2\pi \mathrm{i} Q_\mathcal{B}} = \mathbf{1}$; $Q_{\mathbb{Z}_{N_\phi}} = N_\uparrow$; $Q_\mathcal{B} + Q_{\mathcal{B}'} = Q_{\mathcal{B}\cup\mathcal{B}'}$ for disjoint bands; and $\Theta^* Q_\mathcal{B} \Theta = Q_{\mathcal{B}+1}$.*

Proof. By the anticommutation relations, $n_{j\sigma}^2 = c_{j\sigma}^*(\mathbf{1} - c_{j\sigma}^* c_{j\sigma})c_{j\sigma} = n_{j\sigma}$, because $c_{j\sigma}^2 = 0$, and $n_{j\sigma}$ commutes with $c_{j'\sigma'}$ and $c_{j'\sigma'}^*$ for $(j',\sigma') \neq (j,\sigma)$, being a product of two operators that anticommute with them. So the $n_{j\sigma}$ are mutually commuting orthogonal projections, and the $Q_\mathcal{B}$, $N_\uparrow$ and $N_\downarrow$, which are sums of them, commute. In the occupation-number basis $c_{j_1\sigma_1}^* \cdots c_{j_p\sigma_p}^* \Omega_0$, with Ω_0 the vacuum, every $n_{j\sigma}$ is diagonal with eigenvalues 0 and 1, so $Q_\mathcal{B}$ is diagonal with eigenvalues in $\{0,\dots,|\mathcal{B}|\}$; this gives $\|Q_\mathcal{B}\| \le |\mathcal{B}|$ and $e^{2\pi\mathrm{i}Q_\mathcal{B}} = \mathbf{1}$. The identities $Q_{\mathbb{Z}_{N_\phi}} = N_\uparrow$ and $Q_\mathcal{B} + Q_{\mathcal{B}'} = Q_{\mathcal{B}\cup\mathcal{B}'}$ are immediate from the definitions, and by Lemma 2.6(iii), $\Theta^* Q_\mathcal{B}\Theta = \sum_{j\in\mathcal{B}} n_{j+1,\uparrow} = Q_{\mathcal{B}+1}$. □

Lemma 4.11 (band charges in the slot lattice)**.** *Let the torus be admissible.*

(i) For every j, $\iota(n_{j\uparrow}) = \hat{c}(v_j)^*\hat{c}(v_j)$ with $v_j := \mathcal{S}\psi_j \otimes \uparrow$, which is supported in $\mathrm{Col}(j)\times\{\uparrow\}$. The operators $\iota(n_{j\uparrow})$ are mutually commuting orthogonal projections on $\widehat{\mathcal{H}}_N$, $\iota(n_{j\uparrow}) \in \mathfrak{A}(\mathrm{Col}(j))$, and $\iota(Q_\mathcal{B}) \in \mathfrak{A}(\mathrm{Col}(\mathcal{B}))$ for every band $\mathcal{B}$.

(ii) For $X \subset \Lambda_N$ and a band $\mathcal{B}$ let $\mathcal{B}_X := \{j \in \mathcal{B} : \mathrm{Col}(j) \cap X \neq \emptyset\}$. For $A \in \mathfrak{A}(X)$ and $u \in \mathbb{R}$,

$$e^{\mathrm{i}u\iota(Q_\mathcal{B})} A e^{-\mathrm{i}u\iota(Q_\mathcal{B})} = e^{\mathrm{i}u\iota(Q_{\mathcal{B}_X})} A e^{-\mathrm{i}u\iota(Q_{\mathcal{B}_X})} \in \mathfrak{A}\big(X \cup \mathrm{Col}(\mathcal{B}_X)\big).$$

If $X = \mathrm{Str}(I)$ for an interval I, then $X \cup \mathrm{Col}(\mathcal{B}_X) \subset \mathrm{Str}(I_{2\delta_x})$.

(iii) Let $(\mathbb{T}_N)_{N\in\mathcal{N}}$ be admissible, let $Z_N = \mathrm{Str}(I_N)$ with I_N open intervals of length at least $2\delta_x$, and let $A \in \mathcal{A}_Z$ with prefactor exponent m and $\|A_N\| \le CN^m$. Then for all bands $\mathcal{B}_N$ and $u \in \mathbb{R}$, $e^{\mathrm{i}uQ_{\mathcal{B}_N}} A_N e^{-\mathrm{i}uQ_{\mathcal{B}_N}}$ is almost supported in Z with prefactor exponent m and constants depending only on C, on the constants of A and on δ_0, uniformly in u and in the bands.

Proof. (i) Since ι is a $*$-homomorphism and $\mathcal{S}(\psi_j\otimes\uparrow) = \mathcal{S}\psi_j\otimes\uparrow$, $\iota(n_{j\uparrow}) = \iota(c_{j\uparrow})^*\iota(c_{j\uparrow}) = \hat{c}(v_j)^*\hat{c}(v_j)$. By Lemma 3.6, $\mathcal{S}\psi_j$ is supported in $\mathrm{Col}(j)$, so $\hat{c}(v_j) = \sum_{s\in\mathrm{Col}(j)} \overline{(\mathcal{S}\psi_j)(s)}\hat{c}_{s\uparrow} \in \mathrm{CAR}(\mathrm{Col}(j))$, and $\hat{c}(v_j)^*\hat{c}(v_j)$, a sum of products of two generators, is even; thus $\iota(n_{j\uparrow}) \in \mathfrak{A}(\mathrm{Col}(j))$. The $n_{j\uparrow}$ are mutually commuting orthogonal projections (proof of Lemma 4.10), hence so are their images under the $*$-homomorphism ι, and $\iota(Q_\mathcal{B}) = \sum_{j\in\mathcal{B}} \iota(n_{j\uparrow}) \in \mathfrak{A}(\mathrm{Col}(\mathcal{B}))$.

(ii) By (i), $\iota(Q_\mathcal{B}) = \iota(Q_{\mathcal{B}_X}) + \iota(Q_{\mathcal{B}\setminus\mathcal{B}_X})$ with commuting summands, so

$$e^{\mathrm{i}u\iota(Q_\mathcal{B})} = e^{\mathrm{i}u\iota(Q_{\mathcal{B}_X})}\, e^{\mathrm{i}u\iota(Q_{\mathcal{B}\setminus\mathcal{B}_X})}.$$

The second factor lies in $\mathfrak{A}(Y)$, $Y := \mathrm{Col}(\mathcal{B}\setminus\mathcal{B}_X)$, since $\mathfrak{A}(Y)$ is a finite-dimensional algebra containing $\iota(Q_{\mathcal{B}\setminus\mathcal{B}_X})$ and hence its exponential series; and $Y \cap X = \emptyset$ by the definition of $\mathcal{B}_X$. So this factor commutes with $A \in \mathfrak{A}(X)$, which gives the identity. The right side lies in $\mathfrak{A}(X\cup\mathrm{Col}(\mathcal{B}_X))$, since $e^{\pm\mathrm{i}u\iota(Q_{\mathcal{B}_X})} \in \mathfrak{A}(\mathrm{Col}(\mathcal{B}_X))$. Now let $X = \mathrm{Str}(I)$ and $j \in \mathcal{B}_X$, and let $s \in \mathrm{Col}(j)\cap X$. For every $s' \in \mathrm{Col}(j)$, $|x_{s'} - x_s|_{L_x} \le |x_{s'} - \kappa j|_{L_x} + |\kappa j - x_s|_{L_x} < 2\delta_x$ and $x_s \in I$, so $\mathrm{dist}(x_{s'}, I) < 2\delta_x$ and $s' \in \mathrm{Str}(I_{2\delta_x})$. Hence $\mathrm{Col}(\mathcal{B}_X) \subset \mathrm{Str}(I_{2\delta_x})$, and $X \subset \mathrm{Str}(I_{2\delta_x})$ trivially.

(iii) Let C_k be the constants of A, and $A_{N,r} \in \mathfrak{A}((Z_N)_{(r)})$ with $\|\iota(A_N) - A_{N,r}\| \le C_k N^m (1+r)^{-k}$. Put $\mathcal{U}_N := e^{\mathrm{i}u\iota(Q_{\mathcal{B}_N})}$, which equals $\iota(e^{\mathrm{i}uQ_{\mathcal{B}_N}})$ by Lemma 3.8(ii), so that $\iota(e^{\mathrm{i}uQ_{\mathcal{B}_N}} A_N e^{-\mathrm{i}uQ_{\mathcal{B}_N}}) =$

$\mathcal{U}_N\iota(A_N)\mathcal{U}_N^*$. By Lemma 3.10, $(Z_N)_{(r)} \subset \mathrm{Str}((I_N)_r)$; by (ii), $\mathcal{U}_N A_{N,r}\mathcal{U}_N^* \in \mathfrak{A}(\mathrm{Str}(((I_N)_r)_{2\delta_x}))$; the triangle inequality gives $((I_N)_r)_{2\delta_x} \subset (I_N)_{r+2\delta_x}$; and by Lemma 3.10 again, $\mathrm{Str}((I_N)_{r+2\delta_x}) \subset (Z_N)_{(r+3\delta_x)}$. Hence $\mathcal{U}_N A_{N,r}\mathcal{U}_N^* \in \mathfrak{A}((Z_N)_{(r+3\delta_x)})$, and $\|\mathcal{U}_N\iota(A_N)\mathcal{U}_N^* - \mathcal{U}_N A_{N,r}\mathcal{U}_N^*\| = \|\iota(A_N) - A_{N,r}\| \le C_k N^m(1+r)^{-k}$. Given $r' \ge 0$, if $r' \ge 3\delta_x$, take $r := r' - 3\delta_x$; since $1+r' \le (1+3\delta_x)(1+r)$, the error is at most $C_k(1+3\delta_x)^k N^m(1+r')^{-k}$. If $r' < 3\delta_x$, take the approximant 0; the error is $\|A_N\| \le CN^m \le C(1+3\delta_x)^k N^m(1+r')^{-k}$. As $\delta_x \le 2\delta_0$, the conjugated sequence is almost supported in Z with prefactor exponent m and constants $(1+6\delta_0)^k \max\{C_k, C\}$, independent of u and of the bands. □

Conjugation by a band charge thus preserves strips exactly. A ball, on the other hand, is in general only mapped into its union with full columns: on a column that straddles an edge of the band, the one-particle operator $\mathcal{S}P_{\mathcal{B}}\mathcal{S}^*$, with $P_{\mathcal{B}}$ the orthogonal projection onto the span of the ψ_j, $j \in \mathcal{B}$, is a sharply truncated Fourier sum along the column. For this reason every distance used in Sections 6 and 7 is a transverse distance between strips.

5 Locality estimates

Throughout this section $(\mathbb{T}_N)_{N\in\mathcal{N}}$ are admissible tori and $H = (H_N)$ satisfies (H1) of Definition 4.6, with constants C_h and C_k^H. We write $\widehat{H} = \widehat{H}_N := \iota(H_N) = \int \iota(h_R)\,\mathrm{d}R$ and $\tau_t(A) := e^{\mathrm{i}t\widehat{H}} A e^{-\mathrm{i}t\widehat{H}}$ for operators A on $\widehat{\mathcal{H}}_N$. By Lemma 3.8(ii), $\tau_t(\iota(A)) = \iota(e^{\mathrm{i}tH_N} A e^{-\mathrm{i}tH_N})$ for operators A on $\mathcal{H}_N$, and we use the same symbol τ_t for the dynamics $A \mapsto e^{\mathrm{i}tH_N} A e^{-\mathrm{i}tH_N}$ on $\mathcal{H}_N$. All constants in this section depend only on the indicated orders, on C_{vol} (hence on δ_0), on C_h, on finitely many of the C_k^H, and on the constants of the operators or sequences to which a statement is applied (for Lemma 5.13, those of the sequences in $\mathcal{O}$); in addition, those of Lemma 5.6 depend on F, those of Lemmas 5.9 and 5.10 and of Proposition 5.11 on ϑ, γ, Δ, δ and ζ, and those of Lemma 5.13 on c, C and m_*. None depends on N.

5.1 Conditional expectations and the Lieb–Robinson bound

Lemma 5.1 (conditional expectations)**.** *For $Y \subset \Lambda_N$ let $\mathbb{E}_Y$ be the trace-preserving conditional expectation from the operators on $\widehat{\mathcal{H}}_N$ onto $\mathrm{CAR}(Y)$, that is, the orthogonal projection onto $\mathrm{CAR}(Y)$ for the Hilbert–Schmidt inner product. Then $\mathbb{E}_Y$ is linear, unital, positive, contractive, idempotent and $*$-preserving; it commutes with the parity automorphism and with the gauge transformations of Lemma 3.8(iii); it maps even operators into $\mathfrak{A}(Y)$; and for every even A,*

$$\|A - \mathbb{E}_Y(A)\| \le 2\inf\{\|A - A'\| : A' \in \mathfrak{A}(Y)\}.$$

In particular, if $u \mapsto A_u$ is continuous (respectively Borel) and $u \mapsto Y_u$ takes finitely many values on Borel sets, then $u \mapsto \mathbb{E}_{Y_u}(A_u)$ is Borel; approximants may therefore always be chosen measurable in time and in space, at the cost of a factor 2.

Proof. Write $\widehat{N} := \widehat{N}_\uparrow + \widehat{N}_\downarrow$, so that the parity automorphism is $A \mapsto (-1)^{\widehat{N}} A (-1)^{\widehat{N}}$.

A tensor decomposition adapted to Y. Number the $n := 2|\Lambda_N|$ modes (s,σ) of $\widehat{\mathcal{H}}_N$ as $\mu_1, \dots, \mu_n$ in such a way that the modes with $s \in Y$ are $\mu_1, \dots, \mu_{n_Y}$, $n_Y := 2|Y|$, and write $a_j := \hat{c}_{\mu_j}$. With $\widehat{\Omega}_0$ the vacuum of $\widehat{\mathcal{H}}_N$, the vectors

$$|n_1 \cdots n_n\rangle := (a_1^*)^{n_1} \cdots (a_n^*)^{n_n}\, \widehat{\Omega}_0, \qquad n_j \in \{0,1\},$$

form an orthonormal basis of $\widehat{\mathcal{H}}_N$, and moving a_j to the right with the anticommutation relations gives $a_j|n_1\cdots n_n\rangle = (-1)^{n_1+\cdots+n_{j-1}}\, n_j\, |n_1\cdots n_{j-1}\,0\,n_{j+1}\cdots n_n\rangle$. We identify $\widehat{\mathcal{H}}_N$ with $(\mathbb{C}^2)^{\otimes n}$ by $|n_1\cdots n_n\rangle \mapsto e_{n_1}\otimes\cdots\otimes e_{n_n}$, where (e_0,e_1) is the standard basis of $\mathbb{C}^2$; for an operator b on $\mathbb{C}^2$ we write $b_{(i)}$ for b acting in the i-th factor, and we put $\mathsf{l} := |e_0\rangle\langle e_1|$ and $\mathsf{z} := |e_0\rangle\langle e_0| - |e_1\rangle\langle e_1|$. Then

$$a_j = \mathsf{z}_{(1)}\cdots\mathsf{z}_{(j-1)}\,\mathsf{l}_{(j)}, \qquad a_j^* a_j = |e_1\rangle\langle e_1|_{(j)}.$$

Each a_j with $j \le n_Y$ acts only in the first n_Y factors, so $\mathrm{CAR}(Y) \subset \mathcal{B}_{n_Y}\otimes\mathbf{1}$, where $\mathcal{B}_{n_Y}$ denotes the algebra of all operators on $(\mathbb{C}^2)^{\otimes n_Y}$ and $\mathbf{1}$ the identity of $(\mathbb{C}^2)^{\otimes(n-n_Y)}$. Conversely, for $j \le n_Y$, $\mathsf{z}_{(j)} = \mathbf{1} - 2a_j^*a_j \in \mathrm{CAR}(Y)$, hence $\mathsf{l}_{(j)} = \mathsf{z}_{(1)}\cdots\mathsf{z}_{(j-1)}a_j \in \mathrm{CAR}(Y)$; the operators $\mathsf{l}_{(j)}\mathsf{l}_{(j)}^*$, $\mathsf{l}_{(j)}$, $\mathsf{l}_{(j)}^*$ and $\mathsf{l}_{(j)}^*\mathsf{l}_{(j)}$ are the four matrix units of the j-th factor, and products of matrix units of the factors $1,\dots,n_Y$ span $\mathcal{B}_{n_Y}\otimes\mathbf{1}$. Therefore

$$\mathrm{CAR}(Y) = \mathcal{B}_{n_Y}\otimes\mathbf{1}. \tag{5.1}$$

For $Y=\Lambda_N$ this says that $\mathrm{CAR}(\Lambda_N)$ is the algebra of all operators on $\widehat{\mathcal{H}}_N$, and hence that $\mathfrak{A}(\Lambda_N)$ is the algebra of all even operators on $\widehat{\mathcal{H}}_N$.

The conditional expectation. Let tr' be the normalized trace on the operators on $(\mathbb{C}^2)^{\otimes(n-n_Y)}$, and let $\mathbb{E}'$ be the linear map with $\mathbb{E}'(A_1\otimes A_2) = \mathrm{tr}'(A_2)\,A_1\otimes\mathbf{1}$. It maps onto $\mathrm{CAR}(Y)$ and fixes every element of $\mathrm{CAR}(Y)$, and for $C = C_1\otimes\mathbf{1}\in\mathrm{CAR}(Y)$ and $A = A_1\otimes A_2$,

$$\mathrm{Tr}\big(C^*(A-\mathbb{E}'(A))\big) = \mathrm{Tr}(C_1^*A_1)\big(\mathrm{Tr}\,A_2 - 2^{n-n_Y}\mathrm{tr}'(A_2)\big) = 0.$$

By linearity $A-\mathbb{E}'(A)$ is orthogonal to $\mathrm{CAR}(Y)$ for the Hilbert–Schmidt inner product for every A, so $\mathbb{E}' = \mathbb{E}_Y$. Let $\mathsf{p}_0 := \mathbf{1}$, p_1, p_2 and $\mathsf{p}_3 = \mathsf{z}$ be the identity and the three Pauli matrices on $\mathbb{C}^2$, and for $\alpha\in\{0,1,2,3\}^q$ let $\mathsf{p}_\alpha := \mathsf{p}_{\alpha_1}\otimes\cdots\otimes\mathsf{p}_{\alpha_q}$. For every operator C on $(\mathbb{C}^2)^{\otimes q}$,

$$4^{-q}\sum_{\alpha\in\{0,1,2,3\}^q} \mathsf{p}_\alpha\, C\, \mathsf{p}_\alpha^* = 2^{-q}\,\mathrm{Tr}(C)\,\mathbf{1}: \tag{5.2}$$

for $q=1$ this is checked on the four matrix units $|e_a\rangle\langle e_b|$, and for general q both sides are multiplicative on products $C_1\otimes\cdots\otimes C_q$, which span. Applied in the last $n-n_Y$ factors, (5.2) gives

$$\mathbb{E}_Y(A) = 4^{-(n-n_Y)}\sum_{\alpha\in\{0,1,2,3\}^{n-n_Y}} (\mathbf{1}\otimes\mathsf{p}_\alpha)\,A\,(\mathbf{1}\otimes\mathsf{p}_\alpha)^*,$$

an average of conjugations by unitaries. Hence $\mathbb{E}_Y$ is linear, unital, positive, $*$-preserving and contractive; it is idempotent, being a projection, and it preserves the trace.

Symmetries. The parity operator $(-1)^{\widehat{N}} = \prod_j(\mathbf{1}-2a_j^*a_j) = \mathsf{z}^{\otimes n}$ and the gauge unitaries $e^{\mathrm{i}(\varphi_\uparrow\widehat{N}_\uparrow+\varphi_\downarrow\widehat{N}_\downarrow)} = \bigotimes_j e^{\mathrm{i}\varphi_{\sigma(j)}a_j^*a_j}$, where $\sigma(j)$ is the spin of the mode μ_j, are products $U = U_1\otimes U_2$ of a unitary on the first n_Y factors and one on the last $n-n_Y$ factors. For such U and $A = A_1\otimes A_2$, $\mathbb{E}_Y(UAU^*) = \mathrm{tr}'(U_2A_2U_2^*)\,U_1A_1U_1^*\otimes\mathbf{1} = U\,\mathbb{E}_Y(A)\,U^*$, because tr' is invariant under unitary conjugation; by linearity $\mathbb{E}_Y$ commutes with conjugation by U. If A is even, then $(-1)^{\widehat{N}}\mathbb{E}_Y(A)(-1)^{\widehat{N}} = \mathbb{E}_Y(A)$, so $\mathbb{E}_Y(A)$ is an even element of $\mathrm{CAR}(Y)$, that is, $\mathbb{E}_Y(A)\in\mathfrak{A}(Y)$.

The approximation bound. If $A'\in\mathfrak{A}(Y)$, then $\mathbb{E}_Y(A') = A'$, hence $A-\mathbb{E}_Y(A) = (A-A') - \mathbb{E}_Y(A-A')$ and $\|A-\mathbb{E}_Y(A)\| \le 2\|A-A'\|$.

Measurability. $\mathbb{E}_Y$ is a linear map of a finite-dimensional space, hence continuous, so $u\mapsto\mathbb{E}_Y(A_u)$ is continuous (respectively Borel) when $u\mapsto A_u$ is. If $Y_u = Y_i$ for $u\in\mathcal{U}_i$, where $\mathcal{U}_1,\dots,\mathcal{U}_J$ are Borel sets partitioning the parameter set, then $\mathbb{E}_{Y_u}(A_u) = \sum_i \mathbf{1}_{\mathcal{U}_i}(u)\,\mathbb{E}_{Y_i}(A_u)$ is Borel. □

An *even interaction* on Λ_N is a map Ψ from the subsets of Λ_N to operators on $\widehat{\mathcal{H}}_N$ with $\Psi(Z) = \Psi(Z)^* \in \mathfrak{A}(Z)$; its Hamiltonian is $H_\Psi := \sum_Z \Psi(Z)$, its dynamics is $\tau_t^\Psi(A) := e^{\mathrm{i}tH_\Psi} A e^{-\mathrm{i}tH_\Psi}$, and its *chain matrix* is the $\Lambda_N \times \Lambda_N$ matrix $M^\Psi_{ss'} := \sum_{Z \ni s,s'} \|\Psi(Z)\|$ (diagonal included).

Theorem 5.2 (Lieb–Robinson bound for even interactions)**.** *Let Ψ be an even interaction on Λ_N, let $X, Y \subset \Lambda_N$, and let $A \in \mathrm{CAR}(X)$, $B \in \mathfrak{A}(Y)$ (or $A \in \mathfrak{A}(X)$, $B \in \mathrm{CAR}(Y)$). Then for every $t \in \mathbb{R}$*

$$\|[\tau_t^\Psi(A), B]\| \le 2\|A\|\|B\| \sum_{s \in X,\ s' \in Y} \big[e^{2|t|M^\Psi}\big]_{ss'},$$

and the matrix $e^{2|t|M^\Psi}$ may be replaced by $e^{2|t|M^\Psi} - \mathbf{1}$ if $X \cap Y = \emptyset$. For $\lambda > 0$ let $C(\Psi,\lambda) := \max_s \sum_{s'} M^\Psi_{ss'} e^{d(s,s')/\lambda}$. Then

$$\|[\tau_t^\Psi(A), B]\| \le 2\|A\|\|B\|\, e^{2|t|C(\Psi,\lambda)} \sum_{s \in X} e^{-d(s,Y)/\lambda},$$

and $e^{2|t|C(\Psi,\lambda)}$ may be replaced by $e^{2|t|C(\Psi,\lambda)} - 1$ if $X \cap Y = \emptyset$. Both bounds remain valid, with the same matrix M^Ψ, for the dynamics of any interaction obtained from Ψ by setting some of its terms to zero.

This is the Lieb–Robinson bound of [NSY18, Theorem 3.1] for lattice fermions, in the form that results from bounding the chain sums in its proof by powers of the chain matrix; it is derived from [NSY18, Lemma 3.3] in Appendix B, where the proof is given.

5.2 Propagation from a set

Theorem 5.3 (propagation from a set)**.** *Assume (H1). Let $c_k := 2^k \max\{C_h, 4C_k^H\}$, $r_j := 2^j$ for $j \ge 0$, and*

$$C_1 := e^2 C_{\mathrm{vol}} \sum_{j \ge 0} c_5 (1 + r_j)^{-5}\, \pi r_j^2\, (1 + 2r_j)^2 \ \le\ 192\pi e^2\, C_{\mathrm{vol}} \max\{C_h, 4C_5^H\}.$$

*(a) (*master inequality*) For every dyadic $\lambda = r_{j_0} \ge 1$, every $k \ge 3$, $\rho > 0$, $X \subset \Lambda_N$, $A \in \mathrm{CAR}(X)$ and $t \in \mathbb{R}$,*

$$\big\|\tau_t(A) - \tau_t^{\lambda, X_{(\rho)}}(A)\big\| \le \|A\||X|\Big[\frac{16\pi c_k}{3C_1}\lambda^{-(k-2)}(1 + C_1|t|)^3 + e^{-\rho/\lambda}\big(e^{2C_1|t|} - 1\big)\Big],$$

where, for $Y \subset \Lambda_N$, $\tau^{\lambda,Y}$ is the dynamics generated by a self-adjoint operator $\widehat{H}_{\lambda,Y} \in \mathfrak{A}(Y)$ constructed in the proof; in particular $\tau_t^{\lambda, X_{(\rho)}}(A) \in \mathrm{CAR}(X_{(\rho)})$ has the parity of A.

*(b) (*light cone*) For $\beta \in (0,1)$ and $k' > 0$ put $C_{\mathrm{lc}}(\beta) := \max\{1, (4C_1)^{1/(1-\beta)}\}$, $\lambda := 2^{\lfloor \beta \log_2 \rho \rfloor}$ and $k := 2 + \lceil (k'+3)/\beta \rceil$. Whenever $\rho \ge C_{\mathrm{lc}}(\beta)(1 + |t|)^{1/(1-\beta)}$,*

$$\inf\big\{\|\tau_t(A) - A'\| : A' \in \mathrm{CAR}(X_{(\rho)}) \text{ of the parity of } A\big\} \le C_{k',\beta}\|A\||X|\rho^{-k'},$$

where

$$C_{k',\beta} := \frac{16\pi c_k 2^{k-2}}{3C_1}\Big(\frac{\max\{1, C_1\}}{4C_1}\Big)^3 + \Big(\frac{2k'}{e(1-\beta)}\Big)^{k'/(1-\beta)}.$$

*(c) (*every light-cone exponent above one*) Consequently, for every $\alpha > 1$ and $k > 0$ there are C_{lc} and C_k, depending only on α, k, C_{vol}, C_h and finitely many $C_{k'}^H$, such that for all N, $X \subset \Lambda_N$, even $A \in \mathfrak{A}(X)$, $t \in \mathbb{R}$ and $\rho \geq C_{\mathrm{lc}}(1+|t|)^\alpha$,*

$$\inf\{\|\tau_t(A) - A'\| : A' \in \mathfrak{A}(X_{(\rho)})\} \leq C_k \|A\| |X| \rho^{-k};$$

for $k = 0$ the same bound holds with the constant 1 in place of C_k.

Proof. Fix N and drop it from the notation. Besides (H1), the proof uses only the ball count of Proposition 3.4(ii), the fact that the disc $\mathbb{D}_r(R) := \{R' \in \mathbb{T}_N : d(R,R') \leq r\}$ has measure at most πr^2, being the image of a planar disc of radius r under the covering map, and, in Step 4, the fact that the lattice of periods of $\mathbb{T}_N$ is rectangular. For a statement S we write $[S] := 1$ if S holds and $[S] := 0$ otherwise. We may assume $X \neq \emptyset$: if $X = \emptyset$, then $A \in \mathrm{CAR}(\emptyset) = \mathbb{C}\mathbf{1}$ and every difference estimated below vanishes.

Step 0: canonical approximants. For $R \in \mathbb{T}_N$ and $r \geq 0$ put

$$\tilde{h}_{R,r} := \mathbb{E}_{\Lambda_r(R)}(\iota(h_R)).$$

Since h_R is sector-preserving it is even (Lemma 2.6(ii)), hence so is $\iota(h_R)$ (Lemma 3.8(i)). By Lemma 5.1 and (H1), $\tilde{h}_{R,r} \in \mathfrak{A}(\Lambda_r(R))$ is self-adjoint, $\|\tilde{h}_{R,r}\| \leq \|h_R\| \leq C_h$, and $\|\iota(h_R) - \tilde{h}_{R,r}\| \leq 2C_k^H(1+r)^{-k}$ for every k at once. For fixed r, the map $R \mapsto \Lambda_r(R)$ takes finitely many values, each on a Borel set $\{R : \Lambda_r(R) = Z\} = \bigcap_{s\in Z}\{R : d(s,R) \leq r\} \cap \bigcap_{s\notin Z}\{R : d(s,R) > r\}$, and $R \mapsto \iota(h_R)$ is Borel because $R \mapsto h_R$ is and ι is linear; by Lemma 5.1, $R \mapsto \tilde{h}_{R,r}$ is Borel.

Step 1: the multiscale interaction. Let $j_{\max}$ be the least j with $r_j \geq \operatorname{diam}\mathbb{T}_N$. For $j \geq j_{\max}$ and every R, $\Lambda_{r_j}(R) = \Lambda_N$, so $\tilde{h}_{R,r_j} = \iota(h_R)$ by (5.1). Put

$$\hat{h}_R^{(0)} := \tilde{h}_{R,r_0}, \qquad \hat{h}_R^{(j)} := \tilde{h}_{R,r_j} - \tilde{h}_{R,r_{j-1}} \quad (1 \leq j \leq j_{\max}).$$

Then $\sum_{j=0}^{j_{\max}} \hat{h}_R^{(j)} = \iota(h_R)$, each $\hat{h}_R^{(j)} \in \mathfrak{A}(\Lambda_{r_j}(R))$ is self-adjoint and Borel in R, and, since $1 + r_{j-1} \geq (1+r_j)/2$, $\|\hat{h}_R^{(j)}\| \leq 2C_k^H[(1+r_j)^{-k} + (1+r_{j-1})^{-k}] \leq 2^{k+2}C_k^H(1+r_j)^{-k}$ for $j \geq 1$, while $\|\hat{h}_R^{(0)}\| \leq C_h = 2^k C_h(1+r_0)^{-k}$. Hence, for all j, R and k,

$$\|\hat{h}_R^{(j)}\| \leq c_k(1+r_j)^{-k}. \tag{5.3}$$

For $Z \subset \Lambda_N$ and a dyadic $\lambda = r_{j_0} \geq 1$ let

$$\Psi(Z) := \sum_{j=0}^{j_{\max}} \int_{\mathbb{T}_N} [\Lambda_{r_j}(R) = Z]\, \hat{h}_R^{(j)}\, \mathrm{d}R, \qquad \Psi_\lambda(Z) := \sum_{j=0}^{\min\{j_0, j_{\max}\}} \int_{\mathbb{T}_N} [\Lambda_{r_j}(R) = Z]\, \hat{h}_R^{(j)}\, \mathrm{d}R.$$

The integrands are bounded Borel functions with values in the finite-dimensional space $\mathfrak{A}(Z)$, so Ψ and Ψ_λ are even interactions in the sense of Theorem 5.2, and $\sum_Z \Psi(Z) = \sum_j \int \hat{h}_R^{(j)}\,\mathrm{d}R = \widehat{H}$. Put $\widehat{H}_\lambda := \sum_Z \Psi_\lambda(Z) = \int \tilde{h}_{R,r_{\min\{j_0,j_{\max}\}}}\mathrm{d}R$ and, for $Y \subset \Lambda_N$, $\widehat{H}_{\lambda,Y} := \sum_{Z\subset Y}\Psi_\lambda(Z) \in \mathfrak{A}(Y)$; τ^λ and $\tau^{\lambda,Y}$ denote the dynamics they generate. The operator $\widehat{H}_{\lambda,Y}$ is the Hamiltonian of the interaction obtained from Ψ_λ by setting its terms $\Psi_\lambda(Z)$ with $Z \not\subset Y$ to zero. (Terms with $Z = \emptyset$ are multiples of the identity and play no role.)

Step 2: the chain weights of Ψ_λ. Let $M := M^{\Psi_\lambda}$. Since $s \in \Lambda_r(R)$ if and only if $R \in \mathbb{D}_r(r_s)$, (5.3) with $k = 5$ gives

$$M_{ss'} \leq \sum_{j\leq j_0} \int [s, s' \in \Lambda_{r_j}(R)]\, \|\hat{h}_R^{(j)}\|\, \mathrm{d}R \leq \sum_{j\leq j_0} c_5(1+r_j)^{-5}\, |\mathbb{D}_{r_j}(r_s) \cap \mathbb{D}_{r_j}(r_{s'})|.$$

The intersection is empty unless $d(s,s') \le 2r_j$, and its measure is at most πr_j^2. If $d(s,s') \le 2r_j$ with $j \le j_0$, then $e^{d(s,s')/\lambda} \le e^2$, as $r_j \le \lambda$; and the number of s' with $d(s,s') \le 2r_j$ is $\#\Lambda_{2r_j}(r_s) \le C_{\mathrm{vol}}(1+2r_j)^2$. Hence

$$C(\Psi_\lambda,\lambda) = \max_s \sum_{s'} M_{ss'} e^{d(s,s')/\lambda} \le e^2 C_{\mathrm{vol}} \sum_{j\ge 0} c_5 (1+r_j)^{-5}\, \pi r_j^2\, (1+2r_j)^2 = C_1, \tag{5.4}$$

uniformly in λ and N. The uniformity in λ rests on the summability over the scales j: for a representation of $\widehat{H}_\lambda$ by a single term $\tilde{h}_{R,\lambda}$ per point R, the same argument bounds the weighted row sums only by $e^2 C_h \pi \lambda^2 C_{\mathrm{vol}}(1+2\lambda)^2$, which grows like λ^4. Since $\pi r_j^2 \le \pi(1+r_j)^2$ and $(1+2r_j)^2 \le 4(1+r_j)^2$, the j-th term of the series is at most $4\pi c_5 (1+2^j)^{-1}$, and $\sum_{j\ge 0}(1+2^j)^{-1} < \frac{3}{2}$; this gives $C_1 \le 6\pi e^2 C_{\mathrm{vol}} c_5 = 192\pi e^2 C_{\mathrm{vol}} \max\{C_h, 4C_5^H\}$.

Step 3: localization in $X_{(\rho)}$. We write the time integrals for $t \ge 0$; for $t < 0$ the same estimates hold with $\int_0^t$ replaced by $\int_t^0$, since all bounds below depend on time only through its absolute value. Let $Y := X_{(\rho)} \supset X$. As $A \in \mathrm{CAR}(X) \subset \mathrm{CAR}(Y)$ and $e^{\mathrm{i}t\widehat{H}_{\lambda,Y}}$ is an even element of $\mathfrak{A}(Y)$, the operator $\tau_t^{\lambda,Y}(A)$ lies in $\mathrm{CAR}(Y)$ and has the parity of A. The derivative of $u \mapsto \tau_u^\lambda(\tau_{t-u}^{\lambda,Y}(A))$ is $\tau_u^\lambda(\mathrm{i}[\widehat{H}_\lambda - \widehat{H}_{\lambda,Y}, \tau_{t-u}^{\lambda,Y}(A)])$, so that

$$\tau_t^\lambda(A) - \tau_t^{\lambda,Y}(A) = \int_0^t \tau_u^\lambda\Big(\mathrm{i}\big[\widehat{H}_\lambda - \widehat{H}_{\lambda,Y}, \tau_{t-u}^{\lambda,Y}(A)\big]\Big)\mathrm{d}u, \qquad \widehat{H}_\lambda - \widehat{H}_{\lambda,Y} = \sum_{Z\not\subset Y} \Psi_\lambda(Z).$$

Theorem 5.2, applied to the interaction whose dynamics is $\tau^{\lambda,Y}$ (with the matrix M, by its last sentence), to $A \in \mathrm{CAR}(X)$ and to $B = \Psi_\lambda(Z) \in \mathfrak{A}(Z)$, gives

$$\|[\Psi_\lambda(Z), \tau_{t'}^{\lambda,Y}(A)]\| \le 2\|A\|\|\Psi_\lambda(Z)\| \sum_{s\in X}\sum_{s''\in Z} \big[e^{2|t'|M}\big]_{ss''}.$$

For each $Z \not\subset Y$ choose $s_Z \in Z \setminus Y$. For every nonnegative function f on Λ_N,

$$\sum_{Z\not\subset Y} \|\Psi_\lambda(Z)\| \sum_{s''\in Z} f(s'') \le \sum_{s'\notin Y}\sum_{s''\in\Lambda_N} f(s'') \sum_{Z\ni s'',s'} \|\Psi_\lambda(Z)\| = \sum_{s'\notin Y}\sum_{s''\in\Lambda_N} f(s'') M_{s''s'},$$

because for each Z the right-hand side contains the term with $s' = s_Z$. Hence, as $\frac{\mathrm{d}}{\mathrm{d}t'} e^{2t'M} = 2e^{2t'M}M$,

$$\|\tau_t^\lambda(A) - \tau_t^{\lambda,Y}(A)\| \le 2\|A\| \sum_{s\in X}\sum_{s'\notin Y} \int_0^{|t|} \big[e^{2t'M}M\big]_{ss'}\mathrm{d}t' = \|A\| \sum_{s\in X}\sum_{s'\notin Y} \big[e^{2|t|M} - \mathbf{1}\big]_{ss'}.$$

For $s \in X$, every $s' \notin Y = X_{(\rho)}$ satisfies $d(s,s') > \rho$. By (B.4) of Appendix B and (5.4), $\sum_{s'\notin Y}[e^{2|t|M} - \mathbf{1}]_{ss'} \le e^{-\rho/\lambda}(e^{2C_1|t|} - 1)$, and therefore

$$\|\tau_t^\lambda(A) - \tau_t^{\lambda,X_{(\rho)}}(A)\| \le \|A\||X|\, e^{-\rho/\lambda}\big(e^{2C_1|t|} - 1\big). \tag{5.5}$$

Step 4: restoring the scales above λ. If $j_0 \ge j_{\max}$, then $\widehat{H}_\lambda = \widehat{H}$ and $\tau = \tau^\lambda$. Otherwise $\widehat{H} - \widehat{H}_\lambda = \sum_{j_0<j\le j_{\max}} \int \hat{h}_R^{(j)}\,\mathrm{d}R$ and, as in Step 3,

$$\tau_t(A) - \tau_t^\lambda(A) = \int_0^t \tau_u\Big(\mathrm{i}\big[\widehat{H} - \widehat{H}_\lambda, \tau_{t-u}^\lambda(A)\big]\Big)\mathrm{d}u,$$

$$\|\tau_t(A) - \tau_t^\lambda(A)\| \le \int_0^{|t|} \sum_{j_0<j\le j_{\max}} \int_{\mathbb{T}_N} \|[\hat{h}_R^{(j)}, \tau_{t'}^\lambda(A)]\|\,\mathrm{d}R\,\mathrm{d}t'.$$

Only the truncated dynamics τ^λ enters, since τ_u is isometric; the norm $\|\widehat{H} - \widehat{H}_\lambda\|$, for which only a bound proportional to $N(1+\lambda)^{-k}$ is available, is never used. Fix $j > j_0$, R and t'. Since $\hat{h}_R^{(j)} \in \mathfrak{A}(\Lambda_{r_j}(R))$ and $d(s, \Lambda_{r_j}(R)) \geq (d(s,R) - r_j)_+$ for every slot s, the weighted bound of Theorem 5.2 for τ^λ, (5.4), the trivial bound $\|[B, \tau_{t'}^\lambda(A)]\| \leq 2\|A\|\|B\|$ and $\min\{1, \sum_s a_s\} \leq \sum_s \min\{1, a_s\}$ (for $a_s \geq 0$) give, with (5.3),

$$\begin{aligned}\|[\hat{h}_R^{(j)}, \tau_{t'}^\lambda(A)]\| &\leq 2\|A\| c_k (1+r_j)^{-k} \sum_{s\in X} \min\{1, e^{2C_1|t'| - (d(s,R)-r_j)_+/\lambda}\} \\ &= 2\|A\| c_k (1+r_j)^{-k} \sum_{s\in X} e^{-(d(s,R)-q)_+/\lambda},\end{aligned}$$

where $q := r_j + 2C_1|t'|\lambda$. Keeping the sum over $s \in X$ outside the integral over R is what makes the final bound linear in $|X|$. For a slot s, let $\tilde{r}_s \in \mathbb{R}^2$ be a lift of r_s. The rectangle $\tilde{r}_s + [-L_x/2, L_x/2) \times [-L_y/2, L_y/2)$ is a fundamental domain of $\mathbb{T}_N$, and for $\tilde{R}$ in it the distance from r_s to the image of $\tilde{R}$ equals the Euclidean distance $|\tilde{R} - \tilde{r}_s|$, because the lattice of periods is rectangular. Hence

$$\begin{aligned}\int_{\mathbb{T}_N} e^{-(d(s,R)-q)_+/\lambda} \mathrm{d}R &\leq \int_{\mathbb{R}^2} e^{-(|\xi|-q)_+/\lambda} \mathrm{d}\xi = \pi q^2 + 2\pi \int_q^\infty v\, e^{-(v-q)/\lambda} \mathrm{d}v \\ &= \pi(q^2 + 2q\lambda + 2\lambda^2) \leq 2\pi(q+\lambda)^2.\end{aligned}$$

Since $r_j \geq 2\lambda$, $q + \lambda \leq 2r_j(1 + C_1|t'|)$, and therefore

$$\int_{\mathbb{T}_N} \|[\hat{h}_R^{(j)}, \tau_{t'}^\lambda(A)]\| \, \mathrm{d}R \leq 16\pi c_k \|A\| |X| (1 + C_1|t'|)^2 r_j^{2-k}.$$

As $k \geq 3$, $\sum_{j>j_0} r_j^{2-k} = \lambda^{2-k}/(2^{k-2} - 1) \leq \lambda^{-(k-2)}$, and $\int_0^{|t|} (1 + C_1 t')^2 \mathrm{d}t' \leq (1 + C_1|t|)^3/(3C_1)$. Hence

$$\|\tau_t(A) - \tau_t^\lambda(A)\| \leq \frac{16\pi c_k}{3C_1} \|A\| |X| \lambda^{-(k-2)} (1 + C_1|t|)^3. \tag{5.6}$$

Proof of (a). Add (5.5) and (5.6); the properties of $\tau_t^{\lambda, X_{(\rho)}}(A)$ were shown in Step 3.

Proof of (b). Let $\rho \geq C_{\mathrm{lc}}(\beta)(1 + |t|)^{1/(1-\beta)}$. Then $\rho \geq 1$, so $\lambda = r_{j_0}$ with $j_0 = \lfloor \beta \log_2 \rho \rfloor \geq 0$ and $\rho^\beta/2 < \lambda \leq \rho^\beta$; moreover $\rho^{1-\beta} \geq C_{\mathrm{lc}}(\beta)^{1-\beta}(1 + |t|) \geq 4C_1(1 + |t|)$, and $\rho/\lambda \geq \rho^{1-\beta}$. We apply (a) with this λ, with $k = 2 + \lceil (k' + 3)/\beta \rceil \geq 3$, and with the approximant $A' := \tau_t^{\lambda, X_{(\rho)}}(A)$, which lies in $\mathrm{CAR}(X_{(\rho)})$ and has the parity of A. For the second term of (a), $2C_1|t| \leq \rho^{1-\beta}/2$ gives $e^{-\rho/\lambda}(e^{2C_1|t|} - 1) \leq e^{-\rho^{1-\beta}/2}$, and with $x := \rho^{1-\beta}$, $b := k'/(1-\beta)$ and $\max_{x>0} x^b e^{-x/2} = (2b/e)^b$,

$$e^{-\rho^{1-\beta}/2} \leq \Big(\frac{2k'}{e(1-\beta)}\Big)^{k'/(1-\beta)} \rho^{-k'}.$$

For the first term, $1 + C_1|t| \leq \max\{1, C_1\}(1 + |t|) \leq \max\{1, C_1\}\rho^{1-\beta}/(4C_1)$ and $\lambda^{-(k-2)} < 2^{k-2}\rho^{-\beta(k-2)}$, so it is at most

$$\frac{16\pi c_k 2^{k-2}}{3C_1} \Big(\frac{\max\{1, C_1\}}{4C_1}\Big)^3 \rho^{3(1-\beta) - \beta(k-2)},$$

and $\beta(k-2) \geq k' + 3$ gives $3(1-\beta) - \beta(k-2) \leq -k'$; since $\rho \geq 1$, the power of ρ is at most $\rho^{-k'}$. Adding the two terms gives (b).

Proof of (c). Let $\beta := 1 - 1/\alpha \in (0,1)$, so that $1/(1-\beta) = \alpha$, and $C_{\mathrm{lc}} := C_{\mathrm{lc}}(\beta) = \max\{1, (4C_1)^{\alpha}\}$. If $A \in \mathfrak{A}(X)$ is even, the approximant of (b) is an even element of $\mathrm{CAR}(X_{(\rho)})$, that is, an element of $\mathfrak{A}(X_{(\rho)})$. For $k > 0$, (b) with $k' = k$ gives the claim with $C_k := C_{k,\beta}$, which depends on α, k, C_1, C_h and $C^H_{2+\lceil (k+3)/\beta \rceil}$. For $k = 0$ the claim holds with constant 1: if $X \neq \emptyset$ the approximant 0 gives $\|\tau_t(A)\| = \|A\| \leq \|A\||X|$, and if $X = \emptyset$ then $\tau_t(A) = A \in \mathfrak{A}(\emptyset)$.

Independence of N. The constants C_1, $C_{\mathrm{lc}}(\beta)$, $C_{k',\beta}$, C_{lc} and C_k depend only on β (or α), on the decay orders, on C_{vol}, C_h, C^H_5 and one further $C^H_{k''}$, and not on N. Through C_{vol} (Proposition 3.4(ii)) and the constants of (H1) they depend on the torus only through its admissibility. □

We fix the light-cone exponent $\alpha := 4$, that is $\beta = 3/4$, in the rest of the paper, and write $C_{\mathrm{lc}} := \max\{1, (4C_1)^4\}$. Any fixed $\alpha > 1$ would do.

Remark 5.4 (what the propagation bound says, and what it does not). Theorem 5.3 gives a polynomial light cone with a constant independent of N; for $\alpha > 1$ it does not give a finite propagation velocity, and the argument does not reach $\alpha = 1$. The size $|X|$ of the initial support enters to the first power. The constants are not explicit in any useful sense: the bound on C_1 is about $4.5 \cdot 10^3\, C_{\mathrm{vol}} \max\{C_h, 4C^H_5\}$, where C^H_5 involves high derivatives of the window, and $C_{\mathrm{lc}} = \max\{1, (4C_1)^4\}$. The bound has content only for $\rho \geq C_{\mathrm{lc}}$, so every statement derived from it below is purely asymptotic in N.

5.3 Almost-supported inputs and time filters

Lemma 5.5 (propagation of almost-supported operators)**.** *For every $k \geq 0$ there is C_k such that for every even operator A on $\widehat{\mathcal{H}}_N$, every $X \subset \Lambda_N$ with $\|A\|_{X;\max\{k,3\}} < \infty$, and every $t \in \mathbb{R}$,*

$$\|\tau_t(A)\|_{X;k} \leq C_k (1 + |t|)^{4k} |X| \big(\|A\| + \|A\|_{X;\max\{k,3\}}\big).$$

Proof. Let $\bar{k} := \max\{k, 3\}$ and $n_A := \|A\| + \|A\|_{X;\bar{k}}$, finite by hypothesis. We may assume $X \neq \emptyset$: if $X = \emptyset$, then $X_{(r)} = \emptyset$ for every r and $\mathfrak{A}(\emptyset) = \mathbb{C}\mathbf{1}$, so $n_A < \infty$ forces $A \in \mathbb{C}\mathbf{1}$, and then $\tau_t(A) = A$ and both sides vanish. Let $C_{\mathrm{lc}} = \max\{1, (4C_1)^4\}$ and, for $k > 0$, let C^τ_k be the constant of Theorem 5.3(c) for $\alpha = 4$ and decay order k; for $k = 0$ put $C^\tau_0 := 1$, which is allowed by Theorem 5.3(c).

Dyadic decomposition. For $j \geq 0$ let $B_j \in \mathfrak{A}(X_{(2^j)})$ be a best approximant of A; it exists because $\mathfrak{A}(X_{(2^j)})$ is finite-dimensional, and $\|A - B_j\| \leq \|A\|_{X;\bar{k}} (1 + 2^j)^{-\bar{k}}$. Let j_X be the least j with $X_{(2^j)} = \Lambda_N$; then $B_{j_X} = A$, since A is even and $\mathfrak{A}(\Lambda_N)$ contains every even operator (Lemma 5.1, (5.1)). Put $A^{(0)} := B_0$ and $A^{(j)} := B_j - B_{j-1}$ for $1 \leq j \leq j_X$. Then $A = \sum_{j=0}^{j_X} A^{(j)}$ and $A^{(j)} \in \mathfrak{A}(X_{(2^j)})$. Moreover, as $1 + 2^{j-1} \geq (1 + 2^j)/2$,

$$\|A^{(j)}\| \leq \|A - B_j\| + \|A - B_{j-1}\| \leq 2^{\bar{k}+1} n_A (1 + 2^j)^{-\bar{k}} \qquad (1 \leq j \leq j_X),$$

and the same bound holds for $j = 0$, since $\|A^{(0)}\| \leq \|A\| + \|A - B_0\| \leq n_A$.

Large radii. Let $r \geq 2C_{\mathrm{lc}}(1 + |t|)^4$ and $\rho := r/2$, so that $\rho \geq C_{\mathrm{lc}}(1 + |t|)^4$ and $r \geq 2$. For j with $2^j \leq r/2$, Theorem 5.3(c), applied to the even operator $A^{(j)} \in \mathfrak{A}(X_{(2^j)})$, gives $B'_j \in \mathfrak{A}\big((X_{(2^j)})_{(\rho)}\big)$ with $\|\tau_t(A^{(j)}) - B'_j\| \leq C^\tau_k \|A^{(j)}\| |X_{(2^j)}| \rho^{-k}$, and $(X_{(2^j)})_{(\rho)} \subset X_{(2^j+\rho)} \subset X_{(r)}$ by the triangle inequality. By Proposition 3.4(ii), $|X_{(2^j)}| \leq \sum_{s \in X} \#\Lambda_{2^j}(r_s) \leq C_{\mathrm{vol}}(1 + 2^j)^2 |X|$. With $B' := \sum_{2^j \leq r/2} B'_j \in \mathfrak{A}(X_{(r)})$, and approximating $\tau_t(A^{(j)})$ by 0 when $2^j > r/2$,

$$\|\tau_t(A) - B'\| \leq C^\tau_k 2^k r^{-k}\, C_{\mathrm{vol}} |X| \sum_{j \geq 0} \|A^{(j)}\| (1 + 2^j)^2 + \sum_{2^j > r/2} \|A^{(j)}\|.$$

Since $\bar{k} \geq 3$, $\sum_j (1+2^j)^{2-\bar{k}} \leq \sum_j (1+2^j)^{-1} < \frac{3}{2}$, so the first sum is at most $3 \cdot 2^{\bar{k}} n_A$. If j_* is the least j with $2^j > r/2$, the second sum is at most $2^{\bar{k}+1} n_A \sum_{j \geq j_*} 2^{-j\bar{k}} \leq 2^{\bar{k}+1} \frac{8}{7} n_A (r/2)^{-\bar{k}} \leq 2^{k+\bar{k}+2} n_A r^{-k}$, because $r/2 \geq 1$ and $\bar{k} \geq k$. As $|X| \geq 1$ and $1 + r \leq 2r$,

$$(1+r)^k \inf\{\|\tau_t(A) - A'\| : A' \in \mathfrak{A}(X_{(r)})\} \leq 2^{2k+\bar{k}} (3C_k^\tau C_{\mathrm{vol}} + 4) |X|\, n_A.$$

Small radii. If $r < 2C_{\mathrm{lc}}(1+|t|)^4$, then $1 + r \leq 3C_{\mathrm{lc}}(1+|t|)^4$, and the approximant 0 gives $(1+r)^k \inf\{\|\tau_t(A) - A'\| : A' \in \mathfrak{A}(X_{(r)})\} \leq (1+r)^k \|A\| \leq (3C_{\mathrm{lc}})^k (1+|t|)^{4k} |X|\, n_A$, since $\|\tau_t(A)\| = \|A\|$.

Taking the supremum over $r \geq 0$ proves the lemma with $C_k := \max\{2^{2k+\bar{k}}(3C_k^\tau C_{\mathrm{vol}} + 4), (3C_{\mathrm{lc}})^k\}$, which depends only on k, C_{vol}, C_h and finitely many $C_{k'}^H$. □

Lemma 5.6 (time filters)**.** *Let $F \in L^1(\mathbb{R})$ with $M_j(F) := \int (1+|t|)^j |F(t)|\, \mathrm{d}t < \infty$ for every j. For every k there is C_k such that for every even operator A on $\widehat{\mathcal{H}}_N$ and every $X \subset \Lambda_N$ with $\|A\|_{X;\max\{k,3\}} < \infty$,*

$$\Big\| \int F(t) \tau_t(A)\, \mathrm{d}t \Big\|_{X;k} \leq C_k |X| (\|A\| + \|A\|_{X;\max\{k,3\}}) M_{4k}(F).$$

Consequently, if $A = (A_N) \in \mathcal{A}_Z$ is even, with prefactor exponent m and $\|A_N\| \leq CN^{m'}$, and $|Z_N| \leq CN^{m''}$, then $\big(\int F(t)\tau_t(A_N)\, \mathrm{d}t\big)_N \in \mathcal{A}_Z$ with prefactor exponent $\max\{m, m'\} + m''$.

Proof. Let $C_k^\sharp$ be the constant of Lemma 5.5, $\bar{k} := \max\{k, 3\}$ and $n_A := \|A\| + \|A\|_{X;\bar{k}}$, finite by hypothesis; as in the proof of Lemma 5.5 we may assume $X \neq \emptyset$. The map $t \mapsto \tau_t(A)$ is continuous and bounded by $\|A\|$, and $\tau_t(A)$ is even, because $\widehat{H}$ is even (Lemma 2.6(ii) and Lemma 3.8(i)); so $\int F(t)\tau_t(A)\, \mathrm{d}t$ is defined. Fix $r \geq 0$ and put $B_t := \mathbb{E}_{X_{(r)}}(\tau_t(A))$. By Lemma 5.1, $t \mapsto B_t$ is continuous with values in $\mathfrak{A}(X_{(r)})$, $\|B_t\| \leq \|A\|$, and, by the definition of the set norm and Lemma 5.5,

$$\|\tau_t(A) - B_t\| \leq 2(1+r)^{-k} \|\tau_t(A)\|_{X;k} \leq 2C_k^\sharp (1+|t|)^{4k} |X|\, n_A\, (1+r)^{-k}.$$

Hence $B := \int F(t) B_t\, \mathrm{d}t \in \mathfrak{A}(X_{(r)})$, and

$$\Big\| \int F(t)\tau_t(A)\, \mathrm{d}t - B \Big\| \leq \int |F(t)|\, \|\tau_t(A) - B_t\|\, \mathrm{d}t \leq 2C_k^\sharp |X|\, n_A\, M_{4k}(F) (1+r)^{-k},$$

which is the first claim with $C_k := 2C_k^\sharp$.

For the second claim let C_j^A be the constants of A in Definition 3.11. For each N, the operator $\iota(A_N)$ is even, $\|\iota(A_N)\| = \|A_N\| \leq CN^{m'}$, and $\|\iota(A_N)\|_{Z_N;j} \leq C_j^A N^m$ for every j. Since ι is linear and continuous and intertwines the dynamics (Lemma 3.8(ii)), $\iota(\int F(t)\tau_t(A_N)\, \mathrm{d}t) = \int F(t)\tau_t(\iota(A_N))\, \mathrm{d}t$. By the first claim with $X = Z_N$, for every $r \geq 0$ the distance of this operator to $\mathfrak{A}((Z_N)_{(r)})$ is at most

$$C_k\, CN^{m''} (CN^{m'} + C_{\bar{k}}^A N^m) M_{4k}(F) (1+r)^{-k} \leq C_k' N^{\max\{m,m'\}+m''} (1+r)^{-k},$$

and the distance is attained by an element of $\mathfrak{A}((Z_N)_{(r)})$. This is the second claim. □

Lemma 5.7 (commutator with the Hamiltonian)**.** *If $A = (A_N) \in \mathcal{A}_X$ is even, with prefactor exponent m and $\|A_N\| \leq CN^{m'}$, then $(\mathrm{i}[H_N, A_N])_N \in \mathcal{A}_X$ with prefactor exponent $1 + \max\{m, m'\}$, and $\|[H_N, A_N]\| \leq 2C_h |\mathbb{T}_N| \|A_N\| = 8\pi C_h N \|A_N\|$.*

Proof. By (H1), $\|H_N\| \le \int \|h_R\| \,\mathrm{d}R \le C_h |\mathbb{T}_N| = 4\pi C_h N$, which gives the norm bound. Fix N, drop it from the notation, and fix $r \ge 0$. Let $A_r \in \mathfrak{A}(X_{(r)})$ be a best approximant of $\iota(A)$, so that $\|\iota(A) - A_r\| \le C_k^A N^m (1+r)^{-k}$ for every k, with the constants C_k^A of A; in particular $\|A_r\| \le C N^{m'} + C_0^A N^m =: a_N$. For $R \in \mathbb{T}_N$ let $d_R := \min\{d(s,R) : s \in X_{(r)}\}$, and split $\mathbb{T}_N$ into the Borel sets $\mathcal{D}_{\le} := \{R : d_R \le r\}$ and $\mathcal{D}_{>} := \{R : d_R > r\}$. With the canonical approximants $\tilde{h}_{R,r} = \mathbb{E}_{\Lambda_r(R)}(\iota(h_R))$ of the proof of Theorem 5.3, which are Borel in R, lie in $\mathfrak{A}(\Lambda_r(R))$ and satisfy $\|\iota(h_R) - \tilde{h}_{R,r}\| \le 2C_k^H (1+r)^{-k}$, put

$$B_r := \mathrm{i} \int_{\mathcal{D}_{\le}} [\tilde{h}_{R,r}, A_r] \,\mathrm{d}R.$$

If $R \in \mathcal{D}_{\le}$ and $s \in \Lambda_r(R)$, choose $s' \in X_{(r)}$ with $d(s',R) \le r$; then $d(s,X) \le d(s,s') + d(s',X) \le 2r + r$. Hence $\Lambda_r(R) \cup X_{(r)} \subset X_{(3r)}$ for $R \in \mathcal{D}_{\le}$, and $B_r \in \mathfrak{A}(X_{(3r)})$. Since $\iota(\mathrm{i}[H,A]) = \mathrm{i}[\widehat{H}, \iota(A) - A_r] + \mathrm{i}\int_{\mathbb{T}_N} [\iota(h_R), A_r] \,\mathrm{d}R$,

$$\iota(\mathrm{i}[H,A]) - B_r = \mathrm{i}[\widehat{H}, \iota(A) - A_r] + \mathrm{i}\int_{\mathcal{D}_{\le}} [\iota(h_R) - \tilde{h}_{R,r}, A_r] \,\mathrm{d}R + \mathrm{i}\int_{\mathcal{D}_{>}} [\iota(h_R), A_r] \,\mathrm{d}R.$$

The first term has norm at most $2\|\widehat{H}\| \|\iota(A) - A_r\| \le 8\pi C_h C_k^A N^{1+m} (1+r)^{-k}$, and the second at most $|\mathbb{T}_N| \cdot 2 \cdot 2C_k^H (1+r)^{-k} a_N = 16\pi C_k^H N a_N (1+r)^{-k}$. For the third, let $R \in \mathcal{D}_{>}$. The slots of $\Lambda_{d_R/2}(R)$ are at distance at most $d_R/2 < d_R$ from R, so $\Lambda_{d_R/2}(R) \cap X_{(r)} = \emptyset$, and the operator $h_{R,d_R/2} \in \mathfrak{A}(\Lambda_{d_R/2}(R))$ of (H1) commutes with A_r. Hence $[\iota(h_R), A_r] = [\iota(h_R) - h_{R,d_R/2}, A_r]$, whose norm is at most $2C_k^H (1 + d_R/2)^{-k} a_N \le 2^{k+1} C_k^H (1+r)^{-k} a_N$, as $d_R > r$; the third term is at most $4\pi N \cdot 2^{k+1} C_k^H a_N (1+r)^{-k}$. Altogether

$$\|\iota(\mathrm{i}[H,A]) - B_r\| \le C_k' N^{1+\max\{m,m'\}} (1+r)^{-k},$$

with C_k' depending only on C, C_h, C_k^H and the constants of A. Since $B_{r/3} \in \mathfrak{A}(X_{(r)})$ and $(1 + r/3)^{-k} \le 3^k (1+r)^{-k}$, the sequence $(\mathrm{i}[H_N, A_N])_N$, which is even, is almost supported in X with prefactor exponent $1 + \max\{m, m'\}$. □

5.4 The filters

Definition 5.8 (the filters)**.** Fix an even $\vartheta \in C_c^\infty((-1,1))$ with $0 \le \vartheta \le 1$ and $\vartheta = 1$ near 0. For $\gamma > 0$ let

$$\widehat{W}_\gamma(\omega) := \frac{\mathrm{i}\,(1 - \vartheta(\omega/\gamma))}{\omega}, \qquad W_\gamma(t) := \frac{1}{2\pi} \lim_{\Omega \to \infty} \int_{-\Omega}^{\Omega} \widehat{W}_\gamma(\omega) e^{\mathrm{i}\omega t} \,\mathrm{d}\omega,$$

an improper integral, since $\widehat{W}_\gamma$ is not integrable; the limit exists for every t (Lemma 5.9). For $\Delta \ge 0$ and $\gamma > 2\Delta$ fix $\delta \in (0, \gamma/2 - \Delta)$ and $\zeta \in C_c^\infty((-\delta,\delta))$ with $\zeta \ge 0$ and $\int \zeta = 1$, and let

$$g(\omega) := \int_{\mathbb{R}} \zeta(\omega - \omega') \, \mathbf{1}_{\{\omega' < -\gamma/2\}} \,\mathrm{d}\omega'.$$

For an operator A on $\mathcal{H}_N$ let

$$\mathcal{Q}(A) := g(\mathrm{ad}_{H_N})(A) := \sum_{E,E' \in \sigma(H_N)} g(E - E') \, \mathbf{1}_{\{E\}}(H_N) \, A \, \mathbf{1}_{\{E'\}}(H_N),$$

and define $\mathcal{Q}$ on operators on $\widehat{\mathcal{H}}_N$ in the same way, with $\widehat{H}_N$ in place of H_N.

The function g is smooth and real, with $g(\omega)=0$ for $\omega\ge-\Delta$ and $g(\omega)=1$ for $\omega\le-\gamma+\Delta$, in particular for $\omega\le-\gamma$. By Lemma 3.8(ii), $\mathcal{Q}(\iota(A))=\iota(\mathcal{Q}(A))$. The filter W_γ is that of [BBDF21, Section 4.1], and g and $\mathcal{Q}$ are those of [BBDF21, Appendix A].

Lemma 5.9 (the generator filter)**.** *W_γ is real and bounded, smooth on $\mathbb{R}\setminus\{0\}$, with a jump of -1 at $t=0$, and*

$$W_\gamma(t)=-\tfrac12\operatorname{sign}(t)+\frac1\pi\int_0^\infty\vartheta(\omega/\gamma)\frac{\sin(\omega t)}{\omega}\,\mathrm{d}\omega,\qquad W_\gamma(t)=W_1(\gamma t).$$

Moreover $W_\gamma(t)=O(|t|^{-\infty})$ as $|t|\to\infty$, so $M_j(W_\gamma)<\infty$ for every j; for every n there is C_n with $\int_{|t|>T}|W_\gamma(t)|\,\mathrm{d}t\le C_n\gamma^{-1}(\gamma T)^{-n}$; $\int W_\gamma(t)e^{-\mathrm{i}\omega t}\,\mathrm{d}t=\widehat{W}_\gamma(\omega)=\mathrm{i}/\omega$ for $|\omega|\ge\gamma$; and for every operator A on $\mathcal{H}_N$,

$$\int W_\gamma(t)\,\tau_t(A)\,\mathrm{d}t=\sum_{E,E'\in\sigma(H_N)}\widehat{W}_\gamma(E'-E)\,\mathbf{1}_{\{E\}}(H_N)\,A\,\mathbf{1}_{\{E'\}}(H_N).$$

Proof. We first show that the limit in Definition 5.8 exists for every t. As ϑ is even, $\widehat{W}_\gamma$ is odd, and the part of $e^{\mathrm{i}\omega t}$ even in ω integrates to zero:

$$\frac{1}{2\pi}\int_{-\Omega}^{\Omega}\widehat{W}_\gamma(\omega)e^{\mathrm{i}\omega t}\,\mathrm{d}\omega=-\frac1\pi\int_0^\Omega\frac{\sin(\omega t)}{\omega}\,\mathrm{d}\omega+\frac1\pi\int_0^\Omega\vartheta(\omega/\gamma)\frac{\sin(\omega t)}{\omega}\,\mathrm{d}\omega.\tag{5.7}$$

Let $\operatorname{Si}(x):=\int_0^x\sin v\,\mathrm{d}v/v$; then $0\le\operatorname{Si}(x)\le\operatorname{Si}(\pi)<2$ for $x\ge0$, and $\operatorname{Si}(x)\to\pi/2$ as $x\to\infty$. The first term of (5.7) is $-\pi^{-1}\operatorname{sign}(t)\operatorname{Si}(\Omega|t|)$, which tends to $-\frac12\operatorname{sign}(t)$; the second does not depend on Ω once $\Omega\ge\gamma$, since $\vartheta(\omega/\gamma)=0$ for $\omega\ge\gamma$. This proves the existence of the limit and the formula for W_γ, and the substitution $\omega=\gamma\omega'$ in the formula gives $W_\gamma(t)=W_1(\gamma t)$. For $t\ne0$ and $0<\Omega\le\infty$, the substitution $v=\omega|t|$ and an integration by parts give

$$\int_0^\Omega\vartheta(\omega/\gamma)\frac{\sin(\omega t)}{\omega}\,\mathrm{d}\omega=\operatorname{sign}(t)\Big[\vartheta(\Omega/\gamma)\operatorname{Si}(\Omega|t|)-\int_0^{\Omega|t|}\operatorname{Si}(v)\,\vartheta'\Big(\frac{v}{\gamma|t|}\Big)\frac{\mathrm{d}v}{\gamma|t|}\Big]$$

(with $\vartheta(\infty):=0$), whose absolute value is at most $2+2\int_0^\infty|\vartheta'|$. Hence the truncated integrals (5.7), and W_γ, are bounded uniformly in t and Ω. The second term of the formula for W_γ equals $\pi^{-1}\int_0^\gamma\vartheta(\omega/\gamma)\sin(\omega t)\,\mathrm{d}\omega/\omega$, a smooth function of t which vanishes at $t=0$; so W_γ is real, odd and smooth on $\mathbb{R}\setminus\{0\}$, and $W_\gamma(0\pm)=\mp\frac12$, a jump of -1.

Decay. Let $f(\omega):=(1-\vartheta(\omega))/\omega$ for $\omega>0$. Since $\vartheta=1$ near 0 and $\vartheta(\omega)=0$ for $\omega\ge1$, f is smooth, vanishes near 0 and equals $1/\omega$ for $\omega\ge1$; hence every derivative $f^{(j)}$, $j\ge0$, tends to 0 at infinity, and $f^{(j)}\in L^1(0,\infty)$ for $j\ge1$. For $t>0$, (5.7) with $\gamma=1$ gives $W_1(t)=-\pi^{-1}\lim_{\Omega\to\infty}\int_0^\Omega f(\omega)\sin(\omega t)\,\mathrm{d}\omega$. The integration by parts $\int_0^\Omega f(\omega)\sin(\omega t)\,\mathrm{d}\omega=-f(\Omega)\cos(\Omega t)/t+t^{-1}\int_0^\Omega f'(\omega)\cos(\omega t)\,\mathrm{d}\omega$ shows that $W_1(t)=-(\pi t)^{-1}\int_0^\infty f'(\omega)\cos(\omega t)\,\mathrm{d}\omega$, an absolutely convergent integral, and $n-1$ further integrations by parts, whose boundary terms vanish because each $f^{(j)}$ vanishes near 0 and at infinity, give $|W_1(t)|\le\pi^{-1}\|f^{(n)}\|_{L^1(0,\infty)}\,t^{-n}$ for every $n\ge1$. By oddness the same bound holds for $t<0$. Together with the boundedness of W_1 this gives $|W_1(t)|\le C_n'(1+|t|)^{-n}$ for every n, with C_n' depending only on ϑ. Hence $W_\gamma(t)=W_1(\gamma t)=O(|t|^{-\infty})$, $M_j(W_\gamma)<\infty$ for every j, and for $T>0$ and $n\ge1$

$$\int_{|t|>T}|W_\gamma(t)|\,\mathrm{d}t=\frac1\gamma\int_{|u|>\gamma T}|W_1(u)|\,\mathrm{d}u\le\frac{C'_{n+1}}{\gamma}\int_{|u|>\gamma T}|u|^{-n-1}\,\mathrm{d}u=\frac{2C'_{n+1}}{n}\,\gamma^{-1}(\gamma T)^{-n},$$

while for $n=0$ the same integral is at most $\gamma^{-1}\|W_1\|_{L^1}$.

The Fourier transform. Fix $\omega_0 \in \mathbb{R}$ and $\varepsilon > 0$. The truncated integrals (5.7) are bounded uniformly and converge to $W_\gamma(t)$ for every $t \neq 0$; by dominated convergence and Fubini's theorem on $[-\Omega, \Omega] \times \mathbb{R}$,

$$\int W_\gamma(t) e^{-\mathrm{i}\omega_0 t - \varepsilon t^2} \,\mathrm{d}t = \lim_{\Omega\to\infty} \frac{1}{2\pi} \int_{-\Omega}^{\Omega} \widehat{W}_\gamma(\omega) \int e^{\mathrm{i}(\omega-\omega_0)t - \varepsilon t^2} \,\mathrm{d}t \,\mathrm{d}\omega = \int_{\mathbb{R}} \widehat{W}_\gamma(\omega) \,\frac{e^{-(\omega-\omega_0)^2/4\varepsilon}}{\sqrt{4\pi\varepsilon}} \,\mathrm{d}\omega.$$

As $\varepsilon \to 0$, the left-hand side tends to $\int W_\gamma(t) e^{-\mathrm{i}\omega_0 t} \,\mathrm{d}t$, because $W_\gamma \in L^1$, and the right-hand side tends to $\widehat{W}_\gamma(\omega_0)$, because $\widehat{W}_\gamma$ is bounded and continuous ($1 - \vartheta(\omega/\gamma)$ vanishes near $\omega = 0$) and the Gaussians form an approximate identity. Hence $\int W_\gamma(t) e^{-\mathrm{i}\omega t} \,\mathrm{d}t = \widehat{W}_\gamma(\omega)$ for every ω, and this is i/ω for $|\omega| \geq \gamma$, where $\vartheta(\omega/\gamma) = 0$.

The spectral formula. By the spectral theorem in the finite-dimensional space $\mathcal{H}_N$, $\tau_t(A) = \sum_{E,E'} e^{\mathrm{i}t(E-E')} \mathbf{1}_{\{E\}}(H_N) A \mathbf{1}_{\{E'\}}(H_N)$, and $\int W_\gamma(t) e^{\mathrm{i}t(E-E')} \,\mathrm{d}t = \widehat{W}_\gamma(E' - E)$. □

Lemma 5.10 (time representation of the clustering filter)**.** *Let $\gamma' := -\gamma/2$, $\hat{\zeta}(t) := \int \zeta(u) e^{-\mathrm{i}ut} \,\mathrm{d}u$ and $F(t) := -e^{-\mathrm{i}\gamma' t} \hat{\zeta}(t)/(2\pi\mathrm{i})$, a Schwartz function. Then $g(\omega) = \frac{1}{2} + \mathrm{PV}\!\int F(t) e^{\mathrm{i}\omega t} \,\mathrm{d}t/t$ for every $\omega \in \mathbb{R}$, and for every operator A on $\mathcal{H}_N$*

$$\mathcal{Q}(A) = (\tfrac{1}{2} + c_F) A + \int_{|t|\geq 1} \frac{F(t)}{t} \,\tau_t(A) \,\mathrm{d}t + \int_{-1}^{1} G(u) \,\tau_u\big(\mathrm{i}[H_N, A]\big) \,\mathrm{d}u,$$

where $c_F := \mathrm{PV}\!\int_{-1}^{1} F(t) \,\mathrm{d}t/t$, $G(u) := \int_u^1 F(t) \,\mathrm{d}t/t$ for $0 < u < 1$ and $G(u) := -\int_{-1}^{u} F(t) \,\mathrm{d}t/t$ for $-1 < u < 0$. Here $|G(u)| \leq \|F\|_\infty \log(1/|u|)$, so both filters on the right have finite moments of all orders, and $\|\mathcal{Q}(A)\| \leq C(1 + \|H_N\|)\|A\|$ with C depending only on ζ and γ. The same holds on $\widehat{\mathcal{H}}_N$ with $\widehat{H}_N$.

Proof. Since $\zeta \in C_c^\infty$, $\hat{\zeta}$ is a Schwartz function, and so is F.

The representation of g. For $\mu \in \mathbb{R}$ and $0 < \varepsilon < T$,

$$\int_{\varepsilon<|t|<T} e^{\mathrm{i}\mu t} \,\frac{\mathrm{d}t}{t} = 2\mathrm{i} \int_\varepsilon^T \sin(\mu t) \,\frac{\mathrm{d}t}{t} = 2\mathrm{i} \,\mathrm{sign}(\mu) \big[\mathrm{Si}(T|\mu|) - \mathrm{Si}(\varepsilon|\mu|)\big],$$

with Si as in the proof of Lemma 5.9; this is bounded by 4 and tends to $\mathrm{i}\pi \,\mathrm{sign}(\mu)$ as $\varepsilon \to 0$ and $T \to \infty$ (with $\mathrm{sign}(0) := 0$). Fix ω. By Fubini's theorem on $\{\varepsilon < |t| < T\} \times \mathrm{supp}\,\zeta$ and dominated convergence,

$$\begin{aligned} \mathrm{PV}\!\int F(t) e^{\mathrm{i}\omega t} \,\frac{\mathrm{d}t}{t} &= -\frac{1}{2\pi\mathrm{i}} \lim_{\varepsilon\to 0,\, T\to\infty} \int \zeta(v) \int_{\varepsilon<|t|<T} e^{\mathrm{i}(\omega-\gamma'-v)t} \,\frac{\mathrm{d}t}{t} \,\mathrm{d}v \\ &= -\frac{1}{2} \int \zeta(v) \,\mathrm{sign}(\omega - \gamma' - v) \,\mathrm{d}v. \end{aligned}$$

The substitution $\omega' = \omega - v$ in the definition gives $g(\omega) = \int \zeta(v) \,\mathbf{1}_{\{v>\omega-\gamma'\}} \,\mathrm{d}v$, and $\int \zeta = 1$; so the last expression equals $-\frac{1}{2}\big((1 - g(\omega)) - g(\omega)\big) = g(\omega) - \frac{1}{2}$.

The operator identity. By the spectral theorem in the finite-dimensional space $\mathcal{H}_N$, $\mathcal{Q}(A) = \sum_{E,E'} g(E - E') \mathbf{1}_{\{E\}}(H_N) A \mathbf{1}_{\{E'\}}(H_N)$ and $\tau_t(A) = \sum_{E,E'} e^{\mathrm{i}t(E-E')} \mathbf{1}_{\{E\}}(H_N) A \mathbf{1}_{\{E'\}}(H_N)$. Inserting the representation of g gives $\mathcal{Q}(A) = \frac{1}{2}A + \mathrm{PV}\!\int F(t) \tau_t(A) \,\mathrm{d}t/t$, the principal value existing because the sum is finite. The part $|t| \geq 1$ of the integral converges absolutely. For $0 < |t| < 1$ write $\tau_t(A) = A + \int_0^t \tau_u(B) \,\mathrm{d}u$ with $B := \mathrm{i}[H_N, A]$, so that $\|\tau_t(A) - A\| \leq |t|\|B\|$. Then

$$\mathrm{PV}\!\int_{-1}^{1} F(t) \tau_t(A) \,\frac{\mathrm{d}t}{t} = c_F A + \int_{-1}^{1} \frac{F(t)}{t} \int_0^t \tau_u(B) \,\mathrm{d}u \,\mathrm{d}t,$$

where $c_F = \int_{-1}^{1}(F(t)-F(0))\,\mathrm{d}t/t$ is finite because F is smooth, and the double integral converges absolutely, since $\int_{-1}^{1}|F(t)/t|\,|t|\|B\|\,\mathrm{d}t < \infty$. Exchanging the order of integration, the part $t > 0$ becomes $\int_0^1(\int_u^1 F(t)\,\mathrm{d}t/t)\tau_u(B)\,\mathrm{d}u$ and the part $t<0$ becomes $-\int_{-1}^{0}(\int_{-1}^{u}F(t)\,\mathrm{d}t/t)\tau_u(B)\,\mathrm{d}u$; together they are $\int_{-1}^{1}G(u)\tau_u(B)\,\mathrm{d}u$. This is the formula.

The filters. $|G(u)| \le \|F\|_\infty \int_{|u|}^{1}\mathrm{d}t/t = \|F\|_\infty \log(1/|u|)$, so G, extended by zero outside $(-1,1)$, is integrable with $\int|G| \le 2\|F\|_\infty$, and $M_j(G) \le 2^{j+1}\|F\|_\infty$. The function equal to $F(t)/t$ for $|t|\ge 1$ and to 0 otherwise is bounded by $|F|$, so its moments are bounded by those of the Schwartz function F. Finally

$$\|\mathcal{Q}(A)\| \le \Big(\tfrac{1}{2} + |c_F| + \int|F(t)|\,\mathrm{d}t\Big)\|A\| + 2\|F\|_\infty \cdot 2\|H_N\|\|A\|,$$

and F is determined by ζ and γ. On $\widehat{\mathcal{H}}_N$ the same argument applies verbatim with $\widehat{H}_N$ in place of H_N. □

Proposition 5.11 (locality of the clustering filter)**.** *If $A=(A_N)\in\mathcal{A}_X$ is even, with prefactor exponent m and $\|A_N\| \le CN^{m'}$, and $|X_N| \le CN^{m''}$, then $(\mathcal{Q}(A_N))_N \in \mathcal{A}_X$ with prefactor exponent $1+\max\{m,m'\}+m''$, and $\|\mathcal{Q}(A_N)\| \le CN^{1+m'}$. No spectral assumption on H is used.*

Proof. By Lemma 5.10,

$$\mathcal{Q}(A_N) = (\tfrac{1}{2}+c_F)A_N + \int F_1(t)\,\tau_t(A_N)\,\mathrm{d}t + \int G_1(u)\,\tau_u(B_N)\,\mathrm{d}u, \qquad B_N := \mathrm{i}[H_N, A_N],$$

where $F_1(t) := F(t)/t$ for $|t|\ge 1$, $F_1(t) := 0$ otherwise, and G_1 is G on $(-1,1)$ and 0 otherwise. Both F_1 and G_1 are integrable with finite moments of all orders (Lemma 5.10). The first term is in $\mathcal{A}_X$ with prefactor exponent m. By Lemma 5.6 with $Z=X$, the second is in $\mathcal{A}_X$ with prefactor exponent $\max\{m,m'\}+m''$. By Lemma 5.7, $B=(B_N)$ is even, lies in $\mathcal{A}_X$ with prefactor exponent $1+\max\{m,m'\}$, and $\|B_N\| \le 8\pi C_h N \cdot CN^{m'}$; so, by Lemma 5.6, the third term is in $\mathcal{A}_X$ with prefactor exponent $\max\{1+\max\{m,m'\}, 1+m'\}+m'' = 1+\max\{m,m'\}+m''$. A sum of sequences in $\mathcal{A}_X$ is in $\mathcal{A}_X$, with the largest of the prefactor exponents (the approximants add; Lemma 5.12(b)), which gives the first claim. The norm bound follows from Lemma 5.10 and $\|H_N\| \le C_h|\mathbb{T}_N| = 4\pi C_h N$. Nothing about the spectrum of H_N was used. □

5.5 The locality calculus and the bookkeeping

Lemma 5.12 (locality calculus)**.** *Let $(\mathbb{T}_N)_{N\in\mathcal{N}}$ be admissible.*

*(a) (*far-apart operators almost commute*) If $A\in\mathcal{A}_X$ and $B\in\mathcal{A}_Y$ are even, with prefactor exponents m_A and m_B, then for every k there is C_k, depending only on the constants of A and B, with*

$$\|[A_N,B_N]\| \le C_k N^{m_A+m_B}\big(1+\|A_N\|+\|B_N\|\big)\big(1+d(X_N,Y_N)\big)^{-k}$$

for every N with $X_N\cap Y_N=\emptyset$.

*(b) (*sums, products, adjoints*) Let $A,B\in\mathcal{A}_X$ have prefactor exponents m_A and m_B, and let $\|A_N\|,\|B_N\| \le CN^{m'}$. Then $A+B$, A^* and AB are in $\mathcal{A}_X$, with prefactor exponents $\max\{m_A,m_B\}$, m_A and m_A+m_B+m', respectively.*

*(c) (*ordered exponentials*) Let $G_N(u)$, $u \in [0,u_0]$, be self-adjoint and continuous in u, and let $G = (G_N(u))_N$ be in $\mathcal{A}_X$ uniformly in u. Then the solution of $V_N'(u) = \mathrm{i}V_N(u)G_N(u)$, $V_N(0) = \mathbf{1}$, satisfies $V_N(u_0) \in \mathcal{A}_X$, with the same prefactor exponent and constants multiplied by $2u_0$.*

*(d) (*conjugation by band charges*) Let $Z_N = \mathrm{Str}(I_N)$ with I_N open intervals of length at least $2\delta_x$, let $A = (A_N) \in \mathcal{A}_Z$ be self-adjoint with prefactor exponent m and $\|A_N\| \le CN^m$, and let $\mathcal{B}_N$ be bands. Then $e^{\mathrm{i}\varphi(Q_{\mathcal{B}_N}+A_N)}e^{-\mathrm{i}\varphi Q_{\mathcal{B}_N}} \in \mathcal{A}_Z$ with prefactor exponent m, uniformly in $\varphi \in [0,2\pi]$ and in the bands.*

Proof. Throughout, C_k^A and C_k^B denote the constants of A and B in Definition 3.11, and $A_{N,r}$, $B_{N,r}$ best approximants of $\iota(A_N)$, $\iota(B_N)$ in $\mathfrak{A}((X_N)_{(r)})$ (respectively $\mathfrak{A}((Y_N)_{(r)})$), which exist because these algebras are finite-dimensional and which satisfy the bounds of Definition 3.11 for every k at once. In particular $\|A_{N,r}\| \le \|A_N\| + C_0^A N^{m_A}$.

(a) Let N be such that $X_N \cap Y_N = \emptyset$, and put $d := d(X_N, Y_N)$; since distinct slots have distinct centres, $d > 0$. Let $r := d/3$. If $s \in (X_N)_{(r)} \cap (Y_N)_{(r)}$, there are $s_1 \in X_N$ and $s_2 \in Y_N$ with $d(s,s_1), d(s,s_2) \le r$, whence $d \le 2r < d$; so $(X_N)_{(r)} \cap (Y_N)_{(r)} = \emptyset$, and the even operators $A_{N,r}$ and $B_{N,r}$ commute (Proposition B.1). Since ι is an isometric homomorphism,

$$\begin{aligned}
\|[A_N,B_N]\| &= \|[\iota(A_N),\iota(B_N)]\| \\
&\le \|[\iota(A_N)-A_{N,r},\iota(B_N)]\| + \|[A_{N,r},\iota(B_N)-B_{N,r}]\| \\
&\le 2C_k^A N^{m_A}(1+r)^{-k}\|B_N\| + 2(\|A_N\| + C_0^A N^{m_A})C_k^B N^{m_B}(1+r)^{-k} \\
&\le 2(C_k^A + C_k^B + C_0^A C_k^B)N^{m_A+m_B}(1+\|A_N\|+\|B_N\|)(1+r)^{-k},
\end{aligned}$$

and $(1+d/3)^{-k} \le 3^k(1+d)^{-k}$. This proves (a) with $C_k := 2\cdot 3^k(C_k^A + C_k^B + C_0^A C_k^B)$.

(b) The approximants $A_{N,r} + B_{N,r}$ and $A_{N,r}^*$ lie in $\mathfrak{A}((X_N)_{(r)})$, with errors at most $(C_k^A + C_k^B)N^{\max\{m_A,m_B\}}(1+r)^{-k}$ and $C_k^A N^{m_A}(1+r)^{-k}$, since $\iota(A_N)^* = \iota(A_N^*)$. For the product, $A_{N,r}B_{N,r} \in \mathfrak{A}((X_N)_{(r)})$ and

$$\begin{aligned}
\|\iota(A_NB_N) - A_{N,r}B_{N,r}\| &\le \|\iota(A_N)-A_{N,r}\|\|B_N\| + \|A_{N,r}\|\|\iota(B_N)-B_{N,r}\| \\
&\le (CC_k^A + CC_k^B + C_0^A C_k^B)N^{m_A+m_B+m'}(1+r)^{-k}.
\end{aligned}$$

(c) Let C_k be the constants of G, uniform in u. Each $\iota(G_N(u))$ is even: since $\mathfrak{A}((X_N)_{(r)})$ consists of even operators, the distance of $\iota(G_N(u))$ to the even operators is at most $C_kN^m(1+r)^{-k}$ for every $r \ge 0$, hence zero. Fix N and $r \ge 0$ and put $G_{N,r}(u) := \mathbb{E}_{(X_N)_{(r)}}(\iota(G_N(u)))$. By Lemma 5.1 these are self-adjoint elements of $\mathfrak{A}((X_N)_{(r)})$, continuous in u, with $\|\iota(G_N(u)) - G_{N,r}(u)\| \le 2C_kN^m(1+r)^{-k}$. Let $V_{N,r}$ be the solution of $V_{N,r}'(u) = \mathrm{i}V_{N,r}(u)G_{N,r}(u)$, $V_{N,r}(0) = \mathbf{1}$. As the solution of a linear differential equation with continuous coefficients in the finite-dimensional algebra $\mathfrak{A}((X_N)_{(r)})$, it takes values in that algebra; it is unitary, as is V_N, because the generators are self-adjoint. Since ι is linear, continuous and multiplicative, $\iota(V_N)$ solves the equation with $\iota(G_N)$ in place of G_N. The function $O(u) := \iota(V_N(u))V_{N,r}(u)^*$ satisfies $O(0) = \mathbf{1}$ and $O'(u) = \mathrm{i}\,\iota(V_N(u))(\iota(G_N(u)) - G_{N,r}(u))V_{N,r}(u)^*$, so

$$\|\iota(V_N(u_0)) - V_{N,r}(u_0)\| = \|O(u_0)-\mathbf{1}\| \le \int_0^{u_0} \|\iota(G_N(u)) - G_{N,r}(u)\|\,\mathrm{d}u \le 2u_0C_kN^m(1+r)^{-k}.$$

(d) Write $Q := Q_{\mathcal{B}_N}$ and $V(\varphi) := e^{\mathrm{i}\varphi(Q+A_N)}e^{-\mathrm{i}\varphi Q}$. Then $V(0) = \mathbf{1}$ and

$$V'(\varphi) = e^{\mathrm{i}\varphi(Q+A_N)}\,\mathrm{i}A_N\,e^{-\mathrm{i}\varphi Q} = \mathrm{i}\,V(\varphi)\,G(\varphi), \qquad G(\varphi) := e^{\mathrm{i}\varphi Q}A_N e^{-\mathrm{i}\varphi Q},$$

with $G(\varphi)$ self-adjoint and continuous in φ. By Lemma 4.11(iii), $(G(\varphi))_N$ is almost supported in Z with prefactor exponent m and with constants independent of $\varphi \in \mathbb{R}$ and of the bands. Part (c) with $u_0 = \varphi \in [0, 2\pi]$ shows that $V(\varphi)$ is almost supported in Z with prefactor exponent m and constants at most 4π times those of G, uniformly in φ and in the bands. No separation between Z and the band is needed, because conjugation by a band charge preserves strips exactly. □

Lemma 5.13 (bookkeeping)**.** *Let $(\mathbb{T}_N)_{N\in\mathcal{N}}$ be admissible with $L_x \geq c_*\sqrt{N}$. Let $\mathcal{O}$ be a finite set, independent of N, of sequences of even operators $O = (O_N)$, possibly depending on a parameter, such that for some $m_* \geq 0$ every $O \in \mathcal{O}$ satisfies $\|O_N\| \leq CN^{m_*}$ and $O \in \mathcal{A}_{Z_O}$ with prefactor exponent at most m_*, uniformly in the parameter. If $O, O' \in \mathcal{O}$ and $d(Z_{O,N}, Z_{O',N}) \geq c\sqrt{N} - C$ for all N, with $c > 0$, then $[O_N, O'_N] = O(N^{-\infty})$ uniformly in the parameter. Any expression obtained from finitely many such commutators, independently of N, by sums and by products with operators of norm at most CN^m is $O(N^{-\infty})$ uniformly in the parameter.*

Proof. Let $O, O' \in \mathcal{O}$ be as stated, and let $N_1 := (2C/c)^2$, so that $c\sqrt{N} - C \geq \frac{c}{2}\sqrt{N} > 0$ for $N \geq N_1$. For $N \geq N_1$ the sets $Z_{O,N}$ and $Z_{O',N}$ are at positive distance, hence disjoint, and Lemma 5.12(a), whose constants are those of O and O' and hence independent of the parameter, gives for every k

$$\|[O_N, O'_N]\| \leq C_k N^{2m_*}(1 + 2CN^{m_*})\big(1 + \tfrac{c}{2}\sqrt{N}\big)^{-k} \leq C_k(1 + 2C)(c/2)^{-k}N^{3m_*-k/2}.$$

Given $j \geq 0$, the choice $k := 2(j + 3m_*)$ gives $\|[O_N, O'_N]\| \leq C'_j N^{-j}$ for $N \geq N_1$. For the finitely many $N \in \mathcal{N}$ with $N < N_1$, $\|[O_N, O'_N]\| \leq 2C^2N^{2m_*} \leq 2C^2N_1^{2m_*+j}N^{-j}$. Hence $[O_N, O'_N] = O(N^{-\infty})$, with constants independent of the parameter. An expression of the stated kind is a sum, with a number of terms independent of N, of products each containing one such commutator and a number, independent of N, of operators of norm at most CN^m; each term, and hence the sum, is bounded by a fixed power of N times $C''_j N^{-j}$ for every j, which is $O(N^{-\infty})$ uniformly in the parameter. The hypothesis $L_x \geq c_*\sqrt{N}$ enters only through the assumed separations. □

Remark 5.14. Lemma 5.13 replaces the bookkeeping of [BBDF21], in which almost-support errors carry the factor $\|A\||Z|$: here every operator of the proof carries a fixed power of N, every almost-support estimate is used at a distance of order $\sqrt{N}$, and the number of such estimates does not depend on N. The commutators to which Lemma 5.13 is applied in Section 6 are listed in the proofs of Lemma 6.13(ii) and Corollary 6.15.

6 The index theorem

6.1 Setting and assumptions

Setting 6.1. $\mathcal{N} \subset \mathbb{N}$ is infinite, $(\mathbb{T}_N)_{N\in\mathcal{N}}$ are admissible tori (Definition 3.1) with $L_x \geq c_*\sqrt{N}$ for a constant $c_* > 0$, and $H = (H_N)_{N\in\mathcal{N}}$ satisfies (H1). On each torus let

$$\Gamma := \{0, 1, \dots, N-1\} \subset \mathbb{Z}_{N_\phi}, \qquad \Gamma_- := \{0, \dots, \lfloor N/4 \rfloor - 1\},$$
$$\Gamma_+ := \{\lceil 3N/4 \rceil, \dots, N-1\}, \qquad \Gamma_m := \Gamma \setminus (\Gamma_- \cup \Gamma_+),$$
$$\partial_- := \mathrm{Col}(0), \qquad \partial_+ := \mathrm{Col}(N), \qquad Q := Q_\Gamma, \qquad Q_\pm := Q_{\Gamma_\pm}, \qquad Q_m := Q_{\Gamma_m},$$

and let $\mathbb{T}_N^- := \{R \in \mathbb{T}_N : R_x \in (-L_x/4, L_x/4] \bmod L_x\}$ and $\mathbb{T}_N^+ := \mathbb{T}_N \setminus \mathbb{T}_N^-$ be the half-tori centred on the lines of the two cuts ∂_- and ∂_+. For a Borel set $\mathcal{D} \subset \mathbb{T}_N$ write $H_\mathcal{D} := \int_\mathcal{D} h_R \, \mathrm{d}R$.

Thus $Q = Q_- + Q_m + Q_+$ is the up-spin charge of the half band Γ of N consecutive orbitals, whose two boundaries lie in the columns ∂_- and ∂_+; $Q_\pm$ are the charges of the quarters of Γ adjacent to $\partial_\pm$, and Q_m that of its middle half.

Lemma 6.2 (separations)**.** *In Setting 6.1, for every $N \in \mathcal{N}$,*

$$d(\partial_-, \partial_+) \geq \tfrac{1}{2}L_x - 2\delta_x, \qquad d(\mathrm{Col}\Gamma_-, \mathrm{Col}\Gamma_+) \geq \tfrac{1}{4}L_x - 2\delta_x,$$
$$d(\partial_\pm, \mathrm{Col}\Gamma_m) \geq \tfrac{1}{8}L_x - \kappa - 2\delta_x, \qquad d(\partial_\mp, \mathrm{Col}\Gamma_\pm) \geq \tfrac{3}{8}L_x - 2\delta_x.$$

All four right-hand sides are at least $\frac{c_*}{16}\sqrt{N}$ for $N \geq N_*$, with N_* depending only on c_* and δ_0. Moreover $|\partial_\pm| \leq 2n_y \leq CL_y \leq C'\sqrt{N}$.

Proof. Write $|u|_{L_x}$ for the distance from $u \in \mathbb{R}$ to $L_x\mathbb{Z}$, so that $|u - u'|_{L_x}$ is the distance of two points of the circle $\mathbb{R}/L_x\mathbb{Z}$ of transverse coordinates. The orbital j has transverse position κj, and $L_x = \kappa N_\phi = 2\kappa N$; for representatives $0 \leq j, j' \leq 2N-1$,

$$|\kappa j - \kappa j'|_{L_x} = \kappa \min\{|j - j'|,\, 2N - |j - j'|\}.$$

The flat distance of two points of $\mathbb{T}_N$ is at least the distance of their transverse coordinates, so $d(s, s') \geq |x_s - x_{s'}|_{L_x}$. If $s \in \mathrm{Col}(j)$ and $s' \in \mathrm{Col}(j')$, then $|x_s - \kappa j|_{L_x} < \delta_x$ and $|x_{s'} - \kappa j'|_{L_x} < \delta_x$ (Lemma 3.6), and the triangle inequality on the circle gives

$$d(s, s') > |\kappa j - \kappa j'|_{L_x} - 2\delta_x. \tag{6.1}$$

Hence $d(\mathrm{Col}\mathcal{B}, \mathrm{Col}\mathcal{B}') \geq \min\{|\kappa j - \kappa j'|_{L_x} : j \in \mathcal{B},\ j' \in \mathcal{B}'\} - 2\delta_x$ for any two bands, and it remains to bound the circular distances of the orbital positions. (For $N \leq 3$ the band Γ_- is empty; distances to the empty set are $+\infty$, and the statements involving Γ_- are void.)

The two cuts. For $j = 0$ and $j' = N$ the minimum above is N, so the distance is $\kappa N = \frac{1}{2}L_x$.

The two quarters. For $0 \leq j \leq \lfloor N/4 \rfloor - 1$ and $\lceil 3N/4 \rceil \leq j' \leq N-1$ we have $\frac{N}{2} + 1 \leq j' - j \leq N - 1$, so both $j' - j$ and $2N - (j' - j)$ exceed $\frac{N}{2}$, and the distance is at least $\frac{1}{2}\kappa N = \frac{1}{4}L_x$.

A cut and the middle band. Let $\lfloor N/4 \rfloor \leq j' \leq \lceil 3N/4 \rceil - 1$, so that $j' \leq \frac{3}{4}N$. For the cut $\partial_- = \mathrm{Col}(0)$, $\min\{j', 2N - j'\} \geq \lfloor N/4 \rfloor \geq \frac{N-3}{4}$, and the distance is at least $\frac{\kappa(N-3)}{4} = \frac{1}{8}L_x - \frac{3}{4}\kappa \geq \frac{1}{8}L_x - \kappa$. For the cut $\partial_+ = \mathrm{Col}(N)$, $N - j' \geq N - \lceil 3N/4 \rceil + 1 \geq \frac{N+1}{4}$ and $2N - (N - j') = N + j' \geq N$, so the distance is at least $\frac{\kappa(N+1)}{4} \geq \frac{1}{8}L_x$.

A cut and the opposite quarter. For $\lceil 3N/4 \rceil \leq j' \leq N-1$, $\min\{j', 2N - j'\} \geq \frac{3}{4}N$, so the distance from the position 0 of ∂_- is at least $\frac{3}{4}\kappa N = \frac{3}{8}L_x$. For $0 \leq j \leq \lfloor N/4 \rfloor - 1$, $N - j \geq \frac{3}{4}N + 1$ and $2N - (N - j) \geq N$, so the distance from the position κN of ∂_+ is at least $\frac{3}{8}L_x$.

Together with (6.1) this proves the four inequalities. By Lemma 3.2(ii), $\kappa \leq \delta_x \leq \delta_+ = 2\delta_0$, so every right-hand side is at least $\frac{1}{8}L_x - 3\delta_x \geq \frac{1}{8}c_*\sqrt{N} - 6\delta_0$, which is at least $\frac{1}{16}c_*\sqrt{N}$ for $N \geq N_* := \lceil (96\delta_0/c_*)^2 \rceil$.

Finally, $\partial_\pm$ is a column, which contains the slots of at most two windows (Lemma 3.6), so $|\partial_\pm| \leq 2n_y$. Since $n_y = \lceil 2\delta_x/\kappa \rceil \leq 2\delta_x/\kappa + 1 = \delta_x L_y/\pi + 1$, $\delta_x \leq 2\delta_0$ and $L_y \geq L_0 \geq 4$, we get $2n_y \leq CL_y$ with $C := 4\delta_0/\pi + \frac{1}{2}$; and $L_y = 4\pi N/L_x \leq (4\pi/c_*)\sqrt{N}$. □

Assumption 6.3 (gap in a sector)**.** There are sectors $\mathbf{n}_N = (n_{\uparrow,N}, n_{\downarrow,N})$, $N \in \mathcal{N}$, and $p \in \mathbb{N}$, $\Delta \geq 0$ and $\gamma > 2\Delta$, independent of N, such that the eigenvalues $E_1 \leq E_2 \leq \cdots$ of $H_N|_{\mathcal{H}_N^{\mathbf{n}_N}}$, repeated according to multiplicity, satisfy

$$E_p - E_1 \leq \Delta \qquad \text{and} \qquad E_{p+1} - E_p \geq \gamma \qquad \text{for every } N \in \mathcal{N}.$$

$P = P_N$ denotes the spectral projection of $H_N|_{\mathcal{H}_N^{\mathbf{n}_N}}$ onto $[E_1, E_p]$, extended by zero to $\mathcal{H}_N$; it has rank p, and $P = P\Pi^{\mathbf{n}_N} = \Pi^{\mathbf{n}_N}P$.

Nothing is assumed about the spectrum of H_N outside the sector $\mathcal{H}_N^{\mathbf{n}_N}$. This is the only spectral assumption of the paper; it is weaker than [BBDF21, Assumption 1], which concerns the whole Hilbert space (and in which $\Delta > 0$; the case $\Delta = 0$ is included here).

The generator that dresses Q, and its two halves, are defined in Definition 6.5; they enter the next assumption.

Assumption 6.4 (charge transport)**.** $U = U_N$, $N \in \mathcal{N}$, are unitaries on $\mathcal{H}_N$ such that

(U1) $[U, N_\uparrow] = [U, N_\downarrow] = 0$;

(U2) $\|[U, P]\| = O(N^{-\infty})$;

(U3) $U^* K_\pm U \in \mathcal{A}_{\partial_\pm}$, with $K_\pm$ as in Definition 6.5;

(U4) there are self-adjoint, sector-preserving $T_\pm \in \mathcal{A}_{\partial_\pm}$ with $\|T_\pm\| \le C N^{m_T}$ such that $U^* Q U - Q = T_- + T_+$ and $e^{2\pi \mathrm{i}(Q+T_\pm)} = \mathbf{1} + O(N^{-\infty})$.

Condition (U3) is the only use of locality preservation by U; a locality preservation of the classes $\mathcal{A}_Z$ of Definition 3.11, restricted to strips, implies it. Condition (U4) is what [BBDF21] derive from locality preservation; here it is assumed, because for the magnetic translation it holds exactly (Lemma 7.1), and we do not need the general derivation.

6.2 The dressed charge

Definition 6.5 (current, generator, dressed charge)**.** In Setting 6.1, with γ from Assumption 6.3 and W_γ from Definition 5.8, let

$$J_\pm := \mathrm{i}[H_{\mathbb{T}_N^\pm}, Q], \qquad K := \int W_\gamma(t)\, \tau_t(\mathrm{i}[H_N, Q])\, \mathrm{d}t,$$
$$K_\pm := \int W_\gamma(t)\, \tau_t(J_\pm)\, \mathrm{d}t, \qquad \overline{Q} := Q - K_- - K_+.$$

Lemma 6.6 (the split of the current)**.** *In Setting 6.1:*

(i) $J_- + J_+ = \mathrm{i}[H_N, Q]$*, hence* $K = K_- + K_+$ *and* $\overline{Q} = Q - K$*;* $J_\pm$ *and* $K_\pm$ *are self-adjoint and sector-preserving, and* $\|J_\pm\| \le 4\pi C_h N^2$*.*

(ii) $J_\pm \in \mathcal{A}_{\partial_\pm}$ *with prefactor exponent* 2*.*

(iii) For $j_0 \in \mathbb{Z}$*, the same holds, with the same constants, for the band* $\Gamma + j_0$*, the half-tori* $\mathbb{T}_N^\pm + \kappa j_0 \hat{e}_x$ *and the columns* $\mathrm{Col}(j_0)$*,* $\mathrm{Col}(j_0 + N)$ *in place of* Γ*,* $\mathbb{T}_N^\pm$ *and* $\partial_\pm$*.*

Proof. (i) The half-tori $\mathbb{T}_N^-$ and $\mathbb{T}_N^+$ partition $\mathbb{T}_N$, so $H_{\mathbb{T}_N^-} + H_{\mathbb{T}_N^+} = H_N$ and $J_- + J_+ = \mathrm{i}[H_N, Q]$; integrating against W_γ gives $K = K_- + K_+$, and hence $\overline{Q} = Q - K$. Each $H_\mathcal{D}$ is the integral of a bounded Borel function with values in the finite-dimensional space of operators on $\mathcal{H}_N$ whose values are self-adjoint and sector-preserving; so $H_\mathcal{D}$ is self-adjoint and sector-preserving, with $\|H_\mathcal{D}\| \le C_h|\mathcal{D}|$. The charge Q is self-adjoint and sector-preserving, with $\|Q\| \le N$ (Lemma 4.10). Hence $J_\pm = \mathrm{i}[H_{\mathbb{T}_N^\pm}, Q]$ is self-adjoint, and it is sector-preserving because the sector-preserving operators form an algebra (Lemma 2.6(ii)). The unitaries $e^{\pm \mathrm{i}t H_N}$ are sector-preserving (Lemma 2.6(ii)), so every $\tau_t(J_\pm)$ is self-adjoint and sector-preserving, and so is $K_\pm$, because W_γ is real (Lemma 5.9). Finally $|\mathbb{T}_N^\pm| = \frac{1}{2}|\mathbb{T}_N| = 2\pi N$, so $\|J_\pm\| \le 2\|H_{\mathbb{T}_N^\pm}\|\|Q\| \le 4\pi C_h N^2$.

(ii) For $R \in \mathbb{T}_N$ let $\xi_-(R) := |R_x|_{L_x}$ and $\xi_+(R) := |R_x - \frac{1}{2}L_x|_{L_x}$ be the transverse distances from R to the centre lines $x = 0$ and $x = \frac{1}{2}L_x = \kappa N$ of the cuts $\partial_- = \mathrm{Col}(0)$ and $\partial_+ = \mathrm{Col}(N)$. By the definition of the half-tori, $\xi_\pm(R) \le \frac{1}{4}L_x$ and $\xi_\pm(R) \le \xi_\mp(R)$ for $R \in \mathbb{T}_N^\pm$.

Step 1: the far side of the cut. The orbitals of Γ have transverse positions in the arc $[0, \frac{1}{2}L_x - \kappa]$, those of $\Gamma^c := \mathbb{Z}_{N_\phi} \setminus \Gamma$ in the arc $[\frac{1}{2}L_x, L_x - \kappa]$. For $R \in \mathbb{T}_N$ let $\mathcal{B}_R := \Gamma^c$ if $R_x \in [0, \frac{1}{2}L_x) \bmod L_x$, and $\mathcal{B}_R := \Gamma$ otherwise. In the first case R_x lies in the arc $(-\kappa, \frac{1}{2}L_x)$ complementary to the positions of $\mathcal{B}_R$, at distances $R_x + \kappa \ge \xi_-(R)$ and $\frac{1}{2}L_x - R_x = \xi_+(R)$ from its endpoints; in the second case R_x lies in the arc $(\frac{1}{2}L_x - \kappa, L_x)$, at distances $R_x - \frac{1}{2}L_x + \kappa \ge \xi_+(R)$ and $L_x - R_x = \xi_-(R)$ from its endpoints. In both cases, for $R \in \mathbb{T}_N^\pm$,

$$|\kappa j - R_x|_{L_x} \ge \min\{\xi_-(R), \xi_+(R)\} = \xi_\pm(R) \qquad \text{for every } j \in \mathcal{B}_R. \tag{6.2}$$

Since $[h_R, N_\uparrow] = 0$ and $N_\uparrow = Q + Q_{\Gamma^c}$ (Lemma 4.10), $[h_R, Q] = -[h_R, Q_{\Gamma^c}]$, so in both cases $[h_R, Q] = \pm[h_R, Q_{\mathcal{B}_R}]$. Every slot $s \in \mathrm{Col}(\mathcal{B}_R)$ has $|x_s - \kappa j|_{L_x} < \delta_x$ for some $j \in \mathcal{B}_R$, hence $d(s, R) \ge |x_s - R_x|_{L_x} > \xi_\pm(R) - \delta_x$ by (6.2). Let $R \in \mathbb{T}_N^\pm$ with $\xi_\pm(R) \ge \delta_x$, put $r_R := \xi_\pm(R) - \delta_x$, and let $h_{R,r_R} \in \mathfrak{A}(\Lambda_{r_R}(R))$ be the approximant of (H1). Then $\Lambda_{r_R}(R) \cap \mathrm{Col}(\mathcal{B}_R) = \emptyset$, so h_{R,r_R} commutes with $\iota(Q_{\mathcal{B}_R}) \in \mathfrak{A}(\mathrm{Col}(\mathcal{B}_R))$ (Lemma 4.11(i)), and, ι being an isometric homomorphism (Lemma 3.8),

$$\|[h_R, Q]\| = \|[\iota(h_R) - h_{R,r_R}, \iota(Q_{\mathcal{B}_R})]\| \le 2C_k^H \big(1 + \xi_\pm(R) - \delta_x\big)^{-k} N.$$

If $\xi_\pm(R) < \delta_x$ we use $\|[h_R, Q]\| \le 2C_h N$ instead. Since $1 + \xi \le (1 + \xi - \delta_x)(1 + \delta_x)$ for $\xi \ge \delta_x$, and $1 + \xi < 1 + \delta_x$ for $\xi < \delta_x$, both cases give, for every k,

$$\|[h_R, Q]\| \le 2N \max\{C_h, C_k^H\}(1 + \delta_x)^k \big(1 + \xi_\pm(R)\big)^{-k}, \qquad R \in \mathbb{T}_N^\pm. \tag{6.3}$$

Step 2: near the cut. Fix $\rho \ge 0$. The operators h_R are sector-preserving, hence even (Lemma 2.6(ii)), and so is $\iota(h_R)$ (Lemma 3.8(i)). By Lemma 5.1, $h'_{R,\rho} := \mathbb{E}_{\Lambda_\rho(R)}(\iota(h_R))$ is a self-adjoint element of $\mathfrak{A}(\Lambda_\rho(R))$ with $\|\iota(h_R) - h'_{R,\rho}\| \le 2C_k^H(1 + \rho)^{-k}$ for every k, and $R \mapsto h'_{R,\rho}$ is Borel, because $R \mapsto \iota(h_R)$ is Borel and $R \mapsto \Lambda_\rho(R)$ takes finitely many values, each on a Borel set (the sets $\{R : d(s, R) \le \rho\}$ are closed). Let $\mathcal{B}'_R := \{j \in \Gamma : \mathrm{Col}(j) \cap \Lambda_\rho(R) \ne \emptyset\}$. Then $\iota(Q_{\Gamma \setminus \mathcal{B}'_R}) \in \mathfrak{A}(\mathrm{Col}(\Gamma \setminus \mathcal{B}'_R))$ commutes with $h'_{R,\rho}$, and

$$[h'_{R,\rho}, \iota(Q)] = [h'_{R,\rho}, \iota(Q_{\mathcal{B}'_R})] \in \mathfrak{A}(\Lambda_\rho(R) \cup \mathrm{Col}(\mathcal{B}'_R)).$$

Every $s \in \Lambda_\rho(R)$ has $|x_s - R_x|_{L_x} \le \rho$; every $j \in \mathcal{B}'_R$ has $|\kappa j - R_x|_{L_x} < \rho + \delta_x$, so every $s \in \mathrm{Col}(\mathcal{B}'_R)$ has $|x_s - R_x|_{L_x} < \rho + 2\delta_x$. Hence $[h'_{R,\rho}, \iota(Q)] \in \mathfrak{A}(\mathrm{Str}((R_x - \rho - 2\delta_x, R_x + \rho + 2\delta_x)))$.

Step 3: integration. Let $I_- := (-\delta_x, \delta_x)$ and $I_+ := (\frac{1}{2}L_x - \delta_x, \frac{1}{2}L_x + \delta_x)$, so that $\partial_\pm = \mathrm{Str}(I_\pm)$ (Lemma 3.6). For $R \in \mathbb{T}_N^\pm$ with $\xi_\pm(R) \le \rho$ the interval $(R_x - \rho - 2\delta_x, R_x + \rho + 2\delta_x)$ lies in the closed $(2\rho + \delta_x)$-neighbourhood of $I_\pm$, and $\mathrm{Str}((I_\pm)_{2\rho+\delta_x}) \subset (\partial_\pm)_{(2\rho+2\delta_x)}$ by Lemma 3.10. Hence

$$J_{\pm,\rho} := \int_{\{R \in \mathbb{T}_N^\pm :\ \xi_\pm(R) \le \rho\}} \mathrm{i}[h'_{R,\rho}, \iota(Q)]\, \mathrm{d}R \ \in\ \mathfrak{A}\big((\partial_\pm)_{(2\rho+2\delta_x)}\big),$$

the integrand being a bounded Borel function with values in this finite-dimensional space; $J_{\pm,\rho}$ is self-adjoint. Since ι is linear, $\iota(J_\pm) = \int_{\mathbb{T}_N^\pm} \iota(\mathrm{i}[h_R, Q])\, \mathrm{d}R$, and

$$\iota(J_\pm) - J_{\pm,\rho} = \int_{\{\xi_\pm \le \rho\}} \mathrm{i}[\iota(h_R) - h'_{R,\rho}, \iota(Q)]\, \mathrm{d}R + \int_{\{\xi_\pm > \rho\}} \iota(\mathrm{i}[h_R, Q])\, \mathrm{d}R.$$

The first integral has norm at most $|\mathbb{T}_N| \cdot 2 \cdot 2C_k^H(1+\rho)^{-k}N = 16\pi C_k^H N^2(1+\rho)^{-k}$. Up to a translation, $\mathbb{T}_N^{\pm}$ is the rectangle $\{|R_x| \le \frac{1}{4}L_x\} \times [0, L_y)$, on which $\xi_\pm(R) = |R_x|$; so by (6.3), for $k \ge 2$, the second integral has norm at most

$$2N \max\{C_h, C_k^H\}(1+\delta_x)^k L_y \cdot 2\int_\rho^\infty (1+u)^{-k}\,\mathrm{d}u \le 4\pi \max\{C_h, C_k^H\}(1+2\delta_0)^k N^2(1+\rho)^{1-k},$$

where we used $L_y = 4\pi N/L_x \le \pi N$ (admissibility gives $L_x \ge 4$) and $\delta_x \le 2\delta_0$. Thus $\|\iota(J_\pm) - J_{\pm,\rho}\| \le C'_k N^2(1+\rho)^{1-k}$ with $C'_k := 16\pi C_k^H + 4\pi \max\{C_h, C_k^H\}(1+2\delta_0)^k$.

Now let $r \ge 0$. If $r \ge 2\delta_x$, put $\rho := \frac{1}{2}(r - 2\delta_x)$; then $J_{\pm,\rho} \in \mathfrak{A}((\partial_\pm)_{(r)})$, and since $1 + r \le 2(1+\delta_x)(1+\rho)$, applying the bound with $k+1$ in place of k gives $\|\iota(J_\pm) - J_{\pm,\rho}\| \le (2+4\delta_0)^k C'_{k+1} N^2(1+r)^{-k}$. If $r < 2\delta_x$, the approximant 0 has error $\|J_\pm\| \le 4\pi C_h N^2 \le 4\pi C_h(1+4\delta_0)^k N^2(1+r)^{-k}$. So $J_\pm \in \mathcal{A}_{\partial_\pm}$ with prefactor exponent 2, with constants depending only on k, C_h, C_{k+1}^H and δ_0.

(iii) Replace Γ by $\Gamma + j_0$, the half-tori by their translates by $\kappa j_0 \hat{e}_x$, and the centre lines $x = 0$, $x = \frac{1}{2}L_x$ by $x = \kappa j_0$, $x = \kappa j_0 + \frac{1}{2}L_x$, which are the centre lines of $\mathrm{Col}(j_0)$ and $\mathrm{Col}(j_0 + N)$. The proof above uses the slot lattice only through Lemmas 3.6, 3.10 and 4.11(i) and (H1), which hold for every orbital, every interval and every point R, and otherwise only transverse distances measured from the two centre lines; it therefore applies verbatim, with the same constants. □

Lemma 6.7 (the in-sector identity)**.** *In Setting 6.1, under Assumption 6.3,* $[K, P] = [Q, P]$. *Hence* $[\overline{Q}, P] = 0$ *exactly.*

Proof. Write $\Pi := \Pi^{\mathbf{n}_N}$, and $P_E := \mathbf{1}_{\{E\}}(H_N)$ for $E \in \sigma(H_N)$. Since H_N is sector-preserving, $P_E\Pi = \Pi P_E$ is the eigenprojection of $H_N|_{\mathcal{H}_N^{\mathbf{n}_N}}$ for E, extended by zero, and it vanishes unless E is an eigenvalue of that restriction (Lemma 2.6(ii)); and $P = \sum_{E \in [E_1, E_p]} P_E\Pi$. By Assumption 6.3 the eigenvalues $E_1 \le \cdots \le E_p$ of the restriction lie in $[E_1, E_p]$, and all others are at least $E_{p+1} \ge E_p + \gamma$. Consequently:

$$\text{if } P_E\Pi \ne 0,\ P_{E'}\Pi \ne 0, \text{ and exactly one of } E, E' \text{ lies in } [E_1, E_p], \text{ then } |E - E'| \ge \gamma. \tag{6.4}$$

Since $P_E\,\mathrm{i}[H_N, Q]\,P_{E'} = \mathrm{i}(E - E')P_EQP_{E'}$, Lemma 5.9 gives

$$K = \sum_{E,E' \in \sigma(H_N)} f(E, E')\,P_EQP_{E'}, \qquad f(E, E') := \mathrm{i}\,\widehat{W}_\gamma(E' - E)\,(E - E'),$$

and $f(E, E') = \mathrm{i} \cdot \mathrm{i}(E - E')/(E' - E) = 1$ whenever $|E - E'| \ge \gamma$, because $\widehat{W}_\gamma(\omega) = \mathrm{i}/\omega$ for $|\omega| \ge \gamma$. With $Q = \sum_{E,E'} P_EQP_{E'}$ this gives $A := K - Q = \sum_{E,E'}(f(E, E') - 1)P_EQP_{E'}$. The operator A is sector-preserving (Lemma 6.6(i) and Lemma 4.10), so $AP = \Pi A \Pi P$ and $PA = P\Pi A\Pi$; since Π commutes with Q and with every P_E, and $P_{E'}P = \mathbf{1}_{[E_1,E_p]}(E')P_{E'}\Pi$,

$$[A, P] = \sum_{E,E' \in \sigma(H_N)} \big(f(E, E') - 1\big)\big(\mathbf{1}_{[E_1,E_p]}(E') - \mathbf{1}_{[E_1,E_p]}(E)\big)\,P_E\Pi\,Q\,\Pi P_{E'}.$$

In a nonzero term $P_E\Pi \ne 0$, $\Pi P_{E'} \ne 0$, and exactly one of E, E' lies in $[E_1, E_p]$; then $|E - E'| \ge \gamma$ by (6.4), and $f(E, E') - 1 = 0$. Hence $[K - Q, P] = 0$, which is $[K, P] = [Q, P]$, and $[\overline{Q}, P] = [Q - K, P] = 0$ by Lemma 6.6(i). □

The proof uses only $E_{p+1} - E_p \ge \gamma$ and the sector preservation of H_N and Q; neither Δ, nor $\gamma > 2\Delta$, nor any property of the spectrum of H_N outside the sector enters.

Lemma 6.8 (strip localization of the dressing)**.** *In Setting 6.1, $K_\pm \in \mathcal{A}_{\partial_\pm}$ with prefactor exponent $5/2$, and $\|K_\pm\| \le \|W_\gamma\|_1 \|J_\pm\|$. The same holds for the dressings of the translated currents of Lemma 6.6(iii), with $\mathrm{Col}(j_0)$, $\mathrm{Col}(j_0+N)$ in place of $\partial_\pm$.*

Proof. The norm bound is $\|K_\pm\| \le \int |W_\gamma(t)| \|\tau_t(J_\pm)\| \,\mathrm{d}t = \|W_\gamma\|_1 \|J_\pm\|$, with $\|W_\gamma\|_1 = M_0(W_\gamma) < \infty$ (Lemma 5.9). We prove the locality of K_-; that of K_+ is the same. Write $Z := \partial_- = \mathrm{Str}(I_-)$ with $I_- = (-\delta_x, \delta_x)$, and C_k^J for the constants of J_- in Lemma 6.6(ii). For $\rho \ge 0$ let J_ρ be a best approximant of $\iota(J_-)$ in the finite-dimensional space $\mathfrak{A}(Z_{(\rho)})$; it satisfies $\|\iota(J_-) - J_\rho\| \le C_k^J N^2 (1+\rho)^{-k}$ for every k at once, and we may take $J_\rho = \iota(J_-)$ when $\rho \ge \operatorname{diam} \Lambda_N$, since then $Z_{(\rho)} = \Lambda_N$ and $\mathfrak{A}(\Lambda_N)$ contains every even operator.

Telescoping. With the dyadic radii $r_j = 2^j$, put $J^{(0)} := J_{r_0}$ and $J^{(j)} := J_{r_j} - J_{r_{j-1}}$ for $j \ge 1$. Then $J^{(j)} \in \mathfrak{A}(Z_{(r_j)})$ is even, $J^{(j)} = 0$ once $r_{j-1} \ge \operatorname{diam} \Lambda_N$, $\iota(J_-) = \sum_{j \ge 0} J^{(j)}$ (a finite sum), and

$$\|J^{(0)}\| \le \|J_-\| + C_0^J N^2, \qquad \|J^{(j)}\| \le 2 C_k^J N^2 (1 + r_{j-1})^{-k} \le 2^{k+1} C_k^J N^2 (1+r_j)^{-k} \quad (j \ge 1).$$

With $k = 1$ and $\|J_-\| \le 4\pi C_h N^2$ this gives $\sum_j \|J^{(j)}\| \le C_J N^2$, $C_J := 4\pi C_h + C_0^J + 4C_1^J$. By Lemma 3.10, $Z_{(\rho)} \subset \mathrm{Str}((I_-)_\rho)$, a strip over an interval of length at most $2\delta_x + 2\rho$, so by Proposition 3.4(iii) and Lemma 6.2,

$$|Z_{(\rho)}| \le 2 C_{\mathrm{str}} (1 + 2\delta_x) L_y (1+\rho) \le C_Z N^{1/2} (1+\rho), \qquad C_Z := 8\pi C_{\mathrm{str}} (1 + 4\delta_0)/c_*. \tag{6.5}$$

The approximant. Let C_{lc} and $C_{k'}$ be the constants of Theorem 5.3(c) for the light-cone exponent $\alpha = 4$, let $r \ge r_* := \max\{2, 32 C_{\mathrm{lc}}\}$, and let $t_r := (r/2C_{\mathrm{lc}})^{1/4} - 1$; then $(r/2C_{\mathrm{lc}})^{1/4} \ge 2$, so $t_r \ge \frac{1}{2}(r/2C_{\mathrm{lc}})^{1/4} \ge 1$. Put $Y := Z_{(r)}$ and

$$K_{-,r} := \int_{-t_r}^{t_r} W_\gamma(t) \sum_{j:\ r_j \le r/2} \mathbb{E}_Y(\tau_t(J^{(j)}))\,\mathrm{d}t.$$

Each $\tau_t(J^{(j)})$ is even, because $\widehat{H}_N$ is; so $\mathbb{E}_Y(\tau_t(J^{(j)}))$ lies in $\mathfrak{A}(Y)$ and depends continuously on t (Lemma 5.1), and $K_{-,r} \in \mathfrak{A}(Y)$. By Lemma 3.8(ii), $\iota(K_-) = \int W_\gamma(t) \tau_t(\iota(J_-))\,\mathrm{d}t$, so $\iota(K_-) - K_{-,r} = \mathrm{(a)} + \mathrm{(b)} + \mathrm{(c)}$ with

$$\mathrm{(a)} := \int_{-t_r}^{t_r} W_\gamma(t) \sum_{r_j \le r/2} \Big(\tau_t(J^{(j)}) - \mathbb{E}_Y(\tau_t(J^{(j)}))\Big)\mathrm{d}t, \qquad \mathrm{(b)} := \int_{-t_r}^{t_r} W_\gamma(t) \sum_{r_j > r/2} \tau_t(J^{(j)})\,\mathrm{d}t,$$

$$\mathrm{(c)} := \int_{|t| > t_r} W_\gamma(t)\, \tau_t\big(\iota(J_-)\big)\,\mathrm{d}t.$$

The term (a). For $|t| \le t_r$ we have $C_{\mathrm{lc}}(1+|t|)^4 \le C_{\mathrm{lc}}(1+t_r)^4 = \frac{r}{2}$. For $r_j \le \frac{r}{2}$, Theorem 5.3(c) with $X = Z_{(r_j)}$ and $\rho = \frac{r}{2}$ therefore gives an element of $\mathfrak{A}((Z_{(r_j)})_{(r/2)}) \subset \mathfrak{A}(Y)$ within $C_{k'} \|J^{(j)}\| |Z_{(r_j)}| (r/2)^{-k'}$ of $\tau_t(J^{(j)})$ (note $(Z_{(a)})_{(b)} \subset Z_{(a+b)}$ by the triangle inequality, and $r_j + \frac{r}{2} \le r$). By Lemma 5.1, (6.5), and $\frac{r}{2} \ge \frac{1}{3}(1+r)$ for $r \ge 2$,

$$\|\mathrm{(a)}\| \le \|W_\gamma\|_1 \cdot 2C_{k'}\, C_J N^2 \cdot C_Z N^{1/2} (1+r) \cdot 3^{k'} (1+r)^{-k'},$$

which, with $k' = k+1$, is at most a constant times $N^{5/2}(1+r)^{-k}$.

The term (b). Since $r \ge 2$, only indices $j \ge 1$ occur, and the radii $r_j > \frac{r}{2}$ are at least $\frac{r}{2}$ and double from one to the next; so $\sum_{r_j > r/2} (1+r_j)^{-k} \le 2(r/2)^{-k} \le 2 \cdot 3^k (1+r)^{-k}$ and $\|\mathrm{(b)}\| \le \|W_\gamma\|_1\, 2^{k+2} 3^k C_k^J N^2 (1+r)^{-k}$.

The term (c). Let C_n^W denote the constant C_n of the tail bound in Lemma 5.9. With $n = 4k$ and $t_r \geq \frac{1}{2}(r/2C_{\mathrm{lc}})^{1/4}$,

$$\|(\mathrm{c})\| \leq \|J_-\| \int_{|t|>t_r} |W_\gamma(t)|\,\mathrm{d}t \leq 4\pi C_h N^2 \cdot C_{4k}^W \gamma^{-1} (2/\gamma)^{4k} (2C_{\mathrm{lc}})^k r^{-k},$$

and $r^{-k} \leq 2^k (1+r)^{-k}$.

Small radii. For $0 \leq r < r_*$ the approximant 0 has error $\|K_-\| \leq 4\pi C_h \|W_\gamma\|_1 (1+r_*)^k N^2 (1+r)^{-k}$.

Altogether $K_- \in \mathcal{A}_{\partial_-}$ with prefactor exponent $\frac{5}{2}$, with constants depending only on k, on γ (through W_γ), on C_h and the constants of J_-, on the constants C_{lc}, C_{k+1} of Theorem 5.3(c), on δ_0 and on c_*. The exponent consists of the N^2 of J_- and one power of $L_y \leq (4\pi/c_*)\sqrt{N}$, from the size of a strip of bounded width.

For the translated currents of Lemma 6.6(iii) the same argument applies verbatim with $\mathrm{Col}(j_0)$ or $\mathrm{Col}(j_0 + N)$ in place of ∂_-: these are strips over open intervals of length $2\delta_x$, and the argument used ∂_- only through Lemma 3.10 and Proposition 3.4(iii). □

Remark 6.9 (the crossing time). In the proof of Lemma 6.8 the dynamics is followed up to the time $t_r = (r/2C_{\mathrm{lc}})^{1/4} - 1$ at approximation radius r, a function of r alone, because the constants of Theorem 5.3 do not depend on N. The localization of $K_\pm$ is used only at distances $r \geq \frac{1}{24} L_x - O(1)$, of order $\sqrt{N}$ magnetic lengths (a third of the separations of Lemma 6.2, by the proof of Lemma 5.12(a)), so the dynamics is controlled up to times of order $N^{1/8}$, and the tails of the filter W_γ beyond these times are $O(N^{-\infty})$ (Lemma 5.9). This is the step for which the standard Lieb–Robinson bound in the orbital basis does not suffice: its constants grow with L_y, and the time up to which it controls the dynamics between the two cuts does not grow with N (Section 1). With the light cone $\rho \geq C_{\mathrm{lc}}(1+|t|)^\alpha$ the controlled time is of order $N^{1/(2\alpha)}$. Since C_{lc} is very large (Remark 5.4), t_r is positive only for very large r.

6.3 Clustering in the sector

Proposition 6.10 (in-sector clustering)**.** *In Setting 6.1, under Assumption 6.3, let $\mathcal{Q}$ be the clustering filter of Definition 5.8 for the Δ and γ of the assumption.*

(i) If A is sector-preserving, then $\mathcal{Q}(A)P = 0$ and $P\mathcal{Q}(A) = PA(\mathbf{1} - P)$.

(ii) If A is sector-preserving and B is any operator, then $PA(\mathbf{1}-P)BP = P[\mathcal{Q}(A), B]P$; in particular, for every unit vector $\Omega \in \operatorname{ran} P$,

$$|\langle \Omega, AB\Omega\rangle - \langle \Omega, APB\Omega\rangle| \leq \|[\mathcal{Q}(A), B]\|, \qquad \|PAB\Omega - PAPB\Omega\| \leq \|[\mathcal{Q}(A), B]\|.$$

(iii) If moreover $A \in \mathcal{A}_X$ and $B \in \mathcal{A}_Y$ are even, with $\|A_N\|, \|B_N\| \leq CN^m$, $|X_N| \leq CN^m$ and $d(X_N, Y_N) \geq c\sqrt{N} - C$, then the right-hand sides in (ii) are $O(N^{-\infty})$.

Proof. Write $\Pi := \Pi^{\mathbf{n}_N}$ and $P_E := \mathbf{1}_{\{E\}}(H_N)$ for $E \in \sigma(H_N)$, and let $\sigma_\Pi \subset \sigma(H_N)$ be the set of eigenvalues of $H_N|_{\mathcal{H}_N^{\mathbf{n}_N}}$, so that $P_E \Pi \neq 0$ if and only if $E \in \sigma_\Pi$. As in the proof of Lemma 6.7, $P = \sum_{E \in \sigma_\Pi,\, E \leq E_p} P_E \Pi$; every $E \in \sigma_\Pi$ satisfies $E \geq E_1$, and every $E \in \sigma_\Pi$ with $E > E_p$ satisfies $E \geq E_{p+1} \geq E_p + \gamma$. Recall that $g(\omega) = 0$ for $\omega \geq -\Delta$ and $g(\omega) = 1$ for $\omega \leq -\gamma$.

(i) Let A be sector-preserving. Since A and the P_E commute with Π,

$$P_E A P_{E'} \Pi = P_E \Pi A \Pi P_{E'} \quad \text{and} \quad \Pi P_E A P_{E'} = P_E \Pi A \Pi P_{E'}, \qquad E, E' \in \sigma(H_N), \tag{6.6}$$

and these vanish unless $E, E' \in \sigma_\Pi$. Hence $\mathcal{Q}(A)P = \sum_{E\in\sigma_\Pi}\sum_{E'\in\sigma_\Pi,\,E'\le E_p} g(E-E')P_E\Pi A\Pi P_{E'}$. In every term $E \ge E_1$ and $E' \le E_p \le E_1+\Delta$, so $E-E' \ge -\Delta$ and $g(E-E')=0$. Thus $\mathcal{Q}(A)P=0$. Likewise $P\mathcal{Q}(A) = \sum_{E\in\sigma_\Pi,\,E\le E_p}\sum_{E'\in\sigma_\Pi} g(E-E')P_E\Pi A\Pi P_{E'}$. If $E' \le E_p$, then $|E-E'| \le E_p - E_1 \le \Delta$ and $g(E-E')=0$; if $E' > E_p$, then $E' \ge E_p+\gamma \ge E+\gamma$, so $E-E' \le -\gamma \le -\gamma+\Delta$ and $g(E-E')=1$. Since $\sum_{E'\in\sigma_\Pi,\,E'>E_p}\Pi P_{E'} = \Pi - P$, this gives $P\mathcal{Q}(A) = PA(\Pi-P)$. Finally $PA(\mathbf{1}-\Pi) = PA\Pi(\mathbf{1}-\Pi)=0$, because $P=P\Pi$ and $\Pi A = A\Pi$; so $P\mathcal{Q}(A) = PA(\mathbf{1}-P)$.

(ii) By (i), $PA(\mathbf{1}-P)BP = P\mathcal{Q}(A)BP = P[\mathcal{Q}(A),B]P + PB\,\mathcal{Q}(A)P = P[\mathcal{Q}(A),B]P$. For a unit vector $\Omega \in \operatorname{ran} P$, $\langle\Omega, AB\Omega\rangle - \langle\Omega, APB\Omega\rangle = \langle\Omega, PA(\mathbf{1}-P)BP\Omega\rangle$ and $PAB\Omega - PAPB\Omega = PA(\mathbf{1}-P)BP\Omega$; both are therefore bounded by $\|P[\mathcal{Q}(A),B]P\| \le \|[\mathcal{Q}(A),B]\|$.

(iii) Let m_A, m_B be prefactor exponents of A and B. The operator H_N is even (its local terms are), so its spectral projections are even and $\mathcal{Q}(A_N)$ is even. By Proposition 5.11, $\mathcal{Q}(A) \in \mathcal{A}_X$ with prefactor exponent $1+\max\{m_A,m\}+m$ and $\|\mathcal{Q}(A_N)\| \le CN^{1+m}$. Lemma 5.13, applied to the two sequences $\mathcal{Q}(A)$ and B, gives $[\mathcal{Q}(A_N), B_N] = O(N^{-\infty})$. □

Statement (ii) is the clustering property of [BBDF21, Proposition 3.1] and of the display used in the proof of their Lemma 4.2, here with the gap assumed only in the sector and with no condition on B. The sector preservation of A cannot be dropped: it is what removes the contributions of eigenvalues of H_N outside the sector, where nothing is assumed.

6.4 The localized unitaries and their commutation relations

Definition 6.11 (the objects of the argument)**.** Under Assumptions 6.3 and 6.4, for $\varphi \in [0,2\pi]$ let

$$K_\pm^U := U^*K_\pm U, \qquad \overline{Q}_\pm := Q_\pm - K_\pm, \qquad \overline{Q}_\pm^U := Q_\pm + T_\pm - K_\pm^U, \qquad \overline{Q}^U := U^*\overline{Q}U,$$
$$\mathcal{M}(\varphi) := U^*e^{\mathrm{i}\varphi\overline{Q}}Ue^{-\mathrm{i}\varphi\overline{Q}} = e^{\mathrm{i}\varphi\overline{Q}^U}e^{-\mathrm{i}\varphi\overline{Q}}, \qquad \mathcal{M}_\pm(\varphi) := e^{\mathrm{i}\varphi\overline{Q}_\pm^U}e^{-\mathrm{i}\varphi\overline{Q}_\pm},$$
$$\widetilde{T}_- := \overline{Q}_-^U - \overline{Q}_- = T_- - K_-^U + K_-.$$

These are the objects of [BBDF21, Section 4.2]; note that $\overline{Q}_\pm^U \ne U^*\overline{Q}_\pm U$ in general.

Lemma 6.12 (localization and size)**.** *Under Assumptions 6.3 and 6.4 there is m_*, depending only on m_T and on the prefactor exponents in (U3) and (U4), such that each operator in the table below is even and sector-preserving, has norm at most CN^{m_*}, and is almost supported in the indicated set with prefactor exponent at most m_*, uniformly in $\varphi \in [0,2\pi]$. For $U=\Theta$ (Section 7) the norms and exponents of the table hold, and one may take $m_* = 9/2$; for a general U the rows of $T_\pm$, $K_\pm^U$, $\overline{Q}_\pm^U$, $\widetilde{T}_-$ and the last two rows depend on the exponents in (U3) and (U4).*

operators	*norm*	*almost supported in*	*exponent*
Q, $Q_\pm$, Q_m	$\le N$	$\mathrm{Col}\Gamma$, $\mathrm{Col}\Gamma_\pm$, $\mathrm{Col}\Gamma_m$ *(exactly)*	0
$T_- = -n_{0\uparrow}$, $T_+ = n_{N\uparrow}$	1	$\partial_\pm$ *(exactly)*	0
$J_\pm$	$\le CN^2$	$\partial_\pm$	2
$K_\pm$, $K_\pm^U$	$\le CN^2$	$\partial_\pm$	5/2
$\overline{Q}_\pm$, $\overline{Q}_\pm^U$	$\le CN^2$	$\mathrm{Col}\Gamma_\pm \cup \partial_\pm$	5/2
$\widetilde{T}_-$	$\le CN^2$	∂_-	5/2
$\mathcal{M}_\pm(\varphi)$, $e^{\pm 2\pi\mathrm{i}\overline{Q}_\pm}$ *(unitary)*	1	$\partial_\pm$	5/2
$\mathcal{Q}(V)$, $\mathcal{Q}(V^*)$, $V \in \{\mathcal{M}_\pm(\varphi), e^{2\pi\mathrm{i}\overline{Q}_\pm}\}$	$\le CN$	$\partial_\pm$	$\le 9/2$

Proof. Two elementary facts about the classes $\mathcal{A}_Z$ are used.

(F1) If $A \in \mathcal{A}_X$ and $B \in \mathcal{A}_Y$ have prefactor exponents m_A and m_B, then $A + B \in \mathcal{A}_{X\cup Y}$ with prefactor exponent $\max\{m_A, m_B\}$: since $\mathfrak{A}(X_{(r)})$ and $\mathfrak{A}(Y_{(r)})$ are contained in $\mathfrak{A}((X\cup Y)_{(r)})$, both sequences lie in $\mathcal{A}_{X\cup Y}$ with their constants, and Lemma 5.12(b) applies there.

(F2) If $A, B \in \mathcal{A}_X$ have prefactor exponents m_A and m_B and $\|A_N\|, \|B_N\| \le 1$, then $AB \in \mathcal{A}_X$ with prefactor exponent $\max\{m_A, m_B\}$, with constants depending only on those of A and B. Indeed, let $A_{N,r}, B_{N,r} \in \mathfrak{A}((X_N)_{(r)})$ be approximants with errors at most $C^A_k N^{m_A}(1+r)^{-k}$ and $C^B_k N^{m_B}(1+r)^{-k}$, and replace $A_{N,r}$ by 0 whenever $\|\iota(A_N) - A_{N,r}\| > 1$; this does not increase the error, which becomes $\|A_N\| \le 1$, and it ensures $\|A_{N,r}\| \le 2$. Then

$$\begin{aligned}\|\iota(A_N B_N) - A_{N,r}B_{N,r}\| &\le \|\iota(A_N) - A_{N,r}\|\|B_N\| + \|A_{N,r}\|\|\iota(B_N) - B_{N,r}\| \\ &\le (C^A_k + 2C^B_k) N^{\max\{m_A,m_B\}}(1+r)^{-k}.\end{aligned}$$

Let m_U be the prefactor exponent of $U^*K_\pm U$ in (U3) and m'_U that of $T_\pm$ in (U4), and put $m_\diamond := \max\{\frac{5}{2}, m_T, m_U, m'_U\}$ and $m_* := m_\diamond + 2$. A prefactor exponent may always be replaced by a larger one, and we do so freely. We go through the table.

Band charges. By Lemma 4.11(i), $\iota(Q_\mathcal{B}) \in \mathfrak{A}(\mathrm{Col}\mathcal{B})$ for every band, so $\iota(Q_\mathcal{B})$ itself is an approximant with error 0 for every r: the prefactor exponent is 0. The norms are at most N, and band charges are self-adjoint and sector-preserving (Lemma 4.10).

$T_\pm$. For $U = \Theta$, $T_- = -Q_{\{0\}}$ and $T_+ = Q_{\{N\}}$ (Lemma 7.1(b)) are, up to sign, band charges, with $\iota(T_\pm) \in \mathfrak{A}(\partial_\pm)$ and $\|T_\pm\| = 1$. For a general U, (U4) gives $T_\pm \in \mathcal{A}_{\partial_\pm}$ with prefactor exponent m'_U, self-adjoint and sector-preserving, with $\|T_\pm\| \le CN^{m_T}$.

$J_\pm$, $K_\pm$, $K^U_\pm$. Lemmas 6.6 and 6.8 give the rows of $J_\pm$ and $K_\pm$, with $\|J_\pm\| \le 4\pi C_h N^2$ and $\|K_\pm\| \le 4\pi C_h \|W_\gamma\|_1 N^2$. The operator $K^U_\pm = U^*K_\pm U$ has the norm of $K_\pm$, is self-adjoint, is sector-preserving by (U1), and lies in $\mathcal{A}_{\partial_\pm}$ with prefactor exponent m_U by (U3); for $U = \Theta$, $m_U = \frac{5}{2}$ by Lemma 7.1(e).

$\overline{Q}_\pm$, $\overline{Q}^U_\pm$, $\widetilde{T}_-$. These are sums of the previous operators: $\overline{Q}_\pm = Q_\pm - K_\pm$, $\overline{Q}^U_\pm = Q_\pm + T_\pm - K^U_\pm$ and $\widetilde{T}_- = T_- - K^U_- + K_-$. By (F1) they lie in $\mathcal{A}_{\mathrm{Col}\Gamma_\pm\cup\partial_\pm}$, $\mathcal{A}_{\mathrm{Col}\Gamma_\pm\cup\partial_\pm}$ and $\mathcal{A}_{\partial_-}$, with prefactor exponents $\frac{5}{2}$, $\max\{m'_U, m_U\}$ and $\max\{m'_U, m_U, \frac{5}{2}\}$, all at most $m_\diamond$, and norms at most $N + CN^{m_T} + CN^2 \le CN^{m_\diamond}$. For $U = \Theta$ the exponents are $\frac{5}{2}$ and the norms at most CN^2. They are self-adjoint and sector-preserving.

$\mathcal{M}_\pm(\varphi)$ *and* $e^{\pm 2\pi\mathrm{i}\overline{Q}_\pm}$. Put $\Xi^U_- := T_- - K^U_-$ and $\Xi_- := -K_-$, so that $\overline{Q}^U_- = Q_- + \Xi^U_-$ and $\overline{Q}_- = Q_- + \Xi_-$, and

$$\mathcal{M}_-(\varphi) = \left[e^{\mathrm{i}\varphi(Q_- + \Xi^U_-)} e^{-\mathrm{i}\varphi Q_-}\right]\left[e^{\mathrm{i}\varphi(Q_- + \Xi_-)} e^{-\mathrm{i}\varphi Q_-}\right]^*.$$

The operators Ξ^U_- and Ξ_- are self-adjoint and lie in $\mathcal{A}_{\partial_-}$ with prefactor exponent $m_\diamond$, and $\|\Xi^U_-\|, \|\Xi_-\| \le CN^{m_\diamond}$; and $\partial_- = \mathrm{Str}(I_-)$ with $I_- = (-\delta_x, \delta_x)$ an open interval of length $2\delta_x$. By Lemma 5.12(d), with the band Γ_-, both brackets lie in $\mathcal{A}_{\partial_-}$ with prefactor exponent $m_\diamond$, uniformly in $\varphi \in [0, 2\pi]$, and so does the adjoint of the second (Lemma 5.12(b)). Both are unitary, so by (F2) $\mathcal{M}_-(\varphi) \in \mathcal{A}_{\partial_-}$ with prefactor exponent $m_\diamond$, uniformly in φ. Since $e^{2\pi\mathrm{i}Q_-} = \mathbf{1}$ (Lemma 4.10), $e^{2\pi\mathrm{i}\overline{Q}_-} = e^{2\pi\mathrm{i}(Q_- + \Xi_-)} e^{-2\pi\mathrm{i}Q_-}$ is the operator of Lemma 5.12(d) at $\varphi = 2\pi$, and it lies in $\mathcal{A}_{\partial_-}$ with prefactor exponent $\frac{5}{2}$, as does its adjoint $e^{-2\pi\mathrm{i}\overline{Q}_-}$. The operators with index + are treated in the same way, with the band Γ_+ and $\partial_+ = \mathrm{Str}(I_+)$, $I_+ = (\frac{1}{2}L_x - \delta_x, \frac{1}{2}L_x + \delta_x)$. All these operators are unitary, and they are products of functions of self-adjoint sector-preserving operators, hence sector-preserving (Lemma 2.6(ii)) and even.

$\mathcal{Q}(V)$ and $\mathcal{Q}(V^*)$. Let $V \in \{\mathcal{M}_\pm(\varphi), e^{2\pi\mathrm{i}\overline{Q}_\pm}\}$. Then V and V^* are even, lie in $\mathcal{A}_{\partial_\pm}$ with prefactor exponent $m_\diamond$, have norm 1, and $|\partial_\pm| \le C\sqrt{N}$ (Lemma 6.2). By Proposition 5.11 with $m = m_\diamond$, $m' = 0$, $m'' = \frac{1}{2}$, $\mathcal{Q}(V)$ and $\mathcal{Q}(V^*)$ lie in $\mathcal{A}_{\partial_\pm}$ with prefactor exponent $m_\diamond + \frac{3}{2} \le m_*$ and have norm at most CN; the constants of Proposition 5.11 depend only on those of its input, so these statements hold uniformly in φ. Since the spectral projections of H_N are even and sector-preserving, so are $\mathcal{Q}(V)$ and $\mathcal{Q}(V^*)$.

Every norm in the table is at most $CN^{m_\diamond}$ and every prefactor exponent at most $m_\diamond + \frac{3}{2}$, which proves the lemma with $m_* = m_\diamond + 2$. For $U = \Theta$, $m_T = m'_U = 0$ and $m_U = \frac{5}{2}$, so $m_\diamond = \frac{5}{2}$, the exponent of $\mathcal{Q}(V)$ is 4, and $m_* = \frac{9}{2}$. □

Lemma 6.13 (commutation relations)**.** *Under Assumptions 6.3 and 6.4, uniformly in $\varphi \in [0, 2\pi]$:*

*(i) (*exact relations*) $\overline{Q} = \overline{Q}_- + Q_m + \overline{Q}_+$ and $\overline{Q}^U = \overline{Q}^U_- + Q_m + \overline{Q}^U_+$; $[Q_-, Q_m] = [Q_m, Q_+] = [Q_-, Q_+] = 0$; and for $U = \Theta$ also $[Q_m, T_\pm] = [Q_\mp, T_\pm] = [T_-, T_+] = 0$.*

*(ii) (*separated pairs*) The commutators $[A, Q_m]$ with $A \in \{K_-, K^U_-, K_+, K^U_+\}$, and $[A_-, B_+]$ with $A_- \in \{Q_-, T_-, K_-, K^U_-\}$ and $B_+ \in \{Q_+, T_+, K_+, K^U_+\}$ not covered by (i), are $O(N^{-\infty})$; so are, for a general U, the commutators of $T_\pm$ with Q_m and with $Q_\mp$. Each of these is a commutator of operators almost supported in sets at distance at least $\frac{1}{8}L_x - \kappa - 2\delta_x$.*

*(iii) (*consequences*) $[Q_m, A] \doteq 0$ for $A \in \{\overline{Q}_\pm, \overline{Q}^U_\pm, \widetilde{T}_-\}$; $[A_-, B_+] \doteq 0$ for $A_- \in \{\overline{Q}_-, \overline{Q}^U_-, \widetilde{T}_-\}$ and $B_+ \in \{\overline{Q}_+, \overline{Q}^U_+\}$; $\mathcal{M}(\varphi) \doteq \mathcal{M}_-(\varphi)\mathcal{M}_+(\varphi)$; $e^{2\pi\mathrm{i}\overline{Q}} \doteq e^{2\pi\mathrm{i}\overline{Q}_-} e^{2\pi\mathrm{i}\overline{Q}_+}$; $[P, e^{2\pi\mathrm{i}\overline{Q}}] = 0$; and $[P, \mathcal{M}(\varphi)] \doteq 0$, with equality if $[U, P] = 0$.*

Proof. All statements $\doteq$ below are uniform in $\varphi \in [0, 2\pi]$. Besides the second bound of Lemma A.8, $\|e^{\mathrm{i}(A+B)} - e^{\mathrm{i}A}e^{\mathrm{i}B}\| \le \frac{1}{2}\|[A, B]\|$ for bounded self-adjoint A and B, we use the following identity: for bounded self-adjoint A and any bounded C,

$$[e^{\mathrm{i}\varphi A}, C] = \mathrm{i}\int_0^\varphi e^{\mathrm{i}uA}[A, C]e^{\mathrm{i}(\varphi-u)A}\,\mathrm{d}u, \qquad \text{so} \qquad \|[e^{\mathrm{i}\varphi A}, C]\| \le |\varphi|\,\|[A, C]\|, \tag{6.7}$$

which follows by integrating $\frac{\mathrm{d}}{\mathrm{d}u}(e^{\mathrm{i}uA}Ce^{\mathrm{i}(\varphi-u)A}) = \mathrm{i}e^{\mathrm{i}uA}[A, C]e^{\mathrm{i}(\varphi-u)A}$ from 0 to φ. Applied twice, it gives $\|[e^{\mathrm{i}\varphi A}, e^{\mathrm{i}\psi B}]\| \le |\varphi\psi|\,\|[A, B]\|$ for bounded self-adjoint A, B.

(i) Since Γ is the disjoint union of Γ_-, Γ_m and Γ_+, $Q = Q_- + Q_m + Q_+$ (Lemma 4.10), and so $\overline{Q} = Q - K_- - K_+ = \overline{Q}_- + Q_m + \overline{Q}_+$. By (U4), $\overline{Q}^U = U^*QU - U^*K_-U - U^*K_+U = Q + T_- + T_+ - K^U_- - K^U_+ = \overline{Q}^U_- + Q_m + \overline{Q}^U_+$. Band charges commute with one another (Lemma 4.10); this gives $[Q_-, Q_m] = [Q_m, Q_+] = [Q_-, Q_+] = 0$, and for $U = \Theta$, where $T_- = -Q_{\{0\}}$ and $T_+ = Q_{\{N\}}$ are, up to sign, band charges (Lemma 7.1(b)), also $[Q_m, T_\pm] = [Q_\mp, T_\pm] = [T_-, T_+] = 0$.

(ii) By Lemma 6.12, every operator below is even, has norm at most CN^{m_*}, and is almost supported, with prefactor exponent at most m_* and uniformly in φ, in the following set: Q_m in $\mathrm{Col}\Gamma_m$; $Q_\pm$ in $\mathrm{Col}\Gamma_\pm$; $T_\pm$, $K_\pm$ and $K^U_\pm$ in $\partial_\pm$. By Lemma 6.2 the pairs fall into three classes:

- (A, Q_m) with $A \in \{K_-, K^U_-, K_+, K^U_+\}$, and for a general U also $(T_\pm, Q_m)$: sets $\partial_\pm$ and $\mathrm{Col}\Gamma_m$, at distance at least $\frac{1}{8}L_x - \kappa - 2\delta_x$;
- (Q_-, B_+) with $B_+ \in \{K_+, K^U_+\}$ and (A_-, Q_+) with $A_- \in \{K_-, K^U_-\}$, and for a general U also (Q_-, T_+) and (T_-, Q_+): sets $\mathrm{Col}\Gamma_-$ and ∂_+, or ∂_- and $\mathrm{Col}\Gamma_+$, at distance at least $\frac{3}{8}L_x - 2\delta_x$;
- (A_-, B_+) with $A_- \in \{T_-, K_-, K^U_-\}$ and $B_+ \in \{T_+, K_+, K^U_+\}$, except (T_-, T_+) when $U = \Theta$: sets ∂_- and ∂_+, at distance at least $\frac{1}{2}L_x - 2\delta_x$.

For $U = \Theta$ these are sixteen pairs. All distances are at least $\frac{c_*}{16}\sqrt{N}$ for $N \geq N_*$ (Lemma 6.2), hence at least $\frac{c_*}{16}\sqrt{N} - \frac{c_*}{16}\sqrt{N_*}$ for every N, and Lemma 5.13, applied to the finite set of sequences of Lemma 6.12, shows that each of these commutators is $O(N^{-\infty})$, uniformly in φ.

(iii) The operators $\overline{Q}_\pm$ and $\overline{Q}_\pm^U$ are almost supported in $\mathrm{Col}\Gamma_\pm \cup \partial_\pm$, a set adjacent to $\mathrm{Col}\Gamma_m$, so Lemma 5.13 cannot be applied to their commutators with Q_m directly. We therefore expand every commutator in (iii) bilinearly into the pairs of (i) and (ii) (for those involving $\widetilde{T}_-$, which is almost supported in ∂_-, this is not necessary, but it is equally valid). Thus

$$[Q_m, \overline{Q}_\pm] = -[Q_m, K_\pm], \qquad [Q_m, \overline{Q}_\pm^U] = [Q_m, T_\pm] - [Q_m, K_\pm^U],$$
$$[Q_m, \widetilde{T}_-] = [Q_m, T_-] - [Q_m, K_-^U] + [Q_m, K_-],$$

and each term on the right is 0 by (i) or $O(N^{-\infty})$ by (ii). Likewise, for $A_- \in \{\overline{Q}_-, \overline{Q}_-^U, \widetilde{T}_-\}$ and $B_+ \in \{\overline{Q}_+, \overline{Q}_+^U\}$, $[A_-, B_+]$ is a sum of at most nine commutators $\pm[a_-, b_+]$ with $a_- \in \{Q_-, T_-, K_-, K_-^U\}$ and $b_+ \in \{Q_+, T_+, K_+, K_+^U\}$, each 0 by (i) or $O(N^{-\infty})$ by (ii). This proves the first two statements of (iii).

For the factorization of $\mathcal{M}(\varphi) = e^{\mathrm{i}\varphi\overline{Q}^U} e^{-\mathrm{i}\varphi\overline{Q}}$, insert (i) and use Lemma A.8:

$$\|e^{\mathrm{i}\varphi\overline{Q}^U} - e^{\mathrm{i}\varphi\overline{Q}_-^U} e^{\mathrm{i}\varphi(Q_m+\overline{Q}_+^U)}\| \leq \tfrac{1}{2}\varphi^2 \|[\overline{Q}_-^U, Q_m + \overline{Q}_+^U]\|,$$
$$\|e^{\mathrm{i}\varphi(Q_m+\overline{Q}_+^U)} - e^{\mathrm{i}\varphi Q_m} e^{\mathrm{i}\varphi\overline{Q}_+^U}\| \leq \tfrac{1}{2}\varphi^2 \|[Q_m, \overline{Q}_+^U]\|,$$
$$\|e^{-\mathrm{i}\varphi\overline{Q}} - e^{-\mathrm{i}\varphi\overline{Q}_+} e^{-\mathrm{i}\varphi(Q_m+\overline{Q}_-)}\| \leq \tfrac{1}{2}\varphi^2 \|[\overline{Q}_+, Q_m + \overline{Q}_-]\|,$$
$$\|e^{-\mathrm{i}\varphi(Q_m+\overline{Q}_-)} - e^{-\mathrm{i}\varphi Q_m} e^{-\mathrm{i}\varphi\overline{Q}_-}\| \leq \tfrac{1}{2}\varphi^2 \|[Q_m, \overline{Q}_-]\|.$$

All four right-hand sides are $O(N^{-\infty})$ by what was just proved. Since a product of unitaries changes by at most the sum of the changes of its factors, $\mathcal{M}(\varphi) \doteq e^{\mathrm{i}\varphi\overline{Q}_-^U} e^{\mathrm{i}\varphi Q_m} \mathcal{M}_+(\varphi) e^{-\mathrm{i}\varphi Q_m} e^{-\mathrm{i}\varphi\overline{Q}_-}$. By (6.7),

$$\|[e^{\mathrm{i}\varphi Q_m}, \mathcal{M}_+(\varphi)]\| \leq \varphi^2 (\|[Q_m, \overline{Q}_+^U]\| + \|[Q_m, \overline{Q}_+]\|),$$
$$\|[\mathcal{M}_+(\varphi), e^{-\mathrm{i}\varphi\overline{Q}_-}]\| \leq \varphi^2 (\|[\overline{Q}_+^U, \overline{Q}_-]\| + \|[\overline{Q}_+, \overline{Q}_-]\|),$$

both $O(N^{-\infty})$. Hence $\mathcal{M}(\varphi) \doteq e^{\mathrm{i}\varphi\overline{Q}_-^U} \mathcal{M}_+(\varphi) e^{-\mathrm{i}\varphi\overline{Q}_-} \doteq e^{\mathrm{i}\varphi\overline{Q}_-^U} e^{-\mathrm{i}\varphi\overline{Q}_-} \mathcal{M}_+(\varphi) = \mathcal{M}_-(\varphi)\mathcal{M}_+(\varphi)$. In the same way, with $\varphi = 2\pi$ and $\overline{Q}$ in place of $\overline{Q}^U$, $e^{2\pi\mathrm{i}\overline{Q}} \doteq e^{2\pi\mathrm{i}\overline{Q}_-} e^{2\pi\mathrm{i}Q_m} e^{2\pi\mathrm{i}\overline{Q}_+} = e^{2\pi\mathrm{i}\overline{Q}_-} e^{2\pi\mathrm{i}\overline{Q}_+}$, because $e^{2\pi\mathrm{i}Q_m} = \mathbf{1}$ (Lemma 4.10). Since $[\overline{Q}, P] = 0$ (Lemma 6.7), P commutes with every function of $\overline{Q}$; in particular $[P, e^{2\pi\mathrm{i}\overline{Q}}] = 0$ and $[P, e^{\pm\mathrm{i}\varphi\overline{Q}}] = 0$. Hence, from $\mathcal{M}(\varphi) = U^* e^{\mathrm{i}\varphi\overline{Q}} U e^{-\mathrm{i}\varphi\overline{Q}}$ and $\|[P, U^*]\| = \|[P, U]^*\| = \|[U, P]\|$,

$$\|[P, \mathcal{M}(\varphi)]\| \leq \|[P, U^*]\| + \|[P, U]\| = 2\|[U, P]\|,$$

which is $O(N^{-\infty})$ by (U2) and vanishes if $[U, P] = 0$. □

Lemma 6.14 (Lemma 4.2 of [BBDF21] in the sector)**.** *Under Assumption 6.3, let V_-, V_+ be unitaries on $\mathcal{H}_N$, even and sector-preserving, with $V_\pm \in \mathcal{A}_{\partial_\pm}$ uniformly in a parameter, and $\|[V_- V_+, P]\| = O(N^{-\infty})$ uniformly in the parameter. Then $\|[V_\pm, P]\| = O(N^{-\infty})$ uniformly in the parameter. Quantitatively, with*

$$\varepsilon_0 := \|[V_- V_+, P]\|, \quad \varepsilon_c := \|[V_-, V_+]\|, \quad \varepsilon_1 := \|P V_+ (\mathbf{1} - P) V_- P\|, \quad \varepsilon_2 := \|P V_+^* (\mathbf{1} - P) V_-^* P\|,$$

and $a := \min\{\sqrt{1-\varepsilon_0-2\varepsilon_c}-\varepsilon_1,\ \sqrt{1-\varepsilon_0}-\varepsilon_2\}$ *(defined when* $\varepsilon_0+2\varepsilon_c\le 1$*): if* $a>0$*, then* $\|[V_-,P]\|\le\sqrt{1-a^2}$*; and the same holds for* V_+ *with the roles of* V_- *and* V_+ *exchanged in the definitions of* ε_1*,* ε_2 *and of the products.*

Proof. The quantitative statement. Put $V := V_-V_+$ and fix a unit vector $\Omega\in\operatorname{ran}P$. Since $V_+V_- = V+[V_+,V_-]$ and $\|[[V_+,V_-],P]\|\le 2\varepsilon_c$, we have $\|[V_+V_-,P]\|\le\varepsilon_0+2\varepsilon_c$, and Lemma A.6(i), applied to the unitary V_+V_-, gives $\|PV_+V_-\Omega\|^2\ge 1-\varepsilon_0-2\varepsilon_c$. As $P\Omega=\Omega$, $PV_+V_-\Omega-PV_+PV_-\Omega = PV_+(\mathbf{1}-P)V_-P\Omega$, of norm at most ε_1. Since $\|PV_+P\|\le 1$,

$$\|PV_-\Omega\|\ge\|PV_+P\,PV_-\Omega\|\ge\|PV_+V_-\Omega\|-\varepsilon_1\ge\sqrt{1-\varepsilon_0-2\varepsilon_c}-\varepsilon_1\ge a.$$

Similarly $V^* = V_+^*V_-^*$ is unitary with $\|[V^*,P]\| = \|[V,P]\| = \varepsilon_0$, so $\|PV_+^*V_-^*\Omega\|^2\ge 1-\varepsilon_0$ by Lemma A.6(i); and $\|PV_+^*V_-^*\Omega - PV_+^*PV_-^*\Omega\| = \|PV_+^*(\mathbf{1}-P)V_-^*P\Omega\|\le\varepsilon_2$, whence $\|PV_-^*\Omega\|\ge\sqrt{1-\varepsilon_0}-\varepsilon_2\ge a$. Let $a>0$. By homogeneity, $\|PV_-\psi\|^2\ge a^2\|\psi\|^2$ and $\|PV_-^*\psi\|^2\ge a^2\|\psi\|^2$ for every $\psi\in\operatorname{ran}P$; since $a\le\sqrt{1-\varepsilon_0}\le 1$, Lemma A.6(ii) with $\varepsilon := \sqrt{1-a^2}\in[0,1)$ gives $\|[V_-,P]\|\le\sqrt{1-a^2}$. The statement for V_+ is the statement just proved for the ordered pair (V_+,V_-) in place of (V_-,V_+), that is, with ε_0, ε_1, ε_2 replaced by $\varepsilon_0' := \|[V_+V_-,P]\|\le\varepsilon_0+2\varepsilon_c$, $\varepsilon_1' := \|PV_-(\mathbf{1}-P)V_+P\|$ and $\varepsilon_2' := \|PV_-^*(\mathbf{1}-P)V_+^*P\|$; the hypotheses are symmetric in V_- and V_+.

The asymptotic statement. Since V_+ and V_+^* are sector-preserving, Proposition 6.10(ii) gives $PV_+(\mathbf{1}-P)V_-P = P[\mathcal{Q}(V_+),V_-]P$ and $PV_+^*(\mathbf{1}-P)V_-^*P = P[\mathcal{Q}(V_+^*),V_-^*]P$, so $\varepsilon_1\le\|[\mathcal{Q}(V_+),V_-]\|$ and $\varepsilon_2\le\|[\mathcal{Q}(V_+^*),V_-^*]\|$; likewise $\varepsilon_1'\le\|[\mathcal{Q}(V_-),V_+]\|$ and $\varepsilon_2'\le\|[\mathcal{Q}(V_-^*),V_+^*]\|$. Let m_V be a prefactor exponent of $V_\pm$ (and hence of $V_\pm^*$, by Lemma 5.12(b)), valid uniformly in the parameter. By Proposition 5.11, with $\|V_\pm\|=1$ and $|\partial_\pm|\le C\sqrt{N}$ (Lemma 6.2), $\mathcal{Q}(V_\pm)$ and $\mathcal{Q}(V_\pm^*)$ lie in $\mathcal{A}_{\partial_\pm}$ with prefactor exponent $m_V+\frac{3}{2}$ and have norm at most CN, uniformly in the parameter. The sets ∂_- and ∂_+ are at distance at least $\frac{1}{2}L_x-2\delta_x$, which is at least $\frac{c_*}{16}\sqrt{N}-\frac{c_*}{16}\sqrt{N_*}$ for every N (Lemma 6.2). Lemma 5.13, applied to the sequences $V_\pm$, $V_\pm^*$, $\mathcal{Q}(V_\pm)$, $\mathcal{Q}(V_\pm^*)$, shows that ε_c, ε_1, ε_2, ε_1', ε_2' are $O(N^{-\infty})$ uniformly in the parameter; so is ε_0, by hypothesis. Put $\bar\varepsilon := \varepsilon_0+2\varepsilon_c+\varepsilon_1+\varepsilon_2$, and suppose $\bar\varepsilon\le\frac{1}{2}$. Since $\sqrt{1-x}\ge 1-x$ for $x\in[0,1]$, $a\ge 1-\bar\varepsilon>0$, and $\|[V_-,P]\|\le\sqrt{1-a^2}\le\sqrt{2(1-a)}\le\sqrt{2\bar\varepsilon}$. Given k, choose C_{2k} with $\bar\varepsilon\le C_{2k}N^{-2k}$; then $\|[V_-,P]\|\le\sqrt{2C_{2k}}\,N^{-k}$ whenever $C_{2k}N^{-2k}\le\frac{1}{2}$, and $\|[V_-,P]\|\le 2\le 2\sqrt{2C_{2k}}\,N^{-k}$ otherwise. Hence $\|[V_-,P]\| = O(N^{-\infty})$, uniformly in the parameter, and the same argument with the primed quantities gives $\|[V_+,P]\| = O(N^{-\infty})$. The square root halves the decay exponent, which is harmless for $O(N^{-\infty})$. □

Corollary 6.15 (the localized unitaries commute with P)**.** *Under Assumptions 6.3 and 6.4,* $[P,\mathcal{M}_\pm(\varphi)]\doteq 0$ *uniformly in* $\varphi\in[0,2\pi]$*, and* $[P,e^{2\pi\mathrm{i}\overline{Q}_\pm}]\doteq 0$.

Proof. By Lemma 6.12, $\mathcal{M}_-(\varphi)$ and $\mathcal{M}_+(\varphi)$ are unitary, even and sector-preserving, and lie in $\mathcal{A}_{\partial_-}$ and $\mathcal{A}_{\partial_+}$ uniformly in $\varphi\in[0,2\pi]$. By Lemma 6.13(iii),

$$\|[\mathcal{M}_-(\varphi)\mathcal{M}_+(\varphi),P]\|\le 2\|\mathcal{M}_-(\varphi)\mathcal{M}_+(\varphi)-\mathcal{M}(\varphi)\|+\|[P,\mathcal{M}(\varphi)]\| = O(N^{-\infty})$$

uniformly in φ. Lemma 6.14 with $V_\pm := \mathcal{M}_\pm(\varphi)$ gives $[P,\mathcal{M}_\pm(\varphi)]\doteq 0$ uniformly in φ. In the same way $e^{2\pi\mathrm{i}\overline{Q}_-}$ and $e^{2\pi\mathrm{i}\overline{Q}_+}$ are unitary, even, sector-preserving and lie in $\mathcal{A}_{\partial_-}$ and $\mathcal{A}_{\partial_+}$, and $\|[e^{2\pi\mathrm{i}\overline{Q}_-}e^{2\pi\mathrm{i}\overline{Q}_+},P]\|\le 2\|e^{2\pi\mathrm{i}\overline{Q}_-}e^{2\pi\mathrm{i}\overline{Q}_+}-e^{2\pi\mathrm{i}\overline{Q}}\|+\|[e^{2\pi\mathrm{i}\overline{Q}},P]\| = O(N^{-\infty})$, the last term being zero (Lemma 6.13(iii)); Lemma 6.14 with $V_\pm := e^{2\pi\mathrm{i}\overline{Q}_\pm}$ gives $[P,e^{2\pi\mathrm{i}\overline{Q}_\pm}]\doteq 0$.

In these two applications, the commutators which the proof of Lemma 6.14 shows to be $O(N^{-\infty})$ are those of the six pairs

$$(\mathcal{M}_-(\varphi),\mathcal{M}_+(\varphi)),\qquad (\mathcal{Q}(\mathcal{M}_+(\varphi)),\mathcal{M}_-(\varphi)),\qquad (\mathcal{Q}(\mathcal{M}_-(\varphi)),\mathcal{M}_+(\varphi)),$$
$$(e^{2\pi\mathrm{i}\overline{Q}_-},e^{2\pi\mathrm{i}\overline{Q}_+}),\qquad (\mathcal{Q}(e^{2\pi\mathrm{i}\overline{Q}_+}),e^{2\pi\mathrm{i}\overline{Q}_-}),\qquad (\mathcal{Q}(e^{2\pi\mathrm{i}\overline{Q}_-}),e^{2\pi\mathrm{i}\overline{Q}_+}),$$

and of the corresponding pairs of adjoints, such as $(\mathcal{Q}(\mathcal{M}_+(\varphi)^*),\mathcal{M}_-(\varphi)^*)$. In each of them one operator is almost supported in ∂_- and the other in ∂_+ (Lemma 6.12), at distance at least $\frac{1}{2}L_x-2\delta_x$ (Lemma 6.2), with norms at most CN and prefactor exponents at most m_*, uniformly in φ. □

6.5 The flux insertion and the determinant

For an operator A with $[A,P]=0$ let $\det_P A:=\det(PAP|_{\operatorname{ran}P})$.

Lemma 6.16 (the differential equation and its stability)**.** *Under Assumptions 6.3 and 6.4, for $\varphi\in[0,2\pi]$ let*

$$\mathcal{Z}_-(\varphi):=P\mathcal{M}_-(\varphi)P,\qquad \mathcal{G}(\varphi):=e^{\mathrm{i}\varphi\overline{Q}}P\widetilde{T}_-Pe^{-\mathrm{i}\varphi\overline{Q}},\qquad \mathcal{Y}(\varphi):=Pe^{\mathrm{i}\varphi(P\widetilde{T}_-P+\overline{Q})}e^{-\mathrm{i}\varphi\overline{Q}}P.$$

Then $\mathcal{G}(\varphi)=P\mathcal{G}(\varphi)P$ is self-adjoint; $-\mathrm{i}\mathcal{Z}_-'(\varphi)=\mathcal{Z}_-(\varphi)\mathcal{G}(\varphi)+\mathcal{E}(\varphi)$ with $\sup_{\varphi\in[0,2\pi]}\|\mathcal{E}(\varphi)\|=O(N^{-\infty})$; $-\mathrm{i}\mathcal{Y}'(\varphi)=\mathcal{Y}(\varphi)\mathcal{G}(\varphi)$ exactly; $\mathcal{Z}_-(0)=\mathcal{Y}(0)=P$; and $\|\mathcal{Z}_-(\varphi)-\mathcal{Y}(\varphi)\|\le\int_0^\varphi\|\mathcal{E}(u)\|\,\mathrm{d}u=O(N^{-\infty})$ uniformly in $\varphi\in[0,2\pi]$.

Proof. The generator. Since $[\overline{Q},P]=0$ (Lemma 6.7), $e^{\pm\mathrm{i}\varphi\overline{Q}}$ commute with P, so $\mathcal{G}(\varphi)=Pe^{\mathrm{i}\varphi\overline{Q}}\widetilde{T}_-e^{-\mathrm{i}\varphi\overline{Q}}P=P\mathcal{G}(\varphi)P$. The operator $\widetilde{T}_-=T_--K_-^U+K_-$ is self-adjoint (T_- by (U4), K_- by Lemma 6.6(i), and hence $K_-^U=U^*K_-U$), so $\mathcal{G}(\varphi)$ is self-adjoint; it depends continuously on φ.

The equation for $\mathcal{Z}_-$. For bounded self-adjoint A and B, $\frac{\mathrm{d}}{\mathrm{d}\varphi}(e^{\mathrm{i}\varphi A}e^{-\mathrm{i}\varphi B})=\mathrm{i}e^{\mathrm{i}\varphi A}(A-B)e^{-\mathrm{i}\varphi B}$. With $A=\overline{Q}_-^U$, $B=\overline{Q}_-$ and $A-B=\widetilde{T}_-$, and since $e^{\mathrm{i}\varphi\overline{Q}_-^U}=\mathcal{M}_-(\varphi)e^{\mathrm{i}\varphi\overline{Q}_-}$,

$$-\mathrm{i}\mathcal{Z}_-'(\varphi)=P\mathcal{M}_-(\varphi)\,e^{\mathrm{i}\varphi\overline{Q}_-}\widetilde{T}_-e^{-\mathrm{i}\varphi\overline{Q}_-}P.$$

We write the right-hand side as $\mathcal{Z}_-(\varphi)\mathcal{G}(\varphi)+\mathcal{E}_1(\varphi)+\mathcal{E}_2(\varphi)$, where, inserting $P=\mathbf{1}-(\mathbf{1}-P)$ after $\mathcal{M}_-(\varphi)$ and using $P\mathcal{M}_-(\mathbf{1}-P)=-P[\mathcal{M}_-,P]$,

$$\mathcal{E}_1(\varphi):=-P[\mathcal{M}_-(\varphi),P]\,e^{\mathrm{i}\varphi\overline{Q}_-}\widetilde{T}_-e^{-\mathrm{i}\varphi\overline{Q}_-}P,$$
$$\mathcal{E}_2(\varphi):=\mathcal{Z}_-(\varphi)P(e^{\mathrm{i}\varphi\overline{Q}_-}\widetilde{T}_-e^{-\mathrm{i}\varphi\overline{Q}_-}-e^{\mathrm{i}\varphi\overline{Q}}\widetilde{T}_-e^{-\mathrm{i}\varphi\overline{Q}})P,$$

and where we used $Pe^{\mathrm{i}\varphi\overline{Q}}\widetilde{T}_-e^{-\mathrm{i}\varphi\overline{Q}}P=\mathcal{G}(\varphi)$, which holds because $e^{\pm\mathrm{i}\varphi\overline{Q}}$ commute with P. Thus $-\mathrm{i}\mathcal{Z}_-'=\mathcal{Z}_-\mathcal{G}+\mathcal{E}$ with $\mathcal{E}:=\mathcal{E}_1+\mathcal{E}_2$. First, $\|\mathcal{E}_1(\varphi)\|\le\|[\mathcal{M}_-(\varphi),P]\|\|\widetilde{T}_-\|$. Second, put $B:=Q_m+\overline{Q}_+$, so that $\overline{Q}=\overline{Q}_-+B$ (Lemma 6.13(i)), and compare the unitaries $O_1:=e^{\mathrm{i}\varphi\overline{Q}}$ and $O_2:=e^{\mathrm{i}\varphi\overline{Q}_-}e^{\mathrm{i}\varphi B}$: by Lemma A.8, $\|O_1-O_2\|\le\frac{1}{2}\varphi^2\|[\overline{Q}_-,B]\|$, so $\|O_1\widetilde{T}_-O_1^*-O_2\widetilde{T}_-O_2^*\|\le2\|O_1-O_2\|\|\widetilde{T}_-\|\le\varphi^2\|[\overline{Q}_-,B]\|\|\widetilde{T}_-\|$; and $\|O_2\widetilde{T}_-O_2^*-e^{\mathrm{i}\varphi\overline{Q}_-}\widetilde{T}_-e^{-\mathrm{i}\varphi\overline{Q}_-}\|=\|[e^{\mathrm{i}\varphi B},\widetilde{T}_-]\|\le\varphi\|[B,\widetilde{T}_-]\|$ by (6.7). Since $\|\mathcal{Z}_-\|\le1$,

$$\|\mathcal{E}_2(\varphi)\|\le\varphi^2\|[\overline{Q}_-,Q_m+\overline{Q}_+]\|\|\widetilde{T}_-\|+\varphi\|[Q_m+\overline{Q}_+,\widetilde{T}_-]\|.$$

By Corollary 6.15, $\sup_\varphi \|[\mathcal{M}_-(\varphi), P]\| = O(N^{-\infty})$; by Lemma 6.13(iii), the commutators $[\overline{Q}_-, Q_m]$, $[\overline{Q}_-, \overline{Q}_+]$, $[Q_m, \widetilde{T}_-]$ and $[\widetilde{T}_-, \overline{Q}_+]$ are $O(N^{-\infty})$; and $\|\widetilde{T}_-\| \le CN^{m_*}$ (Lemma 6.12). Hence $\sup_{\varphi\in[0,2\pi]}\|\mathcal{E}(\varphi)\| = O(N^{-\infty})$. Note that $\mathcal{E} = P\mathcal{E}P$, since $\mathcal{Z}_-' = P\mathcal{Z}_-'P$ and $\mathcal{G} = P\mathcal{G}P$; and $\mathcal{E}$ is continuous in φ.

The equation for $\mathcal{Y}$. Let $A := P\widetilde{T}_-P + \overline{Q}$ and $B := \overline{Q}$. Both commute with P, hence so does $e^{\mathrm{i}\varphi A}e^{-\mathrm{i}\varphi B}$, and $\mathcal{Y}(\varphi) = Pe^{\mathrm{i}\varphi A}e^{-\mathrm{i}\varphi B}$. Then

$$-\mathrm{i}\mathcal{Y}'(\varphi) = Pe^{\mathrm{i}\varphi A}(A-B)e^{-\mathrm{i}\varphi B} = Pe^{\mathrm{i}\varphi A}e^{-\mathrm{i}\varphi B}\,e^{\mathrm{i}\varphi B}P\widetilde{T}_-Pe^{-\mathrm{i}\varphi B} = \mathcal{Y}(\varphi)\mathcal{G}(\varphi)$$

exactly. Clearly $\mathcal{Z}_-(0) = \mathcal{Y}(0) = P$.

Stability. Let $\mathcal{U}(u,\varphi)$, $0 \le u \le \varphi \le 2\pi$, be the solution of $\partial_\varphi \mathcal{U}(u,\varphi) = \mathrm{i}\mathcal{U}(u,\varphi)\mathcal{G}(\varphi)$ with $\mathcal{U}(u,u) = P$, a linear equation with continuous coefficients on the finite-dimensional space of operators on $\mathcal{H}_N$. Since $\mathcal{G} = P\mathcal{G}P$ is self-adjoint, $\partial_\varphi(\mathcal{U}\mathcal{U}^*) = \mathrm{i}\mathcal{U}\mathcal{G}\mathcal{U}^* - \mathrm{i}\mathcal{U}\mathcal{G}\mathcal{U}^* = 0$, so $\mathcal{U}(u,\varphi)\mathcal{U}(u,\varphi)^* = P$ and $\|\mathcal{U}(u,\varphi)\| \le 1$. The difference $\mathcal{Z}_- - \mathcal{Y}$ vanishes at $\varphi = 0$ and satisfies $(\mathcal{Z}_- - \mathcal{Y})' = \mathrm{i}(\mathcal{Z}_- - \mathcal{Y})\mathcal{G} + \mathrm{i}\mathcal{E}$. The function $\varphi \mapsto \mathrm{i}\int_0^\varphi \mathcal{E}(u)\,\mathcal{U}(u,\varphi)\,\mathrm{d}u$ satisfies the same equation: it vanishes at $\varphi = 0$, and its derivative $\mathrm{i}\,\mathcal{E}(\varphi)P + \mathrm{i}\int_0^\varphi \mathcal{E}(u)\,\mathrm{i}\mathcal{U}(u,\varphi)\mathcal{G}(\varphi)\,\mathrm{d}u$ is $\mathrm{i}\mathcal{E}(\varphi)$ plus i times the function itself times $\mathcal{G}(\varphi)$, because $\mathcal{E} = \mathcal{E}P$. By uniqueness of solutions of linear equations, $\mathcal{Z}_-(\varphi) - \mathcal{Y}(\varphi) = \mathrm{i}\int_0^\varphi \mathcal{E}(u)\mathcal{U}(u,\varphi)\,\mathrm{d}u$, and

$$\|\mathcal{Z}_-(\varphi) - \mathcal{Y}(\varphi)\| \le \int_0^\varphi \|\mathcal{E}(u)\|\|\mathcal{U}(u,\varphi)\|\,\mathrm{d}u \le \int_0^\varphi \|\mathcal{E}(u)\|\,\mathrm{d}u \le 2\pi \sup_{u\in[0,2\pi]}\|\mathcal{E}(u)\| = O(N^{-\infty}).$$

The size of $\mathcal{G}$, which is of order $\|\widetilde{T}_-\|$, does not enter this bound. □

Lemma 6.17 (the determinant at 2π)**.** *Under Assumptions 6.3 and 6.4:*

(i) $\mathcal{Z}_-(2\pi) \doteq PU^*e^{2\pi\mathrm{i}\overline{Q}_-}Ue^{-2\pi\mathrm{i}\overline{Q}_-}P$, *and* $\det_P \mathcal{Z}_-(2\pi) \doteq 1$;

(ii) $\det_P \mathcal{Y}(2\pi) = e^{2\pi\mathrm{i}\,\mathrm{Tr}(P\widetilde{T}_-)}$ *exactly;*

(iii) consequently $\mathrm{Tr}(P\widetilde{T}_-) \in \mathbb{Z} + O(N^{-\infty})$;

(iv) $\mathrm{Tr}(P\widetilde{T}_-) - \mathrm{Tr}(PT_-) = -\mathrm{Tr}\big((UPU^* - P)K_-\big)$, *and*

$$\big|\mathrm{Tr}\big((UPU^* - P)K_-\big)\big| \le 2p\|[U,P]\|\|K_-\| = O(N^{-\infty});$$

this vanishes if $[U,P] = 0$.

Proof. (i) *Step 1:* $e^{2\pi\mathrm{i}\overline{Q}_-} \doteq e^{2\pi\mathrm{i}(Q-K_-)}$. We have $Q - K_- = \overline{Q}_- + (Q_m + Q_+)$, and $Q_m + Q_+ = Q_{\Gamma_m\cup\Gamma_+}$ has integer spectrum, so $e^{2\pi\mathrm{i}(Q_m+Q_+)} = \mathbf{1}$ (Lemma 4.10). By Lemma A.8,

$$\|e^{2\pi\mathrm{i}(Q-K_-)} - e^{2\pi\mathrm{i}\overline{Q}_-}e^{2\pi\mathrm{i}(Q_m+Q_+)}\| \le 2\pi^2\|[\overline{Q}_-, Q_m+Q_+]\| = 2\pi^2\|[K_-,Q_m] + [K_-,Q_+]\|,$$

because band charges commute; both commutators are $O(N^{-\infty})$ by Lemma 6.13(ii). (The corresponding step of [BBDF21, Section 4.2] is discussed in Remark A.10(a).)

Step 2: $U^*e^{2\pi\mathrm{i}(Q-K_-)}U \doteq e^{2\pi\mathrm{i}\overline{Q}_-^U}$. By (U4), $U^*(Q-K_-)U = Q + T_- + T_+ - K_-^U = \overline{Q}_-^U + Q_m + (Q_+ + T_+)$. By Lemma A.8, applied twice,

$$\begin{aligned}\|e^{2\pi\mathrm{i}(\overline{Q}_-^U + Q_m + (Q_+ + T_+))} &- e^{2\pi\mathrm{i}\overline{Q}_-^U}e^{2\pi\mathrm{i}Q_m}e^{2\pi\mathrm{i}(Q_+ + T_+)}\| \\ &\le 2\pi^2\big(\|[\overline{Q}_-^U, Q_m + Q_+ + T_+]\| + \|[Q_m, Q_+ + T_+]\|\big).\end{aligned}$$

Expanding $\overline{Q}_-^U = Q_- + T_- - K_-^U$, the first commutator is a sum of commutators of $a \in \{Q_-, T_-, K_-^U\}$ with $b \in \{Q_m, Q_+, T_+\}$, and the second equals $[Q_m, T_+]$; each of these is 0 or $O(N^{-\infty})$ by Lemma 6.13(i) and (ii). Moreover $e^{2\pi\mathrm{i}Q_m} = \mathbf{1}$, and $e^{2\pi\mathrm{i}(Q_+ + T_+)} \doteq \mathbf{1}$: for $U = \Theta$ this is exact (Lemma 7.1(c)); in general $Q + T_+ = (Q_- + Q_m) + (Q_+ + T_+)$, so by Lemma A.8 $\|e^{2\pi\mathrm{i}(Q+T_+)} - e^{2\pi\mathrm{i}(Q_- + Q_m)} e^{2\pi\mathrm{i}(Q_+ + T_+)}\| \le 2\pi^2 \|[Q_- + Q_m, T_+]\| = O(N^{-\infty})$, while $e^{2\pi\mathrm{i}(Q_- + Q_m)} = \mathbf{1}$ and $e^{2\pi\mathrm{i}(Q+T_+)} = \mathbf{1} + O(N^{-\infty})$ by (U4).

Since conjugation by U is isometric, Steps 1 and 2 give $U^* e^{2\pi\mathrm{i}\overline{Q}_-} U \doteq e^{2\pi\mathrm{i}\overline{Q}_-^U}$, hence $\mathcal{M}_-(2\pi) = e^{2\pi\mathrm{i}\overline{Q}_-^U} e^{-2\pi\mathrm{i}\overline{Q}_-} \doteq U^* e^{2\pi\mathrm{i}\overline{Q}_-} U e^{-2\pi\mathrm{i}\overline{Q}_-}$; compressing with P gives the first claim.

Step 3: the determinant. Put $O_1 := U^*$, $O_2 := e^{2\pi\mathrm{i}\overline{Q}_-}$, $O_3 := U$, $O_4 := O_2^*$, $\varepsilon_U := \|[U, P]\|$ and $\varepsilon_- := \|[O_2, P]\|$; then $\|[O_1, P]\| = \|[O_3, P]\| = \varepsilon_U$ and $\|[O_4, P]\| = \varepsilon_-$. Since $PV_i(\mathbf{1} - P) = [P, V_i](\mathbf{1} - P)$, inserting P between consecutive factors gives

$$\|PO_1O_2O_3O_4P - (PO_1P)(PO_2P)(PO_3P)(PO_4P)\| \le 2\varepsilon_U + \varepsilon_-.$$

All operators here map $\operatorname{ran} P$ into itself, vanish on its orthogonal complement and have norm at most one. Let $d_i := \det(PV_iP|_{\operatorname{ran} P})$, so $|d_i| \le 1$. By Lemma A.9(i) and the multiplicativity of the determinant on $\operatorname{ran} P$, $|\det_P(PO_1O_2O_3O_4P) - d_1d_2d_3d_4| \le p(2\varepsilon_U + \varepsilon_-)$. By Lemma A.9(iii), applied to U and to O_2, $|d_1d_3 - 1| \le p\varepsilon_U$ and $|d_4d_2 - 1| \le p\varepsilon_-$, so $|d_1d_2d_3d_4 - 1| \le |d_1d_3 - 1||d_2d_4| + |d_2d_4 - 1| \le p(\varepsilon_U + \varepsilon_-)$. Finally $|\det_P \mathcal{Z}_-(2\pi) - \det_P(PO_1O_2O_3O_4P)| \le p\|\mathcal{M}_-(2\pi) - O_1O_2O_3O_4\|$ by Lemma A.9(i). Altogether

$$|\det_P \mathcal{Z}_-(2\pi) - 1| \le p\Big(\|\mathcal{M}_-(2\pi) - U^* e^{2\pi\mathrm{i}\overline{Q}_-} U e^{-2\pi\mathrm{i}\overline{Q}_-}\| + 3\varepsilon_U + 2\varepsilon_-\Big) = O(N^{-\infty}),$$

by Steps 1–2, (U2) and Corollary 6.15.

(ii) The operators $A := 2\pi\mathrm{i}(P\widetilde{T}_-P + \overline{Q})$ and $B := -2\pi\mathrm{i}\overline{Q}$ commute with P, and $\mathcal{Y}(2\pi) = Pe^A e^B P$. By Lemma A.9(ii),

$$\det_P \mathcal{Y}(2\pi) = \det_P(e^A)\det_P(e^B) = e^{\operatorname{Tr}(PA)} e^{\operatorname{Tr}(PB)} = e^{2\pi\mathrm{i}(\operatorname{Tr}(P\widetilde{T}_-P) + \operatorname{Tr}(P\overline{Q}) - \operatorname{Tr}(P\overline{Q}))} = e^{2\pi\mathrm{i}\operatorname{Tr}(P\widetilde{T}_-)}.$$

(Compare Remark A.10(b).)

(iii) The operators $\mathcal{Y}(2\pi)$ and $\mathcal{Z}_-(2\pi)$ map $\operatorname{ran} P$ into itself, vanish on its complement and have norm at most one, so by Lemma A.9(i) and Lemma 6.16, $|\det_P \mathcal{Y}(2\pi) - \det_P \mathcal{Z}_-(2\pi)| \le p\|\mathcal{Y}(2\pi) - \mathcal{Z}_-(2\pi)\| = O(N^{-\infty})$. With (i) and (ii), $|e^{2\pi\mathrm{i}\operatorname{Tr}(P\widetilde{T}_-)} - 1| = O(N^{-\infty})$. The number $\operatorname{Tr}(P\widetilde{T}_-) = \operatorname{Tr}(P\widetilde{T}_-P)$ is real, and by Lemma A.9(iv) its distance from $\mathbb{Z}$ is at most $\frac{1}{4}|e^{2\pi\mathrm{i}\operatorname{Tr}(P\widetilde{T}_-)} - 1| = O(N^{-\infty})$.

(iv) Since $\widetilde{T}_- = T_- - U^*K_-U + K_-$, the cyclicity of the trace gives $\operatorname{Tr}(P\widetilde{T}_-) - \operatorname{Tr}(PT_-) = -\operatorname{Tr}(UPU^*K_-) + \operatorname{Tr}(PK_-) = -\operatorname{Tr}((UPU^* - P)K_-)$. The operator $UPU^* - P = [U, P]U^*$ has norm $\|[U, P]\|$ and, as a difference of two projections of rank p, rank at most $2p$; its trace norm is therefore at most $2p\|[U, P]\|$, and $|\operatorname{Tr}((UPU^* - P)K_-)| \le 2p\|[U, P]\|\|K_-\|$. By (U2) and $\|K_-\| \le CN^2$ (Lemma 6.12) this is $O(N^{-\infty})$, and it vanishes if $[U, P] = 0$. □

6.6 The theorem

Theorem 6.18 (index theorem for lowest-Landau-level Hamiltonians on the torus)**.** *In Setting 6.1, let Assumptions 6.3 and 6.4 hold. Then there are integers n_N such that for every k there is C_k with*

$$\big|\operatorname{Tr}(P_N T_{-,N}) - n_N\big| \le C_k N^{-k} \qquad \text{for all } N \in \mathcal{N}.$$

Equivalently, the transported charge $\langle T_- \rangle_P = p^{-1}\operatorname{Tr}(PT_-)$ lies in $p^{-1}\mathbb{Z} + O(N^{-\infty})$.

Proof. For each $N \in \mathcal{N}$ let n_N be an integer nearest to $\mathrm{Tr}(P_N \widetilde{T}_-)$. By Lemma 6.17(iii) and (iv),

$$|\mathrm{Tr}(PT_-) - n_N| \le |\mathrm{Tr}(P\widetilde{T}_-) - n_N| + |\mathrm{Tr}\big((UPU^* - P)K_-\big)| = O(N^{-\infty}),$$

which is the claim; dividing by p gives the second formulation.

The proof of Lemma 6.17 rests on the chain of the preceding statements: the dressed charge commutes with P exactly (Lemma 6.7); the dressing is localized at the two cuts (Lemmas 6.6 and 6.8, and (U3) for its transport by U); all operators of the argument have polynomially bounded norms and are almost supported near the cuts or in the three parts of the band (Lemma 6.12); operators attached to different cuts, or to a cut and the middle of the band, almost commute (Lemma 6.13); the localized unitaries almost commute with P, by the clustering in the sector (Proposition 6.10) through Lemma 6.14 (Corollary 6.15); and the flux insertion is compared with an exactly solvable equation on $\operatorname{ran} P$ (Lemma 6.16). Every estimate $O(N^{-\infty})$ in this chain comes from finitely many commutators, their number independent of N, of operators almost supported in sets at distance at least $\frac{c_*}{16}\sqrt{N}$ for $N \ge N_*$, multiplied by factors of polynomial size (Lemma 5.13).

The constants. Every bound in Lemmas 6.2–6.17 is built from the following data: the constants C_h and C_k^H of (H1); the constants of Sections 3–5 that these lemmas use, which by their statements depend only on η, δ_0, C_h and finitely many C_k^H, and, for the filters, on γ, Δ and the choices of ϑ, δ and ζ in Definition 5.8; the rank p; the constants in Assumption 6.4; and the following quantities depending on the torus, each controlled by admissibility and c_* alone: $|\mathbb{T}_N| = 4\pi N$; $L_y = 4\pi N/L_x \le \min\{\pi N, (4\pi/c_*)\sqrt{N}\}$, because $L_x \ge 4$ by admissibility and $L_x \ge c_*\sqrt{N}$; $\kappa \le \delta_x \le 2\delta_0$ (Lemma 3.2(ii)); $n_y \le CL_y$ with C depending on δ_0; and the separations of Lemma 6.2, which are at least $\frac{c_*}{16}\sqrt{N}$ for $N \ge N_*$, with N_* depending only on c_* and δ_0. No other property of the torus enters. The rank p enters only through Lemma A.9(i) and (iii) and through the trace bound of Lemma 6.17(iv), each time as a factor p. This proves the dependence stated in Remark 6.19, given the dependence stated in Sections 3–5. □

Remark 6.19 (the constants, and the asymptotic nature of the statement). The proof shows that C_k depends only on k, on p (linearly), on γ and Δ, on η and δ_0, on C_h and finitely many $C_{k'}^H$, on c_*, and on the constants in Assumption 6.4, which for $U = \Theta$ are absent. In particular C_k depends on the torus only through admissibility and c_*. We expect C_k to depend polynomially on $1/\gamma$ and on $1/(\gamma - 2\Delta)$ for each k, by the scaling of the filters ($W_\gamma(t) = W_1(\gamma t)$), but we have not tracked this dependence. The constants are not explicit, and through the constant C_{lc} of Theorem 5.3 they are very large. Theorem 6.18 is therefore a statement about the limit $N \to \infty$ only: the proof yields neither a rate nor a value of N beyond which the conclusion has quantitative force.

Remark 6.20 (the hypothesis $\gamma > 2\Delta$ can be weakened). Every result of Sections 6–8 that is stated under Assumption 6.3, or under the hypothesis $\gamma > 2\Delta$, holds with $\gamma > \Delta$ in place of $\gamma > 2\Delta$; in particular this applies to Theorem 6.18, Corollaries 7.2–7.4 and Theorem 8.1(i). We keep $\gamma > 2\Delta$ in the statements to match [BBDF21, Assumption 1]. The hypothesis enters the proofs only through the clustering step g of Definition 5.8. Of g the proofs use that $g = 0$ on $[-\Delta, \infty)$ and $g = 1$ on $(-\infty, -\gamma]$, both in the proof of Proposition 6.10(i), where the second is applied only at $E - E' \le -\gamma$; and they use the representation and the locality of $\mathcal{Q}$ (Lemma 5.10 and Proposition 5.11), whose proofs use only that g is the convolution of ζ with the indicator of a half-line. The other place where the gap enters, Lemma 6.7, uses only $E_{p+1} - E_p \ge \gamma$. Now let $\gamma > \Delta$, fix $\delta \in (0, (\gamma - \Delta)/2)$ and ζ as in Definition 5.8, and put $g(\omega) := \int_{\mathbb{R}} \zeta(\omega - \omega')\, \mathbf{1}_{\{\omega' < -(\gamma+\Delta)/2\}}\, \mathrm{d}\omega'$. Then $g(\omega) = 0$ for $\omega \ge -(\gamma + \Delta)/2 + \delta$, a half-line that contains $[-\Delta, \infty)$, and $g(\omega) = 1$ for $\omega \le -(\gamma + \Delta)/2 - \delta$, which contains $(-\infty, -\gamma]$; Lemma 5.10 and Proposition 5.11 hold verbatim with $\gamma' := -(\gamma + \Delta)/2$, their constants depending on ζ, γ and Δ; and with this g the proofs of Sections 6–8 go through

unchanged (the proof of Proposition 6.10 recalls of g only the two properties above). Through δ, the constants then depend on $\gamma - \Delta$ where they depended on $\gamma - 2\Delta$ (compare Remark 6.19). In the language of Section 1: the results hold for patches whose width is less than their gap.

Remark 6.21 (the formalization). The proof of Theorem 6.18 for $U = \Theta$, the case the numbered results use, the proofs of Theorem 8.1, Corollary 8.2 and Corollary 8.4, with Theorem 2.10 on the zero modes they use (proved in Appendix C), and the proof of Lemma 8.3 have been formalized and verified in the Lean 4 proof assistant against the Mathlib library. Appendix E states what is machine-checked, in which form and under which conventions, and gives the correspondence between the statements of this paper and the Lean declarations.

7 The magnetic translation

Lemma 7.1 (the magnetic translation transports one orbital)**.** *In Setting 6.1, let H satisfy (H1) and (H2), and let Assumption 6.3 hold for sectors $\mathbf{n}_N = (n_{\uparrow,N}, n_{\downarrow,N})$. Then:*

(a) $[\Theta, N_\uparrow] = [\Theta, N_\downarrow] = 0$, $[\Theta, H_N] = 0$, *and* $[\Theta, P] = 0$.

(b) $\Theta^* Q \Theta - Q = n_{N\uparrow} - n_{0\uparrow}$. *With* $T_- := -n_{0\uparrow}$ *and* $T_+ := n_{N\uparrow}$, *the operators* $T_\pm$ *are self-adjoint and sector-preserving,* $\|T_\pm\| = 1$, $\iota(T_\pm) \in \mathfrak{A}(\partial_\pm)$, *and* $\Theta^* Q \Theta - Q = T_- + T_+$.

(c) $e^{2\pi \mathrm{i}(Q+T_-)} = e^{2\pi \mathrm{i}(Q+T_+)} = \mathbf{1}$; *moreover* $e^{2\pi \mathrm{i}(Q_+ + T_+)} = e^{2\pi \mathrm{i} Q_m} = \mathbf{1}$.

(d) $\mathrm{Tr}(P n_{j\uparrow}) = p\, n_{\uparrow,N}/N_\phi$ *for every* $j \in \mathbb{Z}_{N_\phi}$; *in particular* $\mathrm{Tr}(PT_-) = -p\, n_{\uparrow,N}/N_\phi$.

(e) $\Theta^* K_\pm \Theta \in \mathcal{A}_{\partial_\pm}$, *and* $\mathrm{Tr}(P\,\Theta^* K_- \Theta) = \mathrm{Tr}(PK_-)$; *hence* $\mathrm{Tr}(P\widetilde{T}_-) = \mathrm{Tr}(PT_-)$ *exactly.*

Proof. (a) Θ commutes with $N_\uparrow$ and $N_\downarrow$ by Lemma 2.6(iii). By (H2) and the translation invariance of Lebesgue measure on $\mathbb{T}_N$, for every Borel set $\mathcal{D} \subset \mathbb{T}_N$,

$$\Theta^* H_{\mathcal{D}} \Theta = \int_{\mathcal{D}} h_{R+\kappa\hat{e}_x}\, \mathrm{d}R = \int_{\mathcal{D}+\kappa\hat{e}_x} h_R\, \mathrm{d}R = H_{\mathcal{D}+\kappa\hat{e}_x};$$

with $\mathcal{D} = \mathbb{T}_N$ this gives $[\Theta, H_N] = 0$. By Lemma 2.6(ii), $P = \mathbf{1}_{[E_1,E_p]}(H_N)\Pi^{\mathbf{n}_N}$; the first factor is a function of H_N and the second a function of $N_\uparrow$ and $N_\downarrow$, and Θ commutes with both. Hence $[\Theta, P] = 0$.

(b) By Lemma 2.6(iii), $\Theta^* Q \Theta = \sum_{j=0}^{N-1} n_{j+1,\uparrow} = \sum_{j=1}^{N} n_{j\uparrow} = Q - n_{0\uparrow} + n_{N\uparrow}$, since the classes of $0, \dots, N$ in $\mathbb{Z}_{N_\phi}$ are distinct. The operators $n_{0\uparrow}$ and $n_{N\uparrow}$ are nonzero orthogonal projections commuting with $N_\uparrow$ and $N_\downarrow$, so $T_\pm$ are self-adjoint and sector-preserving with $\|T_\pm\| = 1$, and $\iota(T_-) \in \mathfrak{A}(\mathrm{Col}(0)) = \mathfrak{A}(\partial_-)$, $\iota(T_+) \in \mathfrak{A}(\mathrm{Col}(N)) = \mathfrak{A}(\partial_+)$ by Lemma 4.11(i).

(c) $Q + T_- = Q_{\{1,\dots,N-1\}}$ and $Q + T_+ = Q_{\{0,\dots,N\}}$ are band charges, with spectra in $\mathbb{Z}$ (Lemma 4.10), so their exponentials $e^{2\pi \mathrm{i}(\cdot)}$ equal $\mathbf{1}$. Since $\Gamma_+ \subset \Gamma$ does not contain the class of N, $Q_+ + T_+ = Q_{\Gamma_+ \cup \{N\}}$; this and Q_m are band charges, and the same argument applies.

(d) By (a), $\Theta P \Theta^* = P$, so by the cyclicity of the trace and Lemma 2.6(iii), $\mathrm{Tr}(P n_{j+1,\uparrow}) = \mathrm{Tr}(P \Theta^* n_{j\uparrow} \Theta) = \mathrm{Tr}(\Theta P \Theta^* n_{j\uparrow}) = \mathrm{Tr}(P n_{j\uparrow})$ for every j. Hence $j \mapsto \mathrm{Tr}(P n_{j\uparrow})$ is constant on $\mathbb{Z}_{N_\phi}$, and, because $P = P\Pi^{\mathbf{n}_N}$ and $N_\uparrow \Pi^{\mathbf{n}_N} = n_{\uparrow,N} \Pi^{\mathbf{n}_N}$,

$$N_\phi\, \mathrm{Tr}(P n_{0\uparrow}) = \sum_{j \in \mathbb{Z}_{N_\phi}} \mathrm{Tr}(P n_{j\uparrow}) = \mathrm{Tr}(P N_\uparrow) = \mathrm{Tr}(P N_\uparrow \Pi^{\mathbf{n}_N}) = n_{\uparrow,N}\, \mathrm{Tr}(P) = p\, n_{\uparrow,N}.$$

Thus $\mathrm{Tr}(P n_{j\uparrow}) = p\, n_{\uparrow,N}/N_\phi$ and $\mathrm{Tr}(PT_-) = -p\, n_{\uparrow,N}/N_\phi$.

(e) Since $[\Theta, H_N] = 0$, we have $\Theta^* \tau_t(A) \Theta = \tau_t(\Theta^* A \Theta)$ for every operator A. By the identity in (a) and Lemma 4.10, therefore,

$$\Theta^* K_- \Theta = \int W_\gamma(t)\, \tau_t(\Theta^* J_- \Theta)\, \mathrm{d}t, \qquad \Theta^* J_- \Theta = \mathrm{i}\big[\Theta^* H_{\mathbb{T}_N^-} \Theta, \Theta^* Q \Theta\big] = \mathrm{i}\big[H_{\mathbb{T}_N^- + \kappa \hat{e}_x}, Q_{\Gamma+1}\big],$$

which is the current of Lemma 6.6(iii) for $j_0 = 1$, and $\Theta^* K_- \Theta$ is its dressing. By Lemma 6.8, $\Theta^* K_- \Theta \in \mathcal{A}_{\mathrm{Col}(1)}$ with prefactor exponent $\frac{5}{2}$ and the constants of K_-. Now $\mathrm{Col}(1) = \mathrm{Str}((\kappa - \delta_x, \kappa + \delta_x))$ is contained in $\mathrm{Str}((I_-)_\kappa)$ with $I_- = (-\delta_x, \delta_x)$, hence in $(\partial_-)_{(\kappa+\delta_x)}$ by Lemma 3.10; so $\mathrm{Col}(1)_{(r)} \subset (\partial_-)_{(r+a)}$ for every $r \geq 0$, with $a := \kappa + \delta_x \leq 4\delta_0$. Given $r \geq a$, the approximant of $\Theta^* K_- \Theta$ at radius $r - a$ relative to $\mathrm{Col}(1)$ lies in $\mathfrak{A}((\partial_-)_{(r)})$, and its error is at most $C_k N^{5/2}(1 + r - a)^{-k} \leq (1+a)^k C_k N^{5/2}(1+r)^{-k}$, because $1 + r \leq (1 + r - a)(1 + a)$; for $r < a$ the approximant 0 has error $\|K_-\| \leq CN^2 \leq (1+a)^k C N^2 (1+r)^{-k}$. Hence $\Theta^* K_- \Theta \in \mathcal{A}_{\partial_-}$ with prefactor exponent $\frac{5}{2}$. In the same way $\Theta^* K_+ \Theta$ is the dressing of the current of Lemma 6.6(iii) for $j_0 = 1$ at the cut $\mathrm{Col}(N+1) \subset (\partial_+)_{(\kappa+\delta_x)}$, and $\Theta^* K_+ \Theta \in \mathcal{A}_{\partial_+}$. Finally $\mathrm{Tr}(P \Theta^* K_- \Theta) = \mathrm{Tr}(\Theta P \Theta^* K_-) = \mathrm{Tr}(P K_-)$, and therefore, with $\widetilde{T}_- = T_- - \Theta^* K_- \Theta + K_-$ (Definition 6.11 for $U = \Theta$), $\mathrm{Tr}(P \widetilde{T}_-) = \mathrm{Tr}(P T_-)$. □

Corollary 7.2 (the translation corollary)**.** *In Setting 6.1, let H satisfy (H1) and (H2), and let Assumption 6.3 hold for sectors $\mathbf{n}_N = (n_{\uparrow,N}, n_{\downarrow,N})$. Then $U = \Theta$ satisfies Assumption 6.4, with $T_\pm$ as in Lemma 7.1, and with (U2) and the two relations in (U4) holding exactly; and there are integers n_N such that for every k there is C_k with*

$$\Big| p\, \frac{n_{\uparrow,N}}{N_\phi} - n_N \Big| \leq C_k N^{-k} \qquad \text{for all } N \in \mathcal{N}.$$

Proof. (U1) and (U2), the latter exactly, are Lemma 7.1(a), and (U3) is Lemma 7.1(e). For (U4), the operators $T_\pm$ of Lemma 7.1(b) are self-adjoint and sector-preserving, with $\|T_\pm\| = 1$, so that one may take $m_T = 0$; they lie in $\mathcal{A}_{\partial_\pm}$, since $\iota(T_\pm) \in \mathfrak{A}(\partial_\pm)$ is an approximant with error 0 at every radius; and $\Theta^* Q \Theta - Q = T_- + T_+$ and $e^{2\pi \mathrm{i}(Q + T_\pm)} = \mathbf{1}$ hold exactly by Lemma 7.1(b) and (c). So $U = \Theta$ satisfies Assumption 6.4, and Theorem 6.18 gives integers n'_N with $|\mathrm{Tr}(P T_-) - n'_N| \leq C_k N^{-k}$ for all $N \in \mathcal{N}$. By Lemma 7.1(d), $\mathrm{Tr}(P T_-) = -p\, n_{\uparrow,N}/N_\phi$, and $n_N := -n'_N$ satisfies the claim. □

This is the counterpart, for lowest-Landau-level Hamiltonians, of the fractional Lieb–Schultz–Mattis theorem [BBDF21, Corollary 5.2]: the average up-spin charge $n_{\uparrow,N}/N_\phi$ of an orbital, multiplied by the dimension of a uniformly gapped patch at the bottom of the sector (in the sense of Section 1, that is, Assumption 6.3), is asymptotically an integer.

Corollary 7.3 (half filling with zero spin polarization)**.** *Let $\mathcal{N}$ consist of even integers, let the tori and H be as in Setting 6.1 with (H1) and (H2), and suppose that Assumption 6.3 holds for the sectors $\mathbf{n}_N = (N/2, N/2)$, $N \in \mathcal{N}$. Then $p \in 4\mathbb{Z}$. In particular, no Hamiltonian satisfying (H1) and (H2) has, in the sectors $(N/2, N/2)$ along an infinite set of even N, a uniformly gapped patch at the bottom of the spectrum (in the sense of Section 1) whose dimension is not divisible by four; for instance, of dimension ten.*

Proof. For $\mathbf{n}_N = (N/2, N/2)$, $n_{\uparrow,N}/N_\phi = (N/2)/(2N) = \frac{1}{4}$, so Corollary 7.2 with $k = 1$ gives integers n_N and a constant C with $\mathrm{dist}(p/4, \mathbb{Z}) \leq |p/4 - n_N| \leq CN^{-1}$ for all $N \in \mathcal{N}$. The left-hand side does not depend on N, and $\mathcal{N}$ is infinite, hence unbounded; letting $N \to \infty$ in $\mathcal{N}$ gives $\mathrm{dist}(p/4, \mathbb{Z}) = 0$, that is, $p \in 4\mathbb{Z}$. The second sentence restates the first: a patch of dimension p as described is exactly Assumption 6.3 for the sectors $(N/2, N/2)$. □

Corollary 7.4 (odd N and spin one half)**.** *Let $\mathcal{N}$ consist of odd integers, let the tori and H be as in Setting 6.1 with (H1) and (H2), and let $\mathbf{n}_N^{\pm} := ((N \pm 1)/2, (N \mp 1)/2)$.*

(i) There are no $p \in \mathbb{N}$, $\Delta \geq 0$ and $\gamma > 2\Delta$ for which Assumption 6.3 holds for the sectors $\mathbf{n}_N^+$ along an infinite subset of $\mathcal{N}$; and likewise for $\mathbf{n}_N^-$.

(ii) For every $D \in \mathbb{N}$ and $\gamma > 0$, for all but finitely many $N \in \mathcal{N}$ the restriction of H_N to $\mathcal{H}_N^{\mathbf{n}_N^{\pm}}$ has more than D eigenvalues, counted with multiplicity, in $[E_1(N), E_1(N) + \gamma)$, where $E_1(N)$ is its lowest eigenvalue.

Proof. (i) Suppose that Assumption 6.3 holds for the sectors $\mathbf{n}_N^+$, $N \in \mathcal{N}'$, with $\mathcal{N}' \subset \mathcal{N}$ infinite and some $p \in \mathbb{N}$, $\Delta \geq 0$, $\gamma > 2\Delta$. The tori and H, restricted to $\mathcal{N}'$, are as in Setting 6.1 with (H1) and (H2), so Corollary 7.2 applies along $\mathcal{N}'$. Since $n_{\uparrow,N}/N_\phi = \frac{N+1}{2} \cdot \frac{1}{2N} = \frac{1}{4} + \frac{1}{4N}$, it gives integers n_N with

$$\operatorname{dist}\Big(\frac{p}{4} + \frac{p}{4N}, \mathbb{Z}\Big) \leq C_k N^{-k} \qquad (N \in \mathcal{N}')$$

for every k. If $p \notin 4\mathbb{Z}$, then $\operatorname{dist}(p/4, \mathbb{Z}) \geq \frac{1}{4}$, since $p/4 \in \frac{1}{4}\mathbb{Z} \setminus \mathbb{Z}$; for $N \geq 2p$ we have $p/(4N) \leq \frac{1}{8}$, so the left-hand side is at least $\frac{1}{8}$, which contradicts the bound with $k = 1$ for large $N \in \mathcal{N}'$. If $p \in 4\mathbb{Z}$, then for $N > p/2$ we have $0 < p/(4N) < \frac{1}{2}$, so the left-hand side equals $p/(4N)$, and the bound with $k = 2$ gives $p/(4N) \leq C_2 N^{-2}$, that is $N \leq 4C_2/p$, for all large $N \in \mathcal{N}'$, which is impossible because $\mathcal{N}'$ is unbounded. For the sectors $\mathbf{n}_N^-$ the same argument applies to $n_{\uparrow,N}/N_\phi = \frac{1}{4} - \frac{1}{4N}$, with $\frac{p}{4} - \frac{p}{4N}$ in place of $\frac{p}{4} + \frac{p}{4N}$.

(ii) We treat $\mathbf{n}_N^+$; the case of $\mathbf{n}_N^-$ is identical. Suppose the claim fails for some $D \in \mathbb{N}$ and $\gamma > 0$. Then there is an infinite set $\mathcal{N}_1 \subset \mathcal{N}$ such that for every $N \in \mathcal{N}_1$ the restriction of H_N to $\mathcal{H}_N^{\mathbf{n}_N^+}$ has at most D eigenvalues in $[E_1(N), E_1(N) + \gamma)$. The sector $\mathcal{H}_N^{\mathbf{n}_N^+}$ has dimension $\binom{2N}{(N+1)/2}\binom{2N}{(N-1)/2} \geq 2N$, so, removing finitely many N, we may assume that it has at least $D + 1$ eigenvalues $E_1(N) \leq \cdots \leq E_{D+1}(N)$, counted with multiplicity; then $E_{D+1}(N) - E_1(N) \geq \gamma$ for $N \in \mathcal{N}_1$. By compactness of $[0, \infty]$, there is an infinite $\mathcal{N}_2 \subset \mathcal{N}_1$ such that, for each $j \leq D + 1$, $E_j(N) - E_1(N)$ converges in $[0, \infty]$ as $N \to \infty$ in $\mathcal{N}_2$, to a limit $\bar{E}_j$. Then $0 = \bar{E}_1 \leq \bar{E}_2 \leq \cdots \leq \bar{E}_{D+1}$ and $\bar{E}_{D+1} \geq \gamma > 0$. Let p be the number of indices j with $\bar{E}_j = 0$, so that $1 \leq p \leq D$ and $\bar{E}_{p+1} > 0$, and put $\mu := \min\{\bar{E}_{p+1}, 1\} > 0$. For all large $N \in \mathcal{N}_2$, $E_p(N) - E_1(N) \leq \frac{\mu}{8}$ and $E_{p+1}(N) - E_1(N) \geq \frac{3\mu}{4}$, hence $E_{p+1}(N) - E_p(N) \geq \frac{5\mu}{8}$. With $\Delta := \frac{\mu}{8}$ and the gap $\frac{\mu}{2} > 2\Delta$ in place of γ, Assumption 6.3 holds for the sectors $\mathbf{n}_N^+$ along an infinite subset of $\mathcal{N}$, with p, Δ and the gap independent of N. This contradicts (i). □

Remark 7.5 (scope of the class). Corollaries 7.3 and 7.4 use only (H1), (H2) and the gap in the sector. They do not use frustration-freeness or the existence or number of zero modes. The hollow-core Hamiltonian is the only Hamiltonian whose membership in the class is proved here (Corollary 4.8); which other lowest-Landau-level Hamiltonians satisfy (H1) and (H2) is not examined. The obstruction in Corollary 7.4 comes from the density defect $\pm p/(4N)$ of the sectors $\mathbf{n}_N^{\pm}$, which forces an unpaired spin; it applies to every member of the class and therefore carries no information on whether the system away from that defect is gapped.

8 The hollow-core Hamiltonian

8.1 The main theorem

Theorem 8.1 (no uniform gap above the zero modes)**.** *Fix $\lambda > 0$. For even N let $\mathbb{T}_N$ be the rectangular torus with $L_x/L_y = \lambda$ and $L_x L_y = 4\pi N$, let H_N be the hollow-core Hamiltonian of*

Definition 2.7, and let $E_1(N) \le E_2(N) \le \cdots$ be the eigenvalues of H_N in the sector $\mathcal{H}_N^{(N/2,N/2)}$, repeated according to multiplicity, so that $E_1(N) = \cdots = E_{10}(N) = 0 < E_{11}(N)$.

(i) If for some $p \in \mathbb{N}$, $\Delta \ge 0$ and $\gamma > 2\Delta$ the inequalities $E_p(N) - E_1(N) \le \Delta$ and $E_{p+1}(N) - E_p(N) \ge \gamma$ hold for infinitely many even N, then $p \in 4\mathbb{Z}$ and $p \ge 12$.

(ii) $E_{11}(N) \to 0$ as $N \to \infty$ through the even integers.

Proof. The side lengths are $L_x = (4\pi\lambda N)^{1/2}$ and $L_y = (4\pi N/\lambda)^{1/2}$. Both tend to infinity with N, so there is N_λ such that the torus is admissible, that is $L_y \ge L_0$ and $L_x \ge 4\delta_0 + 4$ (Definition 3.1), for every $N \ge N_\lambda$. Put $c_* := (4\pi\lambda)^{1/2}$, so that $L_x = c_*\sqrt{N}$.

(i) Let $\mathcal{N}$ be the set of even $N \ge N_\lambda$ for which the two inequalities hold; it is infinite. The tori $(\mathbb{T}_N)_{N\in\mathcal{N}}$ are admissible with $L_x \ge c_*\sqrt{N}$, and the hollow-core Hamiltonians $(H_N)_{N\in\mathcal{N}}$ satisfy (H1) and (H2) by Corollary 4.8; so Setting 6.1 holds. The two inequalities say that Assumption 6.3 holds for the sectors $\mathbf{n}_N = (N/2, N/2)$, $N \in \mathcal{N}$, with p, Δ and γ independent of N. Corollary 7.3 gives $p \in 4\mathbb{Z}$. If $p \le 9$, then $E_p(N) = E_{p+1}(N) = 0$ by (2.8), which contradicts $E_{p+1}(N) - E_p(N) \ge \gamma > 0$. Hence $p \ge 10$, and since neither 10 nor 11 lies in $4\mathbb{Z}$, $p \ge 12$.

(ii) Suppose not. Then there are $\gamma > 0$ and infinitely many even N with $E_{11}(N) \ge \gamma$. By (2.8), for these N we have $E_{10}(N) - E_1(N) = 0$ and $E_{11}(N) - E_{10}(N) = E_{11}(N) \ge \gamma$, so the hypothesis of (i) holds with $p = 10$, $\Delta = 0$ and this γ. By (i), $10 \in 4\mathbb{Z}$, which is false. □

The kernel enters the proof only through its dimension, ten, and through $E_1(N) = 0$ (Theorem 2.10); no other property of the Haldane–Rezayi states is used. In the language of Section 6, along a sequence as in the proof of (ii) the transported charge is $\langle T_- \rangle_P = -\langle n_{0\uparrow} \rangle_P = -\frac{1}{4}$ (Lemma 7.1(d)), and $10 \cdot \frac{1}{4} = \frac{5}{2}$ is at distance $\frac{1}{2}$ from $\mathbb{Z}$.

Corollary 8.2 (odd N)**.** *Fix $\lambda > 0$, and for odd $N \ge 3$ let $\mathbb{T}_N$ and H_N be as in Theorem 8.1. In each of the sectors $\mathcal{H}_N^{((N\pm1)/2,(N\mp1)/2)}$ let $E_1(N) \le E_2(N) \le \cdots$ be the eigenvalues of H_N, so that $E_1(N) = E_2(N) = 0 < E_3(N)$.*

(i) $E_3(N) \to 0$ as $N \to \infty$ through the odd integers.

(ii) For every $D \in \mathbb{N}$ and $\gamma > 0$, for all sufficiently large odd N there are more than D eigenvalues in $[0, \gamma)$.

Proof. Let N_λ and c_* be as in the proof of Theorem 8.1, and let $\mathcal{N}$ be the set of odd $N \ge \max\{N_\lambda, 3\}$. As there, the tori $(\mathbb{T}_N)_{N\in\mathcal{N}}$ and the hollow-core Hamiltonians are as in Setting 6.1, with (H1) and (H2). We treat the sectors $\mathbf{n}_N^+ = ((N+1)/2, (N-1)/2)$; the sectors $\mathbf{n}_N^-$ are treated in the same way. By Theorem 2.10(ii) and $H_N \ge 0$, the eigenvalues of H_N in $\mathcal{H}_N^{\mathbf{n}_N^+}$ satisfy $E_1(N) = E_2(N) = 0 < E_3(N)$.

(i) Suppose that there are $\gamma > 0$ and infinitely many odd N with $E_3(N) \ge \gamma$. For these N, $E_2(N) - E_1(N) = 0$ and $E_3(N) - E_2(N) = E_3(N) \ge \gamma$, so Assumption 6.3 holds for the sectors $\mathbf{n}_N^+$ along the infinite set of such N with $N \ge N_\lambda$, with $p = 2$, $\Delta = 0$ and this γ. This contradicts Corollary 7.4(i).

(ii) By Corollary 7.4(ii), for every $D \in \mathbb{N}$ and $\gamma > 0$ and all but finitely many $N \in \mathcal{N}$, the restriction of H_N to $\mathcal{H}_N^{\mathbf{n}_N^+}$ has more than D eigenvalues, counted with multiplicity, in $[E_1(N), E_1(N) + \gamma) = [0, \gamma)$. □

By Remark 7.5, both statements reflect the unpaired spin forced by the density of these sectors; they say nothing about the sector $(N/2, N/2)$ at even N.

8.2 The scope over aspect ratios

The construction of Section 3 distinguishes the two sides of the torus: the Landau-gauge orbitals are localized across the side of length L_x, and admissibility asks $L_y \geq L_0$ and $L_x \geq 4\delta_0 + 4$. The spectrum of the hollow-core Hamiltonian does not distinguish them.

Lemma 8.3 (exchange of the sides)**.** *Let $L_x, L_y > 0$ with $L_x L_y = 4\pi N$, and let $\mathbb{T}'_N$ be the torus of Definition 2.1 with side lengths $L'_x := L_y$ and $L'_y := L_x$. Denote its lowest Landau level, its Fock space, the annihilation operators on it, its number operators and its hollow-core Hamiltonian by $\mathfrak{h}'_N$, $\mathcal{H}'_N$, $c'(\cdot)$, N'_σ and H'_N. For $f \in \mathfrak{h}_N$ let $(\mathcal{R}f)(x,y) := e^{\mathrm{i}xy} f(-y,x)$. Then $\mathcal{R}$ is a unitary from $\mathfrak{h}_N$ onto $\mathfrak{h}'_N$; the unitary $\mathcal{F}(\mathcal{R}\otimes\mathbf{1}) : \mathcal{H}_N \to \mathcal{H}'_N$ that fixes the vacuum and satisfies $\mathcal{F}(\mathcal{R}\otimes\mathbf{1})\, c(f)\, \mathcal{F}(\mathcal{R}\otimes\mathbf{1})^* = c'((\mathcal{R}\otimes\mathbf{1})f)$ for $f \in \mathfrak{k}_N$ maps N_σ to N'_σ, $\sigma \in \{\uparrow,\downarrow\}$, and H_N to H'_N. In particular, for every sector label $\mathbf{n}$, the restrictions of H_N to $\mathcal{H}_N^{\mathbf{n}}$ and of H'_N to the sector of $\mathcal{H}'_N$ with the same label have the same eigenvalues, with multiplicities.*

Proof. The one-particle map. Let $\varrho(x,y) := (-y,x)$, a rotation of $\mathbb{R}^2$. It maps the lattice $L'_x\mathbb{Z}\times L'_y\mathbb{Z}$ onto $L_x\mathbb{Z}\times L_y\mathbb{Z}$, since $\varrho(mL_y, nL_x) = (-nL_x, mL_y)$, and hence induces a measure-preserving bijection of $\mathbb{T}'_N$ onto $\mathbb{T}_N$. Consequently, for every smooth function F on $\mathbb{T}_N$, the function $F\circ\varrho$ is smooth on $\mathbb{T}'_N$, $\int_{\mathbb{T}'_N} F\circ\varrho\,\mathrm{d}r = \int_{\mathbb{T}_N} F\,\mathrm{d}r$, and $\nabla^2(F\circ\varrho) = (\nabla^2 F)\circ\varrho$, because the Laplacian commutes with rotations.

Let $f \in \mathfrak{h}_N$ and $g := \mathcal{R}f$; g is smooth. By the two conditions (2.2) for f,

$$g(x, y+L'_y) = e^{\mathrm{i}x(y+L_x)} f(-y-L_x, x) = e^{\mathrm{i}xy} e^{\mathrm{i}L_x x} e^{-\mathrm{i}L_x x} f(-y,x) = g(x,y),$$
$$g(x+L'_x, y) = e^{\mathrm{i}(x+L_y)y} f(-y, x+L_y) = e^{\mathrm{i}L'_x y} g(x,y),$$

which are the conditions (2.2) for $\mathbb{T}'_N$. With $D_x = \partial_x$ and $D_y = \partial_y - \mathrm{i}x$, and with $\partial_1 f$, $\partial_2 f$ the partial derivatives of f in its two arguments,

$$\begin{aligned} e^{-\mathrm{i}xy}((D_x+\mathrm{i}D_y)g)(x,y) &= \mathrm{i}y\, f(-y,x) + (\partial_2 f)(-y,x) - \mathrm{i}(\partial_1 f)(-y,x) \\ &= -\mathrm{i}(\partial_1 f + \mathrm{i}\partial_2 f + s\, f)(s,t)\Big|_{(s,t)=(-y,x)} = -\mathrm{i}((D_x+\mathrm{i}D_y)f)(-y,x) = 0, \end{aligned}$$

by (2.3) for f. So $g \in \mathfrak{h}'_N$. Moreover $\overline{\mathcal{R}f_1}\,\mathcal{R}f_2 = (\overline{f_1} f_2)\circ\varrho$ for $f_1, f_2 \in \mathfrak{h}_N$, the phases cancelling, so $\langle \mathcal{R}f_1, \mathcal{R}f_2\rangle = \langle f_1, f_2\rangle$ by the first paragraph. Thus $\mathcal{R}$ is an isometry of $\mathfrak{h}_N$ into $\mathfrak{h}'_N$, and it is onto because both spaces have dimension $L_xL_y/(2\pi) = N_\phi$ (Lemma 2.2(ii)). The operator $\mathcal{F}(\mathcal{R}\otimes\mathbf{1})$ is then defined as in Definition 2.5: it maps $c(f_1)^*\cdots c(f_n)^*\Omega_0$ to $c'(f'_1)^*\cdots c'(f'_n)^*\Omega'_0$, with $f'_i := (\mathcal{R}\otimes\mathbf{1})f_i$ and Ω_0, Ω'_0 the vacua, and it is unitary by the anticommutation relations, because $\mathcal{R}\otimes\mathbf{1}$ is unitary.

The Hamiltonian. For $f_1, f_2, g_1, g_2 \in \mathfrak{h}_N$ let

$$\mathcal{I}(f_1,f_2;g_1,g_2) := 4\pi \int_{\mathbb{T}_N} (\overline{f_1} g_1)(\overline{f_2} g_2)\,\mathrm{d}r + 4\pi \int_{\mathbb{T}_N} (\overline{f_2} g_2)\,\nabla^2(\overline{f_1} g_1)\,\mathrm{d}r,$$

the integrands being smooth functions on $\mathbb{T}_N$ by (2.2), and let $\mathcal{I}'$ be defined in the same way on $\mathbb{T}'_N$. Then $V_{abcd} = \mathcal{I}(\psi_a,\psi_b;\psi_c,\psi_d)$ in (2.6), and by the first paragraph $\mathcal{I}'(\mathcal{R}f_1,\mathcal{R}f_2;\mathcal{R}g_1,\mathcal{R}g_2) = \mathcal{I}(f_1,f_2;g_1,g_2)$. For an orthonormal basis $e = (e_a)$ of $\mathfrak{h}_N$ let

$$H[e] := \frac{1}{2}\sum_{\sigma,\sigma'}\sum_{a,b,c,d} \mathcal{I}(e_a,e_b;e_c,e_d)\, c(e_a\otimes\sigma)^* c(e_b\otimes\sigma')^* c(e_d\otimes\sigma') c(e_c\otimes\sigma).$$

If $e_a = \sum_j u_{ja}\psi_j$ with a unitary matrix u, then, since $\mathcal{I}$ is conjugate-linear in its first two arguments and linear in the last two, $c(\cdot)^*$ is linear and $c(\cdot)$ is conjugate-linear,

$$H[e] = \frac{1}{2}\sum_{\sigma,\sigma'}\sum_{j,k,l,m}\sum_{j',k',l',m'}\Big(\sum_{a,b,c,d}\overline{u_{ja}}u_{j'a}\,\overline{u_{kb}}u_{k'b}\,u_{lc}\overline{u_{l'c}}\,u_{md}\overline{u_{m'd}}\Big) \\ \times \mathcal{I}(\psi_j,\psi_k;\psi_l,\psi_m)\,c^*_{j'\sigma}c^*_{k'\sigma'}c_{m'\sigma'}c_{l'\sigma},$$

and the coefficient in parentheses is $\delta_{jj'}\delta_{kk'}\delta_{ll'}\delta_{mm'}$ because u is unitary. So $H[e] = H[\psi] = H_N$ by (2.7). In the same way $H'[e'] = H'_N$ for every orthonormal basis e' of $\mathfrak{h}'_N$, with $H'[e']$ built from $\mathcal{I}'$ and c'. Now $(\mathcal{R}\psi_a)_a$ is an orthonormal basis of $\mathfrak{h}'_N$, conjugation by $\mathcal{F}(\mathcal{R}\otimes\mathbf{1})$ replaces each $c(\psi_a\otimes\sigma)$ in (2.7) by $c'(\mathcal{R}\psi_a\otimes\sigma)$, and $V_{abcd} = \mathcal{I}(\psi_a,\psi_b;\psi_c,\psi_d) = \mathcal{I}'(\mathcal{R}\psi_a,\mathcal{R}\psi_b;\mathcal{R}\psi_c,\mathcal{R}\psi_d)$; hence

$$\mathcal{F}(\mathcal{R}\otimes\mathbf{1})\,H_N\,\mathcal{F}(\mathcal{R}\otimes\mathbf{1})^* = H'[\mathcal{R}\psi] = H'_N.$$

In the same way the number operators, $N_\sigma = \sum_a c(e_a\otimes\sigma)^*c(e_a\otimes\sigma)$ for every orthonormal basis, are mapped to N'_σ. So the unitary maps each sector $\mathcal{H}_N^{\mathfrak{n}}$ onto the sector of $\mathcal{H}'_N$ with the same label and intertwines the restrictions of H_N and H'_N, which therefore have the same eigenvalues with multiplicities. □

Corollary 8.4 (uniformity over aspect ratios)**.** *For even $N \geq L_0^2/(4\pi)$ let $\mathcal{T}_N$ be the set of rectangular tori with $L_xL_y = 4\pi N$ and $\min\{L_x,L_y\} \geq L_0$, and for $\mathbb{T}\in\mathcal{T}_N$ let $E_{11}(N;\mathbb{T})$ be the eleventh eigenvalue of the hollow-core Hamiltonian of $\mathbb{T}$ in the sector $(N/2,N/2)$. Then $\sup_{\mathbb{T}\in\mathcal{T}_N} E_{11}(N;\mathbb{T}) \to 0$ as $N\to\infty$ through the even integers.*

Proof. By Lemma 8.3, $E_{11}(N;\mathbb{T})$ does not change when the two sides of $\mathbb{T}$ are exchanged, so it suffices to consider the tori of $\mathcal{T}_N$ with $L_x \geq L_y$. Suppose the claim fails. Then there are $\gamma > 0$, even integers $N_1 < N_2 < \cdots$ and tori $\mathbb{T}_{N_i}\in\mathcal{T}_{N_i}$ with $L_x \geq L_y$ and $E_{11}(N_i;\mathbb{T}_{N_i}) \geq \gamma$. Since $L_x \geq L_y$ and $L_xL_y = 4\pi N$, we have $L_x \geq (4\pi N)^{1/2}$; so $\mathbb{T}_{N_i}$ is admissible as soon as $(4\pi N_i)^{1/2} \geq 4\delta_0 + 4$, and we discard the finitely many i for which this fails. Let $\mathcal{N}$ be the set of the remaining N_i. Then $(\mathbb{T}_N)_{N\in\mathcal{N}}$ are admissible tori with $L_x \geq c_*\sqrt{N}$ for $c_* := (4\pi)^{1/2}$, and the hollow-core Hamiltonians on them satisfy (H1) and (H2) by Corollary 4.8, with $C_h = 7/\pi$ and constants C_k^H that depend only on k, η and δ_0 (Proposition 4.4); so Setting 6.1 holds. By (2.8) on each of these tori, Assumption 6.3 holds for the sectors $(N/2,N/2)$, $N\in\mathcal{N}$, with $p = 10$, $\Delta = 0$ and γ. Corollary 7.3 gives $10\in 4\mathbb{Z}$, which is false. □

In this proof the aspect ratios L_x/L_y of the tori $\mathbb{T}_N$, $N\in\mathcal{N}$, are arbitrary numbers in $[1,\infty)$ and may be unbounded. This is allowed: Setting 6.1 asks only that the tori be admissible with $L_x \geq c_*\sqrt{N}$, and Theorem 6.18 and Corollaries 7.2 and 7.3 are proved under exactly these hypotheses. They hold in this generality because every constant of their proofs depends on the torus only through admissibility and c_* (Remark 6.19). The constants of Sections 3 and 4 depend only on η, δ_0 and the order of decay (C_{vol} and C_{str} in Proposition 3.4, the constants C_k of Lemma 3.5, and C_h, C_k^H in Proposition 4.4); those of Section 5 depend only on these, on the data of the filters, and, in Lemma 5.13, on the separation that is assumed there; and in Sections 6 and 7 the torus enters only through $|\mathbb{T}_N| = 4\pi N$, $L_y \leq \min\{\pi N, (4\pi/c_*)\sqrt{N}\}$, $\kappa \leq \delta_x \leq 2\delta_0$, $n_y \leq CL_y$ and the separations of Lemma 6.2, as recorded in the paragraph on the constants at the end of the proof of Theorem 6.18.

Remark 8.5 (thin tori). Corollary 8.4 says nothing about tori with $\min\{L_x,L_y\} < L_0$. By Lemma 8.3 one may take L_y to be the shorter side; then the slot frame of Section 3 is not

available, because a window of width of order δ_0 in the transverse direction need not contain the centre of any orbital (Lemma 3.2(ii) uses $L_y \geq 3\pi/\delta_0$), and exchanging the sides instead violates $L_x \geq 4\delta_0 + 4$ and $L_x \geq c_*\sqrt{N}$. In particular the index argument proves nothing for sequences of tori whose circumference L_y stays below L_0 while $N \to \infty$. For these tori a variational argument, which uses neither the slot frame nor the index theorem, bounds $E_{11}(N)$ by an explicit $C(L_y)/N$ (Theorem D.1); with Corollary 8.4 this gives $E_{11}(N) \to 0$ uniformly over all rectangular tori (Corollary D.2). Section 8.4 comments on this regime.

Remark 8.6 (oblique tori). The slot frame, and with it Theorems 6.18 and 8.1, is constructed here for rectangular tori only, although Theorem 2.10 holds for every modular parameter (Appendix C). For an oblique torus the Landau-gauge orbitals carry phases quadratic in the orbital index; within a window these split into a linear part, which shifts the centres of the slots along the cuts and produces a sheared slot lattice, and a remainder that varies slowly on the scale of a window. We expect the localization estimates of Lemma 3.5 to hold for the resulting frame, with constants depending on the modular parameter, but this is not established here.

8.3 What is not excluded

Remark 8.7 (a larger gapped ground-state space). Theorem 8.1 does not exclude the following situation: there are $p \in 4\mathbb{Z}$ with $p \geq 12$, $\Delta \geq 0$ and $\gamma > 2\Delta$ such that, for infinitely many (or for all large) even N, $E_p(N) - E_1(N) \leq \Delta$ and $E_{p+1}(N) - E_p(N) \geq \gamma$. This is a uniformly gapped patch of dimension p containing the ten zero modes and $p - 10$ further states of energy at most Δ, with a gap above it; in such a phase the convergence $E_{11}(N) \to 0$ of Theorem 8.1(ii) would be a splitting inside the ground-state space, not the closing of a gap above it. We do not know whether such a phase exists.

The method of this paper cannot exclude it. With a magnetic translation as the transport unitary, the conclusion of Theorem 6.18 is that $p\, r_N$ lies within $O(N^{-\infty})$ of an integer, where $r_N := \langle T_- \rangle_P$ is the transported charge per state, which translation covariance determines exactly, as in Lemma 7.1(d). When $r_N = r$ does not depend on N, the conclusion is $p\, r \in \mathbb{Z}$, and this holds for every multiple p of the denominator of the rational number r. The choices available here give the following values in the sector $(N/2, N/2)$. The up-spin band charge with $U = \Theta$ gives $r = -n_{\uparrow,N}/N_\phi = -\frac{1}{4}$ (Corollary 7.3). The proofs of Sections 4–7 use only that the charge is a sum over the orbitals j of the band of mutually commuting, self-adjoint, sector-preserving operators n_j with integer spectrum, norms bounded independently of N and $\iota(n_j) \in \mathfrak{A}(\mathrm{Col}(j))$, whose sum over all $j \in \mathbb{Z}_{N_\phi}$ commutes with every local term h_R, and which satisfy $\Theta^* n_j \Theta = n_{j+1}$; they therefore apply to the down-spin band charge $\sum_{j\in\Gamma} n_{j\downarrow}$, which gives $r = -n_{\downarrow,N}/N_\phi = -\frac{1}{4}$, and to the total band charge $\sum_{j\in\Gamma}(n_{j\uparrow} + n_{j\downarrow})$, which gives $r = -(n_{\uparrow,N} + n_{\downarrow,N})/N_\phi = -\frac{1}{2}$. The up-spin charge with $U = \Theta^2$, for which $T_- = -n_{0\uparrow} - n_{1\uparrow}$, gives $r = -\frac{1}{2}$ as well. All of these conditions are met by $p = 12$. Excluding the situation above requires different information: for instance, a lower bound on $E_{11}(N)$ that excludes a splitting inside a gapped ground-state space, or an identification of every uniformly gapped patch at the bottom of the sector with the space of zero modes.

8.4 Interpretation

What is excluded. Theorem 8.1 excludes a gapped phase whose ground-state space on the torus is the space of zero modes of the hollow-core Hamiltonian: on rectangular tori of any fixed aspect ratio, and uniformly over aspect ratios as in Corollary 8.4, the ten zero modes in the sector $(N/2, N/2)$ are not separated from the rest of the spectrum of this sector by a gap that is

uniform in N. This is the scenario in which the count of ten would be the topological ground-state degeneracy of a gapped Haldane–Rezayi phase. The count was first published, from explicit torus wave functions, by Keski-Vakkuri and Wen [KVW93], who took it as evidence that the state is non-Abelian; Gurarie, Flohr and Nayak [GFN97] described the Haldane–Rezayi state by the conformal field theory of central charge $c = -2$ and attributed the ten-fold degeneracy to a logarithmic partner of the identity field, assuming that the properties of the zero-energy states of the hollow-core Hamiltonian are stable; and Ino [Ino99] obtained, within the same description, $5p$ bulk sectors and a torus degeneracy $5p$ at filling $1/p$, hence ten at filling one half, an argument that presupposes an incompressible state. Read and Rezayi [RR96, Section V], who wrote down the ten states, held instead that the underlying theory has eight sectors at filling one half, the count of ten arising because some sectors carry two conformal blocks. None of these works establishes a gap; Theorem 8.1 shows that, for the hollow-core Hamiltonian, the gap above the ten states that the count of ten as a degeneracy presupposes does not exist.

Nor are the gapped phases with eight states on the torus that have been proposed near the Haldane–Rezayi point ground-state spaces of H_N: the weak- and strong-pairing phases of spin-singlet d-wave pairing in the picture of Read and Green, each with eight states on the torus at filling one half [RG00], and the unitary description with eight states of [CGT99]. A uniformly gapped patch at the bottom of the sector contains the ten zero modes, so its dimension is at least ten and, by Theorem 8.1(i), at least twelve. These proposals satisfy the divisibility condition of Theorem 8.1(i), but their dimension is too small; the count ten is the one that violates the condition.

What is not excluded. Theorem 8.1 does not exclude a gapped phase whose ground-state space strictly contains the zero modes and has dimension 12, 16, … (Remark 8.7); in such a phase $E_{11}(N) \to 0$ would be a splitting inside the ground-state space. The conclusions that the index argument yields here are divisibility conditions, and such dimensions meet them. The expectation since Read and Green [RG00] is stronger than Theorem 8.1: that the hollow-core Hamiltonian has no uniformly gapped patch at the bottom of the sector at all. Read and Green argued, in a mean-field description of spin-singlet d-wave pairing, that the Haldane–Rezayi state describes the critical point between the weak- and strong-pairing phases, with bulk fermion excitations whose energy vanishes at zero momentum; their count of states on the torus at this critical point, five per centre of mass at even N and one spin-$\frac{1}{2}$ doublet per centre of mass at odd N, is the count of Theorem 2.10. Read [Rea09a, Rea09b] gave general arguments that trial states built from non-unitary conformal field theories do not describe gapped phases, and named the Haldane–Rezayi state as critical. Exact diagonalization of small systems on the sphere is consistent with a vanishing neutral gap [CRE19, NPBG23], and in the limit of a thin torus Seidel and Yang [SY11] constructed excitations whose energy tends to zero. None of this is proved; Theorem 8.1 establishes the part of the picture that concerns the ten zero modes.

The statement is asymptotic. Theorem 8.1 concerns the limit $N \to \infty$ only. The constants in Theorem 6.18 exist but are not explicit, and through the light-cone constant $C_{\mathrm{lc}} = \max\{1, (4C_1)^4\}$ of Theorem 5.3, with C_1 bounded by about $4.5 \cdot 10^3\, C_{\mathrm{vol}} \max\{C_h, 4C_5^H\}$ (Remark 5.4), they are astronomically large. The proof gives neither a rate for $E_{11}(N) \to 0$ nor a value N_0 beyond which $E_{11}(N)$ is below a prescribed number, and it says nothing about any fixed system size, in particular about the sizes accessible to numerical diagonalization.

Thin tori. Tori whose circumference L_y stays below L_0 while $N \to \infty$ lie outside Corollary 8.4 (Remark 8.5). For them, and more generally at every circumference, Appendix D proves a variational bound: a zero mode in which the orbitals of the electrons of one spin species are multiplied by a phase that winds once around the torus gives, for a suitable choice of the zero mode, a state in the sector $(N/2, N/2)$ orthogonal to all zero modes with energy at most of order L_y^3/N (for L_y bounded below). Hence, with $\kappa = 2\pi/L_y$,

$$E_{11}(N) \leq 0.40\,(1 + 0.42\,\kappa)^2\,\frac{L_y^3}{N} + 0.032\,(1 + 0.41\,\kappa)^2\,\frac{L_y^5}{N^3}$$

for every even N and every rectangular torus, and the same bound holds for $E_3(N)$ at odd N (Theorem D.1). At fixed aspect ratio L_y^3/N grows like $N^{1/2}$, and this bound says nothing; the two results overlap when $L_0 \leq L_y \leq C$, and the variational bound is needed only for $L_y < L_0$. Together with Lemma 8.3 and Corollary 8.4 it gives $E_{11}(N) \to 0$ uniformly over all rectangular tori (Corollary D.2). Unlike Theorem 8.1, the variational bound is explicit and gives a rate, but only at bounded or slowly growing circumference; like Theorem 8.1, it does not bear on the situation of Remark 8.7.

The class. Corollary 7.3 is not specific to the hollow-core interaction. For every sequence of Hamiltonians satisfying (H1) and (H2), the dimension of a uniformly gapped patch at the bottom of the sector $(N/2, N/2)$, in the sense of Section 1 (Assumption 6.3 for these sectors), is divisible by four: this is a fractional Lieb–Schultz–Mattis condition for the up-spin filling $\frac{1}{4}$ per orbital, in the sense of [BBDF21, Corollary 5.2], and it uses neither frustration-freeness nor zero modes (Remark 7.5). In particular no Hamiltonian of the class has, in this sector, a uniformly gapped ground-state space of dimension ten, the dimension that the conformal-field-theory description assigns to a gapped Haldane–Rezayi phase. Which lowest-Landau-level Hamiltonians other than H_N belong to the class is not examined here.

What remains. (a) *A rate at fixed aspect ratio.* Take $p = 10$, $\Delta = 0$ and $\gamma = E_{11}(N)$. The estimates of Sections 5 and 6, made explicit as functions of N and γ at a single N, would bound $|e^{2\pi \mathrm{i}\,\mathrm{Tr}(P\widetilde{T}_-)} - 1|$ by a quantity $\varepsilon(N, \gamma)$ that tends to zero as $N \to \infty$ for each $\gamma > 0$. Since $\mathrm{Tr}(P\widetilde{T}_-) = \mathrm{Tr}(PT_-) = -\frac{10}{4}$ exactly (Lemma 7.1(d) and (e)), the left-hand side equals 2, so $\varepsilon(N, E_{11}(N)) \geq 2$, an explicit upper bound on $E_{11}(N)$ once ε is explicit and decreasing in γ. A rate obtained in this way would not exclude the situation of Remark 8.7. (b) *Explicit constants*, without which no statement about a fixed system size follows. (c) *Oblique tori* (Remark 8.6). (d) *The stronger statement* that H_N has no uniformly gapped patch at the bottom of the sector $(N/2, N/2)$ of any dimension; it requires excluding the situation of Remark 8.7, for which the index argument alone does not suffice. (e) *Other members of the class* of Definition 4.6.

A The index theorem of Bachmann, Bols, De Roeck and Fraas

This appendix states the setting and the main results of [BBDF21], which Section 6 adapts; reproduces with proofs the operator-theoretic lemmas of [BBDF21, Section 4.1] together with the elementary inequalities used in Section 6; matches the steps of [BBDF21] with those of Sections 5–7; and records two misprints in [BBDF21, Section 4.2]. We follow version 3 of the arXiv e-print; its Lemma 4.1 replaces an incorrect lemma of the published version, and the proof of its Lemma 4.2 changes accordingly, while the statement does not [BBDF21, Section 4.1]. Numbers such as Lemma 4.1 or (4.13) refer to that version.

A.1 The setting and the theorem

In [BBDF21, Sections 2–3], Λ is a finite graph with the graph distance, of diameter L, with $\sup_x |\{x\}_{(r)}| \le C(1+r)^d$, carrying a copy of $\mathbb{C}^n$ at each vertex (a spin system, or a fermion system with $n = 2^f$, whose algebra of observables is the even part of the canonical anticommutation algebra). The Hamiltonian is $H = \sum_{Z\subset\Lambda} h_Z$ with $\operatorname{supp} h_Z = Z$, $h_Z = 0$ unless $\operatorname{diam} Z \le R_H$, and $\|h_Z\| \le m_H$. The charges q_Z satisfy $\operatorname{supp} q_Z = Z$, $q_Z = 0$ unless $\operatorname{diam} Z \le R_Q$, $\|q_Z\| \le m_Q$, $\sigma(q_Z) \subset \mathbb{Z}$ and $[q_Z, q_{Z'}] = 0$; $Q_S := \sum_{Z\subset S} q_Z$, and $[Q_\Lambda, H] = 0$. A set $\Gamma \subset \Lambda$ has boundary $\partial\Gamma = \partial_- \cup \partial_+$ with $\operatorname{dist}(\partial_-, \partial_+) \ge cL$, and $Q := Q_\Gamma$. All of R_H, R_Q, m_H, m_Q, γ and p are independent of L. A sequence $A = A_L$ is almost supported in $Z = Z_L$ if there are A_r supported in $Z_{(r)}$ with $\|A - A_r\| = \|A\||Z|\, O(r^{-\infty})$, and a sequence of unitaries U is locality-preserving if $U^* \mathcal{A}_Z U \subset \mathcal{A}_Z$ for all Z, where $\mathcal{A}_Z$ denotes the sequences almost supported in Z.

Assumption A.1 (gap [BBDF21, Assumption 1])**.** The eigenvalues $E_1 \le E_2 \le \cdots$ of H on the whole Hilbert space, repeated according to multiplicity, satisfy $E_{p+1} - E_p \ge \gamma$ and $E_p - E_1 \le \Delta$, with L-independent $\gamma > 0$, $\Delta > 0$ and p, and $\gamma > 2\Delta$. P is the spectral projection onto $\{E_1, \dots, E_p\}$.

Proposition A.2 (clustering [BBDF21, Proposition 3.1])**.** *Under Assumption A.1, for $A \in \mathcal{A}_X$, $B \in \mathcal{A}_Y$ with $X \cap Y = \emptyset$ and every normalized $\Omega \in \operatorname{ran} P$, $|\langle\Omega, AB\Omega\rangle - \langle\Omega, APB\Omega\rangle| = \|A\|\|B\||X||Y|\, O(d(X,Y)^{-\infty})$.*

Assumption A.3 (charge- and locality-preserving unitary [BBDF21, Assumption 2])**.** There is a locality-preserving unitary U with $[U, P] = O(L^{-\infty})$ and $[U, Q_\Lambda] = 0$.

Under Assumption A.3, $U^*QU - Q = T_- + T_+$ with $T_\pm \in \mathcal{A}_{\partial_\pm}$ and $\|T_\pm\| \le C|\Lambda|$ [BBDF21, (3.3)], and $T_\pm$ can be chosen so that $e^{2\pi\mathrm{i}(Q+T_\pm)} = \mathbf{1} + O(L^{-\infty})$ [BBDF21, (3.4)].

Theorem A.4 ([BBDF21, Theorem 3.2])**.** *Under Assumptions A.1 and A.3, $\operatorname{Tr}(PT_-)$ lies within $O(L^{-\infty})$ of an integer.*

Corollary A.5 (fractional Lieb–Schultz–Mattis theorem [BBDF21, Corollary 5.2])**.** *Let $\Lambda = \mathbb{T} \square \Lambda'$ be the cartesian product of a discrete circle $\mathbb{T} = \mathbb{Z}/L_1\mathbb{Z}$ and a graph Λ', let $\mathcal{T}$ be the shift along $\mathbb{T}$ with $[H, \mathcal{T}] = 0$, and let $\Gamma = \{(x_1, x') : 0 \le x_1 < L_1/2\}$. Then $T_- = -Q_{[0]}$, the charge of the slab $[0] = \{0\} \times \Lambda'$, and under Assumption A.1, $p\langle Q_{[0]}\rangle_P$ lies within $O(L^{-\infty})$ of an integer.*

A.2 Operator-theoretic lemmas

Lemma A.6 ([BBDF21, Lemma 4.1], with a sharpened conclusion)**.** *Let V be a unitary and P an orthogonal projection on a Hilbert space.*

(i) For every unit vector $\psi \in \operatorname{ran} P$, $\|PV\psi\|^2 = 1 + \langle\psi, V^*[P,V]\psi\rangle \ge 1 - \|[V,P]\|$, and likewise $\|PV^*\psi\|^2 \ge 1 - \|[V,P]\|$.

(ii) Let $\varepsilon \in [0,1]$. If $\|PV\psi\|^2 \ge (1-\varepsilon^2)\|\psi\|^2$ and $\|PV^*\psi\|^2 \ge (1-\varepsilon^2)\|\psi\|^2$ for every $\psi \in \operatorname{ran} P$, then $\|(\mathbf{1}-P)VP\| \le \varepsilon$, $\|PV(\mathbf{1}-P)\| \le \varepsilon$, and $\|[V,P]\| \le \varepsilon$.

Proof. (i) Since $PV = VP + [P,V]$, we have $V^*PV = P + V^*[P,V]$. For a unit vector ψ with $P\psi = \psi$, therefore,

$$\|PV\psi\|^2 = \langle V\psi, PV\psi\rangle = \langle\psi, V^*PV\psi\rangle = \langle\psi, P\psi\rangle + \langle\psi, V^*[P,V]\psi\rangle = 1 + \langle\psi, V^*[P,V]\psi\rangle,$$

which is at least $1 - \|V^*[P,V]\| = 1 - \|[V,P]\|$. In the same way $VPV^* = (PV + [V,P])V^* = P + [V,P]V^*$ gives $\|PV^*\psi\|^2 = 1 + \langle\psi, [V,P]V^*\psi\rangle \geq 1 - \|[V,P]\|$.

(ii) Let ϕ be any vector and $\psi := P\phi \in \operatorname{ran} P$. Since V is unitary and P and $\mathbf{1} - P$ are complementary orthogonal projections,

$$\|\psi\|^2 = \|V\psi\|^2 = \|(\mathbf{1}-P)V\psi\|^2 + \|PV\psi\|^2 \geq \|(\mathbf{1}-P)V\psi\|^2 + (1-\varepsilon^2)\|\psi\|^2,$$

so $\|(\mathbf{1}-P)VP\phi\|^2 \leq \varepsilon^2\|P\phi\|^2 \leq \varepsilon^2\|\phi\|^2$, that is, $\|(\mathbf{1}-P)VP\| \leq \varepsilon$. The same argument with V^* in place of V gives $\|(\mathbf{1}-P)V^*P\| \leq \varepsilon$, and $PV(\mathbf{1}-P) = ((\mathbf{1}-P)V^*P)^*$ has the same norm. Finally $[V,P] = VP - PV = (\mathbf{1}-P)VP - PV(\mathbf{1}-P)$. For $\phi = \phi_1 + \phi_2$ with $\phi_1 \in \operatorname{ran} P$ and $\phi_2 \in \operatorname{ran}(\mathbf{1}-P)$ we get $[V,P]\phi = (\mathbf{1}-P)VP\phi_1 - PV(\mathbf{1}-P)\phi_2$, and the two terms lie in the orthogonal subspaces $\operatorname{ran}(\mathbf{1}-P)$ and $\operatorname{ran} P$; hence $\|[V,P]\phi\|^2 \leq \varepsilon^2(\|\phi_1\|^2 + \|\phi_2\|^2) = \varepsilon^2\|\phi\|^2$. □

In [BBDF21, Lemma 4.1(ii)] the two hypotheses are strict inequalities with $\|P\psi\|^2$ on the right-hand side, the second is printed without the square on $\|PV^*\psi\|$, and the conclusion is $\|[V,P]\| < 2\varepsilon$. The proof given there yields the bounds on $(\mathbf{1}-P)VP$ and $PV(\mathbf{1}-P)$ stated here; the block structure of $[V,P]$ used in the last step replaces 2ε by ε.

Lemma A.7 ([BBDF21, Lemma 4.3])**.** *Let A be a bounded self-adjoint operator, P an orthogonal projection, and $\varphi \in \mathbb{R}$. Then $[e^{-\mathrm{i}\varphi A}, P] = -\mathrm{i}\int_0^\varphi e^{-\mathrm{i}uA}[A,P]e^{-\mathrm{i}(\varphi-u)A}\,\mathrm{d}u$; in particular $\|[e^{\mathrm{i}\varphi A}, P]\| \leq |\varphi|\|[A,P]\|$.*

Proof. The function $u \mapsto e^{-\mathrm{i}uA}Pe^{-\mathrm{i}(\varphi-u)A}$ is differentiable in norm, with derivative $e^{-\mathrm{i}uA}(-\mathrm{i}AP + \mathrm{i}PA)e^{-\mathrm{i}(\varphi-u)A} = -\mathrm{i}e^{-\mathrm{i}uA}[A,P]e^{-\mathrm{i}(\varphi-u)A}$. Its values at $u = \varphi$ and at $u = 0$ are $e^{-\mathrm{i}\varphi A}P$ and $Pe^{-\mathrm{i}\varphi A}$, so the fundamental theorem of calculus gives the identity. The exponentials are unitary, so the integrand has norm at most $\|[A,P]\|$, and $\|[e^{-\mathrm{i}\varphi A}, P]\| \leq |\varphi|\|[A,P]\|$ for every $\varphi \in \mathbb{R}$; replacing φ by $-\varphi$ gives the stated bound. □

Lemma A.7 is reproduced for completeness; it is not used in the proofs of this paper, which use instead the identity (6.7) and Lemma A.8.

Lemma A.8 (exponentials of almost commuting operators)**.** *Let A, B, A' be bounded self-adjoint operators. Then*

$$\|[e^{\mathrm{i}\varphi A}, B]\| \leq |\varphi|\|[A,B]\|, \qquad \|e^{\mathrm{i}(A+B)} - e^{\mathrm{i}A}e^{\mathrm{i}B}\| \leq \tfrac{1}{2}\|[A,B]\|, \qquad \|e^{\mathrm{i}A} - e^{\mathrm{i}A'}\| \leq \|A - A'\|.$$

None of these bounds depends on $\|A\|$ or $\|B\|$.

Proof. For the first bound, $u \mapsto e^{\mathrm{i}uA}Be^{\mathrm{i}(\varphi-u)A}$ has derivative $\mathrm{i}e^{\mathrm{i}uA}[A,B]e^{\mathrm{i}(\varphi-u)A}$, so $[e^{\mathrm{i}\varphi A}, B] = \mathrm{i}\int_0^\varphi e^{\mathrm{i}uA}[A,B]e^{\mathrm{i}(\varphi-u)A}\,\mathrm{d}u$, whose norm is at most $|\varphi|\|[A,B]\|$; this part does not use the self-adjointness of B. For the second, let $O(s) := e^{\mathrm{i}sA}e^{\mathrm{i}sB}e^{-\mathrm{i}s(A+B)}$, $s \in [0,1]$. Then $O(0) = \mathbf{1}$ and, since A commutes with $e^{\mathrm{i}sA}$ and B with $e^{\mathrm{i}sB}$,

$$O'(s) = e^{\mathrm{i}sA}\big(\mathrm{i}Ae^{\mathrm{i}sB} + e^{\mathrm{i}sB}\,\mathrm{i}B - e^{\mathrm{i}sB}\,\mathrm{i}(A+B)\big)e^{-\mathrm{i}s(A+B)} = \mathrm{i}\,e^{\mathrm{i}sA}[A, e^{\mathrm{i}sB}]e^{-\mathrm{i}s(A+B)}.$$

By the first bound, with the roles of A and B exchanged, $\|O'(s)\| \leq \|[e^{\mathrm{i}sB}, A]\| \leq s\|[A,B]\|$, and so

$$\|e^{\mathrm{i}A}e^{\mathrm{i}B} - e^{\mathrm{i}(A+B)}\| = \|(O(1)-\mathbf{1})e^{\mathrm{i}(A+B)}\| = \|O(1)-\mathbf{1}\| \leq \int_0^1 s\|[A,B]\|\,\mathrm{d}s = \tfrac{1}{2}\|[A,B]\|.$$

For the third, $u \mapsto e^{\mathrm{i}uA}e^{\mathrm{i}(1-u)A'}$ has derivative $\mathrm{i}e^{\mathrm{i}uA}(A - A')e^{\mathrm{i}(1-u)A'}$, so $e^{\mathrm{i}A} - e^{\mathrm{i}A'} = \mathrm{i}\int_0^1 e^{\mathrm{i}uA}(A - A')e^{\mathrm{i}(1-u)A'}\,\mathrm{d}u$, of norm at most $\|A - A'\|$. All exponentials of self-adjoint operators that occur are unitary, so no bound depends on $\|A\|$ or $\|B\|$. □

Lemma A.9 (determinants and integers)**.** *Let P be an orthogonal projection of rank p, and for A with $[A,P]=0$ let $\det_P A := \det(PAP|_{\operatorname{ran} P})$.*

(i) If A and B act on a p-dimensional space and $\|A\|,\|B\|\le 1$, then $|\det A - \det B| \le p\|A-B\|$.

(ii) If $[A,P]=0$, then $\det_P(e^A) = e^{\operatorname{Tr}(PA)}$; and $\det_P(AB) = \det_P(A)\det_P(B)$ when $[A,P]=[B,P]=0$.

(iii) If V is unitary, then $PV^*PVP - P = PV^*[P,V]P$; hence $|\det(PV^*P|_{\operatorname{ran} P})\det(PVP|_{\operatorname{ran} P}) - 1| \le p\|[P,V]\|$.

(iv) $|e^{2\pi i x}-1| \ge 4\operatorname{dist}(x,\mathbb{Z})$ for every $x\in\mathbb{R}$.

Proof. (i) Fix an orthonormal basis of the p-dimensional space, and let a_k and b_k be the k-th columns of the matrices of A and B. By the multilinearity of the determinant in the columns,

$$\det A - \det B = \sum_{k=1}^{p} \det(b_1,\dots,b_{k-1},\, a_k - b_k,\, a_{k+1},\dots,a_p).$$

By Hadamard's inequality, $|\det(v_1,\dots,v_p)| \le \prod_k |v_k|$ for the Euclidean norms of the columns: in the Gram–Schmidt factorization $(v_1,\dots,v_p) = OR$, with O unitary and R upper triangular, $|R_{kk}| \le |v_k|$ and $|\det(v_1,\dots,v_p)| = \prod_k |R_{kk}|$. The columns of A and B are images of unit vectors under contractions, so they have norm at most one, and $|a_k - b_k| \le \|A-B\|$. Hence each term has modulus at most $\|A-B\|$.

(ii) If $[A,P]=0$, then A maps $\operatorname{ran} P$ into itself, and the restriction of e^A to $\operatorname{ran} P$ is e^{A_P}, with $A_P := PAP|_{\operatorname{ran} P}$. A matrix M with eigenvalues $\mu_1,\dots,\mu_p$, counted with algebraic multiplicity, is similar to an upper triangular matrix with these diagonal entries, and e^M is then similar to an upper triangular matrix with diagonal entries $e^{\mu_1},\dots,e^{\mu_p}$; so $\det e^M = e^{\mu_1+\cdots+\mu_p} = e^{\operatorname{Tr} M}$. Hence $\det_P(e^A) = e^{\operatorname{Tr}(A_P)} = e^{\operatorname{Tr}(PAP)} = e^{\operatorname{Tr}(PA)}$. If $[A,P]=[B,P]=0$, then $PA(\mathbf{1}-P) = AP(\mathbf{1}-P) = 0$, so $PABP = PAPBP + PA(\mathbf{1}-P)BP = PAPBP$; the restriction of $PABP$ to $\operatorname{ran} P$ is the product of those of PAP and PBP, and the determinant is multiplicative.

(iii) Since $PV = VP + [P,V]$, $PV^*PVP = PV^*(VP+[P,V])P = P + PV^*[P,V]P$. The operators PV^*P and PVP map $\operatorname{ran} P$ into itself, and the restriction of their product is that of PV^*PVP; so $\det(PV^*P|_{\operatorname{ran} P})\det(PVP|_{\operatorname{ran} P}) = \det\big((P+PV^*[P,V]P)|_{\operatorname{ran} P}\big)$. By (i), applied to $A := PV^*PVP|_{\operatorname{ran} P}$ and $B := \mathbf{1}_{\operatorname{ran} P}$, both of norm at most one, this differs from $\det B = 1$ by at most $p\|PV^*[P,V]P\| \le p\|[P,V]\|$.

(iv) $|e^{2\pi i x}-1| = 2|\sin(\pi x)|$. With $d := \operatorname{dist}(x,\mathbb{Z}) \in [0,\tfrac12]$ we have $|\sin(\pi x)| = \sin(\pi d) \ge 2d$, because $d \mapsto \sin(\pi d)$ is concave on $[0,\tfrac12]$ and equals $2d$ at $d=0$ and $d=\tfrac12$. □

A.3 Correspondence with Sections 5–7

The following table matches the objects and steps of [BBDF21] with those of this paper. The main differences are two. The locality of [BBDF21] is that of a graph and a finite-range Hamiltonian; here it is that of the slot lattice, into which the lowest Landau level is embedded by Lemma 3.8, and of a Hamiltonian that is rapidly local there (Definition 4.6); every error $O(L^{-\infty})$ of [BBDF21] becomes an error $O(N^{-\infty})$, controlled by Lemma 5.13. And the gap is assumed only in a sector (Assumption 6.3), which is what Lemma 6.7 and Proposition 6.10 accommodate.

[BBDF21]	this paper
graph Λ of diameter L; errors $O(L^{-\infty})$ (Section 2, Appendix C)	slot lattice Λ_N and the embedding ι (Definitions 3.1 and 3.7, Lemma 3.8); errors $O(N^{-\infty})$ (Remark 3.12)
local charges q_Z and Q_S (Section 2.2)	occupation numbers $n_{j\uparrow}$ and band charges $Q_{\mathcal{B}}$, exactly supported in columns (Definition 4.9, Lemmas 4.10 and 4.11)
finite-range Hamiltonian $H = \sum_Z h_Z$ (Section 2.1)	$H_N = \int h_R \, \mathrm{d}R$ with (H1) and (H2) (Definition 4.6, Corollary 4.8)
almost supported sequences $\mathcal{A}_Z$ (Section 2.3.1, Appendix C)	Definition 3.11, Lemma 5.12
the Lieb–Robinson bound, used in Section 4.1 and Appendix A	Theorems 5.2 and 5.3, Lemmas 5.5–5.7
Assumption 1 (gap on the whole Hilbert space)	Assumption 6.3 (gap in a sector, $\Delta \geq 0$)
Assumption 2 (locality-preserving U) and the choice (3.4) of $T_\pm$	Assumption 6.4, (U1)–(U4)
K, $\overline{Q}$ and $[\overline{Q}, P] = 0$, (4.1)–(4.3)	Definition 6.5, Lemma 6.7
$\mathrm{i}[H, Q] = J_- + J_+$ and $K_\pm \in \mathcal{A}_{\partial_\pm}$ (Section 4.1)	Lemmas 6.6 and 6.8, Remark 6.9
Proposition 3.1 and its proof in Appendix A (clustering)	Definition 5.8, Lemma 5.10, Propositions 5.11 and 6.10
Lemma 4.1	Lemma A.6
Lemma 4.2	Lemma 6.14
Lemma 4.3	Lemma A.7
the objects of Section 4.2, (4.5)–(4.7)	Definition 6.11, Lemma 6.12
the splitting and the commutation relations (4.8), (4.9)	Lemma 6.13
(4.10) and (4.11)	Corollary 6.15
the differential equation for $\mathcal{Z}_-$ and (4.12)	Lemma 6.16
(4.13), (4.14) and the determinant at 2π	Lemma 6.17
Theorem 3.2	Theorem 6.18
the translation $\mathcal{T}$ and Corollary 5.2	the magnetic translation Θ, Lemma 7.1 and Corollary 7.2

Remark A.10 (two misprints in [BBDF21, Section 4.2]). (a) In the display that precedes (4.13), the first equality, $U^* e^{2\pi \mathrm{i} \overline{Q}_-} U = U^* e^{2\pi \mathrm{i}(\overline{Q}_- + Q_m + Q_+)} U$, holds only up to $O(L^{-\infty})$: $Q_m + Q_+$ has integer spectrum, but it commutes with $\overline{Q}_- = Q_- - K_-$ only approximately, because K_- commutes with Q_m and Q_+ only approximately. The argument is unaffected; Step 1 of the proof of Lemma 6.17(i) gives the corresponding estimate. (b) In the display that follows (4.14), the two exponents are printed as $\mathrm{Tr}(PD_-P + \overline{Q})$ and $\mathrm{Tr}(\overline{Q})$, where D_- is our $\widetilde{T}_-$. By Lemma A.9(ii) they are $\mathrm{Tr}(PD_-P) + \mathrm{Tr}(P\overline{Q})$ and $\mathrm{Tr}(P\overline{Q})$; the product of the two exponentials, and hence the conclusion $e^{2\pi \mathrm{i} \, \mathrm{Tr}(PD_-)}$, is unaffected (Lemma 6.17(ii)).

B The Lieb–Robinson input

The only input from the theory of Lieb–Robinson bounds is the one-step estimate of [NSY18] for lattice fermions with even interactions, which we state in the special case needed here: a finite set of fermionic modes and a time-independent interaction. In [NSY18] each site carries one mode; spin is included by taking as the set of sites the set $\Lambda_N \times \{\uparrow, \downarrow\}$ of modes, and the support of an operator in $\mathrm{CAR}(X)$, $X \subset \Lambda_N$, is then $X \times \{\uparrow, \downarrow\}$.

Let $\mathfrak{M}$ be a finite set, $\mathrm{CAR}(Y)$ for $Y \subset \mathfrak{M}$ the algebra generated by the annihilation operators a_y,

$y \in Y$, and $\mathrm{CAR}(Y)^+$ its even part. An even interaction is a map Φ from subsets of $\mathfrak{M}$ to $\mathrm{CAR}(\mathfrak{M})$ with $\Phi(Z) = \Phi(Z)^* \in \mathrm{CAR}(Z)^+$; for $X \subset \mathfrak{M}$ let $H_X := \sum_{Z \subset X} \Phi(Z)$ and $\tau_t^X(A) := e^{\mathrm{i}tH_X} A e^{-\mathrm{i}tH_X}$, and let $S(X) := \{Z \subset \mathfrak{M} : Z \cap X \neq \emptyset,\ Z \cap (\mathfrak{M} \setminus X) \neq \emptyset\}$.

Proposition B.1 (disjoint supports [NSY18, Proposition 2.1])**.** *Let $X, Y \subset \mathfrak{M}$ be disjoint. If $A \in \mathrm{CAR}(X)^+$ and $B \in \mathrm{CAR}(Y)$, then $[A, B] = 0$. If $A \in \mathrm{CAR}(X)$ and $B \in \mathrm{CAR}(Y)$ are both odd, then $\{A, B\} = 0$.*

Lemma B.2 ([NSY18, Lemma 3.2])**.** *Let $A(t)$, $B(t)$ be norm-continuous families of bounded operators on a Hilbert space, with $A(t)$ self-adjoint. Then the solution of $f'(t) = \mathrm{i}[A(t), f(t)] + B(t)$, $f(t_0) = f_0$, satisfies $\|f(t)\| \le \|f(t_0)\| + \int_{\min\{t,t_0\}}^{\max\{t,t_0\}} \|B(u)\|\,\mathrm{d}u$.*

Lemma B.3 (the one-step estimate [NSY18, Lemma 3.3])**.** *Let Φ be an even interaction on $\mathfrak{M}$, $X \subset \mathfrak{M}$, $A \in \mathrm{CAR}(X)$, $B \in \mathrm{CAR}(\mathfrak{M})$ and $t \ge 0$. Then*

$$\|[\tau_t^{\mathfrak{M}}(A), B]\| \le \|[\tau_t^X(A), B]\| + 2\|A\| \sum_{Z \in S(X)} \int_0^t \|[\tau_u^{\mathfrak{M}}(\Phi(Z)), B]\|\,\mathrm{d}u.$$

In [NSY18] these statements are proved for time-dependent interactions and for all times $s \le t$; negative times are reduced to positive ones in the proof below. Iterating Lemma B.3 gives [NSY18, Theorem 3.1], a Lieb–Robinson bound in terms of a decay function on $\mathfrak{M} \times \mathfrak{M}$. Theorem 5.2 is obtained by the same iteration, with the sums over chains of sets bounded by powers of the chain matrix instead of by convolutions of that decay function.

B.1 Proof of Theorem 5.2

The bounds of Theorem 5.2 are not stated in [NSY18]. We derive them here from [NSY18, Lemma 3.3], which is Lemma B.3, by iterating it as in the proof of [NSY18, Theorem 3.1], and we bound the resulting sums over chains of sets by powers of the chain matrix where [NSY18] use the convolution property of a decay function. For a statement S we write $[S] := 1$ if S holds and $[S] := 0$ otherwise.

The one-step estimate in terms of slots. Let $\mathfrak{M} := \Lambda_N \times \{\uparrow, \downarrow\}$ be the set of modes, with $a_{(s,\sigma)} := \hat{c}_{s\sigma}$, and for $X \subset \Lambda_N$ let $\widetilde{X} := X \times \{\uparrow, \downarrow\}$. The algebra $\mathrm{CAR}(X)$ of Definition 3.7 is the algebra $\mathrm{CAR}(\widetilde{X})$ of this appendix, and $\mathfrak{A}(X) = \mathrm{CAR}(\widetilde{X})^+$: for $A \in \mathrm{CAR}(\widetilde{X})$, conjugation by $(-1)^{\widehat{N}_\uparrow + \widehat{N}_\downarrow}$ coincides with conjugation by the parity of the modes in $\widetilde{X}$, because the number operator of the remaining modes commutes with every a_y, $y \in \widetilde{X}$. By (5.1), $\mathrm{CAR}(\mathfrak{M})$ is the algebra of all operators on $\widehat{\mathcal{H}}_N$. Given the even interaction Ψ on Λ_N, an even interaction Φ on $\mathfrak{M}$ in the sense of this appendix is defined by $\Phi(\widetilde{Z}) := \Psi(Z)$ for $Z \subset \Lambda_N$ and $\Phi(W) := 0$ for the subsets $W \subset \mathfrak{M}$ that are not of the form $\widetilde{Z}$. For $X \subset \Lambda_N$, $H_{\widetilde{X}} = \sum_{Z \subset X} \Psi(Z) =: H_X$; in particular $H_{\mathfrak{M}} = H_\Psi$. The sets of $S(\widetilde{X})$ on which Φ does not vanish are the $\widetilde{Z}$ with Z in

$$S_\Lambda(X) := \{Z \subset \Lambda_N : Z \cap X \neq \emptyset,\ Z \not\subset X\}.$$

Write τ_t^X for the dynamics generated by H_X, which is the dynamics $\tau_t^{\widetilde{X}}$ above; in particular $\tau_t^{\Lambda_N} = \tau_t^\Psi$. Then Lemma B.3 says: for $X \subset \Lambda_N$, $A \in \mathrm{CAR}(X)$, every operator B on $\widehat{\mathcal{H}}_N$ and every $t \ge 0$,

$$\|[\tau_t^\Psi(A), B]\| \le \|[\tau_t^X(A), B]\| + 2\|A\| \sum_{Z \in S_\Lambda(X)} \int_0^t \|[\tau_u^\Psi(\Psi(Z)), B]\|\,\mathrm{d}u. \tag{B.1}$$

Reduction to $t \geq 0$. For $t < 0$, $\tau_t^{\Psi} = \tau_{|t|}^{-\Psi}$, and $-\Psi$ is an even interaction with $M^{-\Psi} = M^{\Psi}$ and $C(-\Psi, \lambda) = C(\Psi, \lambda)$. It therefore suffices to prove the theorem for $t \geq 0$, which we assume from now on.

The first term. Let $W \subset \Lambda_N$ and $u \geq 0$. Since H_W is an even element of $\mathrm{CAR}(W)$, conjugation by $e^{\mathrm{i}uH_W}$ maps $\mathrm{CAR}(W)$ onto itself and preserves parity, so τ_u^W maps $\mathfrak{A}(W)$ onto itself. Hence, by Proposition B.1, if either $A' \in \mathfrak{A}(W)$ and $B \in \mathrm{CAR}(Y)$, or $A' \in \mathrm{CAR}(W)$ and $B \in \mathfrak{A}(Y)$, then $[\tau_u^W(A'), B] = 0$ whenever $W \cap Y = \emptyset$; and $\|[\tau_u^W(A'), B]\| \leq 2\|A'\|\|B\|$ in all cases. Thus

$$\|[\tau_u^W(A'), B]\| \leq 2\|A'\|\|B\| \, [W \cap Y \neq \emptyset]. \tag{B.2}$$

Under either hypothesis of the theorem, (B.2) applies to $W = X$ and $A' = A$; and since every $\Psi(Z)$ is even, it applies to $W = Z$ and $A' = \Psi(Z)$ for every $B \in \mathrm{CAR}(Y)$.

Iteration. For $n \geq 1$, a *chain of length n from X* is a sequence $(Z_1, \ldots, Z_n)$ of subsets of Λ_N with $Z_1 \in S_\Lambda(X)$ and $Z_i \in S_\Lambda(Z_{i-1})$ for $2 \leq i \leq n$. Put $a_0 := [X \cap Y \neq \emptyset]$ and, for $n \geq 1$,

$$a_n := \sum_{(Z_1,\ldots,Z_n)} [Z_n \cap Y \neq \emptyset] \prod_{i=1}^{n} \|\Psi(Z_i)\|, \qquad b_n := \sum_{(Z_1,\ldots,Z_n)} \prod_{i=1}^{n} \|\Psi(Z_i)\|,$$

the sums running over the chains of length n from X. We claim that for every $n \geq 0$

$$\|[\tau_t^{\Psi}(A), B]\| \leq 2\|A\|\|B\| \sum_{j=0}^{n} \frac{(2t)^j}{j!} \, a_j + R_n(t), \tag{B.3}$$

where, with the sum over the chains $(Z_1, \ldots, Z_{n+1})$ of length $n+1$ from X,

$$R_n(t) := 2^{n+1}\|A\| \sum_{(Z_1,\ldots,Z_{n+1})} \prod_{i=1}^{n} \|\Psi(Z_i)\| \int_0^t \int_0^{u_1} \cdots \int_0^{u_n} \|[\tau_{u_{n+1}}^{\Psi}(\Psi(Z_{n+1})), B]\| \, \mathrm{d}u_{n+1} \cdots \mathrm{d}u_1.$$

For $n = 0$, (B.3) is (B.1) with its first term bounded by (B.2). If (B.3) holds for some n, apply (B.1) to $A' = \Psi(Z_{n+1}) \in \mathrm{CAR}(Z_{n+1})$ at time u_{n+1}, with its first term bounded by (B.2):

$$\begin{aligned} \|[\tau_{u_{n+1}}^{\Psi}(\Psi(Z_{n+1})), B]\| &\leq 2\|\Psi(Z_{n+1})\|\|B\| \, [Z_{n+1} \cap Y \neq \emptyset] \\ &\quad + 2\|\Psi(Z_{n+1})\| \sum_{Z_{n+2} \in S_\Lambda(Z_{n+1})} \int_0^{u_{n+1}} \|[\tau_{u_{n+2}}^{\Psi}(\Psi(Z_{n+2})), B]\| \, \mathrm{d}u_{n+2}. \end{aligned}$$

Inserted in $R_n(t)$, the first term contributes $2^{n+1}\|A\| \cdot 2\|B\| \, a_{n+1} \, t^{n+1}/(n+1)!$, because the simplex $\{t \geq u_1 \geq \cdots \geq u_{n+1} \geq 0\}$ has volume $t^{n+1}/(n+1)!$; this is the term $j = n+1$ of (B.3). The second term contributes exactly $R_{n+1}(t)$. This proves (B.3) for every n. Bounding the integrand of $R_n(t)$ by $2\|\Psi(Z_{n+1})\|\|B\|$ gives

$$R_n(t) \leq 2\|A\|\|B\| \, \frac{(2t)^{n+1}}{(n+1)!} \, b_{n+1}.$$

Chains and powers of the chain matrix. Let $M := M^{\Psi}$ and $\mu := \max_s \sum_{s'} M_{ss'}$. We claim that for every $n \geq 0$

$$a_n \leq \sum_{s \in X, \; s' \in Y} (M^n)_{ss'}, \qquad b_n \leq \sum_{s \in X, \; s' \in \Lambda_N} (M^n)_{ss'} \leq |X|\mu^n.$$

For $n=0$, $a_0 \le |X\cap Y| = \sum_{s\in X,\,s'\in Y}[s=s']$. Let $n\ge 1$, and let $(Z_1,\dots,Z_n)$ be a chain from X with $Z_n\cap Y\neq\emptyset$. Choose $s_1\in Z_1\cap X$, $s_i\in Z_{i-1}\cap Z_i$ for $2\le i\le n$ (these intersections are nonempty because $Z_i\in S_\Lambda(Z_{i-1})$), and $s_{n+1}\in Z_n\cap Y$. Then $s_i, s_{i+1}\in Z_i$ for $1\le i\le n$, so the term $\prod_i\|\Psi(Z_i)\|$ of the chain is one of the nonnegative terms obtained by expanding

$$\sum_{s_1\in X}\ \sum_{s_2,\dots,s_n\in\Lambda_N}\ \sum_{s_{n+1}\in Y}\ \prod_{i=1}^{n}\Big(\sum_{Z_i\ni s_i,s_{i+1}}\|\Psi(Z_i)\|\Big) = \sum_{s\in X,\ s'\in Y}(M^n)_{ss'},$$

and distinct chains give distinct terms. This proves the bound on a_n. The bound on b_n follows in the same way, with $s_{n+1}\in Z_n$ arbitrary, and $\sum_{s'}(M^n)_{ss'}\le\mu^n$.

The first bound. By the last two steps, $R_n(t)\le 2\|A\|\|B\||X|(2t\mu)^{n+1}/(n+1)!\to 0$ as $n\to\infty$. Letting $n\to\infty$ in (B.3),

$$\|[\tau_t^\Psi(A),B]\|\le 2\|A\|\|B\|\sum_{j\ge 0}\frac{(2t)^j}{j!}\,a_j\le 2\|A\|\|B\|\sum_{s\in X,\ s'\in Y}\big[e^{2tM}\big]_{ss'}.$$

If $X\cap Y=\emptyset$, then $a_0=0$, and the term $j=0$, which contributes the identity matrix, can be omitted; this replaces e^{2tM} by $e^{2tM}-\mathbf{1}$.

The weighted bound. Let $\lambda>0$, $C:=C(\Psi,\lambda)$ and $M^{\mathrm{w}}_{ss'}:=M_{ss'}e^{d(s,s')/\lambda}$, so that every row of M^{w} sums to at most C. For slots $s_1,\dots,s_{n+1}$ the triangle inequality $d(s_1,s_{n+1})\le\sum_i d(s_i,s_{i+1})$ gives $\prod_{i=1}^n M_{s_is_{i+1}}\le e^{-d(s_1,s_{n+1})/\lambda}\prod_{i=1}^n M^{\mathrm{w}}_{s_is_{i+1}}$; summing over $s_2,\dots,s_n$, $(M^n)_{ss'}\le e^{-d(s,s')/\lambda}\big((M^{\mathrm{w}})^n\big)_{ss'}$. Hence, for every slot s, every $W\subset\Lambda_N$ and every $n\ge 1$, $\sum_{s'\in W}(M^n)_{ss'}\le e^{-d(s,W)/\lambda}\sum_{s'\in\Lambda_N}((M^{\mathrm{w}})^n)_{ss'}\le e^{-d(s,W)/\lambda}C^n$, while $\sum_{s'\in W}[s=s']=[s\in W]\le e^{-d(s,W)/\lambda}$. Summing the exponential series, for every $t\in\mathbb{R}$,

$$\sum_{s'\in W}\big[e^{2|t|M}\big]_{ss'}\le e^{2|t|C}e^{-d(s,W)/\lambda},\qquad \sum_{s'\in W}\big[e^{2|t|M}-\mathbf{1}\big]_{ss'}\le(e^{2|t|C}-1)e^{-d(s,W)/\lambda}. \tag{B.4}$$

Summing (B.4) with $W=Y$ over $s\in X$ and inserting it in the first bound gives the weighted bound, with $e^{2|t|C}-1$ in place of $e^{2|t|C}$ when $X\cap Y=\emptyset$.

Deleting terms. If Ψ' is obtained from Ψ by setting some of its terms to zero, then $0\le M^{\Psi'}\le M^\Psi$ entrywise, hence $(M^{\Psi'})^n\le(M^\Psi)^n$ entrywise for every n, and $C(\Psi',\lambda)\le C(\Psi,\lambda)$. The right-hand sides of both bounds, and of (B.4), are nondecreasing functions of the entries of the matrix, so the bounds for the dynamics of Ψ' hold with the matrix M^Ψ. □

C The zero modes: proof of Theorem 2.10

This appendix proves Theorem 2.10, for tori of arbitrary modular parameter. The proof is written in its own coordinates and notation, which are adapted to theta functions; we first relate them, and the Hamiltonian, to Sections 2–4.

Coordinates and orbitals. The appendix uses the complex coordinate $z=X+iY$, the torus $\mathbb{C}/\Lambda$ with $\Lambda=L_1\mathbb{Z}+L_1\tau\mathbb{Z}$ and $\operatorname{Im}\tau=\tau_2>0$, the vector potential $(-Y,0)$, whose magnetic field is 1, and $L:=N_\phi$ flux quanta, with $L_1^2\tau_2=2\pi L$. For the torus of Definition 2.1 take $L_1:=L_y$ and $\tau:=\mathrm{i}L_x/L_y$, so that $L_1\tau_2=L_x$ and the number $\kappa=2\pi/L_1$ of this appendix is the κ of Definition 2.1. The rotation $(x,y)\mapsto(X,Y):=(y,-x)$ maps $\mathbb{R}^2/(L_x\mathbb{Z}\times L_y\mathbb{Z})$ onto $\mathbb{C}/\Lambda$, preserves the area, and carries the vector potential $(0,x)$ of Definition 2.1 to $(-Y,0)$, because $x\,\mathrm{d}y=-Y\,\mathrm{d}X$. Accordingly, for $f\in\mathfrak{h}_N$ the function $\psi(X,Y):=f(-Y,X)$ satisfies $\psi(X+L_1,Y)=\psi(X,Y)$ and

$\psi(X, Y + L_x) = e^{-\mathrm{i}L_x X}\psi(X, Y)$, by (2.2). Moreover the image of the orbital ψ_j of Lemma 2.2 is $(L_y\sqrt{\pi})^{-1/2}\sum_{k\in\mathbb{Z}} e^{\mathrm{i}\kappa(j+kL)X}e^{-(Y+\kappa(j+kL))^2/2}$, which is the normalized orbital $\phi_j e^{-Y^2/2}/(L_1\sqrt{\pi})^{1/2}$ of (C.2), because $e^{\mathrm{i}\kappa mX}e^{-(Y+\kappa m)^2/2} = e^{\mathrm{i}\kappa mz-\kappa^2m^2/2}e^{-Y^2/2}$ and $\kappa^2/2 = \pi\tau_2/L$ by (C.1). Since the ψ_j span $\mathfrak{h}_N$, $e^{Y^2/2}\psi$ is an entire function of z for every $f \in \mathfrak{h}_N$, and ψ lies in the space $\mathcal{L}$ of Section C.1.1. Composition with the rotation is therefore a unitary map from $\mathfrak{h}_N$ onto $\mathcal{L}$ that takes ψ_j to the j-th normalized orbital of this appendix for every j; its second quantization, with the spin unchanged, maps each sector $\mathcal{H}_N^{(n,m)}$ onto the space of states with n electrons of spin up and m of spin down, and it maps $c_{j\sigma}$ to the annihilation operator of the j-th normalized orbital with spin σ. Under this map the one-particle translation t_2 defined below is θ_x.

The Hamiltonian. The appendix defines the hollow-core Hamiltonian as $H = \sum_{i<j}\mathcal{P}_{ij}$, where $\mathcal{P}$, defined in (C.7), is the sum over the lattice of periods of the translates of the planar projector onto relative angular momentum one. Lemma C.6 shows that, with orbitals normalized to one, $H = \frac{c'_L}{2}\sum_{n\in\mathbb{Z}_{2L}}\sum_{\sigma,\sigma'}(B_n^{\sigma\sigma'})^\dagger B_n^{\sigma\sigma'}$, where $c'_L = c_L/(L_1\sqrt{\pi})^2$ and $c_L = 4\pi\kappa^2 L_1\sqrt{\pi/2}$ (Lemma C.4); since $L_1 = 2\pi/\kappa$, $c'_L/2 = \kappa^3/\sqrt{2\pi}$. For $\tau = \mathrm{i}L_x/L_y$ the weight in (C.8) is $\mathfrak{q}^{d^2} = e^{-\kappa^2 d^2/4}$, and pairing the terms (A, B) and (B, A) of (C.8) gives $B_n^{\uparrow\uparrow} = b_n^+$, $B_n^{\downarrow\downarrow} = b_n^-$ and $B_n^{\uparrow\downarrow} = B_n^{\downarrow\uparrow} = b_n^0/\sqrt{2}$, with the centre pair operators of Definition 4.1: for $d := A - B > 0$, the two terms are $d\,e^{-\kappa^2 d^2/4}(c_{[B]\sigma'}c_{[A]\sigma} - c_{[A]\sigma'}c_{[B]\sigma})$, and the term $d = 0$ vanishes. Hence $\sum_{\sigma,\sigma'}(B_n^{\sigma\sigma'})^\dagger B_n^{\sigma\sigma'} = \sum_\nu (b_n^\nu)^* b_n^\nu$, and by Theorem 4.2(c) the second quantization of the rotation carries H_N, restricted to the states with N electrons, to the operator H of this appendix. It preserves the numbers of electrons of each spin. So Theorem C.19 below, for $\tau = \mathrm{i}L_x/L_y$, is Theorem 2.10, and this proves the identification stated in Remark 2.11.

Notation. Every symbol of this appendix is defined in it, and where a symbol also occurs in the rest of the paper, its meaning here is the one given here. In particular: Λ is the lattice of periods; $L = N_\phi$; $\Theta_\ell[a, b]$ and $\Theta_2[\varepsilon]$ are spaces of theta functions; $\mathcal{G}(n, m; \varepsilon)$ are spaces of meromorphic functions; $\mathcal{P}$ is the torus pair interaction and P_1 the planar one; $\mathsf{D}_{st} := \partial_s - \partial_t$; U is a sum of coordinates and J a product of theta functions; T_K and T_1 are many-body translations, with $T_1 = \Theta^*$ under the map above, and K is the corresponding momentum label; n and m are the numbers of electrons of spin up and spin down; μ_N is the group of N-th roots of unity; $\dagger$ denotes the adjoint; and i is the imaginary unit.

C.1 Setting and notation

C.1.1 The torus lowest Landau level

Magnetic length 1; coordinates $z = X + iY$ on the plane; Landau gauge $\mathbf{A} = (-Y, 0)$. Planar lowest-Landau-level (LLL) one-particle states are $\psi(z) = f(z)e^{-Y^2/2}$ with f entire [RR96, eq. (44)]. The torus is $\mathbb{C}/\Lambda$, $\Lambda = L_1\mathbb{Z} + L_1\tau\mathbb{Z}$, $\tau = \tau_1 + i\tau_2$, $\tau_2 > 0$, threaded by $L := N_\phi$ flux quanta, so $L_1^2\tau_2 = 2\pi L$. Put $\kappa := 2\pi/L_1$; then

$$\tau_2 = \frac{\kappa^2 L}{2\pi}, \qquad L_1\tau_2 = \kappa L. \tag{C.1}$$

The torus LLL is

$$\mathcal{L} := \left\{\psi = fe^{-Y^2/2} :\ f \text{ entire},\ f(z + L_1) = f(z),\ f(z + L_1\tau) = e^{-i\pi L(2z/L_1+\tau)}f(z)\right\}$$

([RR96, eq. (45)] with $\phi_1 = \phi_2 = 0$), with inner product $\langle \phi, \psi \rangle = \int_D \bar{\phi}\psi$, D a fundamental domain of Λ. We use the rectangle $D = \{X + iY : \ 0 \le Y < L_1\tau_2, \ X_0(Y) \le X < X_0(Y) + L_1\}$ with $X_0(Y) = Y\tau_1/\tau_2$ (a fundamental domain for any τ). Write $u := z/L_1$ when convenient (periods $1, \tau$).

Orbitals. For $j \in \mathbb{Z}$ let

$$\phi_j(z) := \sum_{k\in\mathbb{Z}} \exp\Big(i\kappa(j+kL)z + \frac{i\pi\tau}{L}(j+kL)^2\Big) = \vartheta\begin{bmatrix} j/L \\ 0 \end{bmatrix}(Lz/L_1 \,|\, L\tau), \tag{C.2}$$

with $\vartheta[a,b](z|\tau) = \sum_n \exp(i\pi\tau(n+a)^2 + 2\pi i(n+a)(z+b))$ [RR96, eq. (65)]. Then $\phi_{j+L} = \phi_j$, and $\phi_0, \dots, \phi_{L-1}$ is an orthogonal basis of the holomorphic parts of $\mathcal{L}$ with equal norms $\|\phi_j e^{-Y^2/2}\|^2 = L_1\sqrt{\pi}$. *Proof.* Each ϕ_j satisfies the two boundary conditions (term by term, using $e^{i\kappa(j+kL)L_1} = 1$ and $\kappa L L_1 \tau = 2\pi L\tau$). By Lemma C.8(a) below ($\ell = L$) the space of such f has dimension L. At fixed Y the X-integral over a full period kills all products of exponentials with different frequencies $\kappa(a+kL) \neq \kappa(b+lL)$, so $\langle \phi_a, \phi_b \rangle = 0$ for $a \not\equiv b$; and, by (C.1), $|e^{i\kappa Az + i\pi\tau A^2/L}|^2 e^{-Y^2} = e^{-(Y+\kappa A)^2}$, so $\|\phi_a\|^2 = L_1 \sum_k \int_0^{\kappa L} e^{-(Y+\kappa(a+kL))^2} dY = L_1\sqrt{\pi}$. □

Translations. On holomorphic parts define

$$(t_1 f)(z) := f(z + L_1/L), \qquad (t_2 f)(z) := e^{i\pi(2z/L_1 + \tau/L)} f(z + L_1\tau/L).$$

Both preserve $\mathcal{L}$; $t_1\phi_j = e^{2\pi ij/L}\phi_j$, $t_2\phi_j = \phi_{j+1}$, $t_1 t_2 = e^{2\pi i/L} t_2 t_1$, $t_1^L = t_2^L = 1$. (Direct from (C.2): shifting z by $L_1\tau/L$ in the k-th term and completing the square gives $e^{-i\pi(2z/L_1+\tau/L)}$ times the term of ϕ_{j+1}.) On wave functions these are the magnetic translations by L_1/L and $L_1\tau/L$. In second quantization t_2 is the orbital shift $j \mapsto j+1$ and t_1 is $e^{2\pi ij/L}$.

Magnetic translations by lattice vectors. For $a = a_1 + ia_2 \in \mathbb{C}$ let

$$(\tau_a \psi)(z) := e^{-ia_2(X - a_1)} \psi(z - a);$$

τ_a is unitary on $L^2(\mathbb{C})$ and maps planar LLL functions to planar LLL functions (on holomorphic parts: $f(z) \mapsto e^{-ia_2(z-a_1) - a_2^2/2} f(z-a)$; we need only the following consequence). For every $\nu \in \Lambda$ there is $c(\nu) \in U(1)$ with

$$\tau_\nu \psi = c(\nu)\psi \qquad \text{for all } \psi \in \mathcal{L}. \tag{C.3}$$

Proof. For $\nu = -L_1$: $\tau_\nu\psi(z) = \psi(z+L_1) = \psi(z)$. For $\nu = -L_1\tau$: with $a_1 = -L_1\tau_1$, $a_2 = -L_1\tau_2$ and the boundary condition, $\tau_\nu\psi(z) = e^{iL_1\tau_2(X+L_1\tau_1)} f(z + L_1\tau) e^{-(Y+L_1\tau_2)^2/2}$; inserting $f(z+L_1\tau) = e^{-iL_1\tau_2 z - i\pi L\tau} f(z)$ (by (C.1), $2\pi L/L_1 = L_1\tau_2$) all z-dependence cancels and $\tau_\nu\psi = e^{i\pi L\tau_1}\psi$, a phase because $L_1^2\tau_2^2/2 = \pi L\tau_2$. General ν: the τ_a compose up to constant phases. □

C.1.2 Many electrons, canonical components, quantum numbers

N electrons, n up and m down, $N = n+m$; throughout $L = N_\phi = 2N$. A state Ψ is determined by its *canonical component* $F(x_1, \dots, x_n; y_1, \dots, y_m)$ (the holomorphic part of $\Psi(\cdot; \uparrow^n \downarrow^m)$), antisymmetric in the x's and in the y's, each variable in the holomorphic class of $\mathcal{L}$; every other spin component is $\pm F$ with permuted arguments. In terms of second quantization, $\Psi = \sum_{I,I'} c_{I,I'}\, c^\dagger_{I\uparrow} c^\dagger_{I'\downarrow} |0\rangle$, with I and I' the sets of occupied up and down orbitals (up block first, increasing orbitals),

corresponds to $F = s_{n,m}\sum c_{I,I'}\det[\phi_{I_\alpha}(x_\beta)]\det[\phi_{I'_\alpha}(y_\beta)]$ with one sign $s_{n,m}$ per sector. We write $u=(x,y)$ for all coordinates in u-units, $U:=\sum_i u_i$, $\mathbf{1}=(1,\dots,1)$, and e_k for the k-th unit vector.

The many-body translations $T_K := t_1^{\otimes N}$, $T_1 := t_2^{\otimes N}$ act on canonical components. In u-units,

$$T_K^2 F(u) = F(u+\mathbf{1}/N), \qquad T_1^2 F(u) = e^{4\pi i U + 2\pi i\tau} F(u+\tau\mathbf{1}/N), \tag{C.4}$$

$T_K T_1 = e^{2\pi i N/L} T_1 T_K = -T_1 T_K$, and T_K^2, T_1^2 commute with T_K, T_1 and with each other. Since $t_1^L = t_2^L = 1$, $(T_K^2)^N = (T_1^2)^N = 1$. The joint eigenvalues $(\lambda_1,\lambda_2)\in\mu_N^2$ of (T_K^2, T_1^2) are the *relative quantum numbers*. In second quantization $T_K = e^{2\pi i K/L}$ with $K=\sum_j j\, n_j$, and T_1 maps the sector K to $K+N$.

C.1.3 The hollow-core Hamiltonian

The planar projector. For two particles put $Z=(z_1+z_2)/2$, $v=z_1-z_2$ (real Jacobian 1) and $\chi := (X_1Y_1+X_2Y_2)/2 = X_Z Y_Z + X_v Y_v/4$. The gauge factor $e^{i\chi}$ maps Landau-gauge LLL functions to symmetric-gauge ones: for $\psi = f(z_1,z_2)e^{-(Y_1^2+Y_2^2)/2}$,

$$\Psi_s := e^{i\chi}\psi = g(Z+\tfrac{v}{2}, Z-\tfrac{v}{2}) e^{-|Z|^2/2} e^{-|v|^2/8}, \qquad g := f\, e^{(z_1^2+z_2^2)/4} = f\, e^{Z^2/2+v^2/8}, \tag{C.5}$$

by the one-particle identity $e^{iXY/2}e^{-Y^2/2} = e^{z^2/4}e^{-|z|^2/4}$. The symmetric-gauge two-body LLL is $\{\text{centre}\}\otimes\{\text{relative}\}$, and the normalized relative state of angular momentum 1 is $\eta_1(v) = v\, e^{-|v|^2/8}/\sqrt{16\pi}$. The planar hollow-core pair interaction (the V_1 pseudopotential) is the projector onto relative angular momentum 1; in the Landau gauge it is $P_1 := e^{-i\chi}(1_Z\otimes|\eta_1\rangle\langle\eta_1|_v)e^{i\chi}$. For any ψ with $|\psi|$ bounded we use the absolutely convergent formula

$$(P_1\psi)(Z,v) := e^{-i\chi}\eta_1(v)\,\alpha_\psi(Z), \qquad \alpha_\psi(Z) := \int_{\mathbb{C}} \overline{\eta_1(v')}\,\Psi_s(Z,v')\,d^2v'. \tag{C.6}$$

The torus pair operator (image sum). Let $\mathcal{H}_2 := \mathcal{L}\otimes\mathcal{L}$ (two distinguishable particles) and $\tau_\nu^{(1)}$ the magnetic translation of particle 1 by ν. Define

$$\mathcal{P}\psi := \sum_{\nu\in\Lambda} \tau_\nu^{(1)} P_1 (\tau_\nu^{(1)})^{-1}\psi = \sum_{\nu\in\Lambda} c(\nu)^{-1}\,\tau_\nu^{(1)}(P_1\psi) \qquad (\psi\in\mathcal{H}_2), \tag{C.7}$$

the second form by (C.3). This is the torus V_1 pair interaction: the planar interaction periodized over the images of the relative coordinate. (By Poisson summation it equals the usual reciprocal-lattice form; that equivalence is not used here.) The hollow-core Hamiltonian is $H := \sum_{i<j}\mathcal{P}_{ij}$, spin-independent, on the antisymmetric N-electron space.

The orbital pair operators. Put $\mathfrak{q}^{d^2} := e^{i\pi\tau d^2/(2L)}$ (for rectangular $\tau = i\tau_2$ this is $e^{-\kappa^2 d^2/4}$, by (C.1)). For $n\in\mathbb{Z}$ and spins σ,σ' define

$$B_n^{\sigma\sigma'} := \sum_{A+B=n,\ A,B\in\mathbb{Z}} (A-B)\,\mathfrak{q}^{(A-B)^2}\, c_{[B]\sigma'} c_{[A]\sigma}, \qquad [A] := A \bmod L. \tag{C.8}$$

Replacing (A,B) by $(A+L,B+L)$ shows $B_{n+2L}=B_n$: the images at fixed covering centre $n/2$ run over $A-B\mapsto A-B+2L$. The four spin channels reduce to three: $B_n^{\downarrow\uparrow} = B_n^{\uparrow\downarrow}$ (substitute $A\leftrightarrow B$), the $S_z=0$ triplet. We write B_n^μ, $\mu\in\{+1,0,-1\}$, for $B_n^{\uparrow\uparrow}, B_n^{\uparrow\downarrow}, B_n^{\downarrow\downarrow}$.

Remark C.1 (the period-L grouping is a different operator). For fixed orbitals (a, b) the covering pairs $(a + kL, b + lL)$ with centre $n/2$ have $k + l$ fixed, so their relative coordinates form one class mod $2L$. The relative coordinate $d + L$ belongs to the centre $(n + L)/2$: indeed $B_{n+L} = \sum_{d\equiv n(2)}(d + L)\mathsf{q}^{(d+L)^2} c_{[(n-d)/2]}c_{[(n+d)/2]}$, a *different* linear combination of the same pair operators as $B_n = \sum_d d\,\mathsf{q}^{d^2} c_{[(n-d)/2]}c_{[(n+d)/2]}$. Grouping images with period L, $F_L(d) = \sum_k (d + kL)\mathsf{q}^{(d+kL)^2}$, produces $B_n + B_{n+L}$ (summing d over a window of length $2L$) or a truncation of it (window of length L). Neither equals $\sum_n B_n^\dagger B_n$.

C.2 The formulation lemma: the image sum unfolds to local pair conditions

Write $\mathsf{D}_{st} := \partial_s - \partial_t$ and $[\,\cdot\,]_{s=t}$ for restriction. For a two-body $\psi = fe^{-(Y_1^2+Y_2^2)/2} \in \mathcal{H}_2$ write $\mathsf{D}f := \mathsf{D}_{z_1 z_2} f$.

Lemma C.2 (the $m = 1$ amplitude). *For $\psi \in \mathcal{H}_2$ and every $Z \in \mathbb{C}$,*

$$\alpha_\psi(Z) = 2\sqrt{\pi}\; e^{(Z^2-|Z|^2)/2}\,(\mathsf{D}f)(Z, Z),$$
$$(P_1\psi)(z_1, z_2) = \tfrac{1}{2}\,(z_1 - z_2)\,e^{-(z_1-z_2)^2/8}\,(\mathsf{D}f)(Z, Z)\,e^{-(Y_1^2+Y_2^2)/2}.$$

Proof. By (C.5), $\Psi_s(Z, v) = h_Z(v)e^{-|Z|^2/2}e^{-|v|^2/8}$ with $h_Z(v) := g(Z + v/2, Z - v/2)$, an entire function of v. Since $|\psi|$ is continuous and Λ-periodic in each variable it is bounded, $|\psi| \le M$, and $|\Psi_s| = |\psi|$; hence $|h_Z(v)| \le Me^{|Z|^2/2}e^{|v|^2/8}$. Write $h_Z(v) = \sum_k a_k v^k$. On the disc $|v| < R$, rotational orthogonality gives $\int_{|v|<R} \bar v\, v^k e^{-|v|^2/4} d^2v = \delta_{k1} \int_{|v|<R} |v|^2 e^{-|v|^2/4} d^2v$, and the series converges uniformly there; so $\int_{|v|<R} \bar v\, h_Z(v) e^{-|v|^2/4} d^2v = a_1 \int_{|v|<R} |v|^2 e^{-|v|^2/4} d^2v \to 16\pi a_1$ as $R \to \infty$. The full integral converges absolutely, since the integrand is bounded by $Me^{|Z|^2/2}|v|e^{-|v|^2/8}$. Therefore $\alpha_\psi(Z) = e^{-|Z|^2/2}(16\pi)^{-1/2}\,16\pi\, h'_Z(0) = 4\sqrt{\pi}\, e^{-|Z|^2/2} h'_Z(0)$. Finally $h_Z(v) = f(Z + \frac{v}{2}, Z - \frac{v}{2})e^{Z^2/2}e^{v^2/8}$ and $e^{v^2/8}$ is even, so $h'_Z(0) = \frac{1}{2}e^{Z^2/2}(\mathsf{D}f)(Z, Z)$. This gives α_ψ. For $P_1\psi$ insert α_ψ in (C.6) and undo the gauge factor with $e^{-i\chi}e^{-(|z_1|^2+|z_2|^2)/4} = e^{-(z_1^2+z_2^2)/4}e^{-(Y_1^2+Y_2^2)/2}$ and $(z_1^2 + z_2^2)/4 = Z^2/2 + v^2/8$. □

Lemma C.3 (unfolding). *For $\phi, \psi \in \mathcal{H}_2$ the series (C.7) converges absolutely and locally uniformly, $\mathcal{P}\psi \in \mathcal{H}_2$, and*

$$\langle \phi, \mathcal{P}\psi\rangle = \int_D \overline{\alpha_\phi(Z)}\,\alpha_\psi(Z)\,d^2Z.$$

Hence $\mathcal{P}$ is self-adjoint and positive semidefinite on $\mathcal{H}_2$, and

$$\mathcal{P}\psi = 0 \iff (\mathsf{D}f)(Z, Z) = 0 \text{ for all } Z \in \mathbb{C}.$$

Proof. Convergence and membership. $|\alpha_\psi(Z)| \le \|\eta_1\|_{L^1} M$, so $|(P_1\psi)(z_1, z_2)| \le \|\eta_1\|_{L^1} M\,|\eta_1(z_1 - z_2)|$. The ν-th term of (C.7) has modulus $|(P_1\psi)(z_1 - \nu, z_2)|$, a Gaussian in $z_1 - z_2 - \nu$; the series converges absolutely and locally uniformly. Each term is (a unimodular factor times) a planar LLL function, i.e. by Lemma C.2 of the form (entire) $\cdot\, e^{-(Y_1^2+Y_2^2)/2}$, and the holomorphic parts converge locally uniformly, so the limit is again of this form. The boundary conditions: for $\mu \in \Lambda$, $\tau_\mu^{(1)}\tau_\nu^{(1)} = \omega(\mu, \nu)\tau_{\mu+\nu}^{(1)}$ with constant phases ω, and applying both sides to ψ gives $c(\mu)c(\nu) = \omega(\mu, \nu)c(\mu + \nu)$; hence $\tau_\mu^{(1)}\mathcal{P}\psi = \sum_\nu c(\nu)^{-1}\omega(\mu, \nu)\tau_{\mu+\nu}^{(1)}P_1\psi = c(\mu)\mathcal{P}\psi$. For particle 2: the simultaneous translation $S_\nu := \tau_\nu^{(1)}\tau_\nu^{(2)}$ acts on $\mathcal{H}_2$ by the phase $c(\nu)^2$ and commutes with P_1. Indeed, for *any* function, $e^{i\chi}S_\nu e^{-i\chi}$ acts as $\Psi_s(Z, v) \mapsto e^{i(-\nu_2 X_Z + \nu_1 Y_Z + \nu_1\nu_2)}\Psi_s(Z - \nu, v)$ (from $\chi = X_Z Y_Z + X_v Y_v/4$ and the definition of τ_a): a translation of Z times a phase depending on Z

only, while P_1 acts on v only. Combined with the boundary conditions, this gives (C.9) below. Since $\tau^{(2)}_{-\nu}$ equals $\tau^{(1)}_{\nu} S_{-\nu}$ up to a constant phase, $\tau^{(2)}_{-\nu} P_1 (\tau^{(2)}_{-\nu})^{-1} = \tau^{(1)}_{\nu} P_1 (\tau^{(1)}_{\nu})^{-1}$. So (C.7) is also the image sum over translations of particle 2, and the same cocycle argument gives the boundary conditions in z_2. So $\mathcal{P}\psi \in \mathcal{H}_2$.

Unfolding. By dominated convergence, $\langle \phi, \mathcal{P}\psi \rangle = \sum_\nu c(\nu)^{-1} \int_{D\times D} \bar{\phi}\ \tau^{(1)}_{\nu}(P_1\psi)$. Since $\tau^{(1)}_{\nu}$ is a translation of z_1 by ν followed by multiplication by a unimodular function, $\int_{D\times D} \bar{a}\, \tau^{(1)}_{\nu} b = \int_{(D-\nu)\times D} \overline{(\tau^{(1)}_{\nu})^{-1} a}\, b$, and $(\tau^{(1)}_{\nu})^{-1}\phi = c(\nu)^{-1}\phi$, so $\overline{(\tau^{(1)}_{\nu})^{-1}\phi} = c(\nu)\bar{\phi}$. The phases cancel and the translates $D - \nu$ tile $\mathbb{C}$:

$$\langle \phi, \mathcal{P}\psi \rangle = \int_{z_1\in\mathbb{C},\ z_2\in D} \bar{\phi}\ P_1\psi\ d^2z_1 d^2z_2,$$

absolutely convergent. With $\Phi_s := e^{i\chi}\phi$ the gauge factors cancel: $\bar{\phi}\, P_1\psi = \overline{\Phi_s}\, \eta_1(v)\alpha_\psi(Z)$. In the coordinates (Z, v) (Jacobian 1) the domain is $\{v \in \mathbb{C},\ Z - v/2 \in D\}$.

Periodicity in Z. Let $\lambda \in \Lambda$. From (C.3) for each particle,

$$\phi(z_1 - \lambda, z_2 - \lambda) = c(\lambda)^2 e^{i\lambda_2(X_1 + X_2 - 2\lambda_1)} \phi(z_1, z_2),$$

and with $\chi = X_Z Y_Z + X_v Y_v/4$,

$$\Phi_s(Z - \lambda, v) = c(\lambda)^2\, e^{i(\lambda_2 X_Z - \lambda_1 Y_Z - \lambda_1\lambda_2)}\, \Phi_s(Z, v), \tag{C.9}$$

a phase that does not depend on v. The same holds for Ψ_s, hence for α_ψ with the same phase. Therefore $\Phi(Z, v) := \overline{\Phi_s(Z, v)}\eta_1(v)\alpha_\psi(Z)$ is Λ-periodic in Z at fixed v, and $|\Phi| \le \sup|\phi|\, \|\eta_1\|_{L^1} M\, |\eta_1(v)|$ is integrable on $D \times \mathbb{C}$. For a periodic integrable function the integral over $D + v/2$ equals the integral over D, so by Fubini

$$\langle \phi, \mathcal{P}\psi \rangle = \int_{\mathbb{C}} d^2v \int_D d^2Z\, \Phi(Z, v) = \int_D \alpha_\psi(Z) \overline{\int_{\mathbb{C}} \overline{\eta_1(v)}\, \Phi_s(Z, v)\, d^2v}\, d^2Z = \int_D \overline{\alpha_\phi}\, \alpha_\psi.$$

Consequences. The right side is Hermitian and nonnegative, so $\mathcal{P} = \mathcal{P}^\dagger \ge 0$ on the finite-dimensional $\mathcal{H}_2$, and $\mathcal{P}\psi = 0 \iff \langle \psi, \mathcal{P}\psi \rangle = 0 \iff \alpha_\psi = 0$ a.e. on D. By Lemma C.2 this means $(\mathsf{D}f)(Z, Z) = 0$ on D, hence on $\mathbb{C}$, by the identity theorem for the entire function $Z \mapsto (\mathsf{D}f)(Z, Z)$. □

Lemma C.4 (orbital form; the period is $2L$)**.** *Let $\psi = \sum_{a,b\in\mathbb{Z}_L} \psi_{ab}\, \phi_a(z_1)\phi_b(z_2) e^{-(Y_1^2 + Y_2^2)/2} \in \mathcal{H}_2$ and*

$$\beta_n(\psi) := \sum_{A+B=n,\ A,B\in\mathbb{Z}} (A - B)\, \mathsf{q}^{(A-B)^2}\, \psi_{[A][B]} \qquad (n \in \mathbb{Z}).$$

Then $\beta_{n+2L} = \beta_n$,

$$(\mathsf{D}f)(Z, Z) = i\kappa \sum_{n\in\mathbb{Z}} e^{i\kappa n Z + i\pi\tau n^2/(2L)}\, \beta_n(\psi),$$

$$\langle \psi, \mathcal{P}\psi \rangle = c_L \sum_{n\in\mathbb{Z}_{2L}} |\beta_n(\psi)|^2, \qquad c_L = 4\pi\kappa^2 L_1 \sqrt{\pi/2}.$$

In particular $\mathcal{P}\psi = 0 \iff \beta_n(\psi) = 0$ for all $n \in \mathbb{Z}_{2L}$.

Proof. Differentiating (C.2) term by term (absolutely convergent, Gaussian in A, B), $(\mathsf{D}f)(Z, Z) = \sum_{a,b} \psi_{ab} \sum_{A\equiv a, B\equiv b} i\kappa(A - B) e^{i\kappa(A+B)Z} e^{i\pi\tau(A^2+B^2)/L}$. Group by $n = A + B$ and use $A^2 + B^2 = \frac{1}{2}(n^2 + (A - B)^2)$. Periodicity in n: $(A, B) \mapsto (A + L, B + L)$ keeps $A - B$ and the residues. For

the norm, take the rectangular domain D of Section C.1.1. At fixed Y the X-integral over a period kills the cross terms $n \neq n'$; with $|e^{i\kappa nZ + i\pi\tau n^2/(2L)}|^2 = e^{-2\kappa nY - \pi\tau_2 n^2/L}$ and $\pi\tau_2/L = \kappa^2/2$ (by (C.1)), and $|\alpha_\psi(Z)|^2 = 4\pi e^{-2Y^2}|(\mathsf{D}f)(Z,Z)|^2$ (Lemma C.2), Lemma C.3 gives

$$\langle \psi, \mathcal{P}\psi\rangle = 4\pi \int_D e^{-2Y^2}|(\mathsf{D}f)(Z,Z)|^2 = 4\pi\kappa^2 L_1 \sum_{n\in\mathbb{Z}} |\beta_n|^2 \int_0^{\kappa L} e^{-2(Y+\kappa n/2)^2} dY.$$

Since $L_1\tau_2 = \kappa L$, summing the integrals over $n \equiv n_0 \pmod{2L}$ shifts the interval by κL each time and tiles $\mathbb{R}$, giving $\int_{\mathbb{R}} e^{-2Y^2} dY = \sqrt{\pi/2}$. Finally, $(\mathsf{D}f)(Z,Z) \equiv 0$ forces every $\beta_n = 0$: at fixed Y the series is a Fourier series in X with coefficients $i\kappa e^{-\kappa nY} e^{i\pi\tau n^2/(2L)}\beta_n$. □

Lemma C.5 (exchange and spin)**.** *$\mathcal{P}$ commutes with the exchange $E : (z_1, z_2) \mapsto (z_2, z_1)$ and vanishes on E-even functions. Hence, on the antisymmetric two-electron space, $\mathcal{P} \otimes 1_{\text{spin}} = \mathcal{P} \otimes \Pi^{S=1}$.*

Proof. $\mathsf{D}(f \circ E)(Z,Z) = -(\mathsf{D}f)(Z,Z)$, so $\alpha_{E\psi} = -\alpha_\psi$; the form of Lemma C.3 is E-invariant, so $[\mathcal{P}, E] = 0$, and $\alpha_\psi = 0$ for E-even ψ, so $\mathcal{P}\psi = 0$. The antisymmetric two-electron space is the direct sum of the tensor products of the E-odd functions with the symmetric spin states, which span the triplet $S = 1$, and of the E-even functions with the antisymmetric spin state, the singlet $S = 0$. Since $[\mathcal{P}, E] = 0$, $\mathcal{P} \otimes 1_{\text{spin}}$ maps each summand into itself; on the second it vanishes, as does $\mathcal{P} \otimes \Pi^{S=1}$, and on the first $\Pi^{S=1}$ acts as the identity. Hence $\mathcal{P} \otimes 1_{\text{spin}} = \mathcal{P} \otimes \Pi^{S=1}$ on the antisymmetric space. □

Lemma C.6 (formulation lemma on the torus)**.** *Let Ψ have $N_\uparrow = n$, $N_\downarrow = m$ and canonical component F. The following are equivalent.*

(1) $H\Psi = 0$.

(2) (C1) *For all $a < b$, F vanishes to order ≥ 3 along the divisor $\{x_a - x_b \in \Lambda\}$, and likewise for the y's.* (C2) *For all a, k, $[\mathsf{D}_{x_a y_k} F]_{x_a = y_k} = 0$ identically, on the covering diagonal $x_a = y_k$ of $\mathbb{C}^N$.*

(3) $B_n^\mu \Psi = 0$ *for all $n \in \mathbb{Z}_{2L}$ and $\mu \in \{+1, 0, -1\}$.*

Moreover $H = \frac{c'_L}{2} \sum_{n\in\mathbb{Z}_{2L}} \sum_{\sigma,\sigma'} (B_n^{\sigma\sigma'})^\dagger B_n^{\sigma\sigma'}$, *with $c'_L = c_L/(L_1\sqrt{\pi})^2$ and orbitals normalized to* 1.

Proof. (1) $\iff$ ($\mathcal{P}_{ij}\Psi = 0$ for all $i < j$), because H is a sum of positive operators. $\mathcal{P}_{ij}$ acts on the dependence on (z_i, z_j) of each spin component. Write $\Psi(\cdot;\sigma) = \sum_\alpha f_\alpha(z_i, z_j) g_\alpha(\text{rest})$ with linearly independent g_α. By Lemma C.3, $\mathcal{P}_{ij}\Psi = 0$ iff every $\mathsf{D}f_\alpha$ vanishes on the diagonal, i.e. iff $[\mathsf{D}_{z_i z_j}\Psi(\cdot;\sigma)]_{z_i = z_j} \equiv 0$. As σ and (i, j) range, (z_i, z_j) runs over all pairs of arguments of F: every spin component is $\pm F$ with permuted arguments (Section C.1.2), and the component $\Psi(\cdot; \uparrow^n \downarrow^m)$, whose holomorphic part is F, already produces every pair. So (1) holds if and only if $[\mathsf{D}_{st}F]_{s=t} \equiv 0$ for every pair (s, t) of arguments of F.

Same-spin pairs. F is antisymmetric in (x_a, x_b). Near a point of the diagonal, $F = (x_a - x_b)G$ in the local ring with G symmetric, and $[\mathsf{D}F]_{x_a = x_b} = 2G|$. So the condition says $G = (x_a - x_b)G'$ with G' antisymmetric, hence $G' = (x_a - x_b)G''$: order ≥ 3 along the covering diagonal. Conversely order ≥ 3 gives $[\mathsf{D}F] = 0$. The components $\{x_a - x_b = \lambda\}$, $\lambda \in \Lambda$, are translates under $x_a \mapsto x_a + \lambda$. That map multiplies F by the nonvanishing entire boundary-condition multiplier, so vanishing orders agree on all components. This is (C1).

Up–down pairs. The condition is (C2) verbatim.

(2) $\Longleftrightarrow$ (3). Expand F in its first up and first down variables, $F = \sum_{a,b} \phi_a(x_1)\phi_b(y_1)F_{ab}(x';y')$. By Lemma C.4, with the rest as spectators, $[\mathsf{D}_{x_1y_1}F]_{x_1=y_1} \equiv 0$ iff $\sum_{A+B=n}(A-B)\mathsf{q}^{(A-B)^2}F_{[A][B]} \equiv 0$ for every n. The Laplace expansion gives $F_{ab} = (-1)^{\mathrm{pos}_I(a)+\mathrm{pos}_{I'}(b)}\, s_{n,m} \sum c_{I,I'} \det[\phi_{I\setminus a}] \det[\phi_{I'\setminus b}]$. Moving $c_{b\downarrow}c_{a\uparrow}$ through $c^\dagger_{I\uparrow}c^\dagger_{I'\downarrow}$ produces the sign $(-1)^{\mathrm{pos}_I(a)+\mathrm{pos}_{I'}(b)+n}$. So $F_{ab} = \pm$ (canonical component of $c_{b\downarrow}c_{a\uparrow}\Psi$), with a sign that depends only on (n,m). The condition is therefore that the canonical component of $B_n^{\uparrow\downarrow}\Psi$ vanishes, i.e. $B_n^{\uparrow\downarrow}\Psi = 0$; the other up–down pairs follow by antisymmetry. Same-spin pairs work the same way with $B_n^{\uparrow\uparrow}$, $B_n^{\downarrow\downarrow}$, and Lemma C.5 identifies the up–down channel with the triplet $\mu = 0$.

The operator identity. In the orthonormal basis $\phi_a/\|\phi_a\|$, Lemma C.4 says

$$\mathcal{P} = c'_L \sum_{n\in\mathbb{Z}_{2L}} |b_n\rangle\langle b_n|,$$

where $\langle b_n|\psi\rangle = \beta_n(\psi)$. The second quantization of the spin-independent two-body operator $\sum_{i<j}\mathcal{P}_{ij}$ is the stated sum. □

Remark C.7 (image coincidences). The torus coincidence $x_a = y_k + \lambda$, $\lambda \in \Lambda \setminus 0$, imposes nothing beyond (C2). Its condition is the image of (C2) under the magnetic translation $x_a \mapsto x_a + \lambda$, and by (C.3) that image is $c(\lambda)$ times (C2) itself. It is *not* the naive condition $[\mathsf{D}_{x_a y_k}F]_{x_a=y_k+\lambda} = 0$: the boundary-condition multiplier $e^{-i\pi L(2x_a/L_1+\tau)}$ depends on x_a, and its derivative mixes the value of F into the derivative. In the variables of Lemma C.12 below the multipliers are constants, and the condition at images becomes literally the zero-residue condition there.

C.3 Theta functions and the centre-of-mass factorization

C.3.1 Facts about theta functions

In this section and the next we use u-units (periods 1 and τ; $\Lambda_u := \mathbb{Z} + \tau\mathbb{Z}$). From the series, for all a, b:

$$\begin{gathered}\vartheta[a,b](u+1) = e^{2\pi i a}\vartheta[a,b](u), \qquad \vartheta[a,b](u+\tau) = e^{-i\pi\tau-2\pi i(u+b)}\vartheta[a,b](u),\\ \vartheta[a,b](-u) = \vartheta[-a,-b](u),\end{gathered} \tag{C.10}$$

and $\vartheta[a+1,b] = \vartheta[a,b]$, $\vartheta[a,b+1] = e^{2\pi i a}\vartheta[a,b]$. (For the second relation, shift $n \mapsto n-1$ after completing the square; for the third, substitute $n \mapsto -n-2a$ when $2a \in \mathbb{Z}$, and $n \mapsto -n$ in general after rewriting $(n+a) \mapsto -(n+a)$.)

Lemma C.8. *(a) For $\ell \geq 1$ and $a, b \in \mathbb{R}$, the space $\Theta_\ell[a,b]$ of entire F with $F(U+1) = e^{2\pi i a}F(U)$ and $F(U+\tau) = e^{-2\pi i b}e^{-i\pi\ell(2U+\tau)}F(U)$ has dimension ℓ.*

(b) A nonzero $F \in \Theta_\ell[a,b]$ has exactly ℓ zeros, counted with multiplicity, in every period parallelogram whose boundary avoids its zeros. (This part is used only in the budget, Lemma C.13(e), and is not formalized; Appendix E.)

(c) $\vartheta_1 := \vartheta[\frac{1}{2},\frac{1}{2}](\cdot|\tau)$ is odd and satisfies $\vartheta_1(u+1) = -\vartheta_1(u)$, $\vartheta_1(u+\tau) = -e^{-i\pi\tau-2\pi iu}\vartheta_1(u)$. Its zero set is exactly Λ_u, all zeros are simple, and $\vartheta_1'(0) \neq 0$.

(d) $\wp(u) := -\frac{d^2}{du^2}\log\vartheta_1(u|\tau)$ is meromorphic, even and doubly periodic, with poles exactly at Λ_u and $\wp(u) = u^{-2} + O(1)$ at 0 (no u^{-1} term).

(e) *For $\varepsilon \in \{\pm 1\}^2 \setminus \{(1,1)\}$ let $\chi_\varepsilon(p+q\tau) := \varepsilon_1^p \varepsilon_2^q$ on Λ_u. There is an even meromorphic $\wp_\varepsilon$ with $\wp_\varepsilon(u+\lambda) = \chi_\varepsilon(\lambda)\wp_\varepsilon(u)$ ($\lambda \in \Lambda_u$), poles exactly at Λ_u, and $\wp_\varepsilon(u) = u^{-2} + O(1)$ at 0.*

(f) *$J := \prod_{i<j} \vartheta_1(u_i - u_j)$ on $\mathbb{C}^N$: J^2 is symmetric under all permutations and invariant under $u \mapsto u + c\mathbf{1}$, with $J^2(u+e_k) = J^2(u)$ and $J^2(u+\tau e_k) = e^{-2\pi i(N-1)\tau - 4\pi i(Nu_k - U)} J^2(u)$. It vanishes to order exactly 2 along each divisor $\{u_i - u_j \in \Lambda_u\}$ and nowhere else.*

Proof. (a) $G(U) := F(U)e^{-2\pi i a U}$ is entire and 1-periodic, so $G(U) = H(e^{2\pi i U})$ with H holomorphic on $\mathbb{C}^\times$. Its Laurent series gives $F(U) = \sum_{n\in\mathbb{Z}} c_n e^{2\pi i(n+a)U}$, converging absolutely and locally uniformly. Comparing coefficients in the second relation (Laurent coefficients are unique) gives $c_{n+\ell} = c_n \exp(2\pi i b + i\pi\ell\tau + 2\pi i(n+a)\tau)$. Hence $|c_{n+k\ell}| = |c_n| \exp(-\pi\tau_2\ell k^2 - 2\pi\tau_2 k(n+a))$ for all $k \in \mathbb{Z}$, by the recursion and its inverse. Conversely, any $c_0, \dots, c_{\ell-1}$ extended this way have Gaussian decay, so the series converges to an element of $\Theta_\ell[a,b]$. So $\dim = \ell$. (b) Argument principle on $P = U_0 + [0,1] + [0,1]\tau$. By the relations, $F'/F(U+1) = F'/F(U)$ and $F'/F(U+\tau) = F'/F(U) - 2\pi i\ell$. The two sides parallel to τ cancel; the two sides parallel to 1 contribute $\frac{1}{2\pi i}\int_{U_0}^{U_0+1} 2\pi i\ell\, dU = \ell$. (c) The relations follow from (C.10), and oddness from $\vartheta_1(-u) = \vartheta[-\frac12,-\frac12](u) = \vartheta[\frac12,-\frac12](u) = e^{-i\pi}\vartheta_1(u)$. So $\vartheta_1 \in \Theta_1[\frac12,\frac12]$, and $\vartheta_1 \not\equiv 0$ (its Fourier coefficients are nonzero). By (b) it has exactly one zero per cell, simple; since it is odd, $\vartheta_1(0) = 0$. So the zeros are Λ_u, all simple. (d) Locally, $\log\vartheta_1(u+1) = \log\vartheta_1(u) + \text{const}$ and $\log\vartheta_1(u+\tau) = \log\vartheta_1(u) - 2\pi i u + \text{const}$, so the second derivative is doubly periodic. Oddness gives evenness. Near 0, $\vartheta_1(u) = \vartheta_1'(0)u(1 + c_2u^2 + \dots)$, so $\log\vartheta_1 = \log u + \text{const} + c_2u^2 + \dots$ and $\wp = u^{-2} - 2c_2 + O(u^2)$. (e) $\Lambda_\varepsilon := \ker\chi_\varepsilon$ has index 2 in Λ_u; pick a positively oriented basis (ω_1,ω_2) of Λ_ε and $\lambda_\varepsilon \in \Lambda_u \setminus \Lambda_\varepsilon$. By (d) for the lattice Λ_ε, $\wp^{(\varepsilon)}(u) := -\frac{d^2}{du^2}\log\vartheta_1(u/\omega_1 \,|\, \omega_2/\omega_1)$ is even and Λ_ε-periodic, with poles exactly at Λ_ε and $\wp^{(\varepsilon)}(u) = u^{-2} + O(1)$ at 0. Put $\wp_\varepsilon(u) := \wp^{(\varepsilon)}(u) - \wp^{(\varepsilon)}(u - \lambda_\varepsilon)$. Since $2\lambda_\varepsilon \in \Lambda_\varepsilon$, $\wp_\varepsilon(u+\lambda_\varepsilon) = \wp^{(\varepsilon)}(u-\lambda_\varepsilon) - \wp^{(\varepsilon)}(u) = -\wp_\varepsilon(u)$, and $\wp_\varepsilon$ is invariant under Λ_ε. Evenness: $\wp_\varepsilon(-u) = \wp^{(\varepsilon)}(u) - \wp^{(\varepsilon)}(u+\lambda_\varepsilon) = \wp_\varepsilon(u)$. The poles are $\Lambda_\varepsilon \cup (\lambda_\varepsilon + \Lambda_\varepsilon) = \Lambda_u$. (f) Symmetry: ϑ_1 is odd, so the squares are invariant. Multipliers: by (c), each factor with k in either slot contributes $e^{-2\pi i\tau - 4\pi i(u_k - u_j)}$ under $u_k \mapsto u_k + \tau$, and there are $N-1$ such factors. Vanishing: by (c). □

Remark C.9. By the uniqueness in Proposition C.16, $\wp_\varepsilon$ is a multiple of Read and Rezayi's twisted lattice sums $\wp_{a,b}$ [RR96, eq. (80)].

C.3.2 The function spaces

Lemma C.10 (evenness: (C2) is a zero-residue condition)**.** *Near a generic point of $\{x_a = y_k\}$ let $\mathsf{a} = S\,\vartheta_1(x_a - y_k)^2\, g$. Here S is holomorphic, invariant under $x_a \leftrightarrow y_k$, and not identically zero on the diagonal; g is meromorphic with at most a double pole along the diagonal. Write $g(R + \frac{v}{2}, R - \frac{v}{2}, \cdot) = \mathcal{A}(R,\cdot)v^{-2} + \mathcal{B}(R,\cdot)v^{-1} + O(1)$ (fixed pair centre R). Then $[\mathsf{D}_{x_a y_k}\mathsf{a}]_{x_a = y_k} \equiv 0$ iff $\mathcal{B} \equiv 0$.*

Proof. At fixed R and fixed other variables, S and $\vartheta_1(v)^2 = \vartheta_1'(0)^2 v^2(1 + O(v^2))$ are even in v, so $S\vartheta_1(v)^2 = e_2v^2 + e_4v^4 + \cdots$ with $e_2 = S|_{v=0}\vartheta_1'(0)^2$. Since $\mathsf{D}_{x_a y_k} = 2\partial_v$ at fixed R, $[\mathsf{Da}]_{v=0} = 2[v^1](S\vartheta_1^2 g) = 2e_2\mathcal{B}$, and $e_2 \not\equiv 0$. □

Definition C.11. For $n, m \geq 0$, $N = n + m$, and $\varepsilon = (\varepsilon_1, \varepsilon_2) \in U(1)^2$, let $\mathcal{G}(n, m; \varepsilon)$ be the space of meromorphic $g(x_1, \dots, x_n; y_1, \dots, y_m)$ on $\mathbb{C}^N$ such that

(i) $g(u + c\mathbf{1}) = g(u)$ for all $c \in \mathbb{C}$;

(ii) $g(u + e_k) = \varepsilon_1 g(u)$ and $g(u + \tau e_k) = \varepsilon_2 g(u)$ for every k;

(iii) g is antisymmetric in the x's and in the y's;

(iv) the polar set of g lies in $\bigcup_{a,k}\{x_a - y_k \in \Lambda_u\}$, with poles of order ≤ 2;

(v) at fixed pair centre, the v^{-1} coefficient of g at $x_a = y_k$ vanishes: in the notation of Lemma C.10, $\mathcal{B} \equiv 0$.

$\mathcal{G}(0,0;\varepsilon) := \mathbb{C}$. By (ii), (v) at the components $x_a - y_k \in \Lambda_u \setminus 0$ follows from (v) at $x_a = y_k$ (translate x_a; the multiplier is a constant). Also, g is regular on same-spin coincidences and vanishes there, by (iii) and (iv).

C.3.3 The factorization

Lemma C.12 (centre-of-mass factorization)**.** *Fix n, m with $N = n + m \ge 1$, $L = 2N$, and let $V_{n,m}$ be the space of canonical components satisfying* (C1), (C2), *i.e.* $\ker H|_{n,m}$ *by Lemma C.6.*

(a) $\mathcal{G}(n,m;\varepsilon) = 0$ unless $\varepsilon_1^N = \varepsilon_2^N = 1$.

(b) For $\varepsilon \in \mu_N^2$ let $\Theta_2[\varepsilon]$ be the space of entire F with $F(U+1) = \varepsilon_1^{-1}F(U)$ and $F(U+\tau) = \varepsilon_2^{-1}e^{-4\pi i U - 2\pi i\tau}F(U)$. It is two-dimensional, and

$$\bigoplus_{\varepsilon\in\mu_N^2} \Theta_2[\varepsilon]\otimes\mathcal{G}(n,m;\varepsilon) \;\longrightarrow\; V_{n,m}, \qquad F\otimes g \mapsto F(U)\,J^2(u)\,g(u),$$

is an isomorphism. In particular $\dim\ker H|_{n,m} = 2\sum_{\varepsilon\in\mu_N^2}\dim\mathcal{G}(n,m;\varepsilon)$.

(c) On the image of $\Theta_2[\varepsilon]\otimes\mathcal{G}(n,m;\varepsilon)$, $(T_K^2, T_1^2) = (\varepsilon_1^{-1}, \varepsilon_2^{-1})$.

Proof. (a) By (i), $g(u+\mathbf{1}) = g(u)$. By (ii) applied to each of the N coordinates, $g(u+\mathbf{1}) = \varepsilon_1^N g(u)$; so $\varepsilon_1^N = 1$ if $g \ne 0$. The same with τ.

$\dim\Theta_2[\varepsilon] = 2$. $F(U+\tau) = \varepsilon_2^{-1}e^{-i\pi\cdot 2(2U+\tau)}F(U)$, so $\Theta_2[\varepsilon] = \Theta_2[a,b]$ with $e^{2\pi i a} = \varepsilon_1^{-1}$ and $e^{-2\pi i b} = \varepsilon_2^{-1}$. Apply Lemma C.8(a).

(b), "the map is well defined". Let $F \in \Theta_2[\varepsilon]$, $g \in \mathcal{G}(n,m;\varepsilon)$ and $f := FJ^2g$. By (iv) and Lemma C.8(f), $J^2 g$ has no poles along any divisor, so it is holomorphic. Boundary conditions: under $u_k \mapsto u_k + 1$ the factors give $\varepsilon_1^{-1}\cdot 1\cdot\varepsilon_1 = 1$; under $u_k \mapsto u_k + \tau$ they give $\varepsilon_2^{-1}e^{-4\pi i U - 2\pi i\tau}\cdot e^{-2\pi i(N-1)\tau - 4\pi i(Nu_k - U)}\cdot\varepsilon_2 = e^{-i\pi L(2u_k+\tau)}$, as required. (C1): J^2 vanishes to order 2 on same-spin divisors and g vanishes there, so f vanishes to order ≥ 3. (C2): apply Lemma C.10 with $S = F(U)\prod_z \vartheta_1(x_a - z)^2\vartheta_1(y_k - z)^2\cdot$(factors without x_a, y_k), which is invariant under $x_a \leftrightarrow y_k$.

"Injective". Suppose $\sum_\varepsilon\sum_\ell F_{\varepsilon,\ell}(U)J^2 g_{\varepsilon,\ell} = 0$, where for each ε the $F_{\varepsilon,\ell}$ ($\ell = 1,2$) are a basis of $\Theta_2[\varepsilon]$. Divide by J^2 and replace u by $u + c\mathbf{1}$. The g's are translation-invariant, so $\sum_{\varepsilon,\ell} F_{\varepsilon,\ell}(U+Nc)g_{\varepsilon,\ell}(u) = 0$ for all c. The $2N^2$ functions $c \mapsto F_{\varepsilon,\ell}(U+Nc)$ are linearly independent: they are independent within each $\Theta_2[\varepsilon]$, and the spaces for different ε have different multipliers under $c \mapsto c + 1/N$ and $c \mapsto c + \tau/N$, i.e. they lie in different joint eigenspaces of two commuting operators. Hence every $g_{\varepsilon,\ell}(u) = 0$.

"Surjective". T_K^2 and T_1^2 preserve $V_{n,m}$: (C1), (C2) and the boundary conditions are invariant under simultaneous translations, and the gauge factor $e^{4\pi i U + 2\pi i\tau}$ in (C.4) depends on U only, which D_{st} annihilates. They commute and have finite order, so $V_{n,m} = \bigoplus V_\lambda$ over joint eigenvalues $\lambda = (\lambda_1, \lambda_2) \in \mu_N^2$. Let $f \in V_\lambda$ and fix u. By (C.4), $\varphi(c) := f(u + c\mathbf{1})$ satisfies $\varphi(c+1/N) = \lambda_1\varphi(c)$ and $\varphi(c+\tau/N) = \lambda_2 e^{-4\pi i(U+Nc) - 2\pi i\tau}\varphi(c)$. So $C \mapsto \varphi((C-U)/N)$ lies in $\Theta_2[\varepsilon]$ with $\varepsilon = (\lambda_1^{-1}, \lambda_2^{-1})$. With a basis F_1, F_2 of $\Theta_2[\varepsilon]$: $f(u + c\mathbf{1}) = \sum_\ell a_\ell(u)F_\ell(U + Nc)$. The coefficients are unique and are

obtained by solving a 2×2 system from two values of c, so they are holomorphic in u. Comparing at u and $u+c'\mathbf{1}$ shows $a_\ell(u+c'\mathbf{1})=a_\ell(u)$. At $c=0$: $f=\sum_\ell F_\ell(U)a_\ell(u)$.

The conditions pass to each a_ℓ. $\mathsf{D}_{x_ax_b}$ and $\mathsf{D}_{x_ay_k}$ annihilate functions of U. Vanishing to order ≥ 3 along $\{x_a=x_b\}$ means $\mathsf{D}^j_{x_ax_b}f=0$ on the divisor for $j=0,1,2$, i.e. $\sum_\ell F_\ell(U)(\mathsf{D}^j_{x_ax_b}a_\ell)(u)=0$ there. The divisor and the functions $\mathsf{D}^j a_\ell$ are translation-invariant, so $\sum_\ell F_\ell(U+Nc)(\mathsf{D}^j a_\ell)(u)=0$ for all c. By independence of the F_ℓ, each $\mathsf{D}^j a_\ell=0$ on the divisor. The same argument gives (C2) and the vanishing at image components.

Multipliers. From $f(u+e_k)=f(u)$ and $F_\ell(U+1)=\varepsilon_1^{-1}F_\ell(U)$, the translation trick gives $a_\ell(u+e_k)=\varepsilon_1 a_\ell(u)$. From $f(u+\tau e_k)=e^{-i\pi L(2u_k+\tau)}f(u)$, $a_\ell(u+\tau e_k)=\varepsilon_2 e^{-4\pi i(Nu_k-U)-2\pi i(N-1)\tau}a_\ell(u)$. By Lemma C.8(f), $g_\ell:=a_\ell/J^2$ then satisfies (ii) with constant multipliers ε, and (i). (iii) is inherited from f.

(iv), (v). g_ℓ is meromorphic with poles only on coincidence divisors, of order ≤ 2 (a_ℓ is holomorphic, and J^2 vanishes to order exactly 2). On same-spin divisors a_ℓ vanishes to order ≥ 3, so g_ℓ is regular there. (v) follows from (C2) for a_ℓ and Lemma C.10. So $g_\ell\in\mathcal{G}(n,m;\varepsilon)$ and f is in the image.

(c) $T_K^2(F(U)a)=F(U+1)a=\varepsilon_1^{-1}F(U)a$, and $T_1^2(F(U)a)=e^{4\pi iU+2\pi i\tau}F(U+\tau)a=\varepsilon_2^{-1}F(U)a$. □

C.4 The merging lemma

For $n,m\geq 1$ write rest $:=(x_2,\dots,x_n;y_2,\dots,y_m)$, $Z':=\sum\text{rest}$, J' for the product of ϑ_1 over pairs of rest, and $M:=\{x_1=y_1\}\cong\mathbb{C}^{N-1}$ with coordinate w for the common value.

Lemma C.13 (torus merging lemma)**.** *Let $n,m\geq 1$ and $g\in\mathcal{G}(n,m;\varepsilon)$. Then*

$$\mathcal{A}(w;\text{rest}):=\vartheta_1'(0)^{-2}\big[\vartheta_1(x_1-y_1)^2 g\big]_{x_1=y_1=w}=\lim_{v\to 0}v^2 g\big(w+\tfrac{v}{2},x';w-\tfrac{v}{2},y'\big)$$

is a well-defined meromorphic function on M, and:

(a) for rest *outside the divisors $\{x_a-y_k\in\Lambda_u\}$ ($a,k\geq 2$), $w\mapsto\mathcal{A}(w;\text{rest})$ is entire;*

(b) $\mathcal{A}(w+1;\cdot)=\varepsilon_1^2\mathcal{A}(w;\cdot)$ and $\mathcal{A}(w+\tau;\cdot)=\varepsilon_2^2\mathcal{A}(w;\cdot)$;

(c) $\mathcal{A}$ does not depend on w, and $\mathcal{A}\equiv 0$ unless $\varepsilon_1^2=\varepsilon_2^2=1$;

(d) $\mu(g):=\mathcal{A}\in\mathcal{G}(n-1,m-1;\varepsilon)$;

(e) (budget) *for $F\in\Theta_2[\varepsilon]$ and $f:=FJ^2g$,*

$$f|_M=\vartheta_1'(0)^2\,F(2w+Z')\prod_{z\in\text{rest}}\vartheta_1(w-z)^4\;J'^2\;\mu(g).$$

> *As a function of w, $f|_M$ satisfies the level-$4N$ relations, so it has $4N$ zeros per cell if it is nonzero. Of these, $4(N-2)$ lie at $w\in\text{rest}+\Lambda_u$ (order 4 each), and the remaining $8=4\cdot 2$ are the zeros of $F(2w+Z')$. The relative part has no w-dependence at all: the budget is 0. Nothing in the sequel uses (e); it is a check on the count, and it is not formalized (Appendix E).*

Proof. Step 0 (set-up). $\mathsf{a}:=J^2g$ is holomorphic and satisfies (C1), (C2) (the argument in the proof of Lemma C.12(b), "well defined", which does not use F). $G:=\vartheta_1(x_1-y_1)^2 g$ is meromorphic, and by (iv) its polar set lies in the up–down divisors other than $\{x_1-y_1\in\Lambda_u\}$. So M is not in its

polar set, and $G|_M$ is meromorphic. Since $\vartheta_1(v)^2 = \vartheta_1'(0)^2 v^2(1+O(v^2))$, $\mathcal{A}$ is the v^{-2} coefficient of g at fixed centre w. Splitting J^2,

$$\mathsf{a}|_M = \vartheta_1'(0)^2\, E\, J'^2\, \mathcal{A}, \qquad E(w;\mathrm{rest}) := \prod_{z\in\mathrm{rest}} \vartheta_1(w-z)^4 \tag{C.11}$$

(squares of ϑ_1 do not see the sign of the argument).

Step 1 (order 4 at $x_a = w$). Fix $a \geq 2$. By (C1), $\mathsf{a} = \vartheta_1(x_1 - x_a)^3 Q$ with Q holomorphic on $\mathbb{C}^N$: $\vartheta_1(x_1 - x_a)$ is a local defining function of the smooth divisor $\{x_1 - x_a \in \Lambda_u\}$ (Lemma C.8(c)), and a vanishes to order ≥ 3 along it. Apply (C2) to the pair (x_a, y_1) on $x_a = y_1 = s$:

$$\begin{aligned} 0 &= \big[\mathsf{D}_{x_a y_1}\big(\vartheta_1(x_1-x_a)^3 Q\big)\big]_{x_a=y_1=s} \\ &= -3\vartheta_1(x_1-s)^2\vartheta_1'(x_1-s)\, Q|_{x_a=y_1=s} + \vartheta_1(x_1-s)^3\big[\mathsf{D}_{x_a y_1} Q\big]_{x_a=y_1=s}. \end{aligned}$$

Divide by $\vartheta_1(x_1-s)^2$, which is nonzero for $x_1 - s \notin \Lambda_u$: $3\vartheta_1'(x_1-s)Q|_{x_a=y_1=s} = \vartheta_1(x_1 - s)[\mathsf{D}Q]_{x_a=y_1=s}$. Let $x_1 \to s$: since $\vartheta_1'(0) \neq 0 = \vartheta_1(0)$, $Q = 0$ on $\{x_1 = x_a = y_1\}$. So $Q_M := Q|_M$ vanishes on $\{x_a = w\} \subset M$, and near generic points of it $Q_M = \vartheta_1(x_a - w)\widetilde{Q}_M$ with $\widetilde{Q}_M$ holomorphic. Then $\mathsf{a}|_M = \vartheta_1(w-x_a)^3 Q_M = -\vartheta_1(w-x_a)^4\widetilde{Q}_M$, which has order ≥ 4.

Step 2 (order 4 at $y_k = w$). The same with $x \leftrightarrow y$: $\mathsf{a} = \vartheta_1(y_1 - y_k)^3 Q'$, and (C2) for (x_1, y_k) on $x_1 = y_k = s$ reads $0 = 3\vartheta_1(y_1-s)^2\vartheta_1'(y_1-s)Q' + \vartheta_1(y_1-s)^3[\mathsf{D}Q']$, so $Q' = 0$ on $\{x_1 = y_1 = y_k\}$.

Step 3 (proof of (b)). Shifting w by 1 shifts x_1 and y_1 by 1; g picks up ε_1^2 by (ii), and $\vartheta_1(x_1 - y_1)^2$ is unchanged. The same holds for τ.

Step 4 (proof of (a)). Near a generic point of $\{w = x_a\}$ the factors $\vartheta_1'(0)^2$, $E/\vartheta_1(w-x_a)^4$ and J'^2 in (C.11) are holomorphic and nonzero, and by Step 1 $\mathsf{a}|_M$ is $\vartheta_1(w-x_a)^4$ times a holomorphic function. So $\mathcal{A}$ is holomorphic there; likewise at $\{w = y_k\}$ by Step 2. By (b), $\mathcal{A}$ is holomorphic near generic points of the translates $\{w - x_a \in \Lambda_u\}$ and $\{w - y_k \in \Lambda_u\}$ as well. The polar set of $\mathcal{A}$ is a union of components of $M \cap$ (up–down divisors): the families $\{w - y_k \in \Lambda_u\}$, $\{x_a - w \in \Lambda_u\}$ and $\{x_a - y_k \in \Lambda_u\}$ with $a, k \geq 2$. We have just excluded the first two. Hence (a).

Step 5 (proof of (c); Liouville). For rest as in (a), $w \mapsto \mathcal{A}(w;\mathrm{rest})$ is entire and, by (b), $|\mathcal{A}(w+\lambda)| = |\mathcal{A}(w)|$ for $\lambda \in \Lambda_u$. So it is bounded and therefore constant. If it is nonzero, $\mathcal{A} = \varepsilon_1^2\mathcal{A} = \varepsilon_2^2\mathcal{A}$ forces $\varepsilon_1^2 = \varepsilon_2^2 = 1$. These rest form a dense open set, so by the identity theorem $\mathcal{A}(w;\mathrm{rest}) - \mathcal{A}(w';\mathrm{rest}) \equiv 0$ for all w, w', and $\mathcal{A} \equiv 0$ when $\varepsilon^2 \neq 1$.

Step 6 (proof of (d)). Write $\mathcal{A}(\mathrm{rest})$. (i) $\mathcal{A}(w+c;\mathrm{rest}+c) = \mathcal{A}(w;\mathrm{rest})$ by (i) for g; with w-independence, $\mathcal{A}(\mathrm{rest}+c) = \mathcal{A}(\mathrm{rest})$. (ii) Shifting a variable of rest by 1 or τ multiplies g, hence $\mathcal{A}$, by ε_1 or ε_2. (iii) Inherited, since the merge involves only x_1, y_1. (iv) Given a point of rest-space, choose $w \notin \mathrm{rest} + \Lambda_u$; then $E \neq 0$ there. By (C.11), $J'^2\mathcal{A} = \mathsf{a}|_M/(\vartheta_1'(0)^2 E)$ is holomorphic near that point. So $J'^2\mathcal{A}$ is holomorphic, the poles of $\mathcal{A}$ lie on coincidence divisors of rest, and they have order ≤ 2. On a same-spin divisor $\{x_a - x_b \in \Lambda_u\}$ $(a, b \geq 2)$ the function a vanishes to order ≥ 3 in $\mathbb{C}^N$. The divisor is transverse to M, so $\mathsf{a}|_M$, and hence $J'^2\mathcal{A}$, vanishes to order ≥ 3 along it, and $\mathcal{A}$ is regular there. So the polar set of $\mathcal{A}$ lies in the up–down divisors of rest. (v) For $a, k \geq 2$, (C2) for a at (x_a, y_k) holds identically in the other variables, in particular on M (the substitution $x_1 = y_1 = w$ commutes with $\mathsf{D}_{x_a y_k}$ and with $x_a = y_k$). In (C.11), $\vartheta_1'(0)^2 E J'^2/\vartheta_1(x_a - y_k)^2$ is invariant under $x_a \leftrightarrow y_k$: E contains $\vartheta_1(w-x_a)^4\vartheta_1(w-y_k)^4$ and factors without x_a, y_k, and $J'^2/\vartheta_1(x_a-y_k)^2$ is a symmetric product over the other pairs. Lemma C.10 then gives (v) for $\mathcal{A}$.

Step 7 (proof of (e)). The formula is (C.11) times $F(U)$ at $U = 2w + Z'$. Shifting w by τ moves x_1 and y_1, giving the factor $e^{-i\pi L(2w+\tau)}$ twice, i.e. the level-$2L = 4N$ relation in w. So $f|_M$ has $4N$ zeros per cell (Lemma C.8(b)). E accounts for $4(N-2)$ of them. $w \mapsto F(2w + Z')$ has $2 \cdot 2^2 = 8$, since $w \mapsto 2w$ covers four U-cells and F has two zeros in each. $\mu(g)$ accounts for none. $\square$

C.5 A vanishing merge forces zero

Lemma C.14. *(a)* (polarized sectors) *For $k \geq 2$, $\mathcal{G}(k,0;\varepsilon) = \mathcal{G}(0,k;\varepsilon) = 0$. Also $\mathcal{G}(1,0;\varepsilon) = \mathcal{G}(0,1;\varepsilon)$, which is $\mathbb{C}$ (the constants) if $\varepsilon = (1,1)$ and 0 otherwise.*

(b) (vanishing merge) *Let $n, m \geq 1$, $g \in \mathcal{G}(n,m;\varepsilon)$ and $\mu(g) = 0$. Then g is constant. Hence $g = 0$ if $\max(n,m) \geq 2$ or $\varepsilon \neq (1,1)$.*

Proof. (b) By antisymmetry (iii), the v^{-2} coefficient $\mathcal{A}_{ak}$ of g at $x_a = y_k$ is $\pm\mu(g)$ with variables renamed (apply the transpositions $x_1 \leftrightarrow x_a$, $y_1 \leftrightarrow y_k$), so it vanishes. Let $G_{ak} := \vartheta_1(x_a - y_k)^2 g$. It is holomorphic near generic points of $\{x_a = y_k\}$, and in the coordinates (R, v, others) its Taylor coefficients of v^0 and v^1 are $\vartheta_1'(0)^2\mathcal{A}_{ak} = 0$ and $\vartheta_1'(0)^2\mathcal{B}_{ak} = 0$, by (v). So $G_{ak} = v^2 \cdot (\text{holomorphic})$, and $g = G_{ak}/\vartheta_1(v)^2$ is holomorphic near generic points of $\{x_a = y_k\}$; by (ii) also near those of $\{x_a - y_k \in \Lambda_u\}$. By (iv) these are the only candidate polar components, so g is entire on $\mathbb{C}^N$. $|g|$ is invariant under $u \mapsto u + e_k$ and $u \mapsto u + \tau e_k$ ($|\varepsilon_i| = 1$), hence bounded, hence constant by Liouville in each variable. A nonzero constant violates antisymmetry in two variables of one species, and violates (ii) unless $\varepsilon = (1,1)$. (a) With $m = 0$ there are no up–down divisors, so by (iv) g is entire, and it is constant as above. For $k \geq 2$ antisymmetry kills it. For $k = 1$ every constant satisfies (i)–(v) with $\varepsilon = (1,1)$, and only 0 does otherwise. □

C.6 The theorem

Proposition C.15 (the inductive inequality)**.** *Let $n, m \geq 1$ with $\max(n,m) \geq 2$. If $\varepsilon_1^2 = \varepsilon_2^2 = 1$ then $\dim \mathcal{G}(n,m;\varepsilon) \leq \dim \mathcal{G}(n-1,m-1;\varepsilon)$; otherwise $\mathcal{G}(n,m;\varepsilon) = 0$.*

Proof. $\mu : \mathcal{G}(n,m;\varepsilon) \to \mathcal{G}(n-1,m-1;\varepsilon)$ is linear (Lemma C.13(d)), identically zero if $\varepsilon^2 \neq 1$ (Lemma C.13(c)), and injective (Lemma C.14(b)). □

Proposition C.16 (base cases)**.** *(a) $\mathcal{G}(0,0;\varepsilon) = \mathbb{C}$; $\mathcal{G}(k,0;\varepsilon) = \mathcal{G}(0,k;\varepsilon) = 0$ for $k \geq 2$; $\mathcal{G}(1,0;\varepsilon) = \mathcal{G}(0,1;\varepsilon) = \mathbb{C}\,\delta_{\varepsilon,(1,1)}$.*

(b) $\mathcal{G}(1,1;(1,1)) = \operatorname{span}\{\wp(x_1 - y_1),\, 1\}$, of dimension 2.

(c) For $\varepsilon \in \{\pm 1\}^2 \setminus \{(1,1)\}$: $\mathcal{G}(1,1;\varepsilon) = \mathbb{C}\,\wp_\varepsilon(x_1 - y_1)$, of dimension 1.

(d) $\mathcal{G}(1,1;\varepsilon) = 0$ if $\varepsilon^2 \neq 1$.

Proof. (a) is Lemma C.14(a). For (b)–(d), consider $\mu : \mathcal{G}(1,1;\varepsilon) \to \mathcal{G}(0,0;\varepsilon) = \mathbb{C}$. Its kernel is the constants in $\mathcal{G}(1,1;\varepsilon)$ (Lemma C.14(b)): $\mathbb{C}$ if $\varepsilon = (1,1)$, 0 otherwise. And $\mu \equiv 0$ if $\varepsilon^2 \neq 1$ (Lemma C.13(c)). So the dimension is at most 2, at most 1, and 0 in cases (b), (c), (d). *Membership.* $\wp(x_1 - y_1)$ satisfies (i); (ii) with $\varepsilon = (1,1)$ by Lemma C.8(d); (iii) vacuously; (iv) with double poles on $x_1 - y_1 \in \Lambda_u$; and (v), because $\wp$ is even, so at fixed centre $\wp(v) = v^{-2} + O(1)$ with no v^{-1} term. $\mu(\wp) = 1$ and $\mu(1) = 0$, so $\wp$ and 1 are independent. For $\wp_\varepsilon$: $\wp_\varepsilon(x_1 + 1 - y_1) = \varepsilon_1\wp_\varepsilon$ and $\wp_\varepsilon(x_1 - y_1 - 1) = \varepsilon_1^{-1}\wp_\varepsilon = \varepsilon_1\wp_\varepsilon$ since $\varepsilon_1 = \pm 1$, and likewise for τ; (iv) and (v) follow from Lemma C.8(e) (evenness about every pole); and $\mu(\wp_\varepsilon) = 1 \neq 0$. □

Corollary C.17 (upper bounds)**.** *(a) If $|n - m| \geq 2$ then $\mathcal{G}(n,m;\varepsilon) = 0$.*

(b) $\dim \mathcal{G}(m+1,m;\varepsilon) \leq \delta_{\varepsilon,(1,1)}$.

(c) $\dim \mathcal{G}(p,p;\varepsilon) \leq \dim \mathcal{G}(1,1;\varepsilon)$, which is 2, 1 or 0 as in Proposition C.16.

Proof. Iterate Proposition C.15 along $(n,m) \to (n-1,m-1) \to \cdots$, ending at $(|n-m|,0)$ or $(0,|n-m|)$, at $(1,0)$, or at $(1,1)$; every intermediate step has max ≥ 2. Then apply Proposition C.16. □

Proposition C.18 (the explicit states)**.** *Write* $\wp_{(1,1)} := \wp$.

(a) $g_{2m+1} := \det\left[\wp(x_a - y_k) \;\middle|\; 1\right]$ *is the* $(m+1)\times(m+1)$ *determinant with entries* $\wp(x_a - y_k)$, $k \leq m$, *and a last column of ones. It lies in* $\mathcal{G}(m+1,m;(1,1))$, *with* $\mu(g_{2m+1}) = g_{2m-1}(x_2,\dots;y_2,\dots)$ *and* $\mu^m(g_{2m+1}) = g_1 = 1$.

(b) For $\varepsilon \in \{\pm 1\}^2$: $d_\varepsilon^{(p)} := \det[\wp_\varepsilon(x_i - y_j)]_{p\times p}$ *lies in* $\mathcal{G}(p,p;\varepsilon)$, *with* $\mu(d_\varepsilon^{(p)}) = d_\varepsilon^{(p-1)}$ *and* $\mu^{p-1}(d_\varepsilon^{(p)}) = \wp_\varepsilon$.

(c) $f_5^{(p)} := \sum_{i,j}(-1)^{i+j}\det M^{(ij)}$, *where* $M = [\wp(x_i - y_j)]$ *and* $M^{(ij)}$ *is* M *with row* i *and column* j *deleted* ($f_5^{(1)} := 1$), *lies in* $\mathcal{G}(p,p;(1,1))$, *with* $\mu(f_5^{(p)}) = f_5^{(p-1)}$ *and* $\mu^{p-1}(f_5^{(p)}) = 1$. *In particular* $d_{(1,1)}^{(p)}$ *and* $f_5^{(p)}$ *are linearly independent.*

Proof. Membership. (i): entries depend on differences. (ii): each entry $\wp_\varepsilon(x_i - y_j)$ has multiplier ε_1 under $x_i \mapsto x_i + 1$ and $\varepsilon_1^{-1} = \varepsilon_1$ under $y_j \mapsto y_j + 1$. Every term of the determinant contains each x_i and each y_j in at most one entry; in (a) the terms with a 1 in row a are free of x_a, which is consistent because $\varepsilon = (1,1)$ there. (iii): rows and columns. (iv): poles come only from the entries. (v): the coefficient of the entry $\wp_\varepsilon(x_i - y_j)$ in the determinant (its cofactor) depends on neither x_i nor y_j, and the remaining terms are regular at generic points of $x_i = y_j$. So at fixed centre the expansion is (cofactor) $\cdot\,(v^{-2} + O(1)) + O(1)$, with no v^{-1} term. For $f_5^{(p)}$ the coefficient of $\wp(x_i - y_j)$ is $\sum_{k\neq i,\, l\neq j}(-1)^{k+l}$ times the cofactor of that entry in $M^{(kl)}$, again free of x_i, y_j.

The merge. μ takes the coefficient of the $(1,1)$ entry. In (a) and (b) this is the cofactor, the same determinant for rest, with sign $+$. In (c), only minors with $i,j \geq 2$ contain the $(1,1)$ entry. In $M^{(ij)}$ its coefficient is $\det M^{(1i,1j)}$ (rows $1,i$ and columns $1,j$ deleted), with sign $+$. With M' the matrix for rest and $i' = i-1$, $j' = j-1$, this gives $\mu(f_5^{(p)}) = \sum_{i',j'}(-1)^{i'+j'}\det M'^{(i'j')} = f_5^{(p-1)}$. Iterating gives the stated values of μ^{p-1}, μ^m. Independence: apply μ^{p-1} to a vanishing combination; $\wp$ and 1 are independent. □

Theorem C.19 (the zero modes on the torus at $N_\phi = 2N$)**.** *For every* τ *with* $\operatorname{Im}\tau > 0$ *and every* $N \geq 1$, *the hollow-core Hamiltonian* H *of Section C.1.3 at* $N_\phi = 2N$ *satisfies:*

(i) $\ker H \cap \{|S_z| \geq 1\} = 0$.

(ii) For odd N, $\dim\ker H|_{S_z=+1/2} = 2$. *The kernel is spanned by*

$$f^{(K)} := \vartheta\begin{bmatrix} K/L \\ 0 \end{bmatrix}(2U|2\tau)\,\prod_{i<j}\vartheta_1(u_i - u_j)^2\,\det\left[\wp(x_a - y_k)\,|\,1\right], \qquad K \in \{0,N\}$$

(canonical components, u-units). These are the two centre-of-mass partners (T_1 *maps one to the other) of the relative state in the sector* $(T_K^2, T_1^2) = (1,1)$. $S_z = -\frac{1}{2}$ *is the same with the species exchanged. So* $\dim\ker H = 4$.

(iii) For even $N = 2p$, $\ker H$ *lies in* $S_z = 0$ *and has dimension* 10. *It is spanned by* $F(U)J^2 g$ *with* $g \in \{d_\varepsilon^{(p)} : \varepsilon \in \{\pm1\}^2\} \cup \{f_5^{(p)}\}$ *and* $F \in \Theta_2[\varepsilon]$, *two for each* g.

(iv) None of these statements depends on τ. *(The main theorems use rectangular tori only, and the formalization of Appendix E is confined to them.)*

Proof. By Lemma C.6, $\ker H|_{n,m} = V_{n,m}$. By Lemma C.12(b), $\dim V_{n,m} = 2\sum_{\varepsilon\in\mu_N^2} \dim \mathcal{G}(n,m;\varepsilon)$, where only $\varepsilon \in \mu_N^2$ occur by Lemma C.12(a). (i) Corollary C.17(a), with $|n-m| = 2|S_z|$. (ii) For $N = 2m+1$, Corollary C.17(b) gives $\dim V_{m+1,m} \le 2$; Proposition C.18(a) gives $g_N \neq 0$, so equality holds. $\Theta_2[(1,1)]$ is spanned by $\vartheta[a,0](2U|2\tau)$, $a \in \{0, \frac{1}{2}\}$ (by (C.10): the $U \mapsto U+1$ multiplier is $e^{4\pi i a} = 1$, the $U \mapsto U+\tau$ multiplier is $e^{-2\pi i\tau - 4\pi i U}$). T_K shifts U by $N/L = \frac{1}{2}$ and multiplies $\vartheta[a,0](2U|2\tau)$ by $e^{2\pi i a}$, so $a = K/L$ with $K \in \{0, N\}$; T_1 exchanges the two (Lemma C.12(c) and $T_K T_1 = -T_1 T_K$). (iii) For $N = 2p$, Corollary C.17(c) and $\{\pm 1\}^2 \subset \mu_N^2$ give $\dim V_{p,p} \le 2(2+1+1+1) = 10$. Proposition C.18(b),(c) gives five independent relative states (independent across ε by Lemma C.12(b)), so equality holds. (iv) Nothing above depends on τ. □

Remark C.20 (spin). H commutes with the total spin, and by (i) the $S_z = \pm 1$ sectors contain no zero mode. So every zero mode at even N has $S = 0$, and at odd N the kernel is a spin-$\frac{1}{2}$ doublet times the two centre-of-mass states of (ii). Nothing in the sequel uses this, and it is not formalized (Appendix E).

Remark C.21 (Read–Rezayi's states). Read and Rezayi's states [RR96, eq. (52)], $\det \wp_a((z_i^{\uparrow} - z_j^{\downarrow})/L_1|\tau)$, are $d_\varepsilon^{(p)}$; their $\wp_a$ is a multiple of $\wp_\varepsilon$ by Proposition C.16(c). Their eq. (53) is $f_5^{(p)}$. Their odd-N states, described in words only (one unpaired fermion in the constant orbital of the $++$ sector), are (ii); the formula $f^{(K)}$ is our reconstruction. Read and Rezayi do not prove that their states exhaust the zero modes [RR96, Section V]; Theorem C.19 supplies this.

Remark C.22 (not covered). Twisted boundary conditions ($\phi_1, \phi_2 \neq 0$ in [RR96, eq. (45)]) change only the characteristics of $\Theta_2[\varepsilon]$. We expect the same proof to apply to them, but we do not write it. Fluxes other than $2N$ (quasihole sectors) are not addressed.

D A variational bound at fixed circumference

Corollary 8.4 concerns tori whose sides both exceed the length L_0, and, like Theorem 8.1, it gives neither a rate nor explicit constants. This appendix proves a bound of a different kind. It holds on every rectangular torus and for every $N \ge 2$, it is explicit in the units of Definition 2.7, and it is of order L_y^3/N for L_y bounded below. It is therefore useful when the circumference L_y is bounded, or grows more slowly than $N^{1/3}$, in particular on the thin tori of Remark 8.5, and it says nothing at fixed aspect ratio. The trial state is a zero mode in which every orbital ψ_j of the electrons of one spin species is multiplied by the phase $e^{2\pi i j/N_\phi}$, which winds once around the torus in the transverse direction (a *spin twist*); the zero mode is chosen so that the twisted state is orthogonal to all zero modes. Apart from elementary facts of Section 2, the argument uses only the orbital form of H_N (Theorem 4.2(c)), the commutation of H_N with Θ (Lemma 4.5) and the dimensions of its kernel (Theorem 2.10). The slot frame is not used, and the tori need not be admissible.

Throughout this appendix $N \in \mathbb{N}$, $L_x, L_y > 0$ with $L_x L_y = 4\pi N$, and $\kappa = 2\pi/L_y$, as in Definition 2.1. For $k \ge 1$ let

$$\mathsf{G}_k(\kappa) := \sum_{d=1}^{\infty} 2d^k\, e^{-\kappa^2 d^2/4}, \tag{D.1}$$

a convergent series.

D.1 The statement

Theorem D.1 (a variational bound at fixed circumference)**.** *Let $N \ge 2$. If N is even, let $\mathbf{n} := (N/2, N/2)$; if N is odd, let $\mathbf{n}$ be $((N+1)/2, (N-1)/2)$ or $((N-1)/2, (N+1)/2)$. Then*

$\mathcal{H}_N^{\mathbf{n}}$ contains a unit vector, orthogonal to $\mathcal{K}_N$, at which the expectation of H_N is at most

$$\frac{\kappa^3}{\sqrt{2\pi}}\Big(\frac{\pi^2\,\mathsf{G}_2(\kappa)^2}{2N}+\frac{\pi^4\,\mathsf{G}_3(\kappa)^2}{32N^3}\Big)\le\frac{1}{\sqrt{2\pi}}\Big(1+\frac{2\kappa}{e\sqrt{\pi}}\Big)^2\frac{L_y^3}{N}+\frac{1}{4\pi\sqrt{2\pi}}\Big(1+\frac{(6/e)^{3/2}\kappa}{8}\Big)^2\frac{L_y^5}{N^3}. \tag{D.2}$$

Consequently, if N is even, $E_{11}(N)$ is at most the right-hand side of (D.2), and if N is odd, so is $E_3(N)$ in each of the two sectors $((N\pm1)/2,(N\mp1)/2)$. In particular both are at most

$$0.40\,(1+0.42\,\kappa)^2\,\frac{L_y^3}{N}+0.032\,(1+0.41\,\kappa)^2\,\frac{L_y^5}{N^3}.$$

Here $E_{11}(N)$ and $E_3(N)$ are the eigenvalues of (2.8) and of the sentence that follows it, that is, the lowest eigenvalue of H_N in the sector above its zero modes, on the torus with the given L_x and L_y. The unit vector of the theorem is $\mathcal{W}_\uparrow\Upsilon$ for a suitable zero mode $\Upsilon\in\mathcal{K}_N\cap\mathcal{H}_N^{\mathbf{n}}$, with the twist $\mathcal{W}_\uparrow$ of Lemma D.3 below.

If $L_y^3/N\to0$ as $N\to\infty$, for instance if L_y stays bounded, the right-hand side of (D.2) tends to zero; for bounded L_y it is $O(1/N)$, with explicit constants. At fixed aspect ratio L_y is of order $N^{1/2}$, and the right-hand side grows like $N^{1/2}$; that regime is the one of Theorem 8.1. Unlike Theorem 8.1, which uses only that the kernel has dimension at least ten, Theorem D.1 uses the exact dimensions of Theorem 2.10, and with them all of Appendix C. Together with Corollary 8.4 it gives the following.

Corollary D.2 (all rectangular tori)**.** *For even N and a rectangular torus $\mathbb{T}$ with $L_xL_y=4\pi N$, let $E_{11}(N;\mathbb{T})$ be the eleventh eigenvalue of the hollow-core Hamiltonian of $\mathbb{T}$ in the sector $(N/2,N/2)$. Then $\sup_{\mathbb{T}}E_{11}(N;\mathbb{T})\to0$ as $N\to\infty$ through the even integers, where the supremum runs over all rectangular tori with $L_xL_y=4\pi N$.*

Proof. Let $N\ge L_0^2/(4\pi)$ be even, and let $\mathbb{T}$ be a rectangular torus with $L_xL_y=4\pi N$. If $\min\{L_x,L_y\}\ge L_0$, then $\mathbb{T}\in\mathcal{T}_N$. Otherwise, by Lemma 8.3 we may exchange the sides and assume that $L_y=\min\{L_x,L_y\}<L_0$. Since $\kappa L_y=2\pi$,

$$\Big(1+\frac{2\kappa}{e\sqrt{\pi}}\Big)^2L_y^3=\Big(L_y+\frac{4\sqrt{\pi}}{e}\Big)^2L_y,\qquad\Big(1+\frac{(6/e)^{3/2}\kappa}{8}\Big)^2L_y^5=\Big(L_y+\frac{\pi(6/e)^{3/2}}{4}\Big)^2L_y^3,$$

both increasing in L_y, and Theorem D.1 gives $E_{11}(N;\mathbb{T})\le C/N$ with $C:=(2\pi)^{-1/2}(L_0+4\sqrt{\pi}/e)^2L_0+(4\pi\sqrt{2\pi})^{-1}(L_0+\pi(6/e)^{3/2}/4)^2L_0^3$, which depends only on L_0. Hence $\sup_{\mathbb{T}}E_{11}(N;\mathbb{T})\le\max\{\sup_{\mathbb{T}\in\mathcal{T}_N}E_{11}(N;\mathbb{T}),C/N\}$, which tends to zero by Corollary 8.4. □

D.2 The twists

Lemma D.3 (the twists)**.** *For $\sigma\in\{\uparrow,\downarrow\}$ let*

$$\mathcal{W}_\sigma:=\exp\Big(\frac{2\pi\mathrm{i}}{N_\phi}\sum_{j=0}^{N_\phi-1}j\,n_{[j]\sigma}\Big).$$

(i) $\mathcal{W}_\sigma$ is unitary and diagonal in the occupation-number basis; it commutes with $N_\uparrow$, $N_\downarrow$ and the other twist, so it maps every sector into itself; and it does not change if the representatives $0,\dots,N_\phi-1$ of the classes in $\mathbb{Z}_{N_\phi}$ are replaced by any other representatives.

(ii) For every $j\in\mathbb{Z}_{N_\phi}$, $\mathcal{W}_\sigma^*c_{j\sigma}\mathcal{W}_\sigma=e^{2\pi\mathrm{i}j/N_\phi}c_{j\sigma}$, and $\mathcal{W}_\sigma^*c_{j\sigma'}\mathcal{W}_\sigma=c_{j\sigma'}$ for $\sigma'\neq\sigma$.

(iii) $\Theta^*\mathcal{W}_\sigma\Theta = e^{-2\pi\mathrm{i}N_\sigma/N_\phi}\mathcal{W}_\sigma$. *Consequently, if* $\Upsilon \in \mathcal{H}_N^{\mathbf{n}}$, $\mathbf{n} = (n_\uparrow, n_\downarrow)$, *and* $\Theta\Upsilon = \mu\Upsilon$, *then*

$$\Theta\mathcal{W}_\uparrow\Upsilon = e^{2\pi\mathrm{i}n_\uparrow/N_\phi}\mu\,\mathcal{W}_\uparrow\Upsilon, \qquad \Theta\mathcal{W}_\uparrow\mathcal{W}_\downarrow^*\Upsilon = e^{2\pi\mathrm{i}(n_\uparrow - n_\downarrow)/N_\phi}\mu\,\mathcal{W}_\uparrow\mathcal{W}_\downarrow^*\Upsilon.$$

(iv) $\mathcal{W}_\sigma = \mathcal{F}(\theta_y \otimes |\sigma\rangle\langle\sigma| + \mathbf{1} \otimes (\mathbf{1} - |\sigma\rangle\langle\sigma|))$*: the twist* $\mathcal{W}_\sigma$ *is the magnetic translation* θ_y *of Lemma 2.4, by* L_y/N_ϕ *along the circumference, applied to the electrons of spin* σ *alone.*

Proof. (i) The $n_{j\sigma}$ are mutually commuting orthogonal projections, diagonal in the occupation-number basis $c^*_{j_1\sigma_1}\cdots c^*_{j_p\sigma_p}\Omega_0$ (proof of Lemma 4.10). So $\mathcal{W}_\sigma$ is the product of the commuting unitaries $\exp(2\pi\mathrm{i}j\,n_{[j]\sigma}/N_\phi)$, it is diagonal in that basis, and it commutes with $N_\uparrow$, $N_\downarrow$ and $\mathcal{W}_{\sigma'}$, which are diagonal there too. Since $e^{2\pi\mathrm{i}n_{j\sigma}} = \mathbf{1}$, replacing a representative j by $j + mN_\phi$, $m \in \mathbb{Z}$, does not change the corresponding factor.

(ii) By the anticommutation relations $c^2_{j\sigma'} = 0$ and $c_{j\sigma'}c^*_{j\sigma'} = \mathbf{1} - n_{j\sigma'}$, so $n_{j\sigma'}c_{j\sigma'} = c^*_{j\sigma'}c^2_{j\sigma'} = 0$ and $c_{j\sigma'}n_{j\sigma'} = (\mathbf{1}-n_{j\sigma'})c_{j\sigma'} = c_{j\sigma'}$. Since $n_{j\sigma'}$ is an orthogonal projection, $e^{\mathrm{i}\alpha n_{j\sigma'}} = \mathbf{1}+(e^{\mathrm{i}\alpha}-1)n_{j\sigma'}$ for $\alpha \in \mathbb{R}$; hence $e^{-\mathrm{i}\alpha n_{j\sigma'}}c_{j\sigma'} = c_{j\sigma'}$ and $c_{j\sigma'}e^{\mathrm{i}\alpha n_{j\sigma'}} = e^{\mathrm{i}\alpha}c_{j\sigma'}$, so $e^{-\mathrm{i}\alpha n_{j\sigma'}}c_{j\sigma'}e^{\mathrm{i}\alpha n_{j\sigma'}} = e^{\mathrm{i}\alpha}c_{j\sigma'}$. The factors of $\mathcal{W}_\sigma$ that belong to the other modes commute with $c_{j\sigma'}$ (proof of Lemma 4.10). With $\alpha = 2\pi\tilde{\jmath}/N_\phi$, $\tilde{\jmath}$ the representative of j, if $\sigma' = \sigma$, and with no factor for the mode (j,σ') if $\sigma' \neq \sigma$, this gives (ii); the phase $e^{2\pi\mathrm{i}\tilde{\jmath}/N_\phi} = e^{2\pi\mathrm{i}j/N_\phi}$ depends only on the class j.

(iii) By Lemma 2.6(iii), $\Theta^* n_{[j]\sigma}\Theta = n_{[j+1]\sigma}$, and conjugation by the unitary Θ commutes with the exponential of a self-adjoint operator, so, with $k = j+1$,

$$\Theta^*\mathcal{W}_\sigma\Theta = \exp\Big(\frac{2\pi\mathrm{i}}{N_\phi}\sum_{j=0}^{N_\phi-1} j\,n_{[j+1]\sigma}\Big) = \exp\Big(\frac{2\pi\mathrm{i}}{N_\phi}\sum_{k=1}^{N_\phi} k\,n_{[k]\sigma}\Big)\exp\Big(-\frac{2\pi\mathrm{i}}{N_\phi}\sum_{k=1}^{N_\phi} n_{[k]\sigma}\Big),$$

the $n_{[k]\sigma}$ commuting. The first factor is $\mathcal{W}_\sigma$ by (i), with the representatives $1,\dots,N_\phi$, and the sum in the second is N_σ. Hence $\Theta^*\mathcal{W}_\sigma\Theta = e^{-2\pi\mathrm{i}N_\sigma/N_\phi}\mathcal{W}_\sigma$, and, Θ commuting with N_σ, $\Theta\mathcal{W}_\sigma\Theta^* = e^{2\pi\mathrm{i}N_\sigma/N_\phi}\mathcal{W}_\sigma$ and $\Theta\mathcal{W}_\sigma^*\Theta^* = e^{-2\pi\mathrm{i}N_\sigma/N_\phi}\mathcal{W}_\sigma^*$. If $\Upsilon \in \mathcal{H}_N^{\mathbf{n}}$ and $\Theta\Upsilon = \mu\Upsilon$, then, as the twists map $\mathcal{H}_N^{\mathbf{n}}$ into itself, $\Theta\mathcal{W}_\uparrow\Upsilon = e^{2\pi\mathrm{i}N_\uparrow/N_\phi}\mathcal{W}_\uparrow\Theta\Upsilon = e^{2\pi\mathrm{i}n_\uparrow/N_\phi}\mu\,\mathcal{W}_\uparrow\Upsilon$, and likewise $\Theta\mathcal{W}_\uparrow\mathcal{W}_\downarrow^*\Upsilon = e^{2\pi\mathrm{i}(N_\uparrow - N_\downarrow)/N_\phi}\mathcal{W}_\uparrow\mathcal{W}_\downarrow^*\Theta\Upsilon = e^{2\pi\mathrm{i}(n_\uparrow-n_\downarrow)/N_\phi}\mu\,\mathcal{W}_\uparrow\mathcal{W}_\downarrow^*\Upsilon$.

(iv) Let u be the unitary on $\mathfrak{k}_N$ in the parentheses. By Definition 2.5 and the linearity of $c(\cdot)^*$, $\mathcal{F}(u)$ fixes Ω_0, and $\mathcal{F}(u)c^*_{j\sigma}\mathcal{F}(u)^* = c(u(\psi_j \otimes \sigma))^* = e^{2\pi\mathrm{i}j/N_\phi}c^*_{j\sigma}$ by Lemma 2.4(i), while $\mathcal{F}(u)c^*_{j\sigma'}\mathcal{F}(u)^* = c^*_{j\sigma'}$ for $\sigma' \neq \sigma$. By (ii), taking adjoints, $\mathcal{W}_\sigma c^*_{j\sigma}\mathcal{W}_\sigma^* = e^{2\pi\mathrm{i}j/N_\phi}c^*_{j\sigma}$ and $\mathcal{W}_\sigma$ fixes $c^*_{j\sigma'}$ for $\sigma' \neq \sigma$; and $\mathcal{W}_\sigma\Omega_0 = \Omega_0$. So the two unitaries agree on the occupation-number basis, and they are equal. □

By (iv), and since $\mathcal{F}$ is multiplicative (by the uniqueness in its definition), $\mathcal{W}_\uparrow\mathcal{W}_\downarrow = \mathcal{F}(\theta_y \otimes \mathbf{1})$ translates all electrons along the circumference, while $\mathcal{W}_\uparrow$ translates the electrons of spin up relative to those of spin down.

D.3 The energy of a twisted zero mode

For $n \in \mathbb{Z}$ and $d \in \mathbb{N}$ with $d \equiv n \pmod 2$ put

$$c^\nu_{n,d} := c^\nu\big(\psi_{[(n-d)/2]}, \psi_{[(n+d)/2]}\big) \quad (\nu \in \{+,0,-\}), \qquad c^{\mathrm{sing}}_{n,d} := c^{\mathrm{sing}}\big(\psi_{[(n-d)/2]}, \psi_{[(n+d)/2]}\big),$$

where $c^{\mathrm{sing}}(f,g) := \sum_{\sigma,\sigma'} \overline{\chi^{\mathrm{sing}}_{\sigma\sigma'}}\,c(f\otimes\sigma')\,c(g\otimes\sigma)$ is the singlet pair annihilator, built from the antisymmetric spinor χ^{sing} with $\chi^{\mathrm{sing}}_{\uparrow\downarrow} = -\chi^{\mathrm{sing}}_{\downarrow\uparrow} = 2^{-1/2}$ and zero diagonal (as in the proof of

Theorem 4.2). With $j := [(n-d)/2]$ and $j' := [(n+d)/2]$,

$$c^{+}_{n,d} = c_{j\uparrow}c_{j'\uparrow}, \qquad c^{-}_{n,d} = c_{j\downarrow}c_{j'\downarrow},$$
$$c^{0}_{n,d} = 2^{-1/2}(c_{j\downarrow}c_{j'\uparrow} + c_{j\uparrow}c_{j'\downarrow}), \qquad c^{\mathrm{sing}}_{n,d} = 2^{-1/2}(c_{j\downarrow}c_{j'\uparrow} - c_{j\uparrow}c_{j'\downarrow}),$$

and $b^{\nu}_{n} = \sum_{d} 2d\, e^{-\kappa^2 d^2/4} c^{\nu}_{n,d}$, the sum running over $d \in \mathbb{N}$ with $d \equiv n \pmod 2$ (Definition 4.1). As in the proof of Proposition 4.4(i), $\|c^{\nu}_{n,d}\| \le \sqrt{2}$ and $\|c^{\mathrm{sing}}_{n,d}\| \le \sqrt{2}$, so all series below converge absolutely in norm.

Lemma D.4 (the energy of a twisted zero mode)**.** *Let Υ be a unit vector with N electrons such that $H_N\Upsilon = 0$. Then*

$$\langle \mathcal{W}_{\uparrow}\Upsilon, H_N \mathcal{W}_{\uparrow}\Upsilon\rangle = \frac{\kappa^3}{\sqrt{2\pi}} \sum_{n\in\mathbb{Z}_{2N_\phi}} \Big\| \sum_{d} 2d\, e^{-\kappa^2 d^2/4} \Big[\Big(\cos\frac{\pi d}{N_\phi} - 1\Big) c^{0}_{n,d} + \mathrm{i}\sin\frac{\pi d}{N_\phi}\, c^{\mathrm{sing}}_{n,d} \Big] \Upsilon \Big\|^2, \tag{D.3}$$

the inner sum running over $d \in \mathbb{N}$ with $d \equiv n \pmod 2$, for any representative of the class n; and

$$\langle \mathcal{W}_{\uparrow}\Upsilon, H_N \mathcal{W}_{\uparrow}\Upsilon\rangle \le \frac{\kappa^3}{\sqrt{2\pi}} \Big(\frac{\pi^2\, \mathsf{G}_2(\kappa)^2}{2N} + \frac{\pi^4\, \mathsf{G}_3(\kappa)^2}{32N^3} \Big). \tag{D.4}$$

For the two-species twist $\mathcal{W}_{\uparrow}\mathcal{W}_{\downarrow}^{*}$ in place of $\mathcal{W}_{\uparrow}$, (D.3) holds with $\pi d/N_\phi$ replaced by $2\pi d/N_\phi$, and (D.4) with its first term multiplied by four and its second term by sixteen.

Proof. Step 1: only the triplet pairs with one electron of each spin feel the twist. By Theorem 4.2(c), $H_N = (\kappa^3/\sqrt{2\pi}) \sum_{n\in\mathbb{Z}_{2N_\phi}} \sum_{\nu} (b^{\nu}_{n})^{*} b^{\nu}_{n}$, a sum of positive operators; so $0 = \langle \Upsilon, H_N\Upsilon\rangle$ gives $b^{\nu}_{n}\Upsilon = 0$ for all n and ν. Since $\mathcal{W}_{\uparrow}$ is unitary,

$$\langle \mathcal{W}_{\uparrow}\Upsilon, H_N \mathcal{W}_{\uparrow}\Upsilon\rangle = \frac{\kappa^3}{\sqrt{2\pi}} \sum_{n\in\mathbb{Z}_{2N_\phi}} \sum_{\nu} \|\mathcal{W}_{\uparrow}^{*} b^{\nu}_{n} \mathcal{W}_{\uparrow}\Upsilon\|^2.$$

Fix a representative $n \in \mathbb{Z}$ and $d \in \mathbb{N}$ with $d \equiv n \pmod 2$, and put $j_1 := (n-d)/2$ and $j_2 := (n+d)/2$, so that $j_1 + j_2 = n$, $j_2 - j_1 = d$, $j = [j_1]$ and $j' = [j_2]$, as in the proof of Theorem 4.2(b). By Lemma D.3(ii),

$$\mathcal{W}_{\uparrow}^{*} c^{+}_{n,d} \mathcal{W}_{\uparrow} = e^{2\pi \mathrm{i}(j_1+j_2)/N_\phi} c^{+}_{n,d} = e^{2\pi \mathrm{i} n/N_\phi} c^{+}_{n,d}, \qquad \mathcal{W}_{\uparrow}^{*} c^{-}_{n,d} \mathcal{W}_{\uparrow} = c^{-}_{n,d},$$

and, with $u := c_{j\downarrow}c_{j'\uparrow}$ and $v := c_{j\uparrow}c_{j'\downarrow}$, so that $c^{0}_{n,d} = 2^{-1/2}(u+v)$ and $c^{\mathrm{sing}}_{n,d} = 2^{-1/2}(u-v)$, $\mathcal{W}_{\uparrow}^{*} u \mathcal{W}_{\uparrow} = e^{2\pi \mathrm{i} j_2/N_\phi} u$ and $\mathcal{W}_{\uparrow}^{*} v \mathcal{W}_{\uparrow} = e^{2\pi \mathrm{i} j_1/N_\phi} v$. Writing $e^{2\pi \mathrm{i} j_2/N_\phi} = e^{\mathrm{i}\pi n/N_\phi} e^{\mathrm{i}\pi d/N_\phi}$ and $e^{2\pi \mathrm{i} j_1/N_\phi} = e^{\mathrm{i}\pi n/N_\phi} e^{-\mathrm{i}\pi d/N_\phi}$, and using $e^{\mathrm{i}x}u + e^{-\mathrm{i}x}v = \cos x\,(u+v) + \mathrm{i}\sin x\,(u-v)$ for $x \in \mathbb{R}$, we get

$$\mathcal{W}_{\uparrow}^{*} c^{0}_{n,d} \mathcal{W}_{\uparrow} = e^{\mathrm{i}\pi n/N_\phi} \Big[\cos\frac{\pi d}{N_\phi}\, c^{0}_{n,d} + \mathrm{i}\sin\frac{\pi d}{N_\phi}\, c^{\mathrm{sing}}_{n,d} \Big].$$

The phases $e^{2\pi \mathrm{i} n/N_\phi}$ and $e^{\mathrm{i}\pi n/N_\phi}$ do not depend on d. Summing over d with the weights $2d\, e^{-\kappa^2 d^2/4}$, we obtain $\mathcal{W}_{\uparrow}^{*} b^{+}_{n} \mathcal{W}_{\uparrow} = e^{2\pi \mathrm{i} n/N_\phi} b^{+}_{n}$, $\mathcal{W}_{\uparrow}^{*} b^{-}_{n} \mathcal{W}_{\uparrow} = b^{-}_{n}$ and

$$\mathcal{W}_{\uparrow}^{*} b^{0}_{n} \mathcal{W}_{\uparrow} = e^{\mathrm{i}\pi n/N_\phi} \Big(b^{0}_{n} + \sum_{d} 2d\, e^{-\kappa^2 d^2/4} \Big[\Big(\cos\frac{\pi d}{N_\phi} - 1\Big) c^{0}_{n,d} + \mathrm{i}\sin\frac{\pi d}{N_\phi}\, c^{\mathrm{sing}}_{n,d} \Big] \Big).$$

Since $b_n^\nu \Upsilon = 0$, only the sum survives when these operators act on Υ, and this proves (D.3). The norms in it depend on n only through its class in $\mathbb{Z}_{2N_\phi}$, because the left-hand sides above do: $b_{n+2N_\phi}^\nu = b_n^\nu$, and the phases are $2N_\phi$-periodic in n. For the two-species twist, Lemma D.3(ii) applied to both factors multiplies $c_{j\uparrow}$ by $e^{2\pi \mathrm{i} j_1/N_\phi}$ and $c_{j\downarrow}$ by $e^{-2\pi \mathrm{i} j_1/N_\phi}$, and likewise for j' and j_2; so $c_{n,d}^\pm$ acquire the phases $e^{\pm 2\pi \mathrm{i} n/N_\phi}$, u and v acquire $e^{2\pi \mathrm{i} d/N_\phi}$ and $e^{-2\pi \mathrm{i} d/N_\phi}$, and the same computation gives (D.3) with $\pi d/N_\phi$ replaced by $2\pi d/N_\phi$.

Step 2: two Cauchy–Schwarz steps. For vectors ξ_1, ξ_2 we have $\|\xi_1 + \xi_2\|^2 \le 2\|\xi_1\|^2 + 2\|\xi_2\|^2$. For absolutely summable coefficients α_d and vectors Φ_d of bounded norm, the triangle inequality and the Cauchy–Schwarz inequality for sums, applied to $\|\sum_d \alpha_d \Phi_d\| \le \sum_d |\alpha_d|^{1/2} \cdot |\alpha_d|^{1/2} \|\Phi_d\|$, give

$$\Big\| \sum_d \alpha_d \Phi_d \Big\|^2 \le \Big(\sum_d |\alpha_d| \Big) \Big(\sum_d |\alpha_d| \|\Phi_d\|^2 \Big).$$

With $|\sin x| \le |x|$ and $|1 - \cos x| \le x^2/2$, and since the sums over $d \equiv n \pmod 2$ are at most the sums over all $d \ge 1$,

$$\sum_d 2d\, e^{-\kappa^2 d^2/4} \Big| \sin \frac{\pi d}{N_\phi} \Big| \le \frac{\pi}{N_\phi}\, \mathsf{G}_2(\kappa), \qquad \sum_d 2d\, e^{-\kappa^2 d^2/4} \Big| 1 - \cos \frac{\pi d}{N_\phi} \Big| \le \frac{\pi^2}{2N_\phi^2}\, \mathsf{G}_3(\kappa).$$

Applying the two inequalities to each term of (D.3) therefore gives

$$\langle \mathcal{W}_\uparrow \Upsilon, H_N \mathcal{W}_\uparrow \Upsilon \rangle \le \frac{\kappa^3}{\sqrt{2\pi}} \Big(\frac{2\pi^2}{N_\phi^2}\, \mathsf{G}_2(\kappa) \sum_{n,d} 2d^2 e^{-\kappa^2 d^2/4} \|c_{n,d}^{\mathrm{sing}} \Upsilon\|^2 + \frac{\pi^4}{2N_\phi^4}\, \mathsf{G}_3(\kappa) \sum_{n,d} 2d^3 e^{-\kappa^2 d^2/4} \|c_{n,d}^0 \Upsilon\|^2 \Big), \quad \text{(D.5)}$$

where $\sum_{n,d}$ runs over $n \in \mathbb{Z}_{2N_\phi}$ and $d \in \mathbb{N}$ with $d \equiv n \pmod 2$. For the two-species twist, π is replaced by 2π, which multiplies the two terms by four and by sixteen.

Step 3: each pair weight is bounded by its occupation. Fix (n, d), with u and v as in Step 1. Then $(c_{n,d}^{\mathrm{sing}})^* c_{n,d}^{\mathrm{sing}} + (c_{n,d}^0)^* c_{n,d}^0 = u^* u + v^* v$, and $u^* u = c_{j'\uparrow}^* n_{j\downarrow} c_{j'\uparrow} = n_{j\downarrow} n_{j'\uparrow}$ and $v^* v = n_{j\uparrow} n_{j'\downarrow}$, because the modes $(j, \downarrow)$ and $(j', \uparrow)$ are distinct (also when $j = j'$). For commuting orthogonal projections p_1, p_2 one has $p_1 p_2 \le \frac{1}{2}(p_1 + p_2)$, since $p_1 + p_2 - 2p_1 p_2 = p_1(\mathbf{1} - p_2) + (\mathbf{1} - p_1) p_2 \ge 0$. Hence

$$\|c_{n,d}^{\mathrm{sing}} \Upsilon\|^2,\ \|c_{n,d}^0 \Upsilon\|^2 \ \le\ \langle \Upsilon, (n_{j\downarrow} n_{j'\uparrow} + n_{j\uparrow} n_{j'\downarrow}) \Upsilon \rangle \ \le\ \tfrac{1}{2} \langle \Upsilon, (n_{j\uparrow} + n_{j\downarrow} + n_{j'\uparrow} + n_{j'\downarrow}) \Upsilon \rangle.$$

For fixed d, the classes $n \in \mathbb{Z}_{2N_\phi}$ with $n \equiv d \pmod 2$ are represented by $n = d + 2k$, $k = 0, \dots, N_\phi - 1$; then $j = [k]$ and $j' = [k + d]$, and each runs once through $\mathbb{Z}_{N_\phi}$. Summing over these n, the right-hand side therefore gives $\frac{1}{2} \cdot 2 \langle \Upsilon, (N_\uparrow + N_\downarrow) \Upsilon \rangle = N$. Consequently

$$\sum_{n,d} 2d^2 e^{-\kappa^2 d^2/4} \|c_{n,d}^{\mathrm{sing}} \Upsilon\|^2 \le N\, \mathsf{G}_2(\kappa), \qquad \sum_{n,d} 2d^3 e^{-\kappa^2 d^2/4} \|c_{n,d}^0 \Upsilon\|^2 \le N\, \mathsf{G}_3(\kappa).$$

Inserted in (D.5), with $N_\phi = 2N$, this gives $\frac{2\pi^2}{4N^2} \mathsf{G}_2 \cdot N \mathsf{G}_2 = \frac{\pi^2 \mathsf{G}_2^2}{2N}$ and $\frac{\pi^4}{2 \cdot 16 N^4} \mathsf{G}_3 \cdot N \mathsf{G}_3 = \frac{\pi^4 \mathsf{G}_3^2}{32 N^3}$, which is (D.4); for the two-species twist the factors four and sixteen of Step 2 carry over. □

D.4 A zero mode twisted off the kernel

For a sector $\mathbf{n}$ and $\mu \in \mathbb{C}$ let $\mathcal{H}_N^{\mathbf{n}}(\mu) := \{\Upsilon \in \mathcal{H}_N^{\mathbf{n}} : \Theta\Upsilon = \mu\Upsilon\}$.

Lemma D.5 (a zero mode twisted off the kernel)**.** *Let $\mathbf{n} = (n_\uparrow, n_\downarrow)$ be a sector with $n_\uparrow + n_\downarrow = N$, and let $\mathcal{W}$ be $\mathcal{W}_\uparrow$ or $\mathcal{W}_\uparrow\mathcal{W}_\downarrow^*$. Suppose that the dimension of $\mathcal{K}_N \cap \mathcal{H}_N^{\mathbf{n}}$ is not divisible by the order of the root of unity $e^{2\pi \mathrm{i} n_\uparrow/N_\phi}$, respectively $e^{2\pi\mathrm{i}(n_\uparrow - n_\downarrow)/N_\phi}$ (the least $k \geq 1$ for which its k-th power is 1). Then there is a unit vector $\Upsilon \in \mathcal{K}_N \cap \mathcal{H}_N^{\mathbf{n}}$, an eigenvector of Θ, such that $\mathcal{W}\Upsilon$ is orthogonal to $\mathcal{K}_N$.*

Proof. Let ω be the root of unity of the statement, and put $\mathcal{K}(\mu) := \mathcal{K}_N \cap \mathcal{H}_N^{\mathbf{n}}(\mu)$. The unitary Θ maps $\mathcal{H}_N^{\mathbf{n}}$ onto itself (Lemma 2.6(iii)) and commutes with H_N (Lemma 4.5), so it maps the finite-dimensional space $\mathcal{K}_N \cap \mathcal{H}_N^{\mathbf{n}}$ onto itself; its restriction is unitary, and $\mathcal{K}_N \cap \mathcal{H}_N^{\mathbf{n}}$ is the orthogonal sum of the spaces $\mathcal{K}(\mu)$, $|\mu| = 1$, of which finitely many are nonzero.

Suppose that $\dim \mathcal{K}(\mu) \leq \dim \mathcal{K}(\omega\mu)$ for every μ. Along an orbit $\{\omega^k\mu : k \in \mathbb{Z}\}$, which has as many elements as the order of ω, the dimensions are then nondecreasing and return to their initial value, so they are constant. Then $\dim(\mathcal{K}_N \cap \mathcal{H}_N^{\mathbf{n}}) = \sum_\mu \dim \mathcal{K}(\mu)$ is a sum, over the finitely many orbits on which the dimensions do not vanish, of the order of ω times the value on the orbit, and so it is divisible by that order, contrary to the hypothesis. Hence there is μ_0 with $\dim \mathcal{K}(\mu_0) > \dim \mathcal{K}(\omega\mu_0)$.

By Lemma D.3(i) and (iii), $\mathcal{W}$ maps $\mathcal{K}(\mu_0)$ into $\mathcal{H}_N^{\mathbf{n}}(\omega\mu_0)$. A vector of $\mathcal{H}_N^{\mathbf{n}}(\omega\mu_0)$ is orthogonal to $\mathcal{K}(\mu)$ for $\mu \neq \omega\mu_0$, as eigenvectors of the unitary Θ for distinct eigenvalues are orthogonal, and to $\mathcal{K}_N \cap \mathcal{H}_N^{\mathbf{n}'}$ for $\mathbf{n}' \neq \mathbf{n}$; since H_N is sector-preserving, $\mathcal{K}_N$ is the orthogonal sum of the spaces $\mathcal{K}_N \cap \mathcal{H}_N^{\mathbf{n}'}$, so such a vector is orthogonal to $\mathcal{K}_N$ as soon as it is orthogonal to $\mathcal{K}(\omega\mu_0)$. The linear map from $\mathcal{K}(\mu_0)$ to $\mathcal{K}(\omega\mu_0)$ that sends Υ' to the orthogonal projection of $\mathcal{W}\Upsilon'$ onto $\mathcal{K}(\omega\mu_0)$ goes to a space of smaller dimension, so it has a nonzero kernel. A unit vector Υ in this kernel lies in $\mathcal{K}(\mu_0)$, so it is a zero mode in $\mathcal{H}_N^{\mathbf{n}}$ and an eigenvector of Θ, and $\mathcal{W}\Upsilon$ is orthogonal to $\mathcal{K}(\omega\mu_0)$, hence to $\mathcal{K}_N$. □

D.5 The Gaussian sums

Lemma D.6 (the Gaussian sums)**.** *For every $\kappa > 0$,*

$$\mathsf{G}_2(\kappa) \leq \frac{4\sqrt{\pi}}{\kappa^3}\Big(1 + \frac{2\kappa}{e\sqrt{\pi}}\Big), \qquad \mathsf{G}_3(\kappa) \leq \frac{16}{\kappa^4}\Big(1 + \frac{(6/e)^{3/2}\kappa}{8}\Big).$$

Proof. Let $f \geq 0$ be continuous and integrable on $[0,\infty)$, nondecreasing on $[0, t_*]$ and nonincreasing on $[t_*, \infty)$. Then $\sum_{d\geq 1} f(d) \leq \int_0^\infty f + \max f$. Indeed, let $d_* := \lfloor t_* \rfloor$. For integers $1 \leq d \leq d_* - 1$, $f(d) \leq \int_d^{d+1} f$, and for $d \geq d_* + 2$, $f(d) \leq \int_{d-1}^d f$. On $[d_*, d_* + 1]$, which contains t_*, f is at least $f(d_*)$ on $[d_*, t_*]$ and at least $f(d_* + 1)$ on $[t_*, d_* + 1]$, hence at least $\min\{f(d_*), f(d_* + 1)\}$; so $f(d_*) + f(d_* + 1) \leq \max f + \int_{d_*}^{d_*+1} f$ if $d_* \geq 1$, while if $d_* = 0$ only $f(1) \leq \max f$ is needed. The intervals of integration used are disjoint and contained in $[0,\infty)$, which proves the claim.

We apply this to $f(t) = 2t^k e^{-\kappa^2 t^2/4}$ with $k = 2$ and $k = 3$. Since $f'(t) = t^{k-1}e^{-\kappa^2 t^2/4}(2k - \kappa^2 t^2)$, f is nondecreasing up to $t_* = \sqrt{2k}/\kappa$ and nonincreasing after it, and $\max f = f(t_*) = 2(2k/\kappa^2)^{k/2}e^{-k/2}$, that is, $8/(e\kappa^2)$ for $k = 2$ and $2(6/e)^{3/2}/\kappa^3$ for $k = 3$. The substitution $t = 2s/\kappa$ and $\int_0^\infty s^2 e^{-s^2}\,\mathrm{d}s = \sqrt{\pi}/4$, $\int_0^\infty s^3 e^{-s^2}\,\mathrm{d}s = \frac{1}{2}$ give $\int_0^\infty f = 4\sqrt{\pi}/\kappa^3$ for $k = 2$ and $16/\kappa^4$ for $k = 3$. Hence $\mathsf{G}_2(\kappa) \leq 4\sqrt{\pi}/\kappa^3 + 8/(e\kappa^2)$ and $\mathsf{G}_3(\kappa) \leq 16/\kappa^4 + 2(6/e)^{3/2}/\kappa^3$, which are the stated bounds. □

D.6 Proof of Theorem D.1

Even N. Let $\mathbf{n} = (N/2, N/2)$. By Theorem 2.10(iii), $\mathcal{K}_N \subset \mathcal{H}_N^{\mathbf{n}}$ and $\dim \mathcal{K}_N = 10$. The root of unity of Lemma D.5 for $\mathcal{W}_\uparrow$ is $e^{2\pi \mathrm{i}(N/2)/(2N)} = \mathrm{i}$, of order four, and ten is not divisible by four. So Lemma D.5 gives a unit vector $\Upsilon \in \mathcal{K}_N$ such that $\mathcal{W}_\uparrow \Upsilon$ is orthogonal to $\mathcal{K}_N$; it is a unit vector in $\mathcal{H}_N^{\mathbf{n}}$ by Lemma D.3(i), and Lemma D.4 bounds $\langle \mathcal{W}_\uparrow \Upsilon, H_N \mathcal{W}_\uparrow \Upsilon \rangle$ by the left-hand side of (D.2). By Lemma D.6 and $\kappa = 2\pi/L_y$,

$$\frac{\kappa^3}{\sqrt{2\pi}} \cdot \frac{\pi^2\, \mathsf{G}_2(\kappa)^2}{2N} \le \frac{\kappa^3}{\sqrt{2\pi}} \cdot \frac{\pi^2}{2N} \cdot \frac{16\pi}{\kappa^6} \Big(1 + \frac{2\kappa}{e\sqrt{\pi}}\Big)^2 = \frac{8\pi^3}{\kappa^3} \cdot \frac{1}{\sqrt{2\pi}\, N} \Big(1 + \frac{2\kappa}{e\sqrt{\pi}}\Big)^2,$$
$$\frac{\kappa^3}{\sqrt{2\pi}} \cdot \frac{\pi^4\, \mathsf{G}_3(\kappa)^2}{32N^3} \le \frac{\kappa^3}{\sqrt{2\pi}} \cdot \frac{\pi^4}{32N^3} \cdot \frac{256}{\kappa^8} \Big(1 + \frac{(6/e)^{3/2}\kappa}{8}\Big)^2 = \frac{8\pi^4}{\kappa^5} \cdot \frac{1}{\sqrt{2\pi}\, N^3} \Big(1 + \frac{(6/e)^{3/2}\kappa}{8}\Big)^2,$$

and $8\pi^3/\kappa^3 = L_y^3$, $8\pi^4/\kappa^5 = L_y^5/(4\pi)$; this is (D.2). The eigenvalue 0 of H_N in $\mathcal{H}_N^{\mathbf{n}}$ has multiplicity ten and eigenspace $\mathcal{K}_N$ (Theorem 2.10 and (2.8)), so, expanding in an orthonormal basis of eigenvectors, every unit vector of $\mathcal{H}_N^{\mathbf{n}}$ orthogonal to $\mathcal{K}_N$ has expectation at least $E_{11}(N)$. This gives the bound on $E_{11}(N)$.

Odd N. Let $N \ge 3$ be odd and $\mathbf{n}$ one of the two sectors; by Theorem 2.10(ii), $\dim(\mathcal{K}_N \cap \mathcal{H}_N^{\mathbf{n}}) = 2$. The root of unity for $\mathcal{W}_\uparrow$ is $\omega = e^{2\pi \mathrm{i} n_\uparrow/(2N)}$. If $\omega^k = 1$, then $2N$ divides $kn_\uparrow$; since $2n_\uparrow - N = \pm 1$, every common divisor of $n_\uparrow$ and N divides 1, so N divides k. Hence the order of ω is at least $N \ge 3$, and two is not divisible by it. Lemma D.5 gives a unit vector $\Upsilon \in \mathcal{K}_N \cap \mathcal{H}_N^{\mathbf{n}}$ such that $\mathcal{W}_\uparrow \Upsilon$ is orthogonal to $\mathcal{K}_N$, and the rest of the argument is the same, with $E_1(N) = E_2(N) = 0 < E_3(N)$ in place of (2.8).

The numbers. $1/\sqrt{2\pi} = 0.39894\ldots$, $2/(e\sqrt{\pi}) = 0.41510\ldots$, $1/(4\pi\sqrt{2\pi}) = 0.031746\ldots$ and $(6/e)^{3/2}/8 = 0.40991\ldots$, so the right-hand side of (D.2) is at most $0.40(1+0.42\kappa)^2 L_y^3/N + 0.032(1+0.41\kappa)^2 L_y^5/N^3$. □

D.7 Remarks

Remark D.7 (the two-species twist). The twist $\mathcal{W}_\uparrow \mathcal{W}_\downarrow^*$, which multiplies the orbitals of the two spins by opposite phases, also moves the zero modes off the kernel at odd N: in the sectors $((N \pm 1)/2, (N \mp 1)/2)$ the root of unity of Lemma D.5 is $e^{\pm \mathrm{i}\pi/N}$, of order $2N$, and two is not divisible by $2N$. By Lemma D.4 it gives the bound on $E_3(N)$ with four times the first and sixteen times the second term of (D.2), which is weaker than Theorem D.1. At even N, in the sector $(N/2, N/2)$, its root of unity is 1: it commutes with Θ there, and Lemma D.5 gives nothing. The single-species twist multiplies the eigenvalues of Θ by i instead, and the count of Lemma D.5 then rests on the fact that ten is not divisible by four, the arithmetic of Corollary 7.3.

Remark D.8 (the eigenvalues of Θ^2 on the kernel). Lemma D.5 uses only the dimension of the kernel. Appendix C gives more. Under the identification of Remark 2.11, Θ is the inverse of the translation T_1 of that appendix, and $\mathcal{K}_N$ at even $N = 2p$ is the sum of the images of the spaces of Lemma C.12(b) for $\varepsilon \in \{\pm 1\}^2$, of dimensions $2 \cdot 2$ for $\varepsilon = (1, 1)$ and $2 \cdot 1$ for each of the other three (Theorem C.19(iii) and Proposition C.18); on the image for ε, $T_1^2 = \varepsilon_2^{-1}$ (Lemma C.12(c)), so $\Theta^2 = \varepsilon_2$. Hence $\Theta^2 = 1$ on a six-dimensional and $\Theta^2 = -1$ on a four-dimensional subspace of $\mathcal{K}_N$. Since $\mathcal{W}_\uparrow$ multiplies the eigenvalues of Θ by i, it maps the eigenspace of Θ^2 for 1 into that for -1, and the argument of Lemma D.5, applied to Θ^2, shows that the zero modes Υ for which $\mathcal{W}_\uparrow \Upsilon$ is orthogonal to $\mathcal{K}_N$ form a subspace of dimension at least two. The proof of Theorem D.1 does not use this.

Remark D.9 (the constants). The constants are not optimal. If Υ has real coefficients in the occupation-number basis, in which the operators $c_{j\sigma}$, and hence $c^0_{n,d}$ and $c^{\mathrm{sing}}_{n,d}$, have real matrix elements, the two vectors whose sum appears in each term of (D.3) are a real vector and i times a real vector; the real part of their inner product then vanishes, the factor two of $\|\xi_1+\xi_2\|^2 \le 2\|\xi_1\|^2+2\|\xi_2\|^2$ in Step 2 of the proof of Lemma D.4 can be dropped, and both terms of (D.4) halve. Lemma D.5 does not provide a real Υ in general. Since $\sin(\pi d/N_\phi)$ and $\cos(\pi d/N_\phi)$ depend on d only modulo $2N_\phi$, d may be replaced in Step 2 by its distance to $2N_\phi\mathbb{Z}$, which is at most d; with the Gaussian weights this gains nothing of importance. For large κ, that is, for thin tori, $\mathsf{G}_k(\kappa)$ is close to its first term $2e^{-\kappa^2/4}$, and the comparison of Lemma D.6 with an integral is crude.

Remark D.10 (what the bound does not give). At fixed circumference Theorem D.1 bounds the lowest eigenvalue above the zero modes by a quantity of order $1/N$. An eigenvalue of this size, or smaller, is compatible with a gapped phase whose ground-state space strictly contains the zero modes, in which it would be a splitting inside the ground-state space; so the theorem does not bear on the situation of Remark 8.7. At fixed aspect ratio it says nothing, and there Theorem 8.1 is the result of this paper.

E The formalization in Lean

The numbered statements of Sections 2–8 and Appendix B — the definitions, lemmas, propositions, theorems and corollaries listed in the table below — have been formalized and proved in the Lean 4 proof assistant against the Mathlib library, with the exceptions of Lemma 2.8 and, in Section 6, the restriction to $U = \Theta$ described in (a); Appendix C, which proves Theorem 2.10, is formalized with the exceptions listed in (c).[1] The project has forty-seven modules and about 56,400 lines. No proof is left open: every declaration depends only on the three standard axioms of Lean's logic (propositional extensionality, choice and quotient soundness), and every module has been replayed through Lean's kernel by `leanchecker`, independently of the elaborator. What the kernel certifies is the Lean text. Whether that text says what this paper says is a matter of reading, and the table is the dictionary for it.

What is proved. Three kinds of statement should be distinguished.

(a) *Unconditional.* Everything in Sections 2–8 and Appendix B except Lemma 2.8 (below) and the general-U clauses of Section 6 named next; and, of Appendix A, Lemmas A.6, A.8 and A.9. In particular Theorem 8.1, Corollary 8.2 and Corollary 8.4 are proved with no hypothesis on the zero modes (`hollow_core_index_unconditional`, `hollow_core_E11_tendsto_unconditional`, `odd_E3_tendsto_unconditional`, `odd_many_small_unconditional`, `uniform_E11_tendsto_uncondition` `uniform_E11_eventually_unconditional`); the divisibility half of Theorem 8.1(i), $4 \mid p$, needs no information about the zero modes at all (`hollow_core_index_four_dvd`), and the rest uses only the lower bound of Theorem 2.10 — at least ten zero modes of H_N in the sector $(N/2, N/2)$ at every even N, on every rectangular torus, and at least two in each of the sectors $((N\pm1)/2, (N\mp1)/2)$ at every odd $N \ge 3$ — which is proved from the explicit states of Proposition C.18 through Lemma C.6 in the direction (2) $\Rightarrow$ (3) (`ten_le_finrank_ker`, `two_le_finrank_ker_odd`). Theorem 2.10 itself, (i)–(iii), is proved for the rectangular tori

[1] Lean 4.34.1 and Mathlib at commit d13f23b7. The formalization is the Lean project `HRTorus` in the directory `lean/` of `https://github.com/bbradlyn/hr-torus-proof`; its `README.md` gives the build, the axiom checks and the kernel replay, and `lean/MAP.md` the complete correspondence: declaration by declaration for the modules written in the formalization's stages 1 and F, by statement for the first commit's.

from the merging induction of Appendix C (`kernel_sector_eq_bot`, `finrank_kernel_odd`, `finrank_kernel_half`, `KN_le_halfSector`, `finrank_KN_odd`, `finrank_KN_even`, with the spanning statements `kernel_half_eq_span`, `kernel_oddPlus_eq_span`, `kernel_oddMinus_eq_span`), through Theorem C.19(i)–(iii) at the level of the spaces $V_{n,m}$ for every τ (`T2_i`, `T2_ii`, `T2_iii`). Theorem 6.18 is formalized for $U = \Theta$, the case this paper's numbered results use, for every Hamiltonian of Definition 4.6 with (H1)–(H2) (`index_theorem_Θ`), the inputs supplied with an explicit $\varepsilon(N) = O(N^{-\infty})$ and $M(N) = (1 + 4\pi C_h \|W_\gamma\|_1)N^2$; for a general U only its finite-dimensional core is formalized, in the following quantitative form: for every finite-dimensional complex inner product space and every choice of the data of Setting 6.1, Assumptions 6.3–6.4 (with $\gamma > \Delta$) and Definitions 6.5 and 6.11 satisfying the exact relations stated there, those of Lemma 6.6(i) and of the filters (Definition 5.8, Lemma 5.9) included, if each of the twenty-seven quantities — the twenty-one separated commutators of Lemma 6.13(ii), (U2), the + half of (U4) and the four clustering commutators of Corollary 6.15 — is at most ε and $\|T_-\|, \|K_-\| \le M$, then there is an integer n with $|\mathrm{Tr}(PT_-) - n| \le 400\, p\, (M+1)\sqrt{\varepsilon}$ (`index_core_400`). The general-U clauses of Lemma 6.12 and Lemma 6.13(ii) are not formalized.

(b) *In two forms.* The second half of Theorem 8.1(i), $12 \le p$, Theorem 8.1(ii), Corollary 8.2 and Corollary 8.4 are also in the tree in the form in which their statements were first reviewed, as implications from the named hypotheses `HasTenZeroModes` and `HasTwoZeroModes` — exactly ten zero modes in the sector $(N/2, N/2)$ at every even N (for Corollary 8.4, on every torus of the family), respectively exactly two in each of the sectors $((N \pm 1)/2, (N \mp 1)/2)$ at every odd $N \ge 3$, the content of Theorem 2.10 together with $H_N \ge 0$ (`hollow_core_index`, `hollow_core_E11_tendsto`, `odd_E3_tendsto`, `odd_many_small`, `uniform_E11_tendsto`, `uniform_E11_eventually`). The unconditional forms are derived from the same statements under the weaker hypotheses `HasAtLeastTenZeroModes` and `HasAtLeastTwoZeroModes` (at least ten, at least two), which the lower bound discharges. The named hypotheses themselves are theorems (`hasTenZeroModes`, `hasTwoZeroModes`, from the exact counts), so the reviewed statements, too, hold for the rectangular tori with no hypothesis on the zero modes.

(c) *Not formalized.* In Appendix C: Lemmas C.2, C.3 and C.5 (the unfolding to the plane and the spin identification; the Lean reaches Lemma C.6 through the local form of Theorem 4.2 and Lemma C.4 instead), Lemma C.8(b) (the general zero count), Lemma C.13(e) (the budget), Theorem C.19(iv) (the Lean's torus is rectangular; the statements at the level of $V_{n,m}$ hold for every τ), and the remarks, Remarks 2.12 and C.20 on the spin of the zero modes among them; Appendix D (Theorem D.1 and Corollary D.2), which no Lean statement uses; and Lemma 2.8, the planar statement about the units of V_1, on which nothing depends: no result of this paper uses it and no Lean declaration does; it fixes the units in which the gaps are measured.

The correspondence. The table lists every numbered statement of Sections 2–8 and Appendices B and C with the Lean module and the declarations that formalize it; a statement with several parts is usually several declarations. The lemmas of Section 6 that the finite-dimensional index core proves (Lemmas 6.14–6.17 and Theorem 6.18 itself) are formalized in the twelve modules `IndexCore.Statement` to `IndexCore.Sanity`, with `IndexCore.Translation` and `IndexCore.Ring` for Section 7; their declarations cite the statements, and the table names the modules and the main declarations. Lemma 3.2(i)–(ii) and Lemma 3.5 are formalized in

`SlotFrame.SlotFrame`, also of the first commit, and the table names its declarations. Where more than four declarations formalize one statement the table names four and counts the rest; `lean/MAP.md` names them all. The Lean statements follow this paper's text under the conventions listed after the table.

Statement	Title	Lean module: declarations
Definition 2.1	the torus and its lowest Landau level	`LLL.Space`: `Space`, `IsLLL`, `l2inner`
Lemma 2.2	Landau-gauge orbitals	`LLL.Space`: `orbBasis`, `finrank_space`, `psi_isLLL`, `l2inner_psi` and 5 more
Lemma 2.3	evaluation vectors	`LLL.Space`: `ev`, `evD`, `inner_ev`, `inner_evD` and 9 more
Lemma 2.4	magnetic translations	`LLL.Space`: `thxFun_psi`, `thyFun_psi`, `toFun_thx`, `toFun_thy` and 11 more
Definition 2.5	Fock space, sectors, the many-body translation	`Fermion.Pilot.JW`: `IsEven`; `Fock.Physical`: `FockSpace`, `cf`, `cjs`, `njs` and 16 more; `LLL.Space`: `Space2`
Lemma 2.6	sectors and the many-body translation	`Fock.Physical`: `orthogonalFamily_sector`, `isInternal_sector`, `Psec_eq_cfc`, `isSectorPreserving_iff_sum` and 11 more
Definition 2.7	the hollow-core Hamiltonian	`Fock.Hamiltonian`: `Vcoef`, `HN`, `Vcoef_latticeSum`
Lemma 2.8	V_1 units	*not formalized: the planar units statement; nothing in the formalization depends on it*
Theorem 2.10	the zero modes	`Assembly.Main`: `hasTenZeroModes_of_kernel`, `hasTwoZeroModes_of_kernel`; `ZeroModes.Lower`: `exists_ten_zero_modes`, `exists_two_zero_modes_plus`, `exists_two_zero_modes_minus`, `ten_le_finrank_ker` and 3 more; `ZeroModes.Upper`: `KN`, `kernel_sector_eq_bot`, `KN_inf_sector_eq_bot`, `finrank_kernel_odd` and 8 more *((i)–(iii) for the rectangular tori; the lower bounds also directly from the explicit states)*
Definition 3.1	admissible tori and the slot frame	`Fock.Slot`: `slotDist`; `LLL.Frame`: `frameVec`, `analysis`, `analysis2`; `Locality.Basic`: `xs`; `Locality.LiebRobinson`: `slotSetDist`
Lemma 3.2	consistency of the construction	`Locality.Basic`: `card_Ia_lt`, `eq_of_mem_Ia_of_intCast_eq`, `wa_add_nx_add_Nφ`; `SlotFrame.SlotFrame`: `window_contDiff`, `window_eq_zero`, `window_sq_tsum` (i); `SlotFrame.SlotFrame`: `consistency_ii`, `Ia_nonempty` (ii)
Proposition 3.3	Parseval	`LLL.Frame`: `sum_rankOne_frameVec`, `adjoint_analysis_comp_analysis`, `norm_analysis`, `analysis_comp_adjoint` and 5 more
Proposition 3.4	slot geometry	`Locality.Basic`: `slotBall`, `strip`, `Cvol`, `Cstr` and 6 more
Lemma 3.5	uniform localization	`LLL.Frame`: `slotProj`, `localization_iii_analysis`; `SlotFrame.SlotFrame`: `localization_i`, `localization_ii`, `localization_iii`, `localization`
Lemma 3.6	exact column support	`Locality.Basic`: `col`, `inner_orb_frameVec`, `inner_orb_frameVec_eq_zero`, `analysis_orb_mem_col` and 2 more
Definition 3.7	slot Fock space, local algebras, the embedding	`Fermion.Pilot.JW`: `chat`, `numberOp`, `CAR`, `Alg`; `Fock.Embedding`: `iota`, `iota_cf`, `exists_starAlgHom_iota`; `Fock.Slot`: `chatVec`, `chatVec_spinTensor_single`, `SlotFock`, `fat`
Lemma 3.8	properties of the embedding	`Fock.Embedding`: `FfA`, `FfA_conjTranspose_mul_self`, `norm_FfA_apply`, `range_FfA` and 14 more; `Fock.Slot`: `gauge_image_CAR`, `gauge_image_Alg`
Definition 3.9	strips, columns	`Locality.Basic`: `colBand`, `cthickening_circImage_Ioo`
Lemma 3.10	strips and fattenings	`Locality.Basic`: `fat_strip_subset`, `strip_cthickening_subset`
Definition 3.11	almost supported operators	`Locality.Basic`: `TorusSeq`, `setNorm`, `ASBound`, `ASWitness` and 4 more
Definition 4.1	triplet pairs, local terms, centre pair operators	`Hamiltonian.Pairs`: `chi`, `cpair`, `cpairHat`, `Bν` and 7 more
Theorem 4.2	the exact local form	`Hamiltonian.LocalForm`: `localForm_a`, `localForm_b`, `localForm_c`
Lemma 4.3	jets on the diagonal	`Hamiltonian.Pairs`: `D1x`, `D1`, `jetF`, `jets_symm` and 1 more

Statement	Title	Lean module: declarations
Proposition 4.4	uniform size and rapid locality	`Hamiltonian.SlotLocal`: `ChHC`, `Bν R`, `hRr`, `norm_Bν_le` and 6 more
Lemma 4.5	covariance	`Hamiltonian.LocalForm`: `commute_Θ_HN`; `Hamiltonian.Pairs`: `HD`, `Θ_conj_Bν`, `Θ_conj_hR`, `Θ_conj_HD`
Definition 4.6	slot-local covariant Hamiltonians	`Hamiltonian.SlotLocal`: `H1Data`, `H1Data.H2`, `SatisfiesH1`, `SatisfiesH1H2`
Corollary 4.8	the hollow-core Hamiltonian is in the class	`Hamiltonian.SlotLocal`: `hollowCore_mem_class`
Definition 4.9	band charges	`Hamiltonian.BandCharges`: `Qband`
Lemma 4.10	elementary properties of band charges	`Hamiltonian.BandCharges`: `commute_Qband`, `Qband_isSectorPreserving`, `spectrum_Qband_subset`, `norm_Qband_le` and 4 more
Lemma 4.11	band charges in the slot lattice	`Hamiltonian.BandCharges`: `vj`, `bandX`, `iota_njs_up`, `vj_support` and 9 more
Lemma 5.1	conditional expectations	`Fermion.Pilot.JW`: `condExp`, `condExp_isLinearMap`, `condExp_one`, `condExp_posSemidef` and 7 more; `Locality.LiebRobinson`: `measurable_condExp`, `measurable_condExp_of_continuous`
Theorem 5.2	Lieb–Robinson bound for even interactions	`Locality.LiebRobinson`: `lieb_robinson`, `lieb_robinson_of_disjoint`, `lieb_robinson_of_terms_zero`, `chainWeight` and 3 more
Theorem 5.3	propagation from a set	`Locality.Propagation`: `ck`, `rj`, `C1`, `C1_le` and 7 more
Lemma 5.5	propagation of almost-supported operators	`Locality.Propagation`: `setNorm_dyn_le`
Lemma 5.6	time filters	`Locality.Filters`: `moment`, `HasFiniteMoments`, `setNorm_timeFilter_le`, `almostSupported_timeFilter`
Lemma 5.7	commutator with the Hamiltonian	`Locality.Filters`: `norm_comm_HN_le`, `almostSupported_comm_HN`
Definition 5.8	the filters	`Locality.Filters`: `IsCutoff`, `What`, `Wfilt`, `ClusterParams` and 3 more
Lemma 5.9	the generator filter	`Locality.Filters`: `Wfilt_im`, `Wfilt_bounded`, `contDiffOn_Wfilt`, `tendsto_Wfilt_zero` and 7 more
Lemma 5.10	time representation of the clustering filter	`Locality.Filters`: `zetaHat`, `Fclust`, `cF`, `Gclust` and 6 more
Proposition 5.11	locality of the clustering filter	`Locality.Filters`: `almostSupported_clusterFilter`
Lemma 5.12	locality calculus	`Locality.Calculus`: `norm_comm_le_of_disjoint`, `almostSupported_add_star_mul`, `ordExp_almostSupported`, `exp_Qband_add_almostSupported`
Lemma 5.13	bookkeeping	`Locality.Calculus`: `bookkeeping_comm`, `IsONegInf.sum_mul_mul`
Setting 6.1		`Assembly.Setting`: `IxSetting`; index core: `IndexCore.Translation`, `IndexCore.Statement`
Lemma 6.2	separations	`Assembly.Setting`: `sep_cuts`, `sep_quarters`, `sep_cut_mid`, `sep_cut_opposite` and 2 more
Assumption 6.3	gap in a sector	`Assembly.Setting`: `GapAssumption`, `ArgData.rank_P`, `ArgData.P_props`; index core: `IndexCore.Statement`, `IndexCore.Translation`
Assumption 6.4	charge transport	`Assembly.Translation`: `TransportAssumption`; index core: `IndexCore.Statement`
Definition 6.5	current, generator, dressed charge	`Assembly.Setting`: `Jminus`, `Jplus`, `Kfull`, `Kminus` and 2 more; index core: `IndexCore.Statement`, `IndexCore.Translation`
Lemma 6.6	the split of the current	`Assembly.Setting`: `Jminus_add_Jplus`, `Kfull_eq_add`, `J_isSelfAdjoint`, `K_isSelfAdjoint` and 3 more; index core: `IndexCore.KP`, `IndexCore.Statement`
Lemma 6.7	the in-sector identity	`Assembly.Setting`: `K_comm_P`; index core: `IndexCore.KP`, `IndexCore.Statement`
Lemma 6.8	strip localization of the dressing	`Assembly.Setting`: `norm_K_le`, `K_almostSupported`, `norm_KShift_le`

Statement	Title	Lean module: declarations
Proposition 6.10	in-sector clustering	`Assembly.Inputs`: `Qf_mul_P`, `P_mul_comm_mul_P`, `clustering_isONegInf`; index core: `IndexCore.Clustering`, `IndexCore.Statement`
Definition 6.11	the objects of the argument	`Assembly.Inputs`: `KUminus`, `KUplus`, `QbarMinus`, `QbarPlus` and 7 more *(for $U = \Theta$)*; index core: `IndexCore.Statement`, `IndexCore.Commutation`
Lemma 6.12	localization and size	`Assembly.Inputs`: `table_Q`, `table_T`, `table_J`, `table_K` and 4 more *(for $U = \Theta$)*; index core: `IndexCore.Statement`
Lemma 6.13	commutation relations	`Assembly.Inputs`: `sepPairs`, `exact_relations`, `sep_isONegInf`, `comm_Qmid_isONegInf` and 3 more *(for $U = \Theta$)*; index core: `IndexCore.Translation`, `IndexCore.Commutation`
Lemma 6.14	Lemma 4.2 of [BBDF21] in the sector	index core: `IndexCore.BBDF42`, `IndexCore.Clustering`
Corollary 6.15	the localized unitaries commute with P	`Assembly.Inputs`: `clu_isONegInf`; index core: `IndexCore.BBDF42`, `IndexCore.Statement`
Lemma 6.16	the differential equation and its stability	index core: `IndexCore.ODE`
Lemma 6.17	the determinant at 2π	index core: `IndexCore.Determinant`, `IndexCore.Det`
Theorem 6.18	index theorem for lowest-Landau-level Hamiltonians on the torus	`Assembly.Inputs`: `index_theorem_Θ` *(for $U = \Theta$)*; `IndexCore.Main`: `index_core`, `index_core_400` (finite-dimensional form, general U)
Lemma 7.1	the magnetic translation transports one orbital	`Assembly.Translation`: `theta_commute`, `theta_transport`, `theta_exp`, `theta_trace` and 1 more; index core: `IndexCore.Translation`, `IndexCore.Ring`
Corollary 7.2	the translation corollary	`Assembly.Translation`: `translation_corollary`; `IndexCore.Translation`: `Orbital.translation_corollary` (finite content)
Corollary 7.3	half filling with zero spin polarization	`Assembly.Translation`: `four_dvd_of_halfFilling`, `four_dvd_hollowCore`, `no_gapped_patch`; `IndexCore.Translation`: `Orbital.four_dvd_p` (finite content)
Corollary 7.4	odd N and spin one half	`Assembly.Translation`: `no_gap_odd`, `many_low_odd`; `IndexCore.Translation`: `Orbital.odd_plus`, `Orbital.odd_minus` (finite content)
Theorem 8.1	no uniform gap above the zero modes	`Assembly.Main`: `rectTorus`, `evenEig`, `hollow_core_index_four_dvd`, `hollow_core_index` and 1 more; `ZeroModes.Lower`: `hollow_core_index'`, `hollow_core_E11_tendsto'`, `hollow_core_index_unconditional`, `hollow_core_E11_tendsto_unconditional`
Corollary 8.2	odd N	`Assembly.Main`: `oddEigPlus`, `oddEigMinus`, `odd_E3_tendsto`, `odd_many_small`; `ZeroModes.Lower`: `odd_E3_tendsto'`, `odd_many_small'`, `odd_E3_tendsto_unconditional`, `odd_many_small_unconditional`
Lemma 8.3	exchange of the sides	`Assembly.Main`: `swapTorus`, `rotFun`, `FfR`, `isLLL_rotFun` and 6 more
Corollary 8.4	uniformity over aspect ratios	`Assembly.Main`: `tauSet`, `E11`, `uniform_E11_tendsto`, `uniform_E11_eventually`; `ZeroModes.Lower`: `uniform_E11_tendsto'`, `uniform_E11_eventually'`, `uniform_E11_tendsto_unconditional`, `uniform_E11_eventually_unconditional`
Proposition B.1	disjoint supports [NSY18, Proposition 2.1]	`Locality.LiebRobinson`: `commute_of_mem_CARmEven_of_mem_CARm`, `anticomm_of_mem_CARm_of_isOdd`
Lemma B.2	[NSY18, Lemma 3.2]	`Locality.LiebRobinson`: `norm_le_of_hasDerivAt_comm`
Lemma B.3	the one-step estimate [NSY18, Lemma 3.3]	`Locality.LiebRobinson`: `norm_comm_dyn_le_oneStep`
Lemma C.2	the $m = 1$ amplitude	*not formalized: the unfolding to the plane; the Lean reaches Lemma C.6 through the local form of Theorem 4.2 and Lemma C.4*
Lemma C.3	unfolding	*not formalized: the unfolding to the plane; the Lean reaches Lemma C.6 through the local form of Theorem 4.2 and Lemma C.4*

Statement	Title	Lean module: declarations
Lemma C.4	orbital form; the period is $2L$	`ZeroModes.Canonical`: `betaC`, `betaC_add_period`, `lemA3`, `betaC_eq_zero_of_diag` *(the first identity; the norm identity not formalized)*
Lemma C.5	exchange and spin	*not formalized: the spin identification, not needed on the Lean's route to Lemma C.6*
Lemma C.6	formulation lemma on the torus	`ZeroModes.Canonical`: `C1`, `C2`, `lemA_bcen`, `lemA_HN`; `ZeroModes.Formulation`: `lemA_HN_conv`, `lemA_bcen_conv`, `lemA_iff`, `lemA_iff_bcen` *(in full: (2) ⇒ (3) ⇒ (1) in the first module, (1) ⇒ (3) ⇒ (2) in the second)*
Lemma C.8		`ZeroModes.Theta`: `theta1`, `theta1_neg`, `theta1_add_one`, `theta1_add_tau` and 26 more *((a), (c)–(f); (b) not formalized)*
Lemma C.10	evenness: (C2) is a zero-residue condition	`ZeroModes.Explicit`: `lemEven`
Definition C.11		`ZeroModes.Spaces`: `Gnum`, `Gnum_zero_zero`, `finrank_Gnum_zero_zero` *(in numerator form, as the space of the numerators $J^2 g$; the conditions (i)–(iii) of the explicit functions are under Proposition C.18)*
Lemma C.12	centre-of-mass factorization	`ZeroModes.Explicit`: `stateFun`; `ZeroModes.Formulation`: `V`, `kerSector_equiv_V`, `finrank_ker_eq_finrank_V`; `ZeroModes.Spaces`: `Θ2g`, `Θ2g_rootPair`, `finrank_Θ2g_rootPair`, `Gnum_eq_bot_of_not_root` and 6 more; `ZeroModes.Theta`: `Θ2`, `finrank_Θ2` *((a)–(c) in numerator form, with the count $\dim V_{n,m} = 2\sum_{\varepsilon\in\mu_N^2} \dim \mathcal{G}(n,m;\varepsilon)$)*
Lemma C.13	torus merging lemma	`ZeroModes.Merge`: `mu`, `lemC_a`, `lemC_b`, `lemC_c_indep` and 2 more *((a)–(d) in numerator form; (e), the budget, not formalized)*
Lemma C.14		`ZeroModes.Merge`: `Gnum_polarized`, `Gnum_one_zero_pp`, `Gnum_one_zero_ne`, `lemD_b` and 1 more *(in numerator form)*
Proposition C.15	the inductive inequality	`ZeroModes.Merge`: `muL`, `muL_injective`, `finrank_Gnum_le_merge`, `Gnum_eq_bot_of_merge` *(in numerator form)*
Proposition C.16	base cases	`ZeroModes.Merge`: `base_a`, `Gnum_one_one_pp`, `finrank_Gnum_one_one_pp`, `Gnum_one_one_eps` and 2 more *(in numerator form)*
Corollary C.17	upper bounds	`ZeroModes.Merge`: `Gnum_eq_bot_of_imbalanced`, `finrank_Gnum_odd_pp`, `Gnum_odd_eq_bot`, `finrank_Gnum_even_le` and 2 more *(in numerator form)*
Proposition C.18	the explicit states	`ZeroModes.Explicit`: `gOdd`, `dEps`, `f5`, `dEps_G` and 15 more; `ZeroModes.Spaces`: `gOdd_numerator_one_exists`, `gOdd_numerator_one_mem`, `dEps_numerator_one_exists`, `dEps_numerator_one_mem` and 2 more
Theorem C.19	the zero modes on the torus at $N_\phi = 2N$	`ZeroModes.Explicit`: `gOddSwap`, `gOddSwap_numerator_exists`, `gOddSwap_numerator_props`, `thetaK_mem` and 4 more; `ZeroModes.Spaces`: `gOddSwap_numerator_one_exists`, `gOddSwap_numerator_one_mem`; `ZeroModes.Upper`: `T2_i`, `T2_ii`, `T2_iii`, `finrank_KN_odd` *((i)–(iii) at the level of $V_{n,m}$ for every τ, the lower halves from the explicit states; (iv) not formalized)*

Where the Lean proof takes another route. The Lean proofs follow this paper's proofs, except in the following places, where they use a different argument; the statements are unaffected. Where this paper names a constant the Lean's is inside it, with one exception: the constant C_k^H of Proposition 4.4(ii), which the paper's proof gives as $55\,C_k$ and the Lean obtains as $192\pi\,C_k$ (the statement is existential).

- The commutation $[\Theta, H_N] = 0$ is proved twice: as Lemma 4.5, by the paper's route, and, for the statement of Section 2, from the covariance of the orbitals through the coefficient identity $V_{a+1,b+1,c+1,d+1} = V_{abcd}$.

- The sector-preserving approximant of Lemma 3.8 by a finite average over gauge phases, in place of the integral.

- Lemma 4.3 by a real-variable argument rather than a holomorphic one; Theorem 4.2(a) by reducing the matrix entries on the whole Fock space to a form on pair amplitudes, in place of Step 6's coefficient matrix on the two-particle space; Lemma 2.3(iv) by geometric majorants.

- The Lieb–Robinson iteration of Appendix B as an induction on a per-slot bound, without the chain sums; Lemma 5.5 by propagating one best approximant at order $k+2$ instead of the dyadic decomposition; Step 4 of Theorem 5.3(a) by shells of width λ with weights e^{-n} in place of the planar integral, which gives $\sum_{n\ge 0} e^{-n}(n+3)^2 \le 26$, that is $6.5\pi r_j^2(1+C_1|t'|)^2$ where the paper has $8\pi r_j^2(1+C_1|t'|)^2$; the master inequality is then stated with the paper's constant; Lemma 5.12(c) by an energy method.

- The completeness half of Lemma 2.2(ii) with the y-Fourier coefficients defined as integrals and their first-order equation obtained by differentiation under the integral sign and an integration by parts, where the paper differentiates the series term by term.

- Lemma 6.7 and Lemma 7.1(d) by applying the index core's lemmas to instances whose unused fields carry trivial data, so that neither statement acquires a hypothesis this paper's lacks.

- Lemma 6.6(ii)–(iii): the far region bounded by its supremum times the volume of the half-torus, in place of the one-dimensional integral.

- Corollary 7.4(ii) by a finite pigeonhole over the scales $\gamma/4^i$, in place of compactness along a subsequence, with the dimension bound $\binom{2N}{(N-1)/2} \ge (N+1)/2$ where the paper uses $\ge 2N$; in Corollary 7.4(i) the paper's two cases are folded into one, $p-4k\neq 0$ or $p-4k=0$.

- Lemma 8.3: the conjugation of H_N by the exchange unitary through the local form of Theorem 4.2(a), $H_N=\int h_R\,\mathrm{d}R$, in place of the change of basis in V_{abcd}; the equality of the sector eigenvalues through the counting function $E\mapsto \#\{k : E_k\le E\}$.

- Appendix C, the lower bound. The canonical components of Section C.1.2 as a sum over tuples of orbitals, with the Slater-determinant form as a theorem, and the step from a function of the canonical shape back to a Fock vector, which this appendix leaves silent, as a theorem (`canon_surj`, by interpolation in every variable); the zero set of ϑ_1 in Lemma C.8(c) from Jacobi's product, without the argument principle; Mathlib's Weierstrass $\wp$, which differs from this appendix's by an additive constant, so that the relative state $d^{(p)}_{(1,1)}$ of the Lean is this appendix's plus a multiple of $f_5^{(p)}$ (the span, and every statement, is unchanged). Mathlib has no analysis of several complex variables, so the explicit states are carried as holomorphic numerators $F(U)\,J(x)^2J(y)^2\det(\cdots)$, the entries $\vartheta_1^2\wp_\varepsilon$ made entire by the one-variable removable singularity, in the class of continuous functions entire in each variable separately; (C1) through the vanishing of the holomorphic Q in $f=\vartheta_1(u_s-u_t)^2Q$, and Lemma C.6 (2) $\Rightarrow$ (3) using (C1) only at order two on the diagonal; (C2) by this appendix's route at the generic points of a divisor $x_a=y_k$ and extended to every point of it along a complex line inside the divisor, on which only countably many points are non-generic, by the continuity of $\partial_v f$.

- Appendix C, the count. Lemma C.6 (1) $\Rightarrow$ (3) from the sum of squares of Theorem 4.2(c) and (3) $\Rightarrow$ (2) from Lemma C.4, without the unfolding to the plane; everything from Definition C.11 on in numerator form, $\mathcal{G}(n,m;\varepsilon)$ being the space of the numerators J^2g (continuous, entire

in each variable) with the conditions (iv)–(v) transported to vanishing orders, so that no meromorphic function of several variables appears, and the multipliers ε as pairs of complex numbers with μ_N^2 parametrized by (j,k); Lemma C.12(b) through the projectors $P_{jk} = N^{-2}\sum_{r,s<N}\omega^{-jr-ks}(T_K^2)^r(T_1^2)^s$ at the function level, without eigenspace theory; the merge μ as the merged numerator at an admissible centre $w \notin \mathrm{rest} + \Lambda_u$, chosen by Hilbert's ε-operator, Lemma C.13(c) making the choice immaterial; in Lemma C.13(a) the polar-set argument of Step 4 replaced by an explicit division by E in w, and the identity theorem of Step 5 by continuity; Lemma C.10 replaced by the direct computation in Lemma C.13(d); the analysis of several variables replaced by a toolkit for the class — Cauchy's formula for the slice derivatives as a parametric integral, division by $\vartheta_1(u_s - u_t)^k$ through one-variable removable singularities, with continuity by a mean value on a circle and holomorphy in the other variables by locally uniform limits, Liouville one variable at a time, and Baire's theorem for the density of the generic points; in Theorem C.19(iii) the count $2(2+1+1+1)$ through the four sign pairs among the N-th roots of unity; and, for Theorem 2.10 on $\mathcal{K}_N$, the grading of the kernel by the sectors componentwise in the occupation basis.

Conventions. The eigenvalues $E_k(N)$ of H_N in a sector are those of Mathlib's spectral theorem for the compression, the index clamped to $1 \le k \le \dim$; Definition 6.11 and Lemmas 6.12 and 6.13 on the Fock space, and Theorem 6.18, are formalized for $U = \Theta$ only, the case this paper's numbered results use (the index core's Lemma 6.13(i) and (iii) are for a general U); $\varepsilon(N)$ is the largest of the twenty input quantities (for $U = \Theta$, (U2) and (U4) hold exactly and the twenty-one separated commutators reduce to sixteen pairs, which with the four clustering commutators make twenty) and $M(N) = (1+4\pi C_h\|W_\gamma\|_1)N^2$; the set norm of Definition 3.11 and the distances to sets take values in $[0,\infty]$. Two sentences of this paper were made precise by the formalization and read accordingly in this version: Lemma 5.5 and Lemma 5.6 state the hypothesis $\|A\|_{X;\max\{k,3\}} < \infty$ that their proofs assumed (without it, with $X = \emptyset$ and the convention $0 \cdot \infty = 0$, their literal transcription is false), and the sentences on odd N in Section 2 and Corollary 8.2 say $N \ge 3$, since at $N = 1$ the sectors in question are two-dimensional and have no E_3. Some Lean statements carry hypotheses that this paper's statements carry and their proofs do not use; they are kept, and `lean/README.md` lists them.

F Notation

The table lists the notation of Sections 1–8 and Appendices A, B and D; the first subsection of Appendix A describes the setting of [BBDF21] in the notation of that paper. Appendix C uses its own notation, which is defined there.

symbol	meaning	where
General		
$\langle f, g\rangle$	inner product, linear in g	§1
$\lvert u\rvert_P$, $(u)_+$	distance from u to $P\mathbb{Z}$; $\max\{u,0\}$	§1
$O(N^{-\infty})$, $\doteq$	bounded by $C_k N^{-k}$ for every k; equality up to $O(N^{-\infty})$	§1
C, C', c; C_k	constants independent of N; depending on an index k	§1
$\mathbf{1}_X$	identity; orthogonal projection onto functions supported in X	§3

symbol	meaning	where
Torus, lowest Landau level, Fock space (Section 2)		
N, $N_\phi = 2N$	electron-number parameter; number of flux quanta and of orbitals	Def. 2.1
$\mathbb{T}_N$, L_x, L_y, κ	torus $\mathbb{R}^2/(L_x\mathbb{Z} \times L_y\mathbb{Z})$, $L_xL_y = 4\pi N$; $\kappa = 2\pi/L_y$, $L_x = \kappa N_\phi$	Def. 2.1
$r = (x, y)$; $R = (R_x, R_y)$	points of $\mathbb{T}_N$ (x transverse, across the cuts); centres of local terms	Def. 2.1
$d(r, r')$, $\lvert\mathbb{T}_N\rvert$, $\hat{e}_x$	flat distance; area; unit vector along x	Def. 2.1
D_x, D_y, D	covariant derivatives ∂_x, $\partial_y - \mathrm{i}x$; $D = \frac{1}{2}(D_x - \mathrm{i}D_y)$	(2.1)
$\mathfrak{h}_N$	lowest Landau level of $\mathbb{T}_N$	Def. 2.1
ϕ_n, ψ_j	Landau-gauge orbitals in the plane ($n \in \mathbb{Z}$) and on the torus ($j \in \mathbb{Z}_{N_\phi}$)	Lem. 2.2
$[j]$	class of $j \in \mathbb{Z}$ in $\mathbb{Z}_{N_\phi}$	Lem. 2.2
e_R, e_R^D	evaluation vectors: $\langle e_R, f\rangle = f(R)$, $\langle e_R^D, f\rangle = (Df)(R)$	Lem. 2.3
$L_{\min}$	$\min\{L_x, L_y\}$	Lem. 2.3
θ_x, θ_y	one-particle magnetic translations, $\theta_x\psi_j = \psi_{j+1}$	Lem. 2.4
$\mathfrak{k}_N$, $\mathcal{H}_N$, $\mathcal{F}(u)$	$\mathfrak{h}_N \otimes \mathbb{C}^2$; Fock space $\mathcal{F}(\mathfrak{k}_N)$; second quantization of u	Def. 2.5
Ω_0, $\widehat{\Omega}_0$	the vacua of $\mathcal{H}_N$ and of $\widehat{\mathcal{H}}_N$	Lem. 3.8
$c(f)$, $c_{j\sigma}$, $n_{j\sigma}$	annihilation operators (conjugate-linear in f); occupation numbers	Def. 2.5
$\sigma, \sigma' \in \{\uparrow, \downarrow\}$	spin labels	Def. 2.5
$N_\uparrow$, $N_\downarrow$, S_z	numbers of electrons of each spin; $\frac{1}{2}(N_\uparrow - N_\downarrow)$	Def. 2.5
$\mathbf{n} = (n_\uparrow, n_\downarrow)$, $\mathcal{H}_N^{\mathbf{n}}$, $\Pi^{\mathbf{n}}$	sector label; sector; projection onto it	Def. 2.5
Θ	many-body magnetic translation $\mathcal{F}(\theta_x \otimes \mathbf{1})^*$, $\Theta^* n_{j\sigma}\Theta = n_{j+1,\sigma}$	Def. 2.5
V_{abcd}	two-body matrix elements	Def. 2.7
H_N	hollow-core Hamiltonian in V_1 units	Def. 2.7
V, $V_{\mathbb{T}}$	planar interaction $4\pi(\delta + \nabla^2\delta)$; its periodization	Def. 2.7
$\mathcal{K}_N$	zero modes with N electrons	Thm. 2.10
$E_j(N)$	eigenvalues in the relevant sector, increasing, with multiplicity	(2.8)
The slot frame (Section 3)		
η, δ_0, w	profile; target slot width; window $\eta/(\sum_k \eta(\cdot - k)^2)^{1/2}$	Def. 3.1
L_0, $\delta_\pm$	$\max\{4, 3\pi/\delta_0\}$ (admissibility); $\frac{2}{3}\delta_0$, $2\delta_0$	Def. 3.1
n_x, δ_x, x_a	number of windows; window spacing L_x/n_x; window centres	Def. 3.1
n_y, δ_y, y_ℓ	slots per window $\lceil 2\delta_x/\kappa\rceil$; spacing L_y/n_y; positions	Def. 3.1
$w_a(j)$, I_a	window a in the orbital index; its support	Def. 3.1
Λ_N, $s = (a, \ell)$	slot lattice $\mathbb{Z}_{n_x} \times \mathbb{Z}_{n_y}$; a slot	Def. 3.1
r_s, x_s	centre of the slot s; its transverse coordinate	Def. 3.1
z_s, $\mathcal{S}$	frame vectors; analysis map $f \mapsto (\langle z_s, f\rangle)_s$ (also $\mathcal{S} \otimes \mathbf{1}$)	Def. 3.1
$d(s, s')$, $d(s, R)$, $d(X, Y)$	distances between slot centres, points, sets of slots; $d(X, Y) := +\infty$ if X or Y is empty	Def. 3.1, Lem. 6.2
$\Lambda_r(R)$, C_{vol}	slots within distance r of R; ball-count constant	Prop. 3.4
$\mathrm{Str}(I)$, I_ρ	strip over a transverse interval I; ρ-neighbourhood of I	Prop. 3.4, Def. 3.9
C_{str}	strip-count constant: $\lvert\mathrm{Str}(I)\rvert \le C_{\mathrm{str}}L_y(1 + \lvert I\rvert)$	Prop. 3.4
$\mathrm{Col}(j)$, $\mathrm{Col}(\mathcal{B})$	column of the orbital j (width $2\delta_x$); union over a band	Lem. 3.6, Def. 3.9
$\widehat{\mathcal{H}}_N$, $\hat{c}(g)$, $\hat{c}_{s\sigma}$, $\widehat{N}_\sigma$	slot Fock space; slot annihilators; slot number operators	Def. 3.7

symbol	meaning	where
$\mathrm{CAR}(X)$, $\mathfrak{A}(X)$	algebra of the modes of X; its even part	Def. 3.7
$X_{(r)}$	r-fattening $\{s : d(s,X) \le r\}$	Def. 3.7
ι, $\widehat{H}$	embedding $\iota(c(f)) = \hat{c}(\mathcal{S}f)$; $\iota(H)$	Def. 3.7
$\mathcal{A}_Z$, m	almost supported in Z; prefactor exponent	Def. 3.11
$\|A\|_{X;k}$	set norm of order k	Def. 3.11
The Hamiltonian in the slot frame (Section 4)		
$\nu \in \{+,0,-\}$, χ^ν	triplet label; triplet spinors	Def. 4.1
$c^\nu(f,g)$, $\hat{c}^\nu(h,h')$	triplet pair annihilators on $\mathcal{H}_N$, on $\widehat{\mathcal{H}}_N$	Def. 4.1
$B_\nu(R)$, h_R	$c^\nu(e_R, e_R^D)$; local term $8\pi \sum_\nu B_\nu(R)^* B_\nu(R)$	Def. 4.1
b_n^ν	orbital pair operators with centre $n/2$, $n \in \mathbb{Z}_{2N_\phi}$	Def. 4.1
$B_\nu^{(r)}(R)$, $h_{R,r}$	slot approximants of $B_\nu(R)$, h_R in $\mathfrak{A}(\Lambda_r(R))$	Prop. 4.4
C_h, C_k^H	bound on $\|h_R\|$; rapid-locality constants	Prop. 4.4, Def. 4.6
$H_\mathcal{D}$	$\int_\mathcal{D} h_R \, \mathrm{d}R$	Lem. 4.5
(H1), (H2)	slot locality of uniform size; covariance	Def. 4.6
$\mathcal{B}$, $Q_\mathcal{B}$	band (set of orbitals); its up-spin charge	Def. 4.9
Locality estimates (Section 5)		
τ_t	$A \mapsto e^{\mathrm{i}t\widehat{H}} A e^{-\mathrm{i}t\widehat{H}}$ (and $e^{\mathrm{i}tH_N} A e^{-\mathrm{i}tH_N}$ on $\mathcal{H}_N$)	§5
$\mathbb{E}_Y$	conditional expectation onto $\mathrm{CAR}(Y)$	Lem. 5.1
Ψ, τ_t^Ψ, M^Ψ, $C(\Psi,\lambda)$	even interaction; its dynamics; chain matrix; weighted row sum	Thm. 5.2
c_k, r_j, C_1	constants and dyadic radii of the propagation theorem	Thm. 5.3
$\tau^{\lambda,Y}$, $\widehat{H}_{\lambda,Y}$	truncated dynamics; its generator, in $\mathfrak{A}(Y)$	Thm. 5.3
ρ, λ, α, β	approximation radius; truncation scale; light-cone exponent; $1 - 1/\alpha$	Thm. 5.3
C_lc	light-cone constant, $\max\{1, (4C_1)^4\}$ for $\alpha = 4$	Thm. 5.3
$M_j(F)$	moments $\int (1+\|t\|)^j \|F(t)\| \, \mathrm{d}t$	Lem. 5.6
W_γ, $\widehat{W}_\gamma$, ϑ	generator filter; its Fourier transform; cutoff	Def. 5.8
g, ζ, δ, $\mathcal{Q}$	clustering step; mollifier and its width; $\mathcal{Q}(A) = g(\mathrm{ad}_H)(A)$	Def. 5.8
γ', $\hat{\zeta}$, F, G, c_F	$-\gamma/2$; Fourier transform; time representation of g	Lem. 5.10
m_*	bound on all prefactor exponents of the proof	Lem. 5.13, 6.12
The index theorem (Sections 6–7)		
$\mathcal{N}$, c_*	infinite index set; constant in $L_x \ge c_* \sqrt{N}$	Set. 6.1
Γ, Γ_-, Γ_+, Γ_m	half band $\{0, \dots, N-1\}$; its quarters; its middle half	Set. 6.1
∂_-, ∂_+	the cuts $\mathrm{Col}(0)$, $\mathrm{Col}(N)$	Set. 6.1
N_*	$\lceil (96\delta_0/c_*)^2 \rceil$; separations $\ge c_* \sqrt{N}/16$ for $N \ge N_*$	Lem. 6.2
$\mathbb{T}_N^\pm$	half-tori centred on the lines of the cuts	Set. 6.1
Q, $Q_\pm$, Q_m	Q_Γ, $Q_{\Gamma_\pm}$, Q_{Γ_m}	Set. 6.1
$\mathbf{n}_N$, p, Δ, γ, P	sectors; rank, spread and gap of the patch; its projection	Ass. 6.3
U, $T_\pm$, m_T	transport unitary; transported charges; their norm exponent	Ass. 6.4
$J_\pm$, K, $K_\pm$, $\overline{Q}$	current halves; generator; its halves; dressed charge	Def. 6.5
$K_\pm^U$, $\overline{Q}_\pm$, $\overline{Q}_\pm^U$, $\overline{Q}^U$	objects of [BBDF21, Section 4.2]	Def. 6.11
φ, $\mathcal{M}(\varphi)$, $\mathcal{M}_\pm(\varphi)$, $\widetilde{T}_-$	flux angle; flux insertions; dressed transport	Def. 6.11
$\mathcal{Z}_-$, $\mathcal{G}$, $\mathcal{Y}$, $\mathcal{E}$, $\det_P$	compressed $\mathcal{M}_-$; generator, solution and error of the comparison equation; determinant on $\operatorname{ran} P$	Lem. 6.16
$\langle A \rangle_P$	$p^{-1} \operatorname{Tr}(PA)$	Thm. 6.18

symbol	meaning	where
$\mathbf{n}_N^{\pm}$	$((N \pm 1)/2, (N \mp 1)/2)$ at odd N	Cor. 7.4
$\mathcal{R}$	rotation exchanging the sides of the torus	Lem. 8.3
$\mathcal{T}_N$	rectangular tori with $L_x L_y = 4\pi N$, $\min\{L_x, L_y\} \geq L_0$	Cor. 8.4
The variational bound (Appendix D)		
$\mathsf{G}_k(\kappa)$	Gaussian sums $\sum_{d \geq 1} 2d^k e^{-\kappa^2 d^2/4}$	(D.1)
$\mathcal{W}_\sigma$	twist of the electrons of spin σ: $c_{j\sigma} \mapsto e^{2\pi \mathrm{i} j/N_\phi} c_{j\sigma}$	Lem. D.3
χ^{sing}, $c^{\mathrm{sing}}(f, g)$	singlet spinor; singlet pair annihilator	App. D
$c^{\nu}_{n,d}$, $c^{\mathrm{sing}}_{n,d}$	pair annihilators of the orbitals $[(n \mp d)/2]$	App. D
$\mathcal{H}^{\mathbf{n}}_N(\mu)$	eigenspace of Θ in $\mathcal{H}^{\mathbf{n}}_N$ for the eigenvalue μ	App. D

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